

Syllabus for Courses of Monsoon 2026

Ver.3: 03-08-2026

S.No.	Course No	Course Name	Credits (L-T-P-C)	Faculty Name(s)
1	CS2.501	Advanced Computer Architecture	3-1-0-4	Suresh Purini + Priyesh Shukla
2	CS3.402	Advanced Computer Networks	3-1-0-4	Ashok Kumar Das
3	CE1.610	Advanced Design of Steel Structures	3-1-0-4	Sunitha Palissery
4	EC2.414	Advanced Memory Circuits and Systems	3-1-0-4	Priyesh Shukla
5	CS7.501	Advanced NLP	3-1-0-4	Manish Shrivastava+ Aparajitha Allamraju
6	CS3.304	Advanced Operating Systems	3-1-0-4	Krishna Reddy P
7	MA6.501	Advanced Probability Theory	3-1-0-4	Khushboo Agarwal
8	GS1.501	Advanced Remote Sensing	3-1-0-4	RC Prasad + Kiran Chand T
9	PD3.401	AI for Product Managers (H2)	3-1-0-2	Aditya T. (Guest Faculty) - Microsoft
10	CS1.301	Algorithm Analysis and Design	3-1-0-4	Girish Varma
11	CS3.306	Algorithms and Operating Systems	3-1-0-4	Vikram Pudi
12	OC2.101	Arts-1 (H1)	2-0-0-2	Saroja T K (Coordinator)
13	CS1.302	Automata Theory (H1)	3-1-0-2	Shantanav Chakraborty
14	CS6.403	Autonomous Software Engineering(H2)	3-1-0-2	Karthik Vaidhyanathan
15	HS0.203a & HS0.203b	Basics of Ethics (H1&H2)	3-1-0-2	Ashwin Jayanti + Saurabh Todariya
16	CG3.402	Behavioral Research & Experimental Design	3-1-0-4	Vinoo Alluri
17	SC3.202	Bioinformatics (H1)	3-1-0-2	Nita Parekh
18	SC3.302	Bioinformatics-II (H2)	3-1-0-2	Nita Parekh
19	SC3.321	Biomolecular Structure Interaction & Dynamics	3-1-0-4	B.Gopalakrishnan
20	PD2.421	Business Fundamentals	3-1-0-4	Himanshu Warudkar

21	SC2.305	Chemical Kinetics and Reaction Dynamics (H2)	3-1-0-2	Tapan Kumar Sau
22	SC2.309	Chemistry Topics for Engineers	3-1-0-4	Prabhakar B + Tapan Kumar Sau
23	HS0.301	Classical Text Readings	3-1-0-4	Ashwin Jayanti
24	EC2.410	CMOS References and Regulators	3-1-0-4	Zia Abbas + Abhishek P
25	MA4.406	Complex Analysis	3-1-0-4	Lakshmi Burra
26	CL3.202	Computational Linguistics II: Comp Semantics and Discourse parsing	3-1-0-4	Rajakrishnan P Rajkumar
27	SC4.415	Computer Aided Drug Design	3-1-0-4	Deva Priyakumar
28	CS0.301	Computer Problem Solving	3-1-3-4	Shatrunjay Rawat
29	CS0.101	Computer Programming	3-1-3-5	Abhishek Deshpande + Vineet Gandhi+ Charu Sharma
30	SC4.101	Computing in Sciences-1(H2)	2-0-3-2	Marimuthu Krishnan
31	GS2.402	Cyber GIS: Geospatial Web and GeoBI	3-1-0-4	Kuldeep Kurte + K S Rajan
32	CS4.405	Data Analytics I	3-1-0-4	Krishna Reddy Polepalli
33	CS4.301	Data and Applications (H2)	3-1-0-2	Kamal Karlapalam
34	CS1.304	Data Structures & Algorithms for Problem Solving	3-0-2-6	Lini Thomas + Kshitij Gajjar
35	CS9.429	Design for Social Innovation	3-1-0-4	Ramesh Loganathan + Arjun Rajashekar
36	PD1.301	Design Thinking	3-1-0-4	Raman Saxena
37	CS7.510	Diffusion Models for Generative AI	3-1-0-4	Naresh Manwani + Saikiran Bulusu
38	CS7.404	Digital Image Processing	3-1-0-4	Raghavendra GS
39	EC2.101	Digital Systems and Microcontrollers	3-1-3-5	Anshu Sarje + Priyesh Shukla + Anil Kumar V + Madhava Krishna + Sourav Garg
40	EC2.408	Digital VLSI Design	3-1-0-4	Zia Abbas
41	MA5.101	Discrete Structures	3-1-0-4	Praveen P + Suryajith Ch
42	CS3.401	Distributed Systems	3-1-0-4	Kishore Kothapalli
43	SC1.440	Dynamical Processes in Complex Networks	3-1-0-4	Chittaranjan Hens
44	PD2.402	Early Stage Funding for a Startup(H2)	3-1-0-2	Sridhar Kalyanasundaram
45	GS3.603	Earth System Science & Modelling	3-1-0-4	Kiran Chand T

46	CE1.607	Earthquake Resistant Design of Masonry Structures	3-1-0-4	P Pravin Kumar Venkat Rao
47	EC2.102	Electronic Workshop-1 (H2)	2-0-3-2	Praful Mankar + Saikiran B
48	EC3.202	Embedded Systems Workshop (H)	1-0-3-3	Deepak Gangadharan + Sachin Chaudhari + Aftab Hussain
49	HS1.304	Environmental Humanities	3-1-0-4	Subha Chakraborty
50	CS9.428	Environmental Science & Technology	3-1-0-4	R C Prasad
51	CE4.501	Finite Element Method	3-1-0-4	Jofin George
52	EC5.412	Foundations for Signal Processing and Communication	3-1-0-4	Santosh Nannuru + Gowtham Kurri
53	CS2.401	FPGA based Accelerator Design	3-1-0-4	Suresh Purini
54	HS8.306	Gender and Society	3-1-0-4	Sushmita Banerji
55	HS7.101	Human Sciences Lab-1 (H2)	3-1-0-2	Radhika Krishnan
56	HS5.402	Identity and Economic Development	3-1-0-4	Angarika Rakshit
57	CS4.406	Information Retrieval & Extraction	3-1-0-4	Anil Nelakanti
58	EC5.415	Information, Computation and Learning	3-1-0-4	Prasad Krishnan
59	CG1.402	Intro to Cognitive Science	3-1-0-4	Bhaktee Dongaonkar + Bapi Raju S
60	SC3.101	Introduction to Biology	3-1-0-4	Vinod PK
61	HS5.202	Introduction to Economics	3-1-0-4	Anirban Dasgupta
62	CS1.408	Introduction to Game Theory	3-1-0-4	Sujit Gujar
63	HS3.201	Introduction to History	3-1-0-4	Aniket Alam
64	CL1.101	Introduction to Linguistics-1	3-1-0-4	Chiranjeevi Yarra + Radhika Mamidi
65	CG3.401	Introduction to Neural and Cognitive Modeling	3-1-0-4	Bapiraju Surampudi
66	CG4.402	Introduction to Neuroeconomics	3-1-0-4	Kavita Vemuri
67	CG1.401	Introduction to Psychology	3-1-0-4	Priyanka Srivastava
68	SC1.421	Introduction to Quantum Field Theory	3-1-0-4	Subhadip Mitra + Monalisa Patra
69	HS2.201	Introduction to Sociology	3-1-0-4	Rajorshi Ray
70	EC4.402	Introduction to UAV Design	3-1-0-4	Harikumar K
71	CE9.609	IoT Workshop	3-1-0-4	Shruthi K + Sachin Chaudhari
72	CL2.203	Language and Society	3-1-0-4	Aditi Mukherjee

73	CL3.410	Language Models and Agents	3-1-0-4	Vasudeva Varma
74	CG1.403	Learning and Memory	3-1-0-4	Bhaktee Dongaonkar
75	HS4.102	Making of the Contemporary India	3-1-0-4	Aniket Alam + Aakansha Natani
76	PD2.404	Market Research and Validation (H1)	3-1-0-2	Ratan K Putla (Guest Faculty)
77	MA6.301	Maths for Computer Science 1- Probability and Statistics (H1)	3-1-0-2	Naresh Manvani
78	MA6.302	Maths for Computer Science 2 - Linear Algebra (H2)	3-1-0-2	Pawan Kumar
79	CL3.412	Mechanistic Interpretability of Language Models	3-1-0-4	Hrishikesh Terdalkar
80	EC2.416	Micro & Nano Fabrication Technology for ICs	3-1-0-4	G. Rajaram (Emeritus Professor) and L. D. Varma Sangani
81	CS7.503	Mobile Robotics	3-1-0-4	K Madhava Krishna
82	CS1.405	Modern Complexity Theory	3-1-0-4	Suryajith Ch
83	HS1.210	Music Workshop	3-1-0-4	Saroja TK
84	EC5.101	Networks, Signals and Systems	3-1-0-4	Prasad Krishnan + Aftab Hussain
85	CL3.411	NLP for Healthcare	3-1-0-4	Parameshwari Krishnamurthy + Dipti M Sharma
86	SC1.410	Open Quantum Systems and Quantum Thermodynamics	3-1-0-4	Samyadeb Bhattacharya
87	CS3.301	Operating Systems and Networks	3-1-0-4	Deepak Gangadharan
88	EC2.415	Optoelectronics & Integrated Photonics	3-1-0-4	Tanmay Bhowmik
89	SC1.415	Physics of Early Universe	3-1-0-4	Diganta Das
90	CS1.402	Principles of Programming Languages	3-1-0-4	Venkatesh Choppella
91	EC2.409	Principles of Semiconductor Devices	3-1-0-4	Aftab Hussain
92	MA6.102	Probability and Random Processes	3-1-0-4	Praful Mankar
93	MA6.101	Probability and Statistics	3-1-0-4	Tejas Bodas
94	PD2.401	Product Management	3-1-0-4	Ramesh Swaminathan+Guest Faculty
95	PD2.501	Product Marketing	3-1-0-4	Ravi Warriar
96	CS3.406	Programming AI Accelerators	3-1-0-4	Girish Varma + P J Narayanan
97	CS1.505	Quantum aspects of cryptography	3-1-0-4	Atul Singh Arora
98	EC5.413	Quantum Error Correcting Codes	3-1-0-4	Lalitha V

99	SC4.430	Quantum Information Science	3-1-0-4	Siddhartha Das
100	SC1.203	Quantum Mechanics	3-1-0-4	Harjinder Singh
101	HS1.401	Readings from Hindi Literature	3-1-0-4	Harjinder Singh
102	MA4.101	Real Analysis	3-1-0-4	Indranil Chakrabarthy + Uttam Singh + Lalitha V
103	CS8.501	Research in Information Security	3-1-0-4	Srinathan Kannan + Ashok Kumar Das
104	CE1.621	Retrofitting of Existing Structures	3-1-0-4	Shubham Singhal
105	EC4.401	Robotics: Dynamics and Control	3-1-0-4	Antony Thomas
106	SC1.110	Science 1	3-1-0-4	Prabhakar Bhimalapuram + Samyadeb Bhattacharya
107	SC4.110	Science Lab I (H1)	0-0-3-2	Chittaranjan Hens + Tapan Kumar Sau
108	CS5.501	Scientific Visualisation	3-1-0-4	Raghavendra GS
109	PD1.402	Service Design (H2)	3-1-0-2	Raman Saxena
110	EC5.406	Signal Detection and Estimation Theory	3-1-0-4	Praful Mankar + Arti Yardi
111	EC5.201	Signal Processing	3-1-3-5	Santosh Nannuru
112	CS9.425	Social Science Perspective on HCI	3-1-0-4	Nimmi Rangaswamy
113	CS6.302	Software Systems Development	3-0-2-4	Abhishek Kr Singh
114	GS2.401	Spatial Informatics	3-1-0-4	RC Prasad + Kuldeep Kurte
115	GS0.101	Spatial Thinking and Practice	3-1-0-4	Shaik Rehana + K S Rajan
116	SC2.304	Spectroscopy(H1)	3-1-0-2	Harjinder Singh
117	CL2.405	Speech Analysis and Linguistics	3-1-0-4	Chiranjeevi Yarra
118	CS7.403	Statistical Methods in AI	3-1-0-4	Sai Kiran B + Anoop M Namboodiri
119	CE1.501	Structural Dynamics	3-1-0-4	Sunitha Palissery
120	CE1.502	Structural Engineering Design Studio	3-1-0-4	Shubham Singhal
121	CE1.625	Structural Safety of Built Infrastructure	3-1-0-4	Jofin George
122	PD2.405	System Thinking and Design (H1)	3-1-0-2	Raman Saxena
123	EC5.202	Systems Thinking	3-1-0-4	Spandan Roy
124	CE0.501	Theory of Elasticity	3-1-0-4	Pravin Kumar Venkat Rao
125	HS0.202	Thinking and Knowing in the Human Sciences - II	3-1-0-4	Rajorshi Ray + Isha Dubey

126	HS0.216	Thinking and Knowing in the Human Sciences - III	3-1-0-4	Anirban Dasgupta + Khaliq Parkar
127	CS9.442	Topics in AI for Healthcare	3-1-0-4	Vinod PK + Bapi Raju S + Raja Poladi
128	SC2.401	Topics in Nanosciences	3-1-0-4	Tapan Kumar Sau
129	CS6.501	Topics in Software Engineering	3-1-0-4	Raghu Reddy Y
130	HS3.402	Understanding Public and Digital History	3-1-0-4	Isha Dubey
131	OC3.101	Value Education-1 (H)	0-2-0-2	Indranil Chakrabarty + Saurabh Todariya (Coordinators)
132	EC2.201	VLSI Design	3-1-0-4	Abhishek Srivastava
133	EC5.407	Wireless Communications	3-1-0-4	Rajeev Gangula

Title of the Course : **Advanced Computer Architecture**
 Name of the Faculty : Suresh Purini + Priyesh Shukla
 Course Code : CS2.501
 L-T-P : 3-1-0.
 Credits : 4
 (L = Lecture Hours, T = Tutorial Hours, P = Practical Hours)
 Name of the Academic Program: B.Tech in Computer Science and Engineering

Prerequisite Course / Knowledge:

Computer Programming. Computer Systems Organization or Introduction to Processor Architecture.

2. Course Outcomes (COs)

After completion of this course successfully, the students will be able to:

CO-1: Explain the principles and practices underlying the design of modern processors (Cognitive Level: **Understand**)

CO-2: Explain the principles and practices underlying the design of memory hierarchies (Cognitive Level: **Understand**)

CO-3: Be able to analyze the different aspects of a processor/memory architecture qualitatively and quantitatively, and further develop strategies to mitigate any observed bottlenecks. (Cognitive Level: **Apply, Analyze, Evaluate and Create**)

CO-4: Explain and identify security vulnerabilities in modern computing systems. Be able to propose techniques to circumvent the identified threats. (Cognitive Levels: **Understand**)

3. Mapping of Course Outcomes (COs) with Program Outcomes (POs) and Program Specific Outcomes (PSOs) – Course Articulation Matrix

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3	PSO4
CO1	3	3	3	3	3	1	2	1	2	2	2	2	3	3	3	3
CO2	3	3	3	3	3	1	2	1	2	2	2	2	3	3	3	3
CO3	3	3	3	3	3	1	2	1	2	2	2	2	3	3	3	3
CO4	3	3	3	3	3	1	-	1	2	2	2	2	3	3	3	3

4.Detailed Syllabus

- **Unit 1: Instruction Level Parallelism and Superscalar Processors**
 - What is Instruction Level Parallelism?
 - Microarchitecture design and analysis of Out-of-Order processors.
 - Compiler and Architecture Interactions
- **Unit 2: Data Level Parallelism and SIMD/GPU Architectures**
 - Vector Architectures
 - GPUs
 - Compiler and Architecture Interactions
- **Unit 3: Memory Hierarchy Design and Emerging Memory Technologies**
 - Design of Cache Memory
 - Distributed Shared Memory
 - Memory Consistency Models
 - Non-volatile memories
- **Unit 4: Emerging and Domain Specific Accelerators**
 - Hardware accelerators for deep neural networks and Genomics
 - AMD AI/ML Engine
 - TPUs
 - Near
- **Unit 5: AI/ML Techniques in Architecture**
 - Reinforcement Learning Techniques in Cache Optimization, Branch Prediction etc.
 - Power Management
- **Unit 6: Security**
 - Side Channel Attacks such as Spectre and Meltdown

Reference Books:

1. Computer Architecture: A Quantitative Approach, Sixth Edition, John L. Hennessy and David A. Patterson.

5.Teaching-Learning Strategies in brief

This is an advanced course which prepares students to read state-of-the-art research papers in computer architecture from conferences such as ISCA, MICRO, HPCA, etc. Tools such as PIN, and other memory hierarchy simulators will be used to facilitate deeper understanding of the concepts.

6.Assessment methods and weightages in brief

1. HomeWorks: 20 percent
2. Quiz 1: 10 percent

3. Quiz 2: 10 percent
4. Midterm: 20 percent
5. Final Exam: 25 percent
6. Course Project: 15 percent

Title of the Course : **Advanced Computer Networks**
Name of the Faculty : Ashok Kumar Das
Course Code : CS3.402
Credits : 4
L-T-P : 3-1-0
 (L= Lecture hours, T=Tutorial hours, P=Practical hours)
Name of the Academic Program : B.Tech. in Computer Science and Engineering

1.Prerequisite Course / Knowledge:

Basic principles of computer networks and algorithms.

2.Course Outcomes (COs) (5 to 8 for a 3 or 4 credit course):

After completion of this course successfully, the students will be able to..

- CO-1 Demonstrate a familiarity with concepts of network management, standards, and protocols
- CO-2 Discuss various privacy-enhancing techniques used in modern computer networks
- CO-3 Apply the knowledge of distance-vector (RIP and IGRP) and link-state (OSPF and IS-IS) routing protocols to find routing paths for a variety of networks
- CO-4 Analyze wireless LAN technologies including IEEE 802.11
- CO-5 Design efficient routing protocols for advanced computer networks (e.g., SDN and ICN)
- CO-6 Develop a framework for building a large-scale enterprise network

3.Mapping of Course Outcomes (COs) with Program Outcomes (POs) and Program Specific Outcomes (PSOs) – Course Articulation Matrix

	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO10	PO11	PO12	PSO 1	PSO 2	PSO 3	PSO 4
CO1	3	1	2	3	2	1	1	1	1	2	3	3	2	3	1	2
CO2	3	2	3	3	3	2	1	2	1	3	2	3	2	2	2	1
CO3	2	2	3	3	1	1	2	1	1	1	2	2	1	2	1	1
CO 4	2	2	3	3	2	2	3	1	2	1	2	2	2	2	3	2
CO5	2	3	2	2	2	1	2	2	1	3	3	2	3	2	2	1
CO 6	3	3	3	3	2	2	2	3	2	2	2	2	2	2	2	1

Note: Each Course Outcome (CO) may be mapped with one or more Program Outcomes (POs) and PSOs. Write '3' in the box for 'High-level' mapping, 2 for 'Medium-level' mapping, 1 for 'Low'-level' mapping

4.Detailed Syllabus:

- Unit 1: Modeling and measurement: Network traffic modeling, network measurement, simulation issues, network coding techniques
- Unit 2: Flow and congestion control, TCP variants, TCP modeling, active queue management
- Unit 3: Routing: Router design, scheduling, QoS, integrated and differentiated services
- Unit 4: Wireless networks: Mobility supports, MAC, multicast

Unit 5: Overlay networks and Emerging applications: SDN, ICN, P2P, CDN, Web caching, cross-layer optimizations, VoIP, SIP, video over P2P

Reference Books:

1. Larry L. Peterson and Bruce S. Davie, Computer Networks: A Systems Approach, 5th edition, Elsevier, 2012
2. James F. Kurose and Keith W. Ross, Computer Networking: A Top-Down Approach, 6th edition, Pearson Education, 2013
3. Jean Walrand and Pravin Varaiya, High-Performance Communication Networks, 2nd edition, Elsevier, 2000
4. Research papers

5.Teaching-Learning Strategies in brief (4 to 5 sentences):

Lectures by integrating ICT into classroom teaching; tutorials involving problem solving; being a systems course, it requires hands-on working as well as critical thinking and active learning by the students to solve practical problems; and finally, project-based learning by implementing semester-long project(s) to solve real-world issues.

6.Assessment methods and weightages in brief (4 to 5 sentences):

Assignments	20 marks
One at home project	30 marks
Mid Semester Examination	20 marks
End Semester Examination	30 marks

Title of the Course : Advanced Design of Steel Structures

Name of the Faculty : Sunitha Palissery

Course Code : CE1.610

Name of the Program: M.Tech CASE

Credits : 4

L - T - P : 3-1-0

Semester, Year : Monsoon 2026

Pre-Requisites : Design of Steel Structures (Undergraduate course content)

Course Outcomes

After completion of this course successfully, the students will be able to:

- CO-1. Develop knowledge and skills to numerically model, analyze and design steel members and low-rise steel frame buildings
- CO-2. Employ computer application skills in developing structural behavior intuition and predict structural response to dynamic loading and buckling
- CO-3. Demonstrate problem solving skills for various scenarios of structural design and work towards a research-based approach to stability design and seismic design of steel structures
- CO-4: Develop critical thinking to help improve and control structural behavior, with focus on seismic loading effects on steel buildings
- CO-5. Analyze ethical and effective structural design practices to preclude instability failure and earthquake loading failure of steel structures
- CO-6. Reorganize inter-personal skills required to manage possible negotiations with structural engineering design practitioners and promote a safe built environment

Course Topics

Unit 1: Design for Stability- Fundamentals

Compression Members, Flexural Members, Systems/Beam-Columns-Behaviour and Design-Key Definitions-Stability, Bifurcation, Instability; Limit States-Yielding, Buckling, flexural buckling, bending; Column Curves; Euler Buckling, Inelastic Buckling, Imperfections, Amplification of loads; Residual Stresses- Stub-columns, Column Curves for Design

Unit 2: Design for Stability- Design & Behaviour

Behaviour of Flexural Members- Laterally Restrained and unrestrained beams, Effective Length of Columns- AISC Alignment Charts; stability index; Design of Beam Columns-Interaction equations, Design for combined axial and bending effects Introduction to Lateral Torsional Buckling, Governing Differential Equations for Plate Buckling, Plates Subjected Loading Actions, Post-buckling Behaviour of Plates

Unit 3: Earthquake Resistant Design- Codes of Practice

Code recommendations in IS1893:2016, IS800:2007, IS18168:2023; ASCE07-10, AISC documents.

Unit 4: Earthquake Resistant Design- Design & Behaviour

Design of steel members, connections-Joint panel zones, prequalified connections; Design of Column Bases; Nonlinear Static Analysis and Behaviour: Lateral Stiffness, Lateral Strength, Ductility Capacity, Collapse Mechanism and Energy Dissipation Capacity of Designed Steel Building

Preferred Textbooks

1. Alexander, C., *Principles of Structural Stability Theory*, Prentice-Hall Inc, New Jersey
2. Chen, W.F., and Lui, E.M., (1987), *Structural Stability: Theory and Implementation*, Elsevier Science Publishing Co., New York
3. Elnashai, A.S., and Di Sarno, L., (2008), *Fundamentals of Earthquake Engineering*, John Wiley and Sons, UK
4. Gioncu, V., and Mazzolani, M., (2002), *Ductility of Seismic Resistant Steel Structures*, SPON Press, Taylor and Francis, New York
5. Park, R., and Paulay, T., (1975b), *Reinforced Concrete Structures*, John Wiley & Sons, UK
6. Paulay, T., and Priestley, M.J.N., (1992), *Seismic Design of Reinforced Concrete and Masonry Buildings*, John Wiley and Sons, New York
7. Salmon, C.G., and Johnson, J.E., (1996), *Steel Structures Design and Behaviour*, Prentice Hall, NJ

Reference Books

1. American Institute of Steel Construction (AISC), (2016), *Seismic Provisions for Structural Steel Buildings*, (ANSI/AISC341-16), Chicago, Illinois, USA
2. American Institute of Steel Construction (AISC), (2016), *Specifications for Structural Steel Buildings*, (ANSI/AISC360-16), Chicago, Illinois, USA
3. American Society of Civil Engineers (ASCE), (2022), *Minimum Design Loads for Buildings and Other Structures* (ASCE 7-10), USA
4. American Society of Civil Engineers (ASCE), (2013), *Seismic Rehabilitation of Existing Buildings*, (ASCE/SEI 41-13), Virginia, USA
5. Bureau of Indian Standards (BIS), (2007), *Indian Standard Code of Practice for General Construction in Steel*, IS800:2007, New Delhi, India
6. Bureau of Indian Standards (BIS), (2016), *Indian Standard Criteria for Earthquake Resistant Design of Structures*, IS:1893:2016-Part 1, New Delhi, India
7. Bureau of Indian Standards (BIS), (2023), *Earthquake Resistant Design and Detailing of Steel Buildings — Code of Practice*, IS:18168, New Delhi, India

8. Bažant, Z.P., and Cedolin, L., (2010), *Stability of Structures- Elastic, Inelastic, Fracture and Damage Theories*, World Scientific Publishing Co. Pvt. Ltd., Singapore
9. Booth, E., (2014), *Earthquake Design Practice for Buildings*, ICE Publishing, London
10. Chen, W.F., and Lui, E.M., (2005), *Earthquake Engineering for Structural Design*, CRC Press, Taylor and Francis, Boca Raton
11. Galambos, T.V., and Surovek, A.E., (2008), *Structural Stability of Steel: Concepts and Applications for Structural Engineers*, John Wiley & Sons, New Jersey
12. Gambhir, M.L., (2004), *Stability Analysis and Design of Structures*, Springer, New York
13. Guide to Stability Design Criteria for Metal Structures, Edited by Ziemian, R.D., (2010)
14. Murty, C.V.R., Goswami, R., Vijayanarayana, A.R., and Mehta, V.V., (2012), *Earthquake Behaviour of Buildings*, GSDMA, Gujrat
15. Timoshenko, S.P., and Gere, J.M., (1985), *Theory of Elastic Stability*, McGraw Hill International Book Company
16. Varghese, P.C., (2010), *Design of Reinforced Concrete Foundation*, PHI Learning Pvt Ltd, New Delhi

E-book/software Links

<https://www.mastan2.com/>

Grading Plan

Type of Evaluation	Weightage (in %)
Quiz-1	5
Mid Sem Exam	20
Quiz-2	5
End Sem Exam	30
Assignments	20
Project	20

Mapping of Course Outcomes to Program Objectives

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PSO1	PSO2	PSO3	PSO4
CO1	3	3	3	2	3	2	2	2	2	2	3	3	3	3
CO2	3	3	3	2	2	1	3	2	2	2	3	3	3	3
CO3	3	3	3	2	3	1	3	2	2	3	3	3	2	2
CO4	3	3	3	2	3	2	2	2	2	3	3	3	2	3
CO5	3	3	3	2	3	2	2	2	2	3	2	2	1	2
CO6	2	2	2	1	3	1	2	2	3	2	2	2	2	2

Teaching-Learning Strategies in brief (4-5 sentences)

1. Lectures by integrating ICT into classroom teaching
2. Tutorials involving numerical modelling of and steel members and moment frame buildings to reinforce linear and nonlinear structural analysis concepts and design methods commonly used in design practice
3. Assignments involving analysing structural data to understand linear and nonlinear responses

4. Critical and active learning through projects, and project-based learning by doing term-projects which involves hands-on use of software tools to investigate and predict behaviour of steel members and buildings.
-

Title of the Course	: Advanced Memory Circuits and Systems
Faculty Name	: Priyesh Shukla
Name of the Program	: B.Tech in ECE, B.Tech in ECE and MS in ECE by Research, MS in ECE, M.Tech. in VLSI and Semiconductor Design, PhD in ECE
Course Code	: EC2.414
Credits	: 4
L - T - P	: 3-1-0
(L - Lecture hours, T-Tutorial hours, P - Practical hours)	
Semester, Year	: Monsoon, 2026
Pre-Requisites	: Basic Electronics or Analog/Digital VLSI Design, Computer Systems Organization or Introduction to Processor Architecture.

Course Outcomes :

After completion of the course, the students should be able to -

- CO1.** Quantify and trade off the core figures of merit of any memory — latency, bandwidth, density, energy per bit, retention, endurance, and error rate — and explain how the memory hierarchy responds to these conflicting constraints.
- CO2.** Analyze and size canonical bitcells (6T SRAM, 1T1C DRAM, floating-gate / charge-trap Flash) and reason about their stability, sensing, and scaling limits.
- CO3.** Explain the device physics and circuit / system trade-offs of emerging non-volatile memories (PCM, RRAM, MRAM, FeRAM / FeFET).
- CO4.** Design and evaluate memory-system components — caches, controllers, scheduling, and address mapping — and reason about disaggregated and tiered memory (CXL, persistent memory) and its system implications, using standard modeling tools.
- CO5.** Analyze why memory dominates AI / ML performance and energy, and evaluate near-memory and compute-in-memory architectures, including their accuracy / efficiency trade-offs.
- CO6.** Critically read primary research literature and propose, prototype, or evaluate a novel memory idea.

Course Topics :

Module I — Foundations: The memory landscape and the “memory wall”; figures of merit and the memory hierarchy; the MOSFET as switch and capacitor, charge storage, retention and leakage; array organization (wordlines, bitlines, decoders), precharge / equalization and array efficiency; sensing and peripheral circuits (voltage- and current-mode sense amplifiers, reference generation, write drivers, signal margin).

Module II — SRAM: The 6T cell — cross-coupled inverters, read / write / hold, static noise margin and transistor sizing; SRAM arrays and peripherals; stability under variability and scaling, half-select disturbs, read / write assist techniques; alternative bitcells (8T / 10T, dual-port), near- / sub-threshold SRAM, and the SRAM scaling crisis.

Module III — DRAM: The 1T1C cell, capacitor technologies, destructive read and charge sharing; DRAM array architecture, sense / restore, refresh, key timing parameters (tRCD, tRP, tRAS, tCAS) and the command protocol; standards and interfaces (DDR / LPDDR / GDDR) and the DRAM PHY; scaling and 3D, RowHammer and reliability, HBM, and near-memory processing.

Module IV — Non-Volatile Memory: Flash fundamentals (floating gate, tunneling, NOR vs. NAND, SLC / MLC / TLC / QLC); 3D NAND and SSD systems (FTL, wear leveling, garbage collection); resistive and magnetic memories (PCM, RRAM, MRAM); ferroelectric memories (FeRAM, FeFET), a comparative NVM analysis, selector devices, and the crossbar sneak-path problem.

Module V — Memory Systems: Caches and the hierarchy (locality, associativity, replacement, coherence, virtual memory / TLBs); memory controllers (request scheduling, address mapping, row-buffer management, fairness / QoS); signaling and persistent memory; disaggregated and composable memory (CXL, pooling and tiering, far memory).

Module VI — Memory for AI / ML & Compute-in-Memory: The memory bottleneck in AI (arithmetic intensity and the roofline model, weights / activations / optimizer state / KV cache, HBM, quantization and sparsity); near-memory and processing-in-memory; SRAM and digital compute-in-memory; analog compute-in-memory with emerging devices (crossbars, ADC / DAC cost, device non-idealities, hardware-aware training).

Module VII — Frontiers & Synthesis: Memory test and design-for-test (March algorithms, BIST), redundancy and repair; reliability (ECC families, chipkill); memory security (RowHammer, cold-boot, side channels, encryption); 3D / monolithic integration and advanced packaging (chiplets, hybrid bonding); research frontiers (photonic, DNA / molecular, neuromorphic, cryogenic memory) and student project presentations.

Preferred Textbooks :

1. J. M. Rabaey, A. Chandrakasan & B. Nikolić, *Digital Integrated Circuits: A Design Perspective* (memory and periphery chapters).
2. N. Weste & D. Harris, *CMOS VLSI Design: A Circuits and Systems Perspective* (memory, datapath, and sensing).
3. K. Itoh, *VLSI Memory Chip Design* (DRAM / SRAM circuit depth).
4. B. Jacob, S. Ng & D. Wang, *Memory Systems: Cache, DRAM, Disk* (systems view).
5. J. L. Hennessy & D. A. Patterson, *Computer Architecture: A Quantitative Approach* (memory hierarchy, roofline).

Reference Books :

1. J. Brewer & M. Gill (eds.), *Nonvolatile Memory Technologies with Emphasis on Flash*.
2. Survey papers on in-memory and near-memory computing and on resistive / analog crossbar accelerators.
3. Primary literature on RowHammer and DRAM reliability / security and its mitigations.

4. Processing-in-memory architecture papers from the DRAM / HBM-PIM line of work.
5. SRAM and digital compute-in-memory macro papers from recent solid-state-circuits venues.
6. CXL and disaggregated / tiered-memory system papers.
7. Vendor data sheets and JEDEC standards excerpts for DDR / LPDDR / HBM (read as engineering documents).

Conferences / journals to track: ISSCC, VLSI Symposia, IEDM (devices / circuits); ISCA, MICRO, HPCA, ASPLOS (systems); DAC / ICCAD (design); and Nature Electronics / Nature Nanotechnology for emerging-device milestones.

E-book Links : E-editions of the listed textbooks are accessible through the IIIT-H digital library and the respective publisher platforms; survey and primary papers, conference proceedings, and standards excerpts are available via IEEE Xplore and the ACM Digital Library. (Specific institute-licensed access links to be added by the instructor.)

Grading Plan :

Type of Evaluation	Weightage (in %)
Quiz-1	10
Mid Sem Exam	20
Quiz-2	10
End Sem Exam	20
Assignments	10
Project+Term Paper	30

Mapping of Course Outcomes to Program Objectives:

(1 – Lowest, 2—Medium, 3 – Highest, or a ‘-’ dash mark if not at all relevant). Program outcomes are posted at

<https://intranet.iiit.ac.in/offices/static/files/PEOs%2CPOs%26PSOs-ofAllProgrammes-Jan2023.pdf>

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3	PSO4
CO1	3	2	2	2	1	2	2	3	3	3	1	3	3	1	2	3
CO2	3	3	3	3	3	3	3	3	2	3	2	3	3	3	2	3
CO3	3	3	3	3	2	3	3	3	3	2	-	2	3	3	3	3
CO4	3	3	2	3	2	3	3	3	3	3	2	3	3	3	3	3
CO5	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3
CO6	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3

Teaching-Learning Strategies in brief (4-5 sentences):

The course is delivered through lectures and tutorials that build understanding bottom-

up — from device and bitcell foundations to full memory subsystems and the compute-in-memory frontier. Each circuit-oriented topic is paired with a hands-on simulation lab (SPICE for bitcell, SNM, and sense-amp analysis; CACTI, NVSim, and DRAM / full-system simulators for array and system studies) so that abstractions stay grounded in quantitative results. A single running thread revisits the same figures of merit — latency, bandwidth, density, energy, and endurance — for every memory studied, helping students build an internal comparison map rather than isolated facts. Primary research literature is read and critiqued throughout, with graduate students leading paper discussions, and a level-differentiated term project lets each student design, model, or prototype a memory idea at an appropriate depth. The difficulty and emphasis of modules are scaled to the audience mix, expanding the foundations or compressing them in favour of seminar-style discussion of the advanced modules as the cohort requires.

Title of the Course	: Advanced Natural Language Processing (NLP)
Name of the Faculty	: Manish Shrivastava and Aparajitha Allamraju
Name of the Program	: CLD/CSE/PG CSE
Course Code	: CS7.501
Credits	: 4
L - T - P	: 3-1-0
(L - Lecture hours, T-Tutorial hours, P - Practical hours)	
Semester, Year	: Monsoon, 2026
(Ex: Spring, 2022)	
Pre-Requisites	: None, though INLP or background in NLP preferable

Course Outcomes :

After completion of this course successfully, the students will be able to –

- CO-1. Demonstrate the knowledge of Advanced building blocks of NLP
- CO-2. Apply NLP machine learning algorithms
- CO-3. Demonstrate the knowledge of Dense and contextual representation for NLP
- CO-4. Explain the concepts behind current Deep Learning models
- CO-5. Discuss the approaches to global and contextual semantic representation
- CO-6. Apply the above concepts for fundamental NLP tasks.

Course Topics:

Unit 1. Transformers

- Current advances in transformer architecture including optimizations
 - Tokenization – WordPiece, SentencePiece, BPE, Byte Latent, No Tokenization
 - Attention – MHA, GQA, MHSA, Flash, Ring, Sliding Window, Sparse, Hybrid
 - Normalization, Activation, Optimizers
 - FFNNs, MoE Layers, Mamba Layers
- Scaling Laws

Unit 2. Decoding

- Standard decoding
- Speculative decoding
- Multi Token Decoding and Parallel Decoding

Unit 3. Optimizing LLMs

- Quantization
- Inference Optimization
- Distributed Training – Data, Model, & Tensor Parallel, ZeRO

Unit 4. Alternate Architectures

- State Space Models
- Current advances in RNN
- Diffusion Language Models

Unit 5. Reasoning

- Reinforcement learning for NLP
- DPO, PPO, GRPO
- Reasoning in LLMs

Preferred Text Books :

None. Recent research papers.

Reference Books :

1. **Statistical Machine Translation** by Philip Koehn
2. **Deep Learning** by Ian Goodfellow
3. Raschka, S. (2024). *Build a large language model (from scratch)*. Simon and Schuster.
4. Zhang, A., Lipton, Z. C., Li, M., & Smola, A. J. (2023). *Dive into deep learning*. Cambridge University Press.

E-book Links :

1. <https://www.deeplearningbook.org/>
2. <http://www.cs.cmu.edu/~tom/mlbook.html>

Grading Plan :

(The table is only indicative)

Type of Evaluation	Weightage (in %)
Mid Sem Exam	10
End Sem Exam	10
Assignments	30
Project	50
Optional Seminar	10

Mapping of Course Outcomes to Program Objectives:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3	PSO4
CO1	1	3	3	2	2	2	1	-	1	1	1	3	3	3	3	2
CO2	3	3	3	3	3	1	1	1	3	3	3	3	3	3	2	3
CO3	3	3	3	3	3	-	-	-	2	2	2	3	3	3	2	3
CO4	3	3	3	3	3	-	-	-	2	2	2	3	3	3	2	3
CO5	3	3	3	3	3	-	-	-	2	2	2	3	3	3	2	3
CO6	3	3	3	3	3	1	1	1	3	3	3	3	3	3	3	3

Teaching-Learning Strategies in brief (4-5 sentences) :

Lectures by integrating ICT into classroom teaching, weekly tutorials involving problem solving and active learning by students and Project-based Learning by doing four assignments and a project. Evaluation based on personal viva to judge deeper understanding.

Title of the Course : Advanced Operating Systems

Name of the Faculty : Krishna Reddy Polepalli

Course Code : CS3.304

Credits : 4

L - T - P : 3-1-1

(L - Lecture hours, T-Tutorial hours, P - Practical hours)

Name of the Program : MTech in CSE

Semester, Year : Monsoon, 2026

1.Prerequisite Course / Knowledge

Digital Logic Design, Computer Organization, Knowledge of a programming language, Introduction to operating systems

2. Course Outcomes (COs)

Objective: The operating system abstracts the machine part of the given computer system in terms of simple services (system calls) by hiding the details of the machine (hardware). Traditionally, operating system concepts were developed for single processor (machine). Currently, shared memory, shared disk and shared-nothing architectures (computer network) are being employed for building cloud computing systems by exploiting multiple processor cores and commodity machines to manage diverse computing loads, such as search engine, weather forecasting and mobile-based services. The objectives of this course to study the key concepts, which were evolved for building modern operating systems.

The course Outcomes (COs) are as follows.

After completion of this course successfully, the students will be able to,

CO-1. Explain the concepts of several modern computer operating systems (SOLARIS, LINUX, WINDOWS, MAC, Adroid,...)

CO-2: Implement the task on the top of given operating system, in an efficient manner, based on process and thread framework.

CO-3. Prescribe the appropriate scheduling/synchronization/memory management/virtual memory/protection module for a given application.

CO-4. Architect the new application by selecting the appropriate system calls of the given operating system services.

CO-5. Develop a distributed application on multi-processor machine or multiple commodity processors connected through a network.

3.Mapping of Course Outcomes (COs) with Program Outcomes (POs) and Program Specific Outcomes (PSOs) – Course Articulation Matrix

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3	PSO4
CO1	3	2	1	1	2	-	-	-	3	2	2	3	3	2	1	2
CO2	3	3	3	2	2	-	1	-	2	2	2	3	2	1	2	2
CO3	3	3	3	2	2	2	1	-	3	2	2	3	3	2	1	2
CO4	2	2	3	2	2	3	2	-	3	2	2	2	3	3	2	3
CO5	3	2	1	1	2	-	-	-	3	2	2	3	3	2	1	2

Note '3' in the box for 'High-level' mapping, 2 for 'Medium-level' mapping, 1 for 'Low'-level' mapping

4.Detailed Syllabus

Unit 1: Introduction, Computer System Hardware Review, Networking (9 hours)

Unit 2: Operating System Structures, Virtual machines, Process and thread management (9 hours);

Unit 3: CPU scheduling, Process Synchronization, Deadlocks (10 hours);

Unit 4: Memory management, Virtual memory (10 hours);

Unit 5: File systems, Hadoop, Map Reduce, Overview of Protection, Security, Multi-media systems, Cloud Computing (6 hours);

- Five mini projects related to the above syllabus will be done by students in the laboratory

References:

1. Silberschatz, A, Galvin, P, Gagne, G. Operating system concepts, Addison-Wesley, 2018
2. Computer Networks (5th Edition) Andrew S. Tanenbaum, David J. Wetherall
Prentice Hall, 2013
3. Distributed and Cloud Computing: From Parallel Processing to the Internet of Things Kai Hwang,
Geoffrey C. Fox, Jack J. Dongarra, Elsevier, 2012
4. William Stallings, Operating Systems, Prentice-Hall, 2018.
5. Charles Crowley, Operating systems: A design oriented approach, Tata McGraw-Hill, 2017.
6. Tanenbaum, A., Modern Operating Systems, Prentice-Hall, Second Edition (latest edition, 2015.
7. Key research papers related to UNIX, Virtual machines, smart phone OSs, distributed OSs.

5. Teaching-Learning Strategies in brief

Lectures by integrating ICT into classroom teaching, weekly tutorials involving problem solving and active learning by students and Project-based Learning by doing 5 mini-projects in laboratory by the students

6.Assessment methods and weightages in brief

Two Class Room tests: 10 marks; Mid Semester Examination in theory: 20 marks, End Semester Examination in Theory: 40 marks, Assessment of 5 mini projects in Laboratory: 30 marks

Title of the Course : Algorithms and Operating Systems
 Name of the Faculty : Vikram Pudi
 Course Code : CS3.306
 Credits : 4
 L - T - P : 3-1-0
 (L - Lecture hours, T-Tutorial hours, P - Practical hours)
 Semester, Year : Monsoon 2026
 Name of the Academic Program: B. Tech. in Electronics and Communication Engineering

1. Prerequisite Course / Knowledge:

A comfortable understanding of data structures, basic algorithm complexity and beginner level algorithms like sorting and searching.

2. Course Outcomes (COs):

The course has two parts – Algorithms and Operating Systems. After completion of this course successfully, the students will be able to understand the basic components of a computer operating system, and the interactions among the various components.

Also, from the topics in algorithms, the students will be able to determine the best most suited data structures and method to solve a problem while also being able to conduct a proper complexity analysis of the same.

CO-1 : Learn the components of an Operating system, its requirement and how the Operating System handles memory management, files, process synchronization and deadlocks.

CO-2 : Apply the concepts taught to create and evaluate algorithms that allow process synchronization, solve problems to evaluate how an Operating system manages a given memory requirement, determine whether a particular scenario leads to a deadlock and how to avoid it.

CO-3 : Implement an application on the top of given operating system in an efficient manner based on process and thread framework available in the given operating system.

CO-4: Familiarize students with algorithmic thinking, design and evaluating the complexity of the algorithm. Students should be able to compare between different data structures and possible algorithms and pick an appropriate one for a design requirement.

CO-5: Apply principles of dynamic programming, divide and conquer and greedy algorithms. Design and analyze them when required. Argue the correctness of algorithms using proofs.

CO-6: Introduce graph algorithms. Students should be able to model a problem into a graph structure when required and use graph concepts to solve the problems.

3. Mapping of Course Outcomes (COs) with Program Outcomes (POs) and Program Specific Outcomes (PSOs) – Course Articulation Matrix

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3	PSO4
CO1	1	1	1	1	1	2	2	2	1	2	1	1				
CO2	2	3	2	2	1	2	1	2	2	3	1	3				
CO3	1	1	3	2	3	1	1	1	3	2	1	3				
CO4	1	2	2	2	1	1	1	1	1	3	1	1				
CO5	2	2	2	2	1	2	1	2	2	2	1	3				
CO6	2	2	2	2	1	2	1	2	2	2	1	3				

Note: '3' in the box for 'High-level' mapping, 2 for 'Medium-level' mapping, 1 for 'Low'-level' mapping

4. Detailed Syllabus:

- Unit 1
 - Introduction to Operating Systems

- Introduction to Algorithms and Complexity
- Introduction to Process, multithreading and process scheduling
- Unit 2
 - Process Synchronization
 - Greedy Algorithms
 - Divide and Conquer Algorithms
- Unit 3
 - Dynamic Programming
 - Memory Management
- Unit 4
 - Deadlocks
 - Graph Algorithms

Reference Books:

1. Galvin, Peter B., Greg Gagne, and Abraham Silberschatz. Operating system concepts. Vol. 10. John Wiley & Sons, 2003.
2. Cormen, Thomas H., et al. Introduction to algorithms. MIT press, 2009.

5. Teaching-Learning Strategies in brief:

Lectures of the class use the active learning methodology and allow students to learn concepts thoroughly in class along with practicing small examples. Assignments are given to reassert and practice topics done in class. Periodical quizzes support summative assessments.

6. Assessment methods and weightages in brief:

- In-class objective tests: 25%
- Homework: 20%
- Midsemester Examination: 20%
- End Semester Exam: 35%

Title of the Course : Advanced Probability Theory

Faculty Name : Khushboo Agarwal

Name of the Program : ECE/CSE

Course Code : MA6.501

Credits : 4

L - T - P : 3 - 1 - 0

(L - Lecture hours, T-Tutorial hours, P - Practical hours)

Semester, Year : Monsoon 2026

Pre-Requisites : MA6.101 (Probability and Statistics), or
MA6.102 (Probability and Random Processes)

Course Outcomes :

1. Construct probability spaces and random variable models using *measure-theoretic tools*.
2. Analyse the mathematical structure behind fundamental limit theorems to examine the asymptotic behaviour of stochastic sequences.
3. Synthesize the core tools of the course — measure theory, martingales, and convergence — to establish the continuum limit connecting (already known) Markov chains to Brownian motion, a result that underpins modern stochastic analysis and its applications.
4. Evaluate rigorous probabilistic arguments and critically assess solution

strategies, building the analytical maturity needed to undertake research-level problems.

Course Topics : Following is the tentative list of topics, broadly divided into 6 modules, to be covered in this course in 24 lectures. (Each lecture is 90 mins)

Module 1: Probability measures and their construction (3 lectures)

- Axioms of probability
- Conditional probability and Independence
- Probabilities on a finite or countable space

Module 2: Random variables (5 lectures)

- Integration with respect to a probability measure
- Independent random variables
- Probability distributions on $\mathbf{R}, \mathbf{R}^n$
- Conditional expectation — general idea, geometric construction and

basic properties

Module 3: Modes of convergence (3 lectures)

- Basic convergence concepts
- Uniform integrability and L^1 -convergence
- Weak convergence

Module 4: Sequences and sums of independent random variables (4 lectures)

- Kolmogorov's zero-one law and the Borel-Cantelli lemma
- The weak law of large numbers
- The strong law of large numbers
- Applications of law of large numbers

Module 5: Martingales (4 lectures)

- Filtrations and adapted processes
- Discrete-time martingales, sub-martingales, super-martingales
- Martingale inequalities

Module 6: Scaling Limits: From Random Walks to Brownian Motion (5 lectures)

- Random walks revisited and scaling heuristics
- CTMCs — exponential holding times, generator, connection to discrete chains
- Donsker's theorem
- Brownian motion
- Introduction to SDEs

Preferred Textbooks :

1. Jacod, J., & Protter, P. (2004). *Probability essentials*. Springer Science & Business Media.
2. Ross, S. M. (1995). *Stochastic processes*. John Wiley & Sons.
3. Karatzas, I., & Shreve, S. (2014). *Brownian motion and stochastic calculus*. springer.

Reference Books :

1. Billingsley, P. (2017). *Probability and measure*. John Wiley & Sons.
2. Norris, J. R. (1998). *Markov chains* (No. 2). Cambridge university press.

3. Durrett, R. (2019). *Probability: theory and examples* (Vol. 49). Cambridge university press.
4. Rogers, L. C. G., & Williams, D. (2000). *Diffusions, Markov processes, and martingales* (Vol. 2). Cambridge university press.

E-book Links : NA

Grading Plan :

(The table is only indicative)

Type of Evaluation	Weightage (in %)
Quiz-1	10
Mid Sem Exam	30
Quiz-2	10
End Sem Exam	40
Project	10

Mapping of Course Outcomes to Program Objectives: (1 – Lowest, 2—Medium, 3 – Highest, or a '-' dash mark if not at all relevant).

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3	PSO4
CO1	1	3	-	2	-	-	-	-	1	1	-	3	2	1	1	3
CO2	1	3	-	2	-	-	-	-	1	1	-	3	2	1	1	3
CO3	1	3	-	2	-	-	-	-	1	1	-	3	2	1	1	3
CO4	1	3	-	2	-	-	-	-	1	1	-	3	2	1	1	3

Teaching-Learning Strategies in brief (4-5 sentences):

- The first four modules build a rigorous foundation in probability theory, emphasising thorough derivations and proofs that go well beyond what an introductory course covers.
 - Concepts that are mathematically demanding will be developed with careful illustrations and examples to make the underlying ideas accessible without sacrificing details.
 - The final two modules will use familiar objects, e.g., random walks, stock models, and queuing systems, as entry points and will aim to ground abstract machinery in contexts students already know before pushing toward rigorous generalisations.
 - A course project requires students to independently study, present, and critically discuss a research paper of their choice that applies probability theory, bridging the gap between coursework and research-level engagement.
-

Title of the Course : Advanced Remote Sensing
Name of the Faculty : Rama Chandra Prasad P & Kiran Chand T
Course Code : GS1.501
Credits : 4
L - T - P : 3-1-0
 (L - Lecture hours, T-Tutorial hours, P - Practical hours)
Semester, Year : Monsoon 2026
Open Elective (Monsoon) for UG and PG Course

Prerequisite Course / Knowledge:

Basic physics

About the course

The course is designed to impart to the students, the unique advantages of advanced remote sensing techniques and their applications in natural resource mapping (forestry, agriculture, land use changes etc), disaster management and relevant areas involving thermal, hyperspectral, microwave and LiDAR remote sensing, to get a comprehensive understanding of different structural and functional aspects of earth features.

Course Outcomes (COs)

After completion of this course successfully, the students will be able to

CO-1: Comprehend concepts of advance remote sensing and analysis

CO-2: Understand concepts of thermal remote sensing and retrieval of its

derivatives CO-3: Describe the significance hyperspectral remote sensing in

resource management CO-4: Understand the concepts of microwave remote sensing and its evolution

CO-5: Explain the basics of LiDAR and its applications

CO-6: Applications of advanced remote sensing concepts in various domains

Mapping of Course Outcomes (COs) with Program Outcomes (POs) and Program Specific Outcomes (PSOs) – Course Articulation Matrix

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3	PSO4
CO1	2	1	2	1	2	1	1	1	3	1	2	1	2	2	2	2
CO2	2	2	2	2	2	2	2	2	2	1	2	2	3	2	2	2
CO3	2	2	2	2	3	2	2	1	1	1	2	2	2	2	2	2
CO4	2	2	2	2	2	1	1	1	1	2	2	2	2	2	2	2
CO5	2	2	2	2	3	2	2	2	2	1	2	3	3	2	2	2
CO6	1	2	2	1	2	3	1	1	1	3	1	2	2	2	2	1

Detailed Syllabus:

Unit-1: Thermal Remote sensing: Introduction, Thermal IR Sensors- Characteristics of images; Physics of TIR: Radiation laws; Thermal Image Processing and Interpretation: Retrieval of Land Surface Emissivity, Land Surface Temperature and Applications.

Unit -2. Hyperspectral Remote sensing: Introduction, Spectral Radiometry, Image acquisition and data characteristics, Information Extraction from Optical Image Data, Hyperspectral and Ultraspectral Information: Extraction Approaches, Applications.

Unit-3: Microwave Remote sensing: Introduction, Characteristics of Microwave RS, Theory of Radiometry - Emission laws, Passive and Active Microwave Remote sensing, Airborne and Space borne radar systems–physics of RADAR waves, geometric characteristics of radar imagery, Applications

Unit-4: LiDAR: Principles of LiDAR remote sensing, Data processing, Products and applications

References

Online sources of reference books as well as e-books will be provided during the course.

Teaching-Learning Strategies in brief:

Teaching, discussing current approaches of information extraction, challenges and limitations with satellite data; Current research papers presentations by students on chosen topic, writing assignments, periodical evaluation of course project implemented with open data and tools; applying remote sensing satellite imagery in different domains, develop an open source tool as part of project or revise algorithms for feature extraction or for any image processing method.

Assessment methods and weightages in brief :

Assignments (max. of 4)	20%
Project	20%
Mid-term Exams (2)	30% [15% + 15%]
End-Semester Exam (1)	30%

Details of Assignments/Projects will be announced during the course

Title of the Course	: AI for Product Managers/Management
Name of the Faculty	: Aditya T, Microsoft Guest Faculty
Course Code	: PD3.401
Credits	: 2 (L-T-P: 1.5-0-3)
<i>(L=Lecture hours, T=Tutorial hours, P=Practical hours)</i>	
Duration	: 12 Weeks (Monsoon/Spring Semester)

1. Prerequisite Course / Knowledge

- **Foundational Statistics:** Understanding of probability, mean/variance, and hypothesis testing.
- **Product Management Basics:** Familiarity with the Product Life Cycle (PLC) and Agile methodologies.
- **Programming Literacy:** Basic ability to read Python code (No expert coding required).

2. Course Objectives & Outcomes (COs)

This course is designed specifically for the Product Design and Management students at **IIIT Hyderabad**, focusing on the strategic integration of Artificial Intelligence and Machine Learning into the product

lifecycle. Unlike a purely technical engineering course, this curriculum emphasizes the **PM's role** in identifying high-value AI opportunities, managing cross-functional technical teams, and ensuring ethical, user-centric AI implementation.

Learning Outcomes (COs)

This course provides knowledge about leveraging AI/ML to solve business problems and deliver customer value.

- **CO-1:** Demonstrate a solid understanding of AI/ML capabilities and how they influence the value proposition of digital products.
- **CO-2:** Identify and define "AI-First" problems, translating user pain points into machine learning objectives (Classification, Prediction, etc.).
- **CO-3:** Develop a data strategy, understanding the lifecycle of data collection, labeling, and the "Data Flywheel" effect.
- **CO-4:** Demonstrate the ability to create and present PM-specific AI deliverables: AI Product Requirement Documents (PRDs), Model Performance Metrics (Precision/Recall), and ROI analyses.
- **CO-5:** Navigate AI ethics and risk management, including algorithmic bias, data privacy, and managing stakeholder expectations regarding model uncertainty.

3. Mapping of Course Outcomes (COs) with Program Outcomes (POs) and Program Specific Outcomes (PSOs) – Course Articulation Matrix

	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO1 0	PO1 1	PSO 1	PSO 2	PSO 3	PSO 4
CO 1	3	3	3	3	3	1	1	2	2	3	2	3	2	2	2
CO 2	3	3	3	3	3	3	1	2	3	2	2	3	2	2	2
CO 3	3	3	3	2	3	3	1	2	3	3	2	3	2	2	2
CO 4	3	3	3	3	3	3	1	2	3	3	2	3	2	2	2
CO 5	2	2	3	3	2	2	1	3	3	3	2	3	2	2	2

4. Detailed Syllabus

Unit	Topic	Week	Content Focus
Unit 1	Introduction to the AI Landscape	Week 1	AI vs. ML vs. Deep Learning; The PM's role in the AI stack. "Evaluating AI-driven products (e.g., Netflix, Spotify)".
Unit 2	Probabilistic Thinking	Week 2	Managing uncertainty; Deterministic vs. ML-based product logic. "Mapping the ""Human-Action Cycle"" in AI interactions."
Unit 3	ML for Product Managers	Week 3	Supervised & Unsupervised Learning: When to use Classification vs. Regression; Clustering for personas. Case study: Predicting user churn using historical data.
Unit 4	Deep Learning & GenAI	Week 4	Understanding LLMs, Diffusion models, and the ""Magic"" of Generative AI. Hands-on: Prompt engineering for product feature ideation.
Unit 5	Data Strategy	Week 5	Data as a Product: Data collection, labelling, and quality; Building a ""Data Flywheel"". ", Designing a data instrumentation plan for a new feature. ML Ops; Technical debt; Inference costs; Scaling AI products.
Unit 6	AI in UX and AI Ethics,	Week 6	Designing for failure; Feedback loops; Transparency in AI decisions. +1,"Creating wireframes for ""AI-first"" user interfaces." Identifying algorithmic bias; GDPR/DPDP compliance; Safety guardrails.

5. Reference Books & Case Studies

Books:

- *Real World AI: A Practical Guide for Responsible Machine Learning* by Alyssa Simpson Rochwerger & Wilson Pang.
- *Prediction Machines: The Simple Economics of Artificial Intelligence* by Ajay Agrawal.
- *Human-Centered AI* by Ben Shneiderman.

Case Studies:

- **Harvard Business Review:** "Competing in the Age of AI."
- **Stratechery:** "The AI Unbundling" by Ben Thompson.
- **Google Design:** "The People + AI Guidebook."

6. Teaching-Learning Strategies

- The course is divided into 12 lectures/practical sessions including classroom exercises and role-play simulations.
- **Case-Based Learning:** Every week begins with a real-world product failure or success (e.g., Zillow's iBuying algorithm failure).

- **Hands-on "Low-Code":** Using tools like Orange, Teachable Machine, or ChatGPT API to build prototypes without heavy coding.
- **Role-Play:** Mock "Engineering vs. Product" syncs to negotiate model accuracy vs. Latency. Students will analyse real-world case studies focused on product failures caused by poor alignment.
- A short-term project offers the opportunity to practice the full influence cycle from mapping to board-level presentation.
- Students are expected to spend 3 hours per week on home/field assignments.

7. Assessment Methods and Weightages

Component	Weightage	Description
Class Participation & Quizzes	40%	Assignments – Class and Home.
AI PRD Assignment	10%	Oral defence of product decisions and conflict resolution strategies.
Capstone Project	40%	A 4-week project where students design an AI-first product, including a technical feasibility analysis and a UX prototype.

Title of the Course : Algorithm Analysis and Design

Name of the Faculty : Girish Varma

Course Code : CS1.301

Credits : 4

L - T - P : 3-1-0

(L - Lecture hours, T-Tutorial hours, P - Practical hours)

Semester, Year : Monsoon 2026

Name of the Program : B.Tech

Pre-Requisites : Discrete Mathematics, and Data Structures and Algorithms

Course Outcomes : After completion of this course successfully, the students will be able to...

CO-1: Demonstrate the ability to fully understand the analysis of various known algorithms.

CO-2: Identify problems where various algorithm design paradigms can possibly be applied.

CO-3: Understand the notions of computational intractability and learn how to cope with hardness.

CO-4: Understand the notion of approximation and randomized algorithms. If time permits, intro to quantum algorithms.

Detailed syllabus:

1. Basic graph algorithms
2. Greedy algorithms
3. Divide and Conquer
4. Dynamic Programming
5. Network flows
6. NP and computational intractability
7. Intro to Approximation and Randomized algorithms

8. Intro to Quantum algorithms

Assessment method and Grading scheme:

- Deep quizzes 1 and 2: 10 + 10 = 20%
- Mid-semester exam = 20%
- End-semester exam = 30%
- In-class quizzes (unannounced) = 15%
- Assignments = 15%

Mapping of Course Outcomes (COs) with Program Outcomes (POs) and Program Specific Outcomes (PSOs)

	PO1	PO2	PO 3	PO 4	PO5	PO6	PO7	PO 8	PO 9	PO 10	PO 11	PO 12	PS O1	PS O 2	PS O 3	PS O 4
CO1	1	2	2	2	1	1	1	2	1	1	1	2	3	1	1	3
CO2	2	3	2	2	1	1	1	2	2	2	1	3	3	2	2	3
CO3	2	3	1	2	1	1	1	2	2	2	1	3	3	2	2	3
CO4	2	3	1	2	2	1	1	2	2	1	1	3	3	1	1	3

Each Course Outcome (CO) may be mapped with one or more Program Outcomes (POs). Write '3' in the box for 'High-level' mapping, 2 for 'Medium-level' mapping, 1 for 'Low'-level' mapping

Teaching-Learning Strategies in brief (4-5 sentences):

The course lectures will include activities that promote the understanding of the lecture content by using small examples that students work out during the class itself and promote active and participatory learning. A good part of the lecture will involve problem solving and finding solutions to problems rather than expositing known material. In class tests that are held periodically are useful as summative assessments. Homework assignments are designed to reiterate the material covered in class lectures and also solve problems that are based on simple extensions of concepts described in the lectures.

Title of the Course : **Arts 1 (H1)**
Name of the Faculty : **Saroja T K**
Course code : **OC2. 101**
L-T-P : **3 Hour a week**
Credits : **2-0-0-2**
 (L= Lecture hours, T=Tutorial hours, P=Practical hours)
Name of the Academic Program: **Arts**
1.Prerequisite Course / Knowledge: None

2.Course Outcomes (COs) (5 to 8 for a 3 or 4 credit course): After completion of this course successfully, the students will be able to.

CO-1 Understands and appreciate art in a deeper sense, and realize the importance of Art

CO-2 Enhances Imagination and aesthetic sensibility

CO-3 Imparts humanities and artistic skills

CO-4 Understands Art as a system of knowledge

CO-5 Understands the effectiveness of informed Art practice

3.Mapping of Course Outcomes (COs) with Program Outcomes (POs) and Program Specific Outcomes (PSOs) – Course Articulation Matrix

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PS O1	PS O2	PS O3	PS O4
CO1	1	1	1	1	0	2	1	2	3	1	0	3	0	0	0	0
CO2	1	1	1	1	0	2	1	2	3	1	0	3	0	0	0	0
CO3	1	1	1	1	0	2	1	2	3	1	0	3	0	0	0	0
CO4	1	1	1	1	0	2	1	2	3	1	0	3	0	0	0	0
CO5	1	1	1	1	0	2	1	2	3	1	0	3	0	0	0	0

Note: Each Course Outcome (CO) may be mapped with one or more Program Outcomes (POs) and PSOs. Write '3' in the box for 'High-level' mapping, 2 for 'Medium-level' mapping, 1 for 'Low'-level' mapping

4.Detailed Syllabus:

Unit 1: Raga and Rhythm: This course emphasizes understanding the nuances of sound and timing, the basic concepts of any system of music in the world. The students are made to learn different songs, melodic exercises, and rhythmic exercises with a focus on concepts of Indian music and are exposed to the logical elements of the art form in general. The unique concepts of Indian music, raga and tala are introduced to them to make them realize the depth of this system of music and its connections to various branches of study. The introduction of these elements through personal demonstrations, presentation of audio, and videos of acclaimed artists, intends to attract their attention towards the artistic sensibilities, creativity, and discipline in life.

Unit 2: Painting: The course's primary focus is to help students express their ideas and feelings through lines and colors. For this basic drawing and painting skills will be taught to the students in the class. The students will also be given different tasks like oral and visual storytelling, creating logos, symbols, and portraiture. Through these tasks, the student will understand different ways of visual thinking.

Unit 3: Dance: The course informs the students about the significance of dance, and the training involved to perform the dance movements. The course instructs about basic stretches and fundamental movements of the dance of various Indian dance forms. The knowledge about various dance forms of India and the significance of the dance forms in the past and present is discussed. The course helps the students to compose movements and dance their individual units of movements they create out of the instructions and assistance received. In the course, the emphasis point on evaluation is not based on the dancing skills of the students but on their participation in the session in progress.

Unit 4: Sculpture: The course deals with understanding three-dimensional form and creativity. Clay modelling is a great activity that helps students develop in many ways, like self-expression and creativity. In this course, students are taught to make sculptures out of clay. Through this, I try to connect them to nature. The students get a personal experience of the texture of clay, which is an important part of understanding nature. In this course, they will learn how to use different materials to make art.

Unit 5 Collage: Collage is not just a compilation of photos that we create to share on social media. It's an art form where one assembles images from a magazine or newspapers or photographic images, maps, diagrams by cutting pasting or painting or drawing over it to create a unique composition. Artists have manipulated mass produced images to comment on or question body images and narrow beauty standards, gender stereotypes, consumerism, racism and much more. The aim of this course is to equip students with

visual tools to explore the possibility this medium offers through a set of exercises. Students will learn to express their ideas or imagination through the process of selection and deduction and addition.

5. Teaching-Learning Strategies in brief (4 to 5 sentences):

The course is on Imagination, aesthetic sensibility, goodness in life and improving humanities skill. This is achieved by offering training on artistic skills and Art Education. The course does not focus on creating artists out of the students which would be intense, but the course is designed on the thought that the end form is secondary, while the means to achieve is primary. The course introduces the students to the thought and the process of Art creation and Art appreciation. The course explains the confluence of art and other popular knowledge systems.

6. Assessment methods and weightages in brief (4 to 5 sentences):

It is a 2-credit course The semester evaluations are based on the participation of students in the sessions.

Title of the Course : Automata Theory
Name of the Faculty : Shantanav Chakraborty
Course Code : CS1.302
Credits : 2
L-T-P : 3-1-0
 (L=Lecturehours, T=Tutorialhours, P=Practicalhours)
Name of the Academic Program : B.Tech in Computer Science and Engineering
1. Prerequisite Course / Knowledge: Data structures, Elementary Formal Logic

2. Course Outcomes (COs)

After completion of this course successfully, the students will be able to

CO-1. Develop an understanding of the core concepts of Automata theory such as Deterministic Finite Automata, Non-deterministic Finite Automata, Regular Languages, Context Free Languages, Push down Automata, the basics of Turing Machines

CO-2. Design grammar sand automata for different languages

CO-3. Identify formal language classes and prove language member ship properties

CO-4. Describe the limitations of the different computation al models

3. Mapping of Course Outcomes (COs) with Program Outcomes (POs) and Program Specific Outcomes (PSOs)–Course Articulation Matrix

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3	PSO4
CO1	2	2	1	1	1	1	-	-	2	2	1	2	3	1	1	3
CO2	2	2	3	1	2	-	-	-	2	2	1	2	3	1	1	3
CO3	2	2	3	1	1	-	-	-	2	2	1	2	3	1	1	3
CO4	1	2	2	1	1	1	-	-	2	2	2	1	2	1	1	3

‘3’ in the box de notes ‘High-level’ mapping, 2 for ‘Medium-level’ mapping, 1 for ‘Low’-level’ mapping

4. Detailed Syllabus:

Unit1: Introduction, Finite State Machines, Deterministic Finite Automata (DFA), Nondeterministic Finite Automata (NFA), Equivalence of NFA and DFA, Regular Expressions, Regular Languages, Closure Properties of regular languages, Pumping Lemma, Grammars, Left and Right-linear grammars

Unit2: Context Free Grammar (CFG), Chomsky Normal Form, Push Down Automata (PDA), Equivalence of

CFG and PDA, Context Free Languages (CFL), Deterministic PDA and Deterministic CFL, Pumping Lemma for context free languages

Unit 3: Introduction to Turing machines, Total Turing Machines, Recursive languages, Recursively enumerable languages, The Halting problem.

References:

- M. Sipser, Introduction to the Theory of Computation, Third Edition, Cengage Learning 2012.
- J.E. Hopcroft, R. Motwani and J. Ullman, Introduction to Automata Theory, Languages and Computation, Third Edition, Pearson, 2006.

5. Teaching-Learning Strategies in brief:

The lectures will be arranged in a manner that facilitates inter-student and faculty-student discussions. Additionally, the lectures will have small exercises that will ensure that the students actively participate in the learning activity and think out of the box. There will be more emphasis on ideas and reproduction of textbook material. There will be small homework problems that would help the student to re-engage with the essential components of the lecture. Assignments will test the student's ability to apply key concepts learnt, and also inform the faculty of the progress being made by the students in acquiring them.

6. Assessment methods and weightages in brief:

Homework: 25%

Quiz 1: 20%

Quiz 2: 20%

Final exam: 35%

Title of the Course	: Autonomous Software Engineering (AutoSE)
Name of the Faculty	: Karthik Vaidhyanathan
Name of the Academic Program	: CSE
Course Code	: CS6.403
Credits	: 2
L-T-P	: 1-0-2
In Take Limit	: 30

1. **Prerequisite Course / Knowledge:** Design and Analysis of Software Systems (Mandatory), Software Engineering (Desired) or Relevant software engineering background
2. **Course Outcomes (COs):**
 - CO1:** Understand the principles of Agentic AI and their application to software engineering workflows.
 - CO2:** Design AI-assisted workflows and agent-centric solutions for software engineering activities.
 - CO3:** Apply open-source models, agentic frameworks, and orchestration strategies to automate software engineering and software operations tasks.
 - CO4:** Critically evaluate AI-generated software artifacts with respect to reliability, performance, trustworthiness, and sustainability.
 - CO5:** Deploy and assess AI-assisted software engineering solutions across development and operational workflows.
3. **Mapping of Course Outcomes (COs) with Program Outcomes (POs) and Program Specific Outcomes (PSOs) – Course Articulation Matrix**

CO / PO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3	PSO4
CO1	2	2	1	1	2	1	1	1	1	1	1	2	3	2	1	2
CO2	2	3	3	2	2	2	1	1	1	1	1	1	2	3	2	1
CO3	2	2	2	1	3	1	1	1	1	1	2	1	3	2	2	1
CO4	1	2	2	3	1	2	3	2	1	1	1	2	2	3	2	2
CO5	2	2	3	2	3	2	2	2	3	2	2	2	2	2	3	2

4. Detailed Syllabus:

Unit1: Foundations of Agentic AI – Introduction to AI agents, reasoning and planning, memory, tool usage and MCP, AI-native software engineering workflows, open-source models and frameworks.

Unit 2: Agentic Requirements Engineering – AI-assisted requirement elicitation, conversational requirements engineering, user story generation, traceability, and workflow automation.

Unit 3: AI-assisted Software Architecture – Architecture knowledge management, Architecture Decision Records (ADRs), MCP for Architecture tasks, Views and Viewpoint generation, architecture evaluation (Agents for ATAM/CBAM) and Architecture Co-pilots.

Unit 4: Code Generation Agents and Code Maintenance– Anatomy of Coding agent, repository-aware coding agents, automated code generation, Efficiency of coding agents, green-field vs brownfield generation, automated code refactoring, Refactoring legacy systems.

Unit 5: Evaluation of AI-assisted SE - Evaluation of AI-generated software artifacts, Trustworthiness of generated outputs, Sustainability considerations, Benchmarking AI-assisted workflows, Maintainability, human oversight and governance, human vs Automated trade-off analysis.

Unit 6. Agentic Testing, Operations and Autonomous CloudOps – AI-assisted testing, test generation, testing of generated artefacts, issue resolution agents, deployment pipelines, Agentic Root Cause Analysis, autonomous CloudOps and Cloudops Workflow

Unit 7. Autonomous Software Engineering and Emerging Directions – Runtime self-adaptation, adaptive orchestration strategies, AI-native IDEs, self-coding information systems and future directions in AI-assisted software engineering.

Tutorials: Covers concepts using open-source agentic harness and co-pilots in 4 lab sessions with graded activity.

5. Reference Materials

1. E. Hassan, A. (2026) *Agentic Software Engineering*. Online. Available: <https://agenticse-book.github.io/>
2. Weyns, D., 2020. *An introduction to self-adaptive systems: A contemporary software engineering perspective*. John Wiley & Sons.
3. Alammar, J. and Grootendorst, M., 2024. *Hands-on large language models: language understanding and generation*. " O'Reilly Media, Inc."

4. Taibi, D., Muccini, H., Vaidhyanathan, K., Kalinowski, M., Albano, M., Alves, A.P.S., Cerqueira, R., Devino, M., Esposito, M., Falcão, R. and Henning, V., 2026. A Research Agenda on Agents and Software Engineering: Outcomes from the Rio A2SE Seminar. *arXiv preprint arXiv:2605.11720*.
5. Research work done by group: Sa4s @ SERC - Research Group, SA4S @ SERC- Software Architecture for Sustainability. Available at: <https://sa4s-serc.github.io/> (Accessed: 26 May 2026).

6. Teaching-Learning Strategies in brief:

This is a project-based learning course that combines interactive lectures, hands-on tutorials, guided experimentation, and collaborative project work. The course emphasizes active participation, critical thinking, and practical problem-solving using AI agents for software engineering tasks. Learning is reinforced through in-class activities, take-home assignments, and hands-on exploration of modern agentic frameworks, open-source models, and AI-assisted software engineering tools. Students progressively design, build, evaluate, deploy, and assess AI-assisted software engineering workflows spanning requirements engineering, software architecture, coding, testing, debugging, and software operations, with particular emphasis on software architecture, software quality, sustainability, and the effective integration of AI agents into software engineering practices. One of the core aspects of the course is a team-based project where students develop and evaluate an autonomous software engineering agentic system addressing a real-world software engineering challenge. Through this experience, students gain practical exposure to workflow design, implementation, deployment, evaluation, and continuous improvement while critically assessing the quality, trustworthiness, efficiency, and sustainability of AI-assisted software engineering solutions. The course also exposes students to emerging topics such as runtime self-adaptation, trustworthy code generation, and autonomous CloudOps through case studies, guest lectures, and guided project work.

7. Assessment methods and weightages in brief:

Component	Weightage
Class Activities (4)	20%
Take-home Assignments (2)	20%
Course Project and Presentations	60%

Title of the Course	: Basics of Ethics
Name of the faculty	: Ashwin Jayanti + Saurabh Todariya
Course code	: HSo.203a & HSo.203b
L-T-P	: 3-1-0
Credits	: 2
Name of the Academic Programs	: B.Tech. in CSE, B.Tech in ECE

1.Prerequisite Course / Knowledge: : Nil

2.Course Outcomes (COs)

After completion of this course successfully students will be able to:

CO1: **Explain** the philosophical nature of the basic concepts and principles of ethics

CO2: Analyze ethical arguments for logical validity, soundness, and informal fallacies

CO3: Demonstrate the **knowledge of** conceptual challenges involved in normative inquiry in the ethical domain

CO4: Develop skills to formulate fundamental nuances in ethical justification and explanations

CO5. Identify the various kinds of normative elements that constitute ethical frameworks

CO6. Discuss the major tenets of normative ethical theories and their scope of application

3. Mapping of Course Outcomes (COs) with Program Outcomes (POs) and Program Specific Outcomes (PSOs) – Course Articulation Matrix'

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3	PSO4
CO1	1	2	3	1	1	3	2	3	1	1	-	3	1	1	2	3
CO2	2	2	3	3	2	3	2	3	1	3	1	3	1	1	2	3
CO3	2	2	2	3	1	3	2	3	1	2	1	1	1	1	2	3
CO4	1	2	2	3	1	2	2	3	2	2	-	2	1	2	1	3
CO5	2	2	3	3	1	2	3	3	1	1	1	3	1	2	2	2
CO6	2	2	3	3	1	3	3	3	2	2	1	2	1	1	3	2

'3' in the box denotes 'High-level' mapping, 2 for 'Medium-level' mapping, 1 for 'Low'-level' mapping

4. Detailed Syllabus:

Unit I – Introduction (3 hours): Distinction between conventional and critical ethics, philosophical tools for argument analysis, intuition, evidence, justification, and explanation.

Unit II – Skepticism (4.5 hours): Intrinsic vs Instrumental value, challenge of egoism, problem of cultural relativity and subjectivism, error theory and nihilism, distinction between being ethical and seeming ethical.

Unit III – Goodness (3.5 hours): the problem of defining 'good', naturalistic fallacy and the open question argument, implications of the experience machine thought experiment.

Unit IV – Responsibility (3.5 hours): challenge of attributing moral responsibility to agents, the control, competence and epistemic conditions of responsibility, moral luck.

Unit V – Normative theories (5 hours): Consequentialism, deontology, and virtue ethics

Reference books:

1) Shafer-Landau, R. 2019. *Living Ethics: An Introduction with Readings*. Oxford University Press.

2) Shafer-Landau, R. 2013. *Ethical Theory: An Anthology 2nd Edition*. Wiley-Blackwell.

3) Stich, S. and Donaldson. T. 2019. *Philosophy: Asking Questions, Seeking Answers*. Oxford University Press.

5. Teaching-Learning Strategies in brief:

The general teaching strategy employed is the use of moral dilemmas and conceptual puzzles to introduce course topics. Lectures make use of this strategy to impress upon students the need to critically reflect on ethical issues and the relevance of doing a careful, philosophical investigation of those issues. Student interaction at this stage is aimed at bringing out conflicting ethical intuitions. This is followed up by introducing proper vocabulary to map out the problems involved in normative moral assessment. Using case studies and toy examples, ethical principles and methods of inquiry are taught so that students develop effective reasoning skills to engage with any real-world ethical matter. Student interaction and discussion at this stage is aimed to give flesh to the intuitions identified in the previous stage. The teaching-learning strategy emphasizes the merits of avoiding simplistic solutions to complex ethical problems and instead ask meaningful questions that enrich moral debates.

6. Assessment methods and weightages in brief:

This is mainly a writing-driven course, and the exercise questions are carefully designed to make students think independently in ethical contexts. Students are assessed for abilities like logically dissecting issues, questioning assumptions, clarifying distinctions, and bringing out nuances. In assignments and exams,

students are expected to demonstrate these abilities by presenting their views clearly, assessing competing positions systematically, anticipating possible objections to a reasoned conclusion and composing cogent responses to those objections. The assessment components and their weightages are as follows. Assignments: 60 marks, class participation: 10 marks, Mid semester exam: 10 marks, End semester exam: 20 marks.

Title of the Course : Behavioral Research & Experimental Design

Course Code : CG3.402

Faculty Name : Vinoo A R

L-T-P : 3-1-0

Credits : 4

Name of the Academic Program : B. Tech.in CSE

Prerequisite Course / Knowledge:

Interest in conducting behavioral experiments is desirable. Open only for DD, MS, and PhD students. BTech and MTech students can be admitted based on specific requirements and instructor permission.

Course Outcomes (COs):

After completion of this course successfully, the students will be able to:

CO-1: To develop understanding of the basic framework of behavioral research process.

CO-2: To identify various sources of information for literature review for operationalization and data collection.

CO-3: To develop an understanding of various experimental designs and analyses techniques and apply in their own final projects.

CO-4: To operationalize a research question and design, deploy behavioral experiments and analyze the data collected thereof using appropriate statistical tests.

CO-5: To develop an understanding and evaluate the ethical dimensions of conducting applied research.

CO-6: Appreciate the components of scholarly writing and evaluate its quality. Create and develop their unique way of writing and presenting their work whilst balancing scientific standards to effective communication

Mapping of Course Outcomes (COs) with Program Outcomes (POs) and Program Specific Outcomes (PSOs)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3	PSO4
CO1	1	1	1	3	1	1	1	1	1	1	-	2	1	1	3	3
CO2	1	2	2	3	1	1	1	1	2	2	-	1	1	1	3	3
CO3	3	3	3	3	2	1	1	1	3	3	-	1	1	2	3	3
CO4	3	3	3	3	3	1	1	1	3	3	-	1	1	2	3	3
CO5	1	1	1	1	1	3	1	3	2	2	-	1	1	2	3	3
CO6	1	1	1	1	1	1	1	1	1	1	-	2	1	1	2	3

Detailed Syllabus:

1. Introduction to Research Methods: Qualitative and Quantitative Approaches; Conducting Behavioural Research; Ethics in Research; Institute Review Board (IRB) Process

2. Starting on Research, Experimental Design: Hypothesis Testing, Type I and II errors, Hypothesis-based vs Exploratory Research, Operationalizing Research, Literature Review; Sampling, Types of variables and levels of Measurements, Designing an Experiment; Validity, Reliability and Cross-

validation in Research

3. Types of Experimental design: Non-Experimental Designs, Pilot Testing; 4. Data Collection: Surveys Questionnaires; Data Representation: Levels of Measurement, Human Annotation, Different types of design: Simple randomized design, Factorial designs, Simple repeated measures design, Randomized blocks design, Latin square type designs, Between-subject and within-subject factors in an experiment; Scaling Behavioural Experiments: web and mobile experiments, crowdsourcing, big data, large-scale experiments, citizen science, online data collection (PsiTurk, Mechanical Turk, etc).

4. Data Visualization and Analysis: Descriptive Statistics, Tests of Normality and Data Transformation, Outliers, Collinearity in Data, Data Summarization vs Data Reduction Techniques: Exploratory Factor Analysis, Principal Component Analysis, Discriminant Factor Analysis

5. Introduction to Statistical Analysis: Inferential Statistics-Tests of Difference and Tests of Association: Multi-level tests (ANOVA): nonparametric and parametric tests of difference – chi-square test, Mann Whitney U test, Binomial Sign test, Wilcoxon's T test, Related and Unrelated t tests; nonparametric and parametric tests of association – correlation, regression; Significance testing [NOTE: While this course emphasizes basic descriptive and inferential statistical analysis, the Second part of the course to be offered in Spring would cover Statistical Analysis of Behavioural and Neuroimaging data in more detail].

6. Communicating and Assessing Research: Writing, Poster and general Presentations (formatting of the research paper using APA and IEEE journal/conference formats)

Reference Books:

- Howell, D.C. (1997). Statistical Methods for Psychology (4th ed). Belmont, CA: Duxbury.
- Salkind, N.J. (2009). Exploring Research (8th Ed.). Upper Saddle River, NJ: Prentice Hall.
- Cozby, P. & Bates, S. (2011). Methods in Behavioural Research (11th Ed.), McGraw Hill.
- Coolican, H. (2014). *Research Methods and Statistics in Psychology*. London: Hodder & Stoughton.
- Passer, M. W. (2017). Research Methods: Concepts and Connections, 2nd ed. New York: MacMillan.

Teaching-Learning Strategies in brief (4 to 5 sentences):

Lectures are highly interactive as the course requires a student to actively participate and think and be creative. Students learn by doing assignments designed to achieve course outcomes and collaboratively working on a final project. The final project wherein students learn by working in teams, especially to devise a research question, identify hypotheses, operationalize it, deploy it, collect and analyze data and present, promotes collaboration, which is very much needed in research. Deploying their experiment and collecting data allows them to appreciate real-world problems that are faced while creating reliable databases.

Assessment methods and weightages in brief (4 to 5 sentences):

Assignments: 30%

Quizzes (2): 20% (10 + 10)

Class Participation & Hand-Written Summary notes: 10%

Final Project (teams of two or individual – apply, design, run experiments, analyze with appropriate statistics): 40%

Title of the Course : **Bioinformatics**
Name of the Faculty : Nita Parekh
Course Code : SC3.202
L-T-P : 3-1-0
(L= Lecture hours, T=Tutorial hours, P=Practical hours)
Credits : 2
Name of the Academic Program : CND

1.Prerequisite Course / Knowledge: Basic Statistics and computing skills

2.Course Outcomes (COs) (5 to 8 for a 3 or 4 credit course):

After completion of this course successfully, the students will be able to

CO-1: Use various web-based bioinformatic resources (databases and tools) judiciously

CO-2: Understand and implement methods for various biological sequence analysis, viz., pattern search and sequence comparison (pairwise and multiple sequence alignments), and phylogenetic reconstruction, gene prediction

CO-3: Familiarize with the probabilistic models in biological sequence analysis

3.Mapping of Course Outcomes (COs) with Program Outcomes (POs) and Program Specific Outcomes (PSOs) – Course Articulation Matrix

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3	PSO4
CO1	1													3		
CO2		3			1										3	1
CO3	2														1	

Note: Each Course Outcome (CO) may be mapped with one or more Program Outcomes (POs) and PSOs. Write '3' in the box for 'High-level' mapping, 2 for 'Medium-level' mapping, 1 for 'Low'-level' mapping

4.Detailed Syllabus:

Unit 1: Overview – Bioinformatics, Gene and Genome structure, Gene Technology – Restriction Endonucleases, Cloning vectors, DNA sequencing – PCR, cDNA and Whole Genome sequencing, NGS and third generation sequencing technologies

Unit 2: Bio Databases: Major Bioinformatics Resources – NCBI, EBI, PubMed, Primary Nucleotide and Proteins Databases - GenBank, Uniport, PDB, Genome Browsers – Ensemble, UCSC, k-mer analysis and their significance in biological sequences

Unit 3: Sequence Alignment: Pairwise Alignment – Types of pairwise alignments – Global, Local and Overlap alignments, Dot Plots, dynamic programming (DP) algorithm, Scoring matrices for nucleotides and proteins and gap penalties, Sequence-based Database Search algorithms – BLAST, FASTA, Multiple Alignment, Algorithms for Global and Local MSA – DP, Progressive based (ClustalX), Iterative methods, motif search-based methods

Unit 4: Modeling Molecular Evolution – Phylogeny: Markov models of base substitution, Computing Phylogenetic Distances, Phylogenetic Tree Construction Methods, PHYLIP

Unit 5: Gene Prediction: Gene Prediction approaches - Open Reading Frames, Homology search, Content-based methods, Markov models

Reference Books:

1. Bioinformatics Sequence and Genome Analysis, David W. Mount, Cold Spring Harbor Laboratory Press, 2001.
2. Biological Sequence Analysis, Probabilistic Models of Proteins and Nucleic Acids, Richard Durbin, Sean R. Eddy, Anders Kroghs and G. Mitchison, Cambridge University Press 1998.
3. Computational Genome Analysis – An Introduction, Richard C. Deonier, Simon Tavae and Michael S. Waterman, Springer 2005.

5. Teaching-Learning Strategies in brief (4 to 5 sentences):

The objective of the course is to familiarize the students with available web-based bioinformatics resources (databases and tools), how to use them for analysis, extract information from them, and learn to build such tools. First by taking an example of an unknown sample, what information about the sample can be obtained starting with DNA sequence by searching through available resources is provided. Next each one is given a gene sequence, and they extract information about it, perform functional annotation, disease association, etc. To get a clear understanding of the methods learned for biological sequence analysis, the students implement algorithms for performing various tasks such as finding k-mers, restriction recognition sites, pairwise alignment, and gene prediction.

6. Assessment methods and weightages in brief (4 to 5 sentences):

1. Assignments – written, a mini-project using online resources, implementation of algorithms (30%), Class Quizzes + Mid-term evaluation (30%), Final exam (40%).

Title of the course : Biomolecular Structure Interactions and Dynamics

Name of the Faculty : B. Gopalakrishnan

Course Code : SC3.321

Credits : 3-1-0- 4

L - T - P : Each week has 2 lectures and 1 Tutorials [28 L + 14 T]

(L - Lecture hours, T-Tutorial hours, P - Practical hours)

Semester, Year : Monsoon, 2026

Name of the Program: Science Stream Elective, Odd semester

Pre-Requisites : Molecular Architecture, Advanced Biomolecular Architecture or equivalent

Course Outcomes :

1. Review of physicochemical principles at the molecular level
2. From molecules to biochemical systems – appreciation of principles of kinetics and thermodynamics for understanding mechanisms of interactions and reactions of biomolecules
3. Appreciation of the experimental methods used for exploring structures of biomolecules
3. Understanding of important structural concepts used for the analysis of protein and nucleic acid structures
4. Learning to use and understanding the principles of molecular modeling, docking and molecular dynamics simulations for inferring structures, functions and interactions from sequences –
5. Familiarity with important structural and functional databases and their usefulness in biological contexts (list about 5 to 6 outcomes for a full 4 credit course)

The action verbs to be used for writing the course outcomes can be found on slide 22 in the following presentation. You may remove this line and the following link after the course outcomes are formulated.

Course Topics :

(Please list the order in which they will be covered, and preferably arrange these as five to six modules.)

Preferred Textbooks: [e-books are available]

1. *Biochemistry* – Voet, Voet and Pratt
2. *Bio-Chemistry* – L. Stryer
3. *Modeling & Simulations* – Tamar Schlick

Reference Books :

1. **Introduction to Protein Structure** by Branden & J Tooze [ISBN-10 : 9780815323051]
2. **Proteins: Structures and Molecular Properties** by TM Creighton [ISBN-10 : 071677030X]
3. **Molecular Modelling: Principles and Applications** by AR Leach [ISBN-10 : 0582382106]

E-book Links:

https://iiitaphd-my.sharepoint.com/:f/g/personal/shweta_kumari_research_iiit_ac_in/EldcZ_Lqj3JBICDryuvU5z4B8twcGifr3l4ovKjixUVqSg?e=LpyvBS

Active links are also available in Moodle.

Grading Plan : (The table is only indicative)

Type of Evaluation	Weightage (in %)
Quiz-1	5
Mid Sem Exam	10
Quiz-2	5
End Sem Exam	30
Assignments	30
Project	10
Other Evaluation	10

Mapping of Course Outcomes to Program Objectives: (1 – Lowest, 2—Medium, 3 – Highest, or a ‘-’ dash mark if not at all relevant). Program outcomes are posted at

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3	PSO4
CO1	2	2	3	1	3	1	2	-	-	-	-	-	3	2	2	2
CO2	2	2	3	1	3	1	2	-	-	-	-	-	3	2	2	2
CO3	2	2	3	1	3	1	2	-	-	-	-	-	3	2	2	2
CO4	2	2	3	1	3	1	2	-	-	-	-	-	3	2	2	2
CO5	2	2	3	1	3	1	2	-	-	-	-	-	3	2	2	2

Teaching-Learning Strategies in brief (4-5 sentences):

Navigating the ‘Sequence – Structure – Function’ Space for Biomolecules.

Understanding the ‘physicochemical’ principles underlying the ‘structure-interaction-dynamics-function’ of biomolecular machines.

Application of the above principles to analyze the relationship of few molecular systems.

Title of the Course : **Business Fundamentals**
Name of the Faculty : Himanshu Warudkar
Course Code : PD2.421
Name of the Program : M.Tech – Product Design and Management
Credits : 4
L - T - P : 3-0-0
(L - Lecture hours, T-Tutorial hours, P - Practical hours)
Semester, Year : Monsoon 2026

Pre-Requisites : None

Course Objectives:

The course introduces business fundamentals, processes and terminology to non-business students. An understanding of the legal structures (incorporation models), business models, marketing methods and practices, principles of accounting and finance. The students are introduced to the process of business model development through case studies. The methods and approaches offered in this Business Fundamentals course are aimed at improving the capacity of cross-functional teams to collaborate and create excellent products. Students who successfully complete the Business Fundamentals course will have a solid understanding of the critical skills needed for success in product management jobs.

Course Outcomes:

CO1: Demonstrate a clear comprehension of basic concepts of management.

CO2: Exhibit ability to understand the different strategic initiatives adopted by a firm to provide more value to its customers.

CO3: Develop capabilities for demonstrating leadership in decision making in product management roles.

CO4: Develop capabilities to undertake appropriate decision-making to optimize cost in developing a product by appropriately managing resources.

CO5: Develop overall capability required for product development and product marketing.

Course Topics:

- Introduction to Foundations of Business
- Forms of Business Ownership
- Business and Economics
- Organization Structure
- Fundamental areas of Management: Operations Management, HR, Marketing, Accounting and Finance, Information Systems, Strategy Management
- Ethics and Social Responsibility
- Total Quality Management
- Product Development Process and Value Creation
- Managing Teams, Leadership and Global Business

Preferred Textbook :

1. Skripak, Stephen J. 2018. Fundamentals of Business, 2nd Edition, Blacksburg, VA: VT Publishing. <http://hdl.handle.net/10919/84848>. Licensed with CC BY-NC-SA 4.0
<https://creativecommons.org/licenses/by-nc-sa/4.0>

Reference Books/Links :

- Introduction to Business, Gitman et al. An Open tax Publication.
<https://d3bxy9euw4e147.cloudfront.net/oscmsqa/media/documents/IntroductionToBusiness-OP.pdf>

- Fundamentals of business organization and management, 2nd Edition, Y K Bhusan, ISBN: 9789351610618.
- Business Fundamentals: How to Become a Successful Businessman, First Edition, Roger W. Babson.
https://vtechworks.lib.vt.edu/bitstream/handle/10919/111385/Fundamentals_of_Business_4e.pdf?sequence=4&isAllowed=y

Grading Plan:(The table is only indicative)

Type of Evaluation	Weightage (in %)
Quiz-1	10
Mid Sem Exam	20
End Sem Exam	30
Assignments	25
Project	15

Mapping of Course Outcomes to Program Objectives: (1 – Lowest, 2—Medium, 3 – Highest, or a ‘-’ dash mark if not at all relevant). Program outcomes are posted at

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PSO1	PSO2	PSO3	PSO4
CO1	1	2	3	2	2	3	3	1	2	1	2	2	3	2
CO2	2	2	2	2	2	2	2	2	2	3	1	3	2	2
CO3	1	1	2	2	2	2	3	2	2	3	2	2	3	3
CO4	2	2	2	1	1	2	3	1	1	3	2	3	3	2
CO5	3	2	3	3	2	2	3	1	1	3	2	3	2	2

Teaching-Learning Strategies in brief (4-5 sentences):

The sessions would be held in a mix of online and offline classes. The course would involve class lectures, power point presentations, videos, and case study discussions. The students will be required to go through the materials before coming to the class. For each session, every student is expected to go through the suggested topic to facilitate classroom discussion. Furthermore, every student is encouraged to participate in class discussions as the course would adopt an integrative pedagogical approach. Periodic assignments would be given that would require students to reflect and respond based on learning.

Title of the Course : Causal Inference

Faculty Name : Naresh Manwani

Course Code : CS7.506

Credits : 4 Credits

L - T - P : 3:1:0

(L - Lecture hours, T-Tutorial hours, P - Practical hours)

Semester, Year : Monsoon 2026

Pre-Requisites : Course: Statistical Methods in AI,
Knowledge: Machine learning, probability theory, linear algebra

Upper Cap on the Number of Students: 30 (Preference will be given to research students)

Course Outcomes: In statistics, we are used to search for the best predictors of some random variable. In many situations, however, we are interested in predicting a system's behaviour under manipulations.

For such an analysis, we require knowledge about the underlying causal structure of the system. In this course, we study concepts and theory behind causal inference. After doing this course, students will be able to

- CO1: explain/describe differences and similarities between randomized and observational studies of effect.
- CO2: explain/describe the concepts of estimands and counterfactuals
- CO3: explain/describe the basic assumptions needed for causal inference ● CO4: explain/describe the basic rules of causal graph (DAG) theory
- CO4: distinguish between different methods for handling confounding
- CO5: learn and apply foundational causal estimation techniques using Pearls Structural (graphical) causal models to real-world data, in the presence of observed and unobserved variables
- CO6: explain different types of causal discovery algorithms, learn their underlying assumptions and short-comings, and be able to apply them to data using available software

Course Topics :

1. Introduction: Simpson's paradox, applications of causal inference, causality versus correlation
2. Potential Outcome: Potential outcomes and individual treatment effects, the fundamental problem of causal inference.
3. Association and Causation in Graphs: Bayesian networks, causal graphs, chains and forks, colliders and their descendants, d-separation.
4. Causal Models: Do operator and interventional distributions, backdoor adjustment, and structural causal models (SCM)
5. Randomized Experiments: Comparability and co-variate balance, exchangeability, no backdoor paths
6. Frontdoor Adjustment: *do*-calculus, Graph-Based Identification
7. Estimation: Conditional outcome modeling, grouped conditional outcome modeling, propensity score, inverse probability weighting (IPW), doubly robust methods.
8. Unobserved Confounding: No-assumption bounds, monotone treatment response, monotone treatment selection, optimal treatment selection, sensitivity analysis in linear settings.
9. Instrumental variables: No-nonparametric identification of ATE, continuous linear setting, nonparametric identification of local ATE, principal stratification, local ATE.
10. The Difference in Differences: Introducing time, identification.
11. Causal Discovery from Observational Data: Independence-based causal discovery, The PC algorithm, semi-parametric causal discovery, no identifiability without parametric assumption, linear non-Gaussian noise.
12. Causal Discovery from Interventions: Structural interventions, parametric interventions, interventional Markov equivalence,

Preferred Textbooks : Introduction to Causal Inference from a Machine Learning Perspective, Brady Neal, December 17, 2020

Reference Books :

1. Judea Pearl. 2009. Causality: Models, Reasoning and Inference (2nd. ed.). Cambridge University Press, USA.
2. Hernán, M.A., and J.M. Robins. 2020. Causal Inference: What If ? Boca Raton: Chapman & Hall / CRC.
3. Jonas Peters, Dominik Janzing, and Bernhard Schölkopf. 2017. Elements of Causal Inference:

Grading Plan :

Type of Evaluation	Weightage (in %)
Mid Sem Exam	20%
End Sem Exam	40%
Scribing	6%
Programming Assignments (2 x 8%)	16%
Paper Presentation	8%
Use Case Development and Report Writing on it	10%

Mapping of Course Outcomes to Program Objectives: (1 – Lowest, 2—Medium, 3 – Highest, or a ‘-’ dash mark if not at all relevant).

	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PO 12	PS O 1	PS O 2	PS O 3	PS O 4
CO 1	3	3	3	3	3	1	1	1	2	2	2	2	3	3	3	3
CO 2	3	3	3	3	3	1	1	1	2	2	2	2	3	3	3	3
CO 3	3	3	3	3	3	1	1	1	2	2	2	2	3	3	3	3
CO 4	3	3	3	3	3	1	1	1	2	2	2	2	3	3	3	3
CO 5	3	3	3	3	3	1	1	1	2	2	2	2	3	3	3	3
CO 6	3	3	3	3	3	1	1	1	2	2	2	2	3	3	3	3
CO 7	3	3	3	3	3	1	1	1	2	2	2	2	3	3	3	3

Teaching-Learning Strategies: This course emphasizes the mathematical foundations of causal inference and its applications in practice. Hence, it will involve a lot of discussion in class. There are two programming assignments that will help students to work in groups. There are also a use case study and paper reading and presentation which have to be done in groups.

Title of the Course : Chemistry Topics for Engineers
Name of the Faculty : Prabhakar B + Tapan Kumar Sau

Course Code : SC2.309

L-T-P : 3-1-0.

Credits : 4

(L= Lecture hours, T=Tutorial hours, P=Practical hours)

PROGRAM : CSE

1.Prerequisite Course / Knowledge: NONE

2.Course Outcomes (COs): After completing this course successfully, the students will be able to

- CO-1** Demonstrate understanding of the chemical principles behind spectroscopy.
CO-2 Apply the various spectroscopies to solve for chemical identity of system under study.
CO-3 Provide students with a conceptual understanding of electronic properties in solid materials.
CO-4 Define terminology used in the fields of modern materials chemistry.
CO-5 Learn and identify various applications of modern materials and their future potential.
CO-6 Learn the importance and various methods of green synthesis.

3.Mapping of Course Outcomes (COs) with Program Outcomes (POs) and Program Specific Outcomes (PSOs) – Course Articulation Matrix

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3	PSO4
CO1	3	2	2	3	2	2	2	1	1	2	1	1	1	1	1	1
CO2	2	3	2	3	2	2	2	1	1	2	1	1	1	1	1	1
CO3	1	3	3	2	2	2	2	1	1	2	1	1	1	1	1	1
CO4	3	2	2	2	2	2	2	1	1	2	1	1	1	1	1	1
CO5	3	2	3	2	2	2	2	1	1	2	1	1	1	1	1	1
CO6	1	1	2	2	1	2	1	1	1	2	1	1	1	1	1	1

Note: Each Course Outcome (CO) may be mapped with one or more Program Outcomes (POs) and PSOs. Write '3' in the box for 'High-level' mapping, 2 for 'Medium-level' mapping, 1 for 'Low'-level' mapping.

Detailed Syllabus:

Unit 1: Introduction to modern atomic theory and its application to explain the atomic spectra and molecular spectra. Origin of electronic, vibrational and rotational levels, and their signatures in spectra. (3L)

Unit 2: Excited states of atom and molecules. Lasers. Fluorescence, phosphorescence, and Foster energy transfer(2L)

Unit 3: Introduction to Nuclear Magnetic Resonance Spectroscopy, and Electron Spin Resonance Spectroscopy (2L)

Unit 4: Modern Electrochemistry. Butler-Volmer equation. Biosensing using electrochemical principles (4L)

UNIT 5: Molecular Orbital Theory (Polar and Ionic compounds, Conductors, Semiconductor); Inorganic Pigments and Phosphors; and Superconductors Chemistry. (5L)

UNIT 6: Modern and Futuristic Materials: Hydrides and Hydrogen Storage Materials; Molecular Materials; Liquid Crystals; Conducting Polymers; Supramolecular and Nanostructured Materials. (6L)

UNIT 7: Green Synthesis: Principles and methods of synthesis with examples. (1L)

Reference Books:

1. Physical Chemistry by Atkins and De Paula
2. Principles of Molecular Spectroscopy by Barnwell
3. Chemical Principles: The Quest for Insight (5e) by P. Atkins and L. Jones
4. Principles of Modern Chemistry (8e) by D.W. Oxtoby, H.P. Gillis, and L. J. Butler

5.Teaching-Learning Strategies in brief:

This is a course covering (a) spectroscopy and their use in determining composition and other properties in system, and (b) chemical principles behind various materials and devices seen in modern life.

6.Assessment methods and weights in brief:

Assignments + Quizzes – (60%), Final exam (40%)

Title of the Course : Classical Text Readings (Philosophy)

Name of the Faculty : Ashwin Jayanti

Course code : HSo.301

L-T-P : 3-1-0

Credits : 4

Name of the Academic Program: CHD Dual Degree

Semester, Year: Monsoon 2025 Prerequisite Course / Knowledge: Nil Course Description:

How do we know anything? Are there objective truths about right and wrong? Does life have meaning? How do we obtain values from facts? Does scientific knowledge have any limits? Is there free-will? These are some of the questions that will be addressed in this course by recourse to grasping the central debates around these topics and the key thinkers in the history of philosophy— Indian and Western. The course intends to illustrate how philosophy as an activity and practice works in addressing some of the most foundational questions that have been discussed, contested, and debated throughout history. The course is organized both thematically as well as historically so that the students can familiarize themselves with reading original texts to see the developments in philosophical methods in thinking through these themes. Selections from original texts having to do with the following questions will be assigned for in-class discussions: knowledge (how do we know?), metaphysics (what is the nature of reality?), philosophy of mind (how is the mind related to the brain?), ethics (what is the right thing to do?), philosophy of language (how does language relate to the world?), phenomenology and existentialism (what does it mean to be?),

and philosophy of science (what is the scientific method and how does it work? Can there be a science of society?). By way of engaging with these questions, the students shall also gain an appreciation for understanding the relevance of the methods and concepts of philosophy and their significance in our contemporary technoscientific world.

3. Course Outcomes (COs)

After completion of this course successfully students will be able to:

CO-1: Identify and recognize key historical epochs, thinkers, and concepts in the history of philosophy

CO-2: Analyze and evaluate the validity of arguments for and against each of the standpoints on a topic.

CO-3: Understand the significance of philosophical methods and concepts as they apply to addressing some of the most significant questions of a particular time and context.

CO-4: Critically evaluate the effectiveness of philosophical methods and concepts as tools to rationally reconstruct and analyze perspectives on a particular question

CO-5: Apply and synthesize the concepts and methods to think through foundational questions of contemporary significance.

Mapping of Course Outcomes (COs) with Program Outcomes (POs) and Program Specific Outcomes (PSOs) – Course Articulation Matrix

PO1 PO2 PO3 PO4 PO5 PO6 PO7 PO8 PO9 PO10										PO11PO12		PS PS		PS PS	
										O1 O2				O3 O4	
CO1 1	2	2	1	1	3	2	3	2	3	1	3	1	3	1	1
CO2 1	2	1	1	1	3	2	2	1	1	1	3	1	3	1	1
CO3 1	2	1	1	1	3	2	2	1	1	1	3	1	3	1	1
CO4 1	2	2	1	1	3	3	3	1	2	1	3	1	3	1	1
CO5 1	2	3	3	1	3	3	3	1	2	1	3	1	3	1	1

‘3’ in the box denotes ‘High-level’ mapping, 2 for ‘Medium-level’ mapping, 1 for ‘Low’-level’ mapping

Course Modules:

Module 1: The Problem of Knowledge

- Doxa and Episteme: Knowledge as Justified True Belief
- Empiricism and Rationalism
- Synthetic vs. Analytic Judgment
- Pramā (Valid Knowledge) and Apramā (Erroneous Knowledge)
- Inductive vs. Deductive Knowledge

Module 2: Metaphysics

- Being and Becoming
- Potentiality and Actuality
- Universals and Particulars
- Realism versus Conceptualism
- Nominalism versus Realism
- Satkāryavāda and Asatkāryavāda
- Metaphysical Realism and Pragmatism

Module 3: Philosophy of Mind

- Cartesian Dualism: The Mind-Body Debate
- Mind and Cognition: Behaviorism vs. Functionalism
- Internalism vs. Externalism in Mental Content
- Eliminative Materialism and Folk Psychology
- The Hard Problem vs. The Easy Problems of Consciousness
- Anomalous Monism and Epiphenomenalism
- Computationalism and Embodied Cognition

Module 4: Ethics

- Moral Judgment and Emotivism
- Motivation: Hume's Sentimentalism vs. Kant's Rationalism
- Deontology and Virtue Ethics
- Normative Ethics and Anticipatory Ethics
- Moral Realism and Anti-Realism: The Status of Moral Fact
- Moral Luck and Responsibility

Module 5: Philosophy of Language

- Language and Reality: The Relationship Between Words and the World
- Sense and Reference
- Knowledge by Description vs. Knowledge by Acquaintance
- Denotation and Connotation: Literal Meaning and Associated Implications
- Proper Names vs. Descriptions: Reference in Language
- Picture Theory of Meaning vs. Use Theory of Meaning

Module 6: Phenomenology and Existentialism

- Epoche vs Facticity
- Lived Body vs Object Body
- Existence and Essence
- Authenticity vs Inauthenticity
- Self and Other
- Being and Nothingness

Module 7: Philosophy of Science

- Positivism, empiricism, and pragmatism
- Verification and falsification
- Problem of induction
- Quine-Duhem thesis
- Scientific realism
- Facts and values

Reading List

This is an exhaustive list of resources for each topic. The instructor will assign specific sections from some of these readings for the students to familiarize themselves with before each class.

Module 1: The Problem of Knowledge Essential Readings

1. B.K. Matilal. *Essay on Classical Indian Theories of Knowledge*, University of Hawaii Press.
2. Plato. *Theaetetus*, Hackett Publishing Company.
3. David Hume. *A Treatise of Human Nature*, Oxford University Press.
4. René Descartes. *Discourse on the Method*, Oxford University Press.

Recommended Readings:

5. René Descartes. *Meditations on First Philosophy*, Cambridge University Press.
6. Immanuel Kant. *Prolegomena to Any Future Metaphysics*, Cambridge University Press.
7. Rasvihary Das. *A Handbook to Kant's Critique of Pure Reason*, Calcutta University Press.
8. Bertrand Russell. *The Problems of Knowledge*, Home University Library.
9. Gilbert Ryle. *The Concept of Mind*, Hutchinson & Co.
10. C.D. Sharma. *A Critical Survey of Indian Philosophy*, Motilal Banarsidass.

Module 2: Metaphysics Essential Readings:

1. Dutta, S. and Chatterjee, D. *An Introduction to Indian Philosophy*.
2. Lehrer, Keith. *Theory of Knowledge*.

Recommended Readings:

3. Aristotle. *Metaphysics*. Penguin Classics.
4. Putnam, Hilary. *Realism with a Human Face*.
5. Raju, P.T. *Structural Depths of Indian Thought*. SUNY Press.
6. Rorty, Richard. *Philosophy and the Mirror of Nature*.
7. Sharma, C.D. *A Critical Survey of Indian Philosophy*.
8. Nagel, Thomas. *The View from Nowhere*.
9. Stace, W.T. *A Critical History of Greek Philosophy*.

Module 3: Philosophy of Mind Essential Readings:

1. John Searle, *Mind: A Brief Introduction*, OUP, 2004
2. Hilary Putnam. *Reason, Truth and History*, Cambridge University Press.

Recommended Readings:

3. Jeff Coulter and Wes Sharrock, *Brain, Mind and Human Behavior in Contemporary Cognitive Science*, 2007, Edwin Mellen Press.
4. B.Paul M. Churchland. *Matter and Consciousness*, MIT Press.
5. David J. Chalmers. *The Conscious Mind: In Search of a Fundamental Theory*, OU
6. Descartes. *Meditations on First Philosophy*, Cambridge University Press
- Donald Davidson. *Essays on Actions and Events*, Oxford University Press.
7. Jaegwon Kim. *Mind in a Physical World*, MIT Press.
8. Andy Clark. *Being There: Putting Brain, Body, and World Together Again*, MIT Press.
9. Jerry A. Fodor. *The Mind Doesn't Work That Way*, MIT Press.
10. Daniel C. Dennett. *Consciousness Explained*, Little, Brown and Company.
11. Ned Block, Owen Flanagan, and Güven Güzelidere (Eds.). *The Nature of Consciousness: Philosophical Debates*, MIT Press.

Module 4: Ethics

Essential Readings:

1. Aristotle. *Nicomachean Ethics*, Oxford University Press.
2. David Hume. *A Treatise of Human Nature*, Oxford University Press.
3. Immanuel Kant. *The Metaphysics of Morals*, Cambridge University Press.

Recommended Readings:

4. Christine Korsgaard. *The Sources of Normativity*, Cambridge University Press.
5. Alasdair MacIntyre. *After Virtue*, Bloomsbury Academic.
6. Mark Nestor and James M. Wilson. *Anticipatory Ethics and the Use of CRISPR in Humans*, Springer.
7. Russ Shafer-Landau. *Moral Realism: A Defence*, Oxford University Press.
8. J.L. Mackie. *Ethics: Inventing Right and Wrong*, Penguin Books.
9. Thomas Nagel. *Mortal Questions*, Cambridge University Press.
10. Susan Wolf. *Freedom Within Reason*, Oxford University Press.
11. David Edmonds (ed.), *Future Morality*, Oxford University Press, 2021

Module 5: Philosophy of Language Essential Readings:

1. B.K. Matilal. *Mind, Language and World*, Oxford University Press.
2. Gottlob Frege. *Sense and Reference*, in *The Foundations of Arithmetic*, Northwestern University Press.
3. Michael Dummett, *Origins of Analytical Philosophy*, Bloomsbury.
4. Michael Devitt and Kim Sterelny. *Language and Reality: An Introduction to the Philosophy of Language*, MIT Press.

Recommended Readings:

5. Ludwig Wittgenstein. *Tractatus Logico-Philosophicus*, Routledge & Kegan Paul.
6. Ludwig Wittgenstein. *Philosophical Investigations*, Blackwell Publishing.
7. Bertrand Russell. *The Problems of Philosophy*, Oxford University Press.
8. Bertrand Russell. *On Denoting*, *Mind* (journal article, 1905).
9. Saul Kripke. *Naming and Necessity*, Harvard University Press.
10. Hilary Putnam. *The Meaning of 'Meaning'*, University of Minnesota Press.
11. Donald Davidson. *Truth and Meaning*, Synthese Library (book collection).

Module 6: Phenomenology and Existentialism Essential Readings:

1. Edmund Husserl. *Ideas: General Introduction to Pure Phenomenology*, Routledge.
2. Martin Heidegger. *Being and Time*, Harper & Row.
3. Matthew Adamson and Jonardon Ganeri. *Classical Indian Philosophy*, Oxford University Press.
4. Jean-Paul Sartre. *Being and Nothingness*, Washington Square Press.

Recommended Readings:

5. Jonardon Ganeri. *The Concealed Art of the Soul*, Oxford University Press.
6. Maurice Merleau-Ponty. *Phenomenology of Perception*, Routledge.
7. Simone de Beauvoir. *The Ethics of Ambiguity*, Open Road Media.
8. Hans-Georg Gadamer. *Truth and Method*, Continuum.
9. Emmanuel Levinas. *Totality and Infinity*, Duquesne University Press.

Module 7: Philosophy of Science Essential Readings:

1. A.J. Ayer, *Language, Truth, and Logic*, London: V. Gollancz
2. Karl Popper, *The Logic of Scientific Discovery*, New York: Basic Books
3. Thomas Kuhn, *The Structure of Scientific Revolutions*, Chicago: University of Chicago Press

Recommended Readings:

4. Quine, *From a Logical Point of View*, Cambridge: Harvard University Press
5. John Dewey, *Experience and Nature*. Rev. ed. La Salle, IL: Open Court.
6. Hilary Putnam, *The Collapse of the Fact-Value Dichotomy and Other Essays*, Harvard University press
7. Hilary Putnam, *Mind, Language, and Reality*, Cambridge: Cambridge University Press.

Reference Books

1. John Hospers, *An Introduction to Philosophical Analysis*, Allied Publishers
2. Baggini and Fosl, *The Philosopher's Toolkit: A Compendium of Philosophical Concepts and Methods*,
3. Timothy Williamson, *Doing Philosophy: From Common Curiosity to Logical Reasoning*, Oxford University Press
4. Joseph Margolis, *Introduction to Philosophical Problems*
5. Stich and Donaldson, *Philosophy: Asking Questions—Seeking Answers*
6. Simon Blackburn, *Think: A Compelling Introduction to Philosophy*
7. Solomon and Higgins, *The Big Questions: A Short Introduction to Philosophy*, Wadsworth
8. John Hospers, *An Introduction to Philosophical Analysis*, Allied Publishersestions A Short Introduction to Philosophy
9. A.C. Grayling (ed.), *Philosophy: A guide through the subject*
10. Stanley Rosen (ed). *The Philosopher's Handbook: Essential readings from Plato to Kant*
11. Kwame Anthony Appiah, *Thinking it through: An introduction to contemporary philosophy* (2003)

Popular Fiction/Non-Fiction Philosophy Books:

1. Jostein Gaarder, *Sophie's World*, New York :Farrar, Straus and Giroux, 1991.
2. WilliamBarrett, *Irrational Man: A Study in Existential Philosophy*, Anchor Books
3. Apostolos Doxiadia, Christos H. Papadimitriou, *Logicomix: An Epic Search for Truth*, Bloomsbury Publishing India Private Limited
4. David Edmonds, *Would You Kill the Fat Man? The Trolley Problem and What Your Answer Tells Us about Right and Wrong*, Princeton University Press
5. Sarah Blackwell, *At the Existentialist Cafe*, Random House
6. James Hibbard, *The Art of Cycling: Philosophy, Meaning, and a Life on Two Wheels*, Pegasus Books
7. Aaron James, *Surfing with Sartre: An Aquatic Inquiry into a Life of Meaning*, Anchor; Reprint edition
8. Nikhil Krishnan, *A Terribly Serious Adventure:Philosophy at Oxford 1900-60*, Profile Books Ltd
9. Karl Sigmund, *Exact Thinking in Demented Times: The Vienna Circle and the Epic Quest for the Foundations of Science*, Basic Books
10. Robert Pirsig, *Zen and the Art of Motorcycle Maintenance*, Vintage
11. Benjamin J.B. Lipscomb, *The Women Are Up to Something: How Elizabeth Anscombe, Philippa Foot, Mary Midgley, and Iris Murdoch Revolutionized Ethics*, Oxford University Press

12. Clare Mac Cumhaill and Rachael Wiseman, *Metaphysical Animals: How Four Women Brought Philosophy Back to Life*, Chatto & Windus

Grading Plan:

Type of Evaluation	Weightage (%)
Quiz 1	10%
Quiz 2	10%
Mid Sem Exam	20%
Project	20%
End Sem Exam	40%

Project:

The project component requires the students to work in groups of two (depending on the class size) and delve deeper into a particular topic of interest from among the modules discussed. This consists of doing focused and more contemporary readings (assigned by the instructor) on the topic and working towards synthesizing their own reasoned and considered perspective on the topic. The project is an exercise in practicing the philosophical method and using the conceptual toolkit to join the dots across multiple historical contexts to arrive at well-reasoned philosophical analyses on the topic and its implications.

Teaching-Learning Strategies in brief:

The general teaching strategy employed is the use of debates in the history of philosophy to introduce the course topics. Lectures make use of this strategy to impress upon students the need for critical thinking by way of following the example set by the scholars and debates within their historical milieu. The students are expected to read the assigned texts before class and the instructor would facilitate discussion around these to impress the significance of the theme by situating it in its historical milieu. Having gained familiarity with the philosophical method, the students are encouraged to form their own well-reasoned perspectives on the debate in question. The emphasis of the discussions would be to encourage an appreciation for the history of philosophy as well as the relevance of the philosophical method to tackle foundational questions outside the realm of the hard sciences. The course is designed for students to think of the various concepts, methods, styles, and tropes of philosophizing as tools with which to understand, evaluate, question, and synthesize their own reasoned perspectives on questions of contemporary relevance.

Assessment methods and weightages in brief:

This is mainly a reading-driven course, and the discussions are designed to make students grasp the concepts discussed and develop the aptitude to think philosophically. Students are assessed for abilities such as situating the text within its historical milieu, grasping the normative significance of the theme, logically dissecting the debate at hand, questioning the assumptions, clarifying distinctions, and bringing out their own perspectives on these. In assignments and exams, students are expected to demonstrate

these abilities by presenting their views clearly, assessing competing positions systematically, anticipating possible objections to a reasoned conclusion and composing cogent responses to those objections.

Title of the Course : CMOS References and Regulators

Name of the Faculty : Zia Abbas + Abhishek Pullela

Course Code : EC2.410

L-T-P : 3-1-0.

Credits : 4

(L= Lecture hours, T=Tutorial hours, P=Practical hours)

Name of the Academic Program: B. Tech/MS in ECE

Prerequisite Course / Knowledge: Circuit theorems, basics of MOS amplifiers, analog circuits

Course Outcomes (COs):

After completion of this course successfully, the students will be able to.

CO-1: Understand the how power supply is generated in real world applications and trade-offs between power, area, and other parameters.

CO-2: Understand Process/Supply/Temperature/Mismatch/Noise variations in circuits.

CO-3: Design voltage and current references for micro-watt/nano-watt/pico-watt power range applications.

CO-4: Perform stability analyses in complex loops and understand compensation techniques.

CO-5: Design a low drop-out voltage regulator by integrating references and op-amps and understand challenges in integration.

CO-6: Design oscillator circuits like ring oscillator, relaxation oscillator and crystal oscillator.

Mapping of Course Outcomes (COs) with Program Outcomes (POs) and Program Specific Outcomes (PSOs)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3	PSO4
CO1	3	2	1	1	1	1	2	2	2	1	1	2	3	2	2	2
CO2	3	3	2	2	2	1	1	2	1	1	1	1	3	1	1	1
CO3	3	3	2	2	2	1	1	2	2	2	1	2	3	1	1	1
CO4	3	2	3	3	3	1	1	2	2	2	1	3	3	1	1	1
CO5	2	3	3	3	3	1	1	2	3	3	3	3	3	1	2	1
CO6	2	3	3	3	3	1	1	2	3	3	3	3	3	2	1	1

Each Course Outcome (CO) may be mapped with one or more Program Outcomes (POs). Write '3' in the box for 'High-level' mapping, 2 for 'Medium-level' mapping, 1 for 'Low'-level' mapping.

Mapping with PSOs, where applicable.

Detailed Syllabus:

Unit 1: Introduction to Internet-of-Things, Various analog systems in IoT, review of important concepts like op-amp design.

Unit 2: PVT variations, resistor and BJT based PTAT and CTAT generating elements, temperature compensation, defining specifications of voltage reference line regulation, PSRR, accuracy, process-spread, etc., Brokaw bandgap reference, variations of Brokaw bandgap reference, op-amp based bandgap references, op-amp less bandgap references, design procedure for bandgap references, sub-bandgap

references, PSRR analysis in bandgap references, applications of micro-watt bandgap references, start-up circuits, capacitor based and non-capacitor based start-up circuits, switched capacitor bandgap references.

Unit 3: Sub-threshold region of operation, need for MOS based and BJT-MOS hybrid voltage references, MOS only PTAT and CTAT generating cells, 2-transistor based voltage references, process spread comparison between bandgap and MOS based voltage references, trimming circuits for voltage references, BJT and MOS hybrid voltage reference architectures, nano-watt and pico-watt design considerations for IoT applications, 2-transistor PTAT cells, analog temperature sensor, constant gm-bias circuitry, resistor-less and op-amp less reference design, extending analog temperature sensor to digital output sensor using saw-tooth generators, current reference design, op-amp considerations in current references, introduction to gate-leakages in lower technology nodes, gate-leakage based reference architectures.

Unit 4: Integrating bandgap reference, two-stage/folded-cascode op-amps and pass transistor stage in LDO, defining LDO specs like accuracy, line regulation, load regulation, PSRR, loop gain, offset, phase margin, start-up time, undershoot, overshoot, etc., different trade-offs for various industrial applications, NMOS and PMOS LDOs, performance comparison between NMOS and PMOS LDOs, stability compensation techniques, feedforward compensation, DC-DC converter, efficiency comparison between LDOs and DC-DC converters, buck regulator, small-signal model for type-1, type-2, type-3 buck regulator.

Unit 5: Design procedure for LDO, other LDO architectures, comparison between different LDOs, introduction to frequency references, deriving relaxation oscillator from scratch, replacing resistors with references in relaxation oscillators, introduction to ring oscillators, 3-stage and multi-stage ring oscillator design trade-offs, introduction to crystal oscillator and design procedure for crystal oscillator, introduction to voltage-controlled oscillators and their design procedure.

Reference Books:

1. Behzad Razavi. Design of analog CMOS integrated circuits. , 2005.
2. P. E. Allen and D. R. Holberg, "CMOS Analog Circuit Design," 2nd Edition, Oxford University Press, New York, 2004.
3. Power Management Techniques for Integrated Circuit Design By Ke-Horng Chen, Wiley-Blackwell, 2016.
4. Gray, Paul R. and Robert G. Meyer. "Analysis and Design of Analog Integrated Circuits." (1993).

Teaching-Learning Strategies in brief (4 to 5 sentences):

The course will commence by exploring the motivations behind analog circuit design, delving into its real-world applications and the utilization of power management circuits within these contexts. Then, fundamental and core topics of the course will be discussed in detail both at block level and transistor level with hands-on with Cadence Virtuoso tool. During tutorial sessions, select circuits discussed in class will be practically implemented and simulated in different technology nodes using Cadence Virtuoso. This hands-on approach provides students with firsthand experience in circuit design under stringent constraints akin to those encountered in industry applications. Assignments will encompass a blend of equation-solving exercises and analytical questions. Mini projects will consist of small-scale analog blocks (based on industry specs) and the final project will be to integrated all these blocks. To encourage innovation in low-power circuit design, students will be tasked with reviewing research papers, preparing project reports, and delivering presentations to hone their research skills. Assignments, including quizzes and exams, will evaluate students' intuitive grasp of the concepts.

Assessment methods and weightages in brief (4 to 5 sentences):

- Assignments: 20%
- Mid Semester Exam: 10%
- End Semester Exam: 30%
- Mini Projects – 20%
- Project: 20%

Title of the Course	: Chemical Kinetics and Reaction Dynamics
Course Code	: SC2.305
Name of the Faculty	: Tapan Kumar Sau
L-T-P	: 3-1-0
Credits	: 2
Name of the Academic Program	: CND

1.Prerequisite Course / Knowledge None

2.Course Outcomes (COs): After completion of this course successfully, the students will be able to..

CO-1. Determine the rate law for a reaction, the overall order of reaction, the integrated rate laws, the rate constants of the reactions and temperature dependence, and the order of the reaction from concentration/time plots and apply the rate equations to determine the concentration of chemical species and order of the chemical reactions.

CO-2. Explain a reaction mechanism, identify the reaction intermediates and catalysts, determine the molecularity of each step, write the overall reaction, and explain how enzymes act as biological catalysts and why enzymatic reactions respond differently to temperature changes.

CO-3. Interpret a potential energy diagram and a reaction coordinate diagram, potential energy profiles and use them to determine the activation energy and potential energy changes for a reaction.

CO-4. Use Collision Theory to explain how reactions occur at the molecular level, the concept of activation energy and how the collision frequency, kinetic energy, temperature, and orientation of colliding reactant molecules affect the rate of a chemical reaction.

CO-5. Apply transition state theory to explain the roles of various physical factors that govern chemical reactivity.

CO-6. Describe the physical principles that govern electron transfer reactions and explain Marcus theory.

CO-7. Solve problems on chemical kinetics and reaction dynamics of unimolecular, bimolecular, and complex reactions

CO-8. Relate experimental observations to theoretical aspects of chemical kinetics and identify applications of chemical kinetics in everyday life and industry.

3.Mapping of Course Outcomes (COs) with Program Outcomes (POs) and Program Specific Outcomes (PSOs) – Course Articulation Matrix

[illegible]

Note: '3' in the box for 'High-level' mapping, 2 for 'Medium-level' mapping, 1 for 'Low'-level' mapping

4.Detailed Syllabus:

Unit 1: Empirical chemical kinetics: Experimental techniques; The rates of reactions; Integrated rate laws; Reactions approaching equilibrium; The temperature dependence of reaction rates; Accounting for the rate laws; Elementary reactions; Consecutive elementary reactions; Impact on biochemistry: The kinetics of the helix-coil transition in polypeptides; Unimolecular reactions. (2L)

Unit 2: Chain reactions; The rate laws of chain reactions; Explosions; Polymerization kinetics; Stepwise polymerization; Chain polymerization; Homogeneous catalysis; Features of homogeneous catalysis; Enzymes. (2L)

Unit 3: Photochemistry, Kinetics of photophysical and photochemical processes; Impact on: The chemistry of stratospheric ozone; Applications: Impact on environmental sciences, biochemistry, and other areas. (1L)

Unit 4: Molecular Reaction Dynamics: Reactive encounters; Collision theory; Diffusion-controlled reactions; The material balance equation. (2L)

Unit 5: Transition state theory; The Eyring equation; Thermodynamic aspects; The dynamics of molecular collisions; Reactive collisions; Potential energy surfaces; Some results from experiments and calculations. (2L)

Unit 6: The investigation of reaction dynamics with ultrafast laser techniques; Electron transfer in homogeneous systems; The rates of electron transfer processes; Theory of electron transfer processes; Experimental results; Impact on biochemistry: Electron transfer in and between proteins. (2L)

Unit 7: Special topics (oscillating reactions, etc.). (1L)

Reference Books:

1. P. Atkins and J. Paula (2014), *Physical Chemistry*, 10th edition, Oxford University Press, Oxford.
2. D. A. McQuarrie and J. D. Simon (1997), *Physical chemistry: a molecular approach*, 1st edition, University Science Books, California.
3. J. Laidler (1998), *Chemical Kinetics*, 3rd edition, Harper & Row, New York.
4. S. Berry, S. A. Rice and J. Ross (2007), *Physical Chemistry*, 2nd edition, Oxford University Press, Oxford.
5. K. Rohatgi-Mukherjee (2014), *Fundamentals of Photochemistry*, 3rd edition, New Age International Pvt. Ltd., New Delhi.
6. I. Steinfeld, J. S. Francisco, W. L. Hase (1989), *Chemical Kinetics and Dynamics*, 2nd edition, Prentice Hall International Inc., New York.

5.Teaching-Learning Strategies in brief (4 to 5 sentences):

The course involves formal lectures, quizzes, assignments and tutorials.

6.Assessment methods and weightages in brief (4 to 5 sentences):

Students' assessment will be on the basis of:

- | | |
|-----------------------------------|-----|
| 1. Assignments: | 30% |
| 2. Quizzes: | 20% |
| 3. End-Sem Exam (WHOLE Syllabus): | 50% |

Title of the Course : **Complex Analysis**
Name of the Faculty : Lakshmi Burra
Name of the Program : Complex Analysis
Course Code : MA4.406
Credits : 4
L - T - P : 3 -1 -0
(L - Lecture hours, T-Tutorial hours, P - Practical hours)

Semester, Year : Monsoon 2026

Pre-Requisites : Basic Calculus

Course Objectives:

- CO-1 To provide a brief overview to the theories of functions of complex variables; analytic functions.
- CO-2: contour integrations and furnish an introduction to their applications
- CO-3: To understand the concept of zeros and poles,
- CO-4: residues and contour integrations of both complex as well as real functions
- CO-5: Conformal mapping , mobius transformations.

Course Outcomes

After successful completion of the course, students will be able to:

1. Represent complex numbers algebraically and geometrically
2. Analyze limit, continuity and differentiation of functions of complex variables.
3. Understand Cauchy-Riemann equations, analytic functions and various properties of analytic functions.
4. Understand Cauchy theorem and Cauchy integral formulas and apply these to evaluate complex contour integrals.
5. Represent functions as Taylor and Laurent series; classify singularities and poles; find residues and evaluate complex integrals using the residue theorem.
6. Understand conformal mapping

Course Topics :

I.Preliminaries: Brief Overview of Functions: limit, continuity, derivative, analytic functions.

II Complex Integration: line integrals, rectifiable curves,

III Cauchy theorem, index of a closed curve, Cauchy's integral formulae, Cauchy's inequality, Liouville's theorem, Taylor series expansion.

IV Singularities: Laurent series expansions, Removable singularities, Poles and essential singularities, zeros of analytic functions,

V Rouché's theorem, Identity theorem

VI Cauchy's residue theorem; Evaluation of real integrals using Cauchy's residue theorem.

Preferred Textbooks

R.V. Churchill and J.W. Brown, "Complex variables and applications", McGraw Hill Education, 8th edition,

2. . Erwin Kreyszig, "Advanced Engineering Mathematics", Wiley.

Reference Books L. V. Ahlfors, "Complex Analysis", McGraw Hill Education

Grading Plan :

Type of Evaluation	Weightage (in %)
Quiz-1	10
Mid Sem Exam	20
Quiz-2	10
End Sem Exam	30
Assignments	10
Project	20

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3	PSO4
CO1	1	1	0	1	1	1							1	1		
CO2	1	1	1	1	0	1							1	1		
CO3	1	1	1	1	1	1							1	1		
CO4	1	1	0	1	1	1							1	1		
CO5	1	1	0	1	1	1							1	1		

Teaching-Learning Strategies in brief (4-5 sentences):

Active Learning Strategies

- **Solve Independently First:** Before consulting solution manuals attempt problems on your own. This builds necessary problem-solving skills.
- **Practice with Concrete Examples:** Work through examples focusing on contour integration, as this is a central application.
- **Use Visual Tools:** Use mapping examples to understand conformal mappings

Title of the Course : Computational Linguistics 2
Name of the Faculty : Rajakrishnan P Rajkumar
Course Code : CL3.202
L-T-P : 3-1-0
Credits : 4
Name of the Academic Program: : CLD

1.Prerequisite Course / Knowledge : Introduction to Linguistics-1 and 2; Computational Linguistics 1

2.Course Outcomes (COs) (5 to 8 for a 3 or 4 credit course):

After completion of this course successfully, the students will be able to:

- CO-1 Use computational methods to analyze language at semantic and pragmatic levels
- CO-2 Develop requisite skills for problem solving at discourse and conversation levels
- CO-3 **Develop** computational resources and tools for handling text, contextual interpretation of text and representation of meaning in context.
- CO-4 **Perform** theoretical research in computational semantics and computational discourse analysis
- CO-5 **Apply** CL/NLP techniques for real world applications by using real time dialog and discourse data

3. Mapping of Course Outcomes (COs) with Program Outcomes (POs) and Program Specific Outcomes (PSOs) – Course Articulation Matrix

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3	PSO4
CO1	3	2	3	1	1	3	1	2	2	1	1	2	3	1	2	3
CO2	3	3	3	3	3	3	1	2	2	1	1	2	3	1	2	3
CO3	3	2	3	3	3	3	3	2	2	3	1	1	3	1	3	3
CO4	3	2	3	3	3	3	1	2	2	1	1	2	3	1	3	3
CO5	2	2	3	1	1	3	1	2	3	3	1	3	3	1	2	3

Note: '3' in the box for 'High-level' mapping, 2 for 'Medium-level' mapping, 1 for 'Low'-level' mapping

4. Detailed Syllabus:

Unit 1: Background for studying word meaning and sentence meaning, sentence meaning and propositional content; sense and reference; referent, extension, prototype, stereotype; deixis and definiteness; predicates, referring expressions, universe of discourse; properties of sentences - analytic, contradiction, entailment; properties of predicates - reflexive, symmetry, transitive. Word meaning and sentence meaning, content word and grammatical word, contextual variation; Speaker meaning vs Sentence meaning. Building resources using Lexical semantic relations - Synonymy, Antonymy, Hyponymy, Troponymy, Meronymy; Metaphor and Metonymy; Polysemy and Homonymy; Semantic fields; Lexical ambiguity; Building dictionaries; Ontologies.

Unit 2: Formal Semantics: Formal representation of natural language - semantic features, case frames, semantic primitives. Logic, notation for simple propositions; connectives – and, or, but, if etc.; Logical expressions for ambiguous sentences

Unit 3: Pragmatics and Discourse: Pragmatics and Discourse analysis as a study of context dependent aspects of meaning; text, co-text, context and relevance. Computational Discourse analysis: Studying Structure of text and coherence; exchange structure and conversational analysis; turn taking; adjacency pairs; preference organization; deixis; anaphora; ellipsis; discourse connectives and relations; Structural analysis of different kinds of texts.

Unit 4: Text classification and generation: Memory and knowledge representation as schemas - frames, scripts and story grammar; Generation and processing of texts: Sentiment Analysis. Humour Analysis.

Unit 5: Computational Pragmatics: Language Understanding; Meaning beyond textual context; speaker's intention and hearer's inference; inference - bridging inferences, causal and spatial inferences, elaborative and restrictive inferences; Application of pragmatic concepts in Dialogue Systems: conversational implicature, conventional implicature, entailment and presupposition; co- operative interaction and Gricean maxims; speech act theory; language as action, performatives, direct and indirect speech acts and felicity conditions; politeness maxims; Austin and Searle's speech acts; Dialogue data annotation: Dialog Acts, Rhetorical Structure Theory

Reference Books:

1. Jurafsky & Martin, 2000; Speech and Language Processing, Pearson Education
2. James R. Hurford & Brendan Heasley (1983). SEMANTICS - a course book. Cambridge University Press.
3. Judith Greene (1986). Language Understanding - a cognitive approach. Open University Press.
4. Lyons, John. (1977). Semantics. Cambridge University Press.
5. Levinson, Stephen C. (1983). Pragmatics. Cambridge University Press. Leech, Geoffrey. (1983). Principles of Pragmatics. Longman.

6. Brown, G and Yule, G. (1983). Discourse Analysis. Cambridge University Press. Cutting, Joan (2002). Pragmatics and Discourse: A resource book for students.
7. Allen, James. (1994). Natural Language Understanding. Pearson.

5. Teaching-Learning Strategies in brief (4 to 5 sentences):

This is a mix of theory and project based. The focus is on using the methods taught in class to extend to real time situations and uses.

6. Assessment methods and weightages in brief (4 to 5 sentences):

How the students are able to connect the linguistic concepts by using computational techniques to analyze and generate data at the level of semantics and pragmatics. The course will have a project content where students will study and solve a problem using real language data. The focus is on individual as well as collaborative learning.

Type of Evaluation	Weightage (in %)
Assignments	15%,
Seminar	10%
Project	25%,
Midsem Exam	15%,
Endsem Exam	35%

Title of the Course : Computer Aided Drug Design (CADD)

FACULTY NAME : Deva Priya Kumar

COURSE CODE : SC4.415

CREDITS : 4

TYPE-WHEN : Elective

PRE-REQUISITE: Molecular architecture or advanced biomolecular architecture or equivalent. Those who are interested in taking the course and do not have one of these prerequisites; please talk to the course instructor.

OBJECTIVE: This course aims to introduce principal concepts in drug design, and specifically the role of computational models. Various methods that are used in computer aided drug design would be discussed, and the basic principles to understand these methods will also be taught.

COURSE TOPICS: Course overview and Introduction (What to expect for the next 15 weeks) Introduction to few computational methods

-Potential energy surface

-Force Field parameters

-Conformational analysis

-Energy calculations using molecular mechanics

-Free energy of binding

- Basics of chemical bonding Drug design
- Drug -Traditional drug design
- Drug-receptor interactions
- Biological activity and various measures
- Importance of computational methods in drug design Databases
- biomolecules & small molecules Ligand based drug design
- Structure activity relationship
- Quantitative structure activity relationship (2D & 3D methods)
- QSAR parameters
- QSAR validation
- Pharmacophore based models Structure based drug design
- Docking
- Discussion of various common algorithms
- Protein flexibility in docking
- Scoring functions
- De novo drug design

Molecular similarity analysis

ADME-T prediction Advanced topics in drug design (Student Seminars)

PREFERRED TEXTBOOKS: Textbook of drug design & discovery (Ed. Krogsgaard-Larsen & Lilijefors and Madsen) 2001, CRC; Pharmacophore perception, development, and use in drug design (Ed. Gunter) 2000, IUL; Molecular modeling: principles and applications (Leach) 2001, Prentice Hall; Molecular modeling and simulations (Schlick) 2002, Springer.

***REFERENCE BOOKS:** Material will be given from time to time

***PROJECT:**

GRADING: Exams-40% (10 + 10 + 20);

Assignments-40%;

Lab Assignment-10%;

Seminars/Quiz-10%

OUTCOME: Understanding of the basic principles involved in computer aided drug design and, to be able to appreciate the utility of the techniques in pharmaceutical industry.

Remarks:

Title of the Course : **Computer Problem Solving**
Name of the Faculty : Shatrunjay Rawat
Course Code : CS0.301
Name of the Academic Program : MTech CASE
L-T-P : 3-1-0
Credits : 4

1.Prerequisite Course / Knowledge:

None

2.Course Outcomes (COs) (5 to 8 for a 3 or 4 credit course):

After completion of this course successfully, the students will be able to..

CO-1: Demonstrate an understanding of computer programming language concepts.

CO-2: Ability to design and develop C programs, implement the concept of pointers, declarations, initialization, operations on pointers and their usage, arrays, functions. Able to define data types and use them.

CO-3: Ability to define and manage data structures based on problem subject domain.

CO-4: Ability to analyze the complexity of the solution offered.

3.Mapping of Course Outcomes (COs) with Program Outcomes (POs) and Program Specific Outcomes (PSOs) – Course Articulation Matrix

Note: Each Course Outcome (CO) may be mapped with one or more Program Outcomes (POs) and PSOs. Write '3' in the box for 'High-level' mapping, 2 for 'Medium-level' mapping, 1 for 'Low'-level' mapping

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3	PSO4
CO1	1	2	3	1	-	-	-	-	-	-	-	-	-	-	3	3
CO2	1	2	3	1	-	-	-	-	-	-	-	-	-	-	3	3
CO3	1	2	3	1	-	-	-	-	-	-	-	-	-	-	3	3
CO4	1	2	3	1	-	-	-	-	-	-	-	-	-	3	3	3

4.Detailed Syllabus:

Unit 1: Basics of C Programming, Variables declaration, Input-Output, Operators

Unit 2: Arrays and Strings and Control Statements

Unit 3: Functions, Pointers

Unit 4: File Handling, Memory management

Unit 5: Stacks, queues, Linked Lists

Unit 6: Sorting Algorithms

Unit 7: Understanding Algorithm Complexity

Unit 8: Problem Solving and Computations Thinking

Reference Books:

1. The C Programming Language -Brian Kernighan and Dennis Ritchie
2. Art of Computer Programming, Vol. I, II, III, D. E. Knuth, Addison Wesley.
3. How to solve it by computer, R. Dromey, Prentice-Hall India

5. Teaching-Learning Strategies in brief (4 to 5 sentences):

Any concept introduced in the lecture is followed by implementation in the lab session and further discussion and practice on the same in the tutorial. Assignments help students to think about implementing the most efficient solution. Project is to be chosen from the domain the student is from. This brings in a practical application of programming to the field of work.

6. Assessment methods and weightages in brief (4 to 5 sentences)

Term Papers (2): 15% + 20 %

Quizzes: 20 %

Lab Exams: 10%

Project and assignments: 35%

Title of the Course : Computer Programming

Faculty Name : Abhishek Deshpande + Vineet Gandhi + Charu Sharma

Course Code : CS0.101

L-T-P : 3-1-3

Credits : 5

(L = Lecture Hours, T = Tutorial Hours, P = Practical Hours)

Name of the Academic Program: B.Tech

Prerequisite Course / Knowledge:

Basic high school mathematics.

Course Outcomes (COs):

After completion of this course successfully, the students will be able to...

CO-1: Demonstrate an understanding of computer programming language concepts like variables, data types, conditional, loop, print statements and functions.

CO-2: Ability to use pointers, arrays, initialization, dynamic memory allocation and use gdb to debug issues with the code.

CO-3: Design and implement algorithms using recursing, iteration with specific use cases like sorting.

CO-4: Design structured data storage using structs, and read/write them to hard disk using file I/O

CO-5: Ability to analyze the complexity of the solution offered in terms of memory and runtime and choose efficient alternatives.

CO-6: Develop large scale programs using multiple files, header files and make.

Mapping of Course Outcomes (COs) with Program Outcomes (POs) and Program Specific Outcomes (PSOs)

	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PO 12	PS O1	PS O2	PS O3	PS O4
CO1	2	3	1	3	1	1	1	1	2	2	1	3	1	1	1	2
CO2	1	2	2	2	1	1	1	1	2	2	1	1	3	1	1	2
CO3	2	3	1	3	1	1	1	1	2	2	1	3	2	1	1	2
CO4	1	3	2	2	1	1	1	1	2	2	1	1	3	1	2	2
CO5	1	3	2	3	1	1	1	1	2	2	1	1	3	1	1	3
CO6	1	2	1	2	2	1	1	1	2	2	1	2	3	2	2	2

'3' for 'High-level' mapping, 2 for 'Medium-level' mapping, 1 for 'Low'-level' mapping.

Detailed Syllabus:

Unit 1: Basic computer organization, High level programming languages, assemble code, binary instructions, compilers. Program editing, compilation and execution cycle. Basic data types and their representation, Expressions, Operators and precedence levels, Basic I/O functions, Writing straight-line sequence of code, Conditional Statements (if-then-else, switch case). Functions

Unit 2: Loops (for, while etc.), Arrays, Multidimensional Arrays, Parameter passing mechanisms, Pointers, Strings, Pointer Arithmetic.

Unit 3: Recursion, Program stack, scope and lifetime of variables., heap memory, dynamic memory management, linked lists and memory leaks.

Unit 4: Structures, and Enumerations, Standard libraries for string manipulation, disk file access etc. Preprocessor directives, multi-file programming and Make files.

Reference Books:

1. Computer Science: A Structured Programming Approach Using C, 3rd Edition by Behrouz A. Forouzan, Richard F. Gilberg.
2. The C Programming Language -Brian Kernighan and Dennis Ritchie.
3. Let Us C by Kanetkar Yashavant P.

Teaching-Learning Strategies in brief (4 to 5 sentences):

Lectures are conducted in a highly interactive fashion. At the end of every class, a small practice problem which helps in enhancing the concepts discussed in the class will be released. Tutorial sessions are used to teach the utilization of tools such as Visual Studio Code, GCC, GDB, GIT, Make files, perf, valgrind etc. Lab sessions are used to solve programming assignments and teaching assistants help students in developing program logic, debugging etc. on an individual basis. Faculty conducts office hours once in week. On the rest of the days, teaching assistants conduct office hours. This ensures continuous support to students. Five to six programming assignments are designed which gives an in-depth understanding of various concepts discussed in the class and their application to new problem scenarios along with proper analysis.

Assessment methods and weightages in brief (4 to 5 sentences):

- Labs: 20%
- Assignments: 30%
- Mid Sem: 20%
- End Exam: 30%

Title of the Course : **Computing in Sciences-1**

Name of the Faculty : Marimuthu Krishnan

Course Code : SC4.101

L-T-P : 2-0-3

Credits : 2

1.Prerequisite Course / Knowledge:

Familiarity with running programs in BASH shell.

2.Course Outcomes (COs):

After completion of this course successfully, the students will be able to..

CO-1 Demonstrate understanding of basic concepts of molecular modelling

CO-2 **Demonstrate the familiarity in operating** prepackaged software commonly used in molecular mechanics, quantum chemistry, and visualization of molecular systems

CO-3 Compute (a) simple thermodynamic properties using quantum chemistry software for small molecules (b) thermodynamic properties using molecular dynamics software

3.Mapping of Course Outcomes (COs) with Program Outcomes (POs) and Program Specific Outcomes (PSOs) – Course Articulation Matrix

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3	PSO4
CO1	2	1	1	1	1	2	1	1	2	1	2	2	2	2	1	1
CO2	2	2	2	2	2	3	2	1	1	2	1	2	1	1	1	1
CO3	2	2	1	2	2	3	2	1	2	2	1	2	1	1	1	1

Note: '3' in the box for 'High-level' mapping, 2 for 'Medium-level' mapping, 1 for 'Low'-level' mapping

4.Detailed Syllabus:

Unit 1: Basic outline of application of quantum mechanics to molecule/s. Variational Theorem, and introduction to density functional theory. Basis sets. Geometry Optimization and Frequency calculation. And other 'simple' properties from output of Quantum Chemistry software

Unit 2: Motivation for Classical mechanical models of molecules. Use of Newtonian equations of motion to model dynamics of molecular systems. Motivation for ergodic hypothesis, and calculation of thermodynamic properties.

Reference Books:

1. Molecular Modeling by Andrew Leach
2. Molecular Modeling for beginners by Alan Hinchliffe
3. Software user manual for the following: GAUSSIAN, MOLDEN, NAMD and VMD.

5.Teaching-Learning Strategies in brief (4 to 5 sentences):

A lecture on a theory concept will be immediately followed by on-hands-practice using appropriate scientific software. Students will be encouraged to read carefully all the log files and become familiar with technical language, in addition to homework and assignments which will be mini projects with specific task.

6.Assessment methods and weightages in brief (4 to 5 sentences):

The course will rely heavily on the submission of work done using scientific software like setting up input files, preparing initial molecular structures; this work will be in-class or tutorial submissions, homework etc.

End semester will carry 40% weightage of which 75% component will be based on demonstrating familiarity with scientific software taught in the course.

Homework and assignments will carry a weightage of 25%, while the in-class and tutorial submission will be another 10%. Finally, the remainder 15% will be theory part.

Title of the Course : Cyber GIS: Geospatial Web and GeoBI

Name of the Faculty : Kuldeep Kurte + K S Rajan

Course Code : GS2.402

Name of the Program: Open to All Programs on Campus at UG, PG/PhD Level

Graduate Level UG4 & Entry MS (MS/MTech/PhD) Credits

Credits : 4

L - T - P : 3-1-0

(L - Lecture hours, T-Tutorial hours, P - Practical hours)

Semester, Year : Monsoon, 2026

Pre-Requisites : Basic understanding of Locational Data and Computing –

Any UG3, UG4, M.Tech., MS, and Ph.D. student should be able to take it.

Prior course work in Spatial Informatics or Intro to Spatial Science and Technology courses may help.

Course Outcomes :

CO-1: Recognize and explain the challenges of the publishing geospatial data on the web CO-2: Employ various geospatial standards and technologies to publish geospatial data on the web

CO-3: Understand and explain the importance of the geospatial analytics in the geospatial web ecosystem

CO-4: Analyze and evaluate various use-cases of WebGIS systems

CO-5: Design research projects that helps synthesize the learning into an application

Course Topics

Module 1: DesktopGIS to WebGIS/MobileGIS: Publishing maps online

This module introduces the fundamentals of geospatial web and the process of publishing maps online by focusing on the following sub-topics

- Basics of spatial data, spatial data handling, spatial data models and formats
- Digital cartography, map elements, coordinate reference systems, projection systems
- WebGIS history and standardization efforts
- Web Map Services (WMS) and its protocols to publish and disseminate maps online – both Web and on Mobile, basic Map based Spatial queries

Module 2: Technologies for online map publication

This module will continue to focus on the standardization efforts and technological stack required for standardized way of online map data collection, collation, editing, and generation including crowd sourced data management

- Addressing syntactic heterogeneity through OGC standards implementing XML remote procedure call over HTTP, including standards related to data models and encoding, subscription management and other syndication purposes, accessing sensor, searching and finding geospatial resources (discovery, cataloguing/CSW and Registry/WRS, Metadata standards), storing and retrieving geospatial data efficiently.
- Issues related to concurrency handling for Big data, , GML and GeoJSON, Geospatial Web Services covering WMS, WFS, WCS, and WPS, and OGC API services and standards.

Module 3: Data-related challenges for map publication

This module will discuss various data-related challenges that occur when publishing existing data to the web or while collecting crowd-sourced data.

- Challenges due to the heterogeneous representation of same information, some examples illustrating semantic heterogeneity challenges in geospatial web.
- Use of Semantic Web technology stack and W3C standards: Resource Description Framework, Web Ontology Language (OWL), SPARQL, OGC GeoSPARQL standards
- Introduction to various Linked Geospatial Data such as geonames, knowwheregraph, dbpedia

Module 4: Spatial OLAP leading to GeoBI

This module will focus on bringing the geospatial analysis capability to the geospatial web in order to extract relevant insights for stakeholders. This includes the aspects of geo- visualization and GeoHPC.

- Choropleth, Dasymetric, Isarithmic, multivariate mapping, terrain visualization
- Data exploration, geo-visualization and analytics
- Large-scale data visualization and processing (WebGIS backend)
- HPC for geospatial web
- Creating analysis ready data using Open Data Cubes (ODC) standards

Module 5: Learning by demonstration

The purpose of this module is to introduce students to various case studies from different domains such as public health, agriculture, location-based services, disaster situational awareness, etc. Further, students will pick their application of choice and implement an end-to-end geospatial web system by incorporating the learnings from other modules.

- Develop MobileGIS App and WebGIS dashboard for analysis and decision making
- Online Data Visualization and Interaction design- Mapping Multi parameter data on the web and developing Map mashups
- Mapping as a service to harness geospatial processing functionalities

Preferred Textbooks:

1. Geospatial Web Services: Advances in Information Interoperability. IGI Global. 2011.
2. Spatial Computing, By Shashi Shekar and Pamela Vold. The MIT Press. 2020.
3. A Semantic Web Primer, second edition, Grigoris Antoniou and Frank van Harmelen. The MIT press. 2008. Foundations of Semantic Web Technologies, By Pascal Hitzler, Markus Krotzsch, Sebastian Rudolph, 1st edition, CRC press. 2009.
4. Thematic Cartography and Geovisualization. By Terry A. Slocum, Robert B. McMaster, Fritz C. Kessler, Hugh H. Howard. CRC Press; 4th edition 2022.
5. Selected Research Papers and Articles (will be shared with the topics taught on the course portal)

Reference Books :

1. Geographical Data Science and Spatial Data Analysis - An Introduction in R. By Lex Comber and Chris Brunsdon. SAGE Publications Ltd. 2020
2. GIS – A computing perspective. By Micheal Worboys and Matt Duckham. CRC Press; 2nd edition 2004

E-book Links Will be provided in Class as appropriate

Grading Plan :

Type of Evaluation	Weightage (in %)
Class Quizzes + Assignments	20.0
Mid Sem Exams	15.0
End Sem Exam	25.0
Project/Term paper demonstrating the Practical applications	40

Mapping of Course Outcomes to Program Objectives:

	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PO 12	PSO 1	PSO 2	PSO 3	PSO 4
CO 1	1	2	1	1	-	1	1	1	-	2	-	1	3	2	1	2
CO 2	3	-	2	-	3	2	-	-	-	-	1	-	3	2	1	1
CO 3	-	1	2	1	3	-	2	1	3	2	1	1	3	2	1	1
CO 4	2	3	2	2	1	2	2	-	3	2	1	1	3	2	-	2
CO 5	3	2	3	2	3	1	3	3	3	3	3	2	3	3	2	3

Teaching-Learning Strategies in brief (4-5 sentences):

Teaching – Learning Lectures

Guest Lectures

Reading research papers

Class participation in Q&A, discussions Online discussions over MS

Teams

Learning by doing

Hands on exercises – in class and assignments

Short Presentation and Discussion led by Student. Course project on conceptualization and implementation Real world applications

Multi-disciplinary approach

Title of the course : Data Analytics-I

Name of the Faculty : Krishna Reddy Polepalli

Course Code : CS4.405

L-T-P : 3-1-0

Credits : 4

(L= Lecture hours, T=Tutorial hours, P=Practical hours)

TYPE-WHEN Fifth semester and onwards

1. Prerequisite Course / Knowledge:

(i) Data and Applications, or equivalent courses that cover Data modelling, normalization, SQL

(ii) First courses on programming, data-structures and algorithms

(iii) Basics of Python language, to be able to use relevant libraries and toolkits for data analytics

2. Course Outcomes (COs)

Objective: In a computerized and networked society, vast amount of data is being collected every day in multiple domains. We are drowning in data but starving for knowledge or actionable insights. Data mining or data analytics constitute a collection of concepts and algorithms, which are being developed to answer “how” questions by extracting interesting and useful knowledge from large data. Data analytics-based platforms are being operated in multiple domains to extract valuable and actionable insights from the data to improve the business performance. The objective of this first level course is to learn the important concepts and algorithms related to data mining functionalities such as summarization, pattern mining,

classification, clustering and outlier analysis.

The Course Outcomes (COs) are as follows:

- After completing the course successfully, the students are able to
 - CO-1. describe the concepts of data summarization, data warehousing, pattern mining, classification and clustering approaches
 - CO-2. perform the task of data summarization, pattern mining, classification and clustering based on the requirement.
 - CO-3. prescribe a single or a combination of data summarization, pattern mining, classification and clustering approaches for the problem scenario of a business/organization.
 - CO-4. construct the improved data analytics methods for existing services.
 - CO-5. formulate new data mining problems for creating new services and design the corresponding solutions

3. Mapping of Course Outcomes (COs) with Program Outcomes (POs) and Program Specific Outcomes (PSOs) – Course Articulation Matrix

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3	PSO4
CO1	3	2	3	3	3	3	2	2	3	2	2	3	2	2	2	3
CO2	3	1	3	1	3	2	1	1	3	3	2	3	2	1	2	3
CO3	1	3	1	3	1	1	1	1	1	1	1	3	3	1	2	3
CO4	1	2	1	3	3	1	1	1	3	1	2	3	3	1	2	3
CO5	2	3	3	1	1	1	2	1	3	3	2	3	2	2	2	3

‘3’ in the box denotes ‘High-level’ mapping, 2 for ‘Medium-level’ mapping, 1 for ‘Low-level’ mapping.

4. Detailed Syllabus

(please list the order in which they will be covered)

Unit 1: Introduction, data summarization through characterization, discrimination and data warehousing techniques (9 hours)

Unit 2: Concepts and algorithms for mining patterns and associations (9 hours) Unit 3:

Concepts and algorithms related to classification and regression (9 hours) Unit 4:

Concepts and algorithms for clustering the data (9 hours)

Unit 5: Outlier analysis and future trends. (3 hours)

- Five mini projects related to the above syllabus will be done by students in the laboratory

Reference Books and materials:

1. Book: Jiawei Han and Micheline Kamber, Data Mining: Concepts and Techniques, Third edition, 2012, Elsevier Inc.
2. Book: Pang-Nong Tan, Michael Steinbach and Vipin Kumar, Introduction to Data Mining, 2006, Pearson Education.
3. Research Papers: About 25 research papers from the proceeding of the conferences and journals related to data summarization, data warehousing, pattern mining, classification, clustering, outlier detection.

5. Teaching-Learning Strategies in brief

Lectures by integrating ICT into classroom teaching, weekly tutorials involving problem solving and active learning by students and Project-based Learning by doing 5 mini-projects in laboratory by the students

6. Assessment methods and weightages in brief

Two Classroom tests: 10 marks; Mid Semester Examination in theory: 20 marks, End Semester Examination in Theory: 40 marks, Assessment of 5 mini projects in Laboratory: 30 marks

Title of the Course : Data and Applications [Half]

Name of the Faculty : Kamal Karlapalem

Course Code : CS4.301

L-T-P : 3-1-0

Credits : 2

(L= Lecture hours, T=Tutorial hours, P=Practical hours)

Name of the Academic Program: B.Tech. in Computer Science and Engineering

1.Prerequisite Course / Knowledge: Data Structures

2.Course Outcomes (COs) After completion of this course successfully, the students will be able to–

CO-1. State data requirements for an application.

CO-2. Develop a conceptual model (such as, Entity Relationship Model and Diagram) for a set of data requirements.

CO-3: Comprehend relational data model and integrity constraints, and relational database design with normalization.

CO-4. Map the conceptual model to a relational data model and create and populate its corresponding relational database

CO-5. Map user queries into correct relational algebra, Structured Query Language (SQL), and tuple relational calculus expressions/statements. And updates using SQL.

CO-6. Implement an application to access, query and update a relational database.

3.Mapping of Course Outcomes (COs) with Program Outcomes (POs) and Program Specific Outcomes (PSOs) – Course Articulation Matrix

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3	PSO4
CO1	3	3	3	3	1	2	1	2	3	3	3	3	3	3	1	1
CO2	3	3	3	1	3	-	-	-	3	3	3	3	3	3	1	1
CO3	3	2	2	1	2	-	-	-	1	1	1	3	3	3	2	3
CO4	3	3	3	3	1	2	1	2	3	3	3	3	3	3	1	3
CO5	3	3	3	3	1	2	1	2	3	3	3	3	3	3	1	3
CO6	3	3	3	3	1	2	1	2	3	3	3	3	3	3	1	3

‘3’ in the box denotes ‘High-level’ mapping, 2 for ‘Medium-level’ mapping, 1 for ‘Low’-level’ mapping

4.Detailed Syllabus:

Unit 1: Data, Database, Database System (3 hours)

Unit 2: Data models, Conceptual Data Modeling, ER Models (5 hours)

Unit 3: Relational Data Model, Relational Algebra, Tuple Relational Calculus (6 hours)

Unit 4: SQL, Constraints, Triggers, Database Connectivity, Applications (3 hours)

Unit 5: Normalization, Relational Database Design (4 hours)

1. Four mini projects related to the above syllabus will be done by students. References:

2. Elmasri & Navathe, Fundamentals of Database Systems, 6th Edition, Pearson Education, 2013

3. Abraham Silberschatz, Henry F. Korth, S. Sudarshan, Database system concepts, fifth edition, Mc Graw Hill, 2006

5. Teaching-Learning Strategies in brief:

Lectures by integrating ICT into classroom teaching, weekly tutorials involving problem solving and active learning by students and Project-based Learning by doing four mini-projects.

6. Assessment methods and weightages in brief:

Assignments in theory: 10 marks, Quizzes in theory: 10 marks, Mid Semester Examination in theory: 20 marks, End Semester Examination in Theory: 30 marks, Assessment of four mini projects: 30 marks

Title of the Course : Data Structures & Algorithms for Problem Solving

Name of the Faculty : Lini Thomas + Kshitij Gajjar

Course Code : CS1.304

Credits : 6

L - T - P : 3-0 -2

(L - Lecture hours, T-Tutorial hours, P - Practical hours)

Semester, Year : Monsoon 2026

Name of the Program: M. Tech in CSE/CSIS

Pre-Requisites : Basic computer programming
Basic Mathematics

Course Outcomes :

After completion of this course successfully, the students will be able to:

CO-1: Understanding of fundamental and advanced Data Structures including linked-lists, trees, binary search trees, AVL trees, stacks, queues, heaps, hash-table, tries and suffix tree.

CO-2: Ability to program data structures and use them in implementations of abstract data types.

CO-3: Ability to devise novel solutions to small scale programming challenges involving data structures and recursion.

CO-4: Understand basic algorithms including recursion, searching, hashing, dynamic programming, and traversal.

CO-5: Understanding of basic algorithmic complexity. Ability to perform simple inductive proofs and proofs by contradiction and reason about program correctness and invariants.

CO-6: Given a real-world problem have ability to sensibly select appropriate data structures and algorithms for solving the problem and be able to implement the solution.

Course Topics :

(Please list the order in which they will be covered, and preferably arrange these as five to six modules.)

Unit 1: Review of basic data structures and algorithms (Linked list, stack, queue, 2D arrays, dynamic programming, recursion etc.)

Unit 2: Algorithms on Trees and Graphs (Binary Search Tree, AVL Tree, Heaps, Graph Traversal, shortest path algorithms etc.)

Unit 3: Problem solving with other data structures and algorithms (Hashing, Tries, Splay Trees, Range Trees, sorting etc.)

Reference Books: Introduction to Algorithms, 3rd Edition (The MIT Press) 3rd Edition

by [Thomas H. Cormen](#) , [Charles E. Leiserson](#), [Ronald L. Rivest](#), and [Clifford Stein](#)

Grading Plan : (The table is only indicative)

Type of Evaluation	Weightage (in %)
Quiz-1	5
Mid Sem Exam	10
Quiz-2	5
End Sem Exam	25
Assignments	20
Weekly Labs	10
Mid Lab exam	10
Final Lab exam	15

Mapping of Course Outcomes to Program Objectives:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3	PSO4
CO1	3	3	3	2	2	2	2	1	2	1	1	3	3	3	2	3
CO2	3	3	3	2	2	1	1	1	2	1	1	3	2	2	2	2
CO3	2	3	3	2	2	2	2	3	3	2	2	2	2	2	2	2
CO4	3	3	3	2	2	2	2	1	2	1	1	3	3	3	2	3
CO5	2	2	2	3	1	1	1	1	2	1	1	3	3	2	2	3
CO6	2	2	3	2	2	2	3	2	3	3	2	2	3	2	3	3

Teaching-Learning Strategies in brief (4-5 sentences) :

The course involves heavy theory and programming components. The strategy is to first thoroughly cover all the basics, with board-work-based teaching. The basics will be covered by solving examples problems, analyzing complexity, writing pseudo codes on board. In second stage, each class will introduce a problem statement, discuss solutions which go beyond the already covered topics. Introduce the novel algorithm or data structure, solve the problem in class and cement the idea and use case. Give them ideas on other problems where the discussed algorithm or data structure can be applied. The course will also have weekly labs, which will allow students to practice and code problems related to the topics covered in the class. There will be regular assignments with focus on the problem-solving aspect.

Title of the Course : Design for Social Innovation

Name of the Faculty : Ramesh Loganathan + Arjun Rajashekar

Course Code : CS9.429

L-T-P : 3-1-0

Credits : 4

(L= Lecture hours, T=Tutorial hours, P=Practical hours)

Name of the Academic Program: Bachelor of Technology in Computer Science and Engineering

1.Prerequisite Course / Knowledge: None specific.

General understanding of systems & design. Any UG3, UG4, M.Tech.& MS/Ph.D. student can take.

2.Course Outcomes (COs) (5 to 8 for a 3 or 4 credit course):

The course will provide a rigorous overview of social sector and the need for technology led innovations in addressing the same. Leading to insights, concepts, and tools for designing solutions for social impact.

A social immersion led course to understand the problems and needs of the grassroots and solve them in creative ways leveraging technology. Work with social organizations or NGOs to get a first-person view of the ground reality, and understand problems that need to be solved.

Understand the tools and frameworks they need to define and design technology for social impact. Explore creative solutions to address these social problems, leveraging emerging technologies. Inspire social enterprises through the technology led journey.

Course Outcomes:

CO1-Understand how to evaluate social needs and solution opportunities

Social Immersion: A social immersion led course to understand the problems and needs of the grassroots and solve them in creative ways leveraging technology.

CO2-Connect solutions with markets and identify requirements & target user segments.

Societal Challenge Identification: Work with social organizations or NGOs to get a first-person view of the ground reality, and understand problems that need to be solved.

CO3-Assess alternatives and evolve unique Value proposition. Define and design the solution.

Design For Change: Understand the tools and frameworks they need to define and design technology for social impact.

CO4-Engineer & Build the solution for the real world. As part of the course project

Technology For Social Innovation: Explore creative solutions to address these social problems, leveraging emerging technologies. Inspire social enterprises through the technology led journey.

3.Mapping of Course Outcomes (COs) with Program Outcomes (POs) and Program Specific Outcomes (PSOs) – Course Articulation Matrix

Mapping to the POs of the **PDM** program.

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PSO1	PSO2	PSO3	PSO4
CO1	3	3	3	2	3	3	3	2	3	3	3	2	3	2
CO2	2	3	3	2	2	3	3	3	2	3	2	2	2	3
CO3	2	3	2	2	3	3	3	2	2	3	3	2	2	2
CO4	3	3	3	2	2	3	3	2	2	3	3	3	3	2

Note: Each Course Outcome (CO) may be mapped with one or more Program Outcomes (POs) and PSOs. Write '3' in the box for 'High-level' mapping, 2 for 'Medium-level' mapping, 1 for 'Low'-level' mapping

PEDAGOGY

Design For Social Innovation (DFSI) is a project-based course that will enable students to do a deep dive into societal challenges being addressed by NGO(s), social enterprises & the government and build solutions to alleviate these complex social problems through immersion, design & technology.

The pedagogy will include interactive lectures, inspiring guest talks, field visits, social immersion and a course project. Applying and synthesizing information from these sources to define the social problem to address and take up the solution as the course project, with your group. Social immersions with NGOs/social sector will be a key part of the course. Will all lead to the course project that will address needs of the social sector.

COURSE TOPICS

The course will introduce social context and various players in the social space, present approaches to discover and understand social needs. Social immersion and inspiring conversations that will culminate in developing an actual, idea for problem-based intervention, based on an in-depth understanding of a key social problem.

The course will comprise of four modules, one in each month. Each module will address few topics through lectures, chance to meet the social change makers through guest lectures, upclose work with NGOs and in parallel work on the course project.

MODULE [I] *Social Immersion*

- What Is Social Innovation
- Interaction With Leading Changemakers
- Identifying Core Developmental Sectors [Agri, Fintech, Healthcare, Edu etc] & Field Visits
- Student Paper [I] Case Study On A Social Innovation Of Choice
- Hackathon #1: Needs Identification (maybe, during a field immersion/at an NGO)

MODULE [II] *Societal Challenge Identification*

- Deep Dive Into Defining The Problem
- Market Research/Competitive Landscape
- Circular Validation From Stakeholders [Organizations/Customers/End Users]
- Student Paper [II] White Paper On The Chosen Social Innovation
- Hackathon #2: Design thinking/Problem definition

MODULE [III] *Design For Change*

- Ideation/Design Thinking On Problem Breakdown
- Build An SRS Document
- Identify Technology Components For Social Innovation
- Project Design & Execution Plan.
- Hackathon #3: Solution design, Design thinking & Prototypes

MODULE [IV] *Technology For Social Innovation*

- Updates On Building The Technology
- Examples of technology used in social innovation
- Teams to build in the use of emerging technology/research.
- Hackathon #4: Final hackathon. Complete the solution

Finale

- Deploy Solution In Stakeholder/End User Environment. Measure Impact
- 15 Minute Presentation Structure To Be Designed
- Term Paper Technology For Social Innovation

PREFERRED TEXT BOOKS: Course material and reference/prescribed reading will be shared.

GRADING PLAN: Type of Evaluation	Weightage (in %)
Quizzes	10
Assignments [Paper I/II]	15
Hackathons (2)	30
Technology Demonstration	15
Stake Holder Presentation	15
Final demos & Term paper (based on social immersion)	15

Reference courses

- Stanford: Center for Social Innovation offers Design for Extreme Affordability (OIT333 [link](#))
 - LSE: Social Innovation design ([MG4G2-link](#))
 - UPenn impact immersion week program (NPLD 585 [link](#))
 - Columbia- Design for social innovation ([link](#))
-

Title of the Course : Design Thinking 101 – Research to Define

Name of the Faculty : Raman Saxena

Course Code : PD1.301

L-T-P : 3 -1-0- 2

(L= Lecture hours, T=Tutorial hours, P=Practical hours)

Semester, Year : 1st Sem – Year 1 (Monsoon, 2026)

1.Prerequisite Course / Knowledge: No prerequisites are required

2.Course Objectives & Outcomes (COs)

The overall goal of design thinking course is to help design better solutions, products, services, systems, processes, strategies, and experiences. This course is aimed at guiding through the Design Thinking Process and will help developing a solid understanding of the overall process, phases and methods in design thinking. Introduce the concept of Human-centred approach, empathy, collaboration, co-creation and product-user & product-market fit. It will provide the theory and operational skills to follow Human (User)-Centred approach and how to implement this knowledge in professional work life.

After completion of this course successfully, the students will be able to...

CO-1 Understand the Human-Centered (HCD) led Empathy (end user advocacy) & Creative Thinking based approach for Problem solving and designing/delivering new products, solutions and services

CO-2 Demonstrate good understanding of various methods and tools used to understand the user's socio-cultural-economic context during the research/empathies and define stages of the Design Thinking Process.

CO-3 Apply hands-on skills, methods and tools for user research including User Research, Empathy, Contextual Inquiries, Shadowing, User Personas, Use and User Journey mapping, etc.

CO-4Create, document and present the various deliverables and communications including Stakeholder Mapping, User Personas, Use Case Scenarios, User Journey Maps, Empathy Maps etc. related to the Design Thinking process and deliverables.

CO-5Demonstrate the ability to collaborate and co-create the design solution and integrate the DT process within the overall product development and management life cycle.

3.Mapping of Course Outcomes (COs) with Program Outcomes (POs) and Program Specific Outcomes (PSOs) – Course Articulation Matrix

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO1	PSO1	PSO2	PSO3	PSO4
CO1	3	3	3	2	3	2	1	2	3	2	3	2	2	2
CO2	3	3	3	2	3	2	1	2	3	2	3	2	2	2
CO3	3	3	3	2	3	2	1	3	3	2	3	2	2	2
CO4	3	2	3	2	3	2	1	3	3	3	3	2	2	2
CO5	2	2	2	2	2	2	1	2	3	3	3	2	2	2

Note: Each Course Outcome (CO) may be mapped with one or more Program Outcomes (POs) and PSOs. Write '3' in the box for 'High-level' mapping, 2 for 'Medium-level' mapping, 1 for 'Low'-level' mapping

3.Detailed Syllabus:

UNIT 1. UNLEARNING (Week 1 - Lecture 1 & 2)

Initial part of the course will emphasize on unlearning and to cultivate a knack for design thinking, and creative problem solving among the students that will work as a good foundation before introducing them to detailed process, methods and tools of DESIGN THINKING.

UNIT 2. UNDERSTANDING DESIGN AND DESIGN DOMAIN (Week 2 - Lecture 3 & 4)

- Understanding Design
- Role & Functions of design and designers.
- Design Elements – (Function, Ergonomics & Aesthetics) + Desirability, Feasibility & Viability)
- Understanding Design Domains and perspectives – Product Design, HCI, Visual Communication, Service Design and User Experience

UNIT 3. INTRODUCTION TO DESIGN THINKING (Week 3 - Lecture 5 & 6)

- What is Design Thinking?
- Why Design Thinking?
- Design Thinking approach in new product development & innovative solutions

UNIT 4. DESIGN THINKING PROCESS (Week 4 - Lecture 7 & 8)

- Design Thinking Process – human-focused, empathy, research, ideation and prototype-driven, innovative design approach.
- User-Centred Design / Double Diamond Process explaining stage of Discovery, Define, Design, Prototype & Test and Implement.
- Introduce/Initiate Design Thinking Pilot Project which is built into course structure and will run parallel to the course content in the DT-Part1 and will conclude in DT-Part 2.

UNIT 5. DISCOVERY PHASE (Week 5 - Lecture 9 & 10)

- What is Discovery and Validation phase and why?
- Understanding User Context? – Why & How to Empathies?
- Understanding the User Needs and Goals through empathy by observing their behavior and drawing conclusions based on qualitative information
- Understanding Business Goals

- Tools and Methods and Deliverables

UNIT 6. DEFINE PHASE (Week 6 - Lecture 11 & 12)

- Analysis and Synthesis of Data and Information.
- Driving Insights (both user and business) and solution directions
- Tools and Deliverables of the Define phase

UNIT 7. DRIVING ACTIONABLE BRIEF (Week 7 - Lecture 13 & 14)

- Through the process of analysis and synthesis, identifying user-business insights, arriving at an actionable brief in form of HMW statement.
- Debriefing and briefing on upcoming course “Design Thinking 101 – Research to Define”

Reference Books:

1. Case1: Design Thinking and Innovation at Apple, Stefan T. & Barbara F. (HBS 9-609-066)
2. Case2: Defining Innovative Mobile Strategies: How Design Thinking Offers an Effective Way to Address the “Wicked Problem” of Enterprise Mobility by SAP
3. Case3: Good Kitchen- Public service delivery Innovation
4. Book: HBR's 10 Must Reads on Design Thinking, by Harvard Business Review
5. Book: Design Thinking for Strategic Innovation by Idris Mootee
6. Book: Design Thinking by Tim Brown (HBR – Ro8o6E)
7. Book: Innovation Through Design by Bill Moggridge
8. Book: 101 Design Methods: A structured approach for driving innovation in your organization by Vijay Kumar
9. Book: Design Thinking and Social Innovation by Tim Brown and Jocelyn Wyatt in Stanford Social Innovation review

4. Teaching-Learning Strategies in brief (4 to 5 sentences):

- The Course will be divided into lectures (around 12 nos.) and hands-on work including assignments, classroom exercises and homework.
- The course will also include fieldwork, hand on activities, learning by doing, to practice the learning from the lectures.
- I will also introduce and discuss a couple of case studies including cases related to the new product development and ICT domain.
- It is supported by the design thinking and research approaches of various design, technology and business schools including Stanford, NID, IIM Bangalore etc. and also prestigious design consulting's including IDEO, FROG Design, Nokia Research, Nokia Design and Siemens etc. to bring both academic and industrial flavor in the content and learning.
- Other than attending lectures and doing classroom exercises & assignments, students need to spend 4 to 6 hours per week on home/field assignments.

5. Assessment methods and weightages in brief (4 to 5 sentences):

1.	Classroom /Home activities & assignments		20%
2.	Project Individual/Group		50%
4.	Final Exam		20%
TOTAL			100%

Title of the Course : Diffusion Models For Generative AI
Name of the Faculty : Naresh Manwani and Saikiran Bulusu
Name of the Program : CSE and ECE
Course Code : CS7.510
Credits : 4
L - T - P : 3-1-0
(L - Lecture hours, T-Tutorial hours, P - Practical hours)
Semester, Year : Monsoon, 2026

Pre-Requisites : Mathematical maturity, good understanding of probability theory, and linear algebra. A background in machine learning will be helpful but should not be necessary.

Course Outcomes : After successful completion of the course, the students will be able to

1. CO1: Understand the mathematical foundations of diffusion models, including stochastic processes, stochastic differential equations, probability theory, and functional inequalities relevant to generative AI.
2. CO2: Analyze forward and reverse diffusion processes, score functions, and reverse-time stochastic dynamics used in diffusion-based generative modeling.
3. CO3: Formulate and apply score matching, denoising score matching, and estimation-theoretic techniques for learning high-dimensional probability distributions.
4. CO4: Derive and interpret continuous-time generative modeling frameworks using stochastic differential equations, probability flow ordinary differential equations, and transport-based formulations.
5. CO5: Evaluate, design, and investigate diffusion-based generative AI models by connecting concepts from optimal transport, flow matching, and modern generative learning methodologies.

Course Topics: [OBJ]

The course is organized into five modules as follows.

1. **Module 0: Mathematical Foundations (Weeks 1–3):** Probability measures on $\mathbb{R}^n$, Weak convergence and Wasserstein distance, Markov semigroups, Itô calculus and stochastic differential equations, Fokker–Planck equation, Functional inequalities: Poincaré inequality, Log-Sobolev inequality.
2. **Module 1: Forward and Reverse Diffusion (Weeks 4–6):** Construction of forward diffusion processes, Time reversal of diffusion processes, Anderson’s theorem (reverse SDE), Score function and Stein identities, Fisher divergence.
3. **Module 2: Score Matching and Estimation Theory (Weeks 7–9):** Score matching objective, Denoising score matching, Consistency of score estimation, Identifiability conditions, High-dimensional behavior.
4. **Module 3: Continuous-Time Generative Models (Weeks 10–12):** Probability flow ODE, Fokker–Planck PDE, Kolmogorov equations, Likelihood computation via instantaneous change of variables, Equivalence between SDE and ODE formulations.
5. **Module 4: Flow Matching and Transport (Weeks 13–14):** Deterministic transport maps, Flow matching objective, Continuity equation, Benamou–Brenier formulation of optimal transport, Connections to normalizing flows.

Note: To provide exposure to recent advancements in Diffusion Models-based Generative AI and its applications in various domains, we have included a project component, which has 20% weightage.

Reference Material: The course content mostly follows the material for the following courses at other universities.

- Diffusion models for Generative AI (ECE 381V; Fall 2025) [sanjay-shakkottai](#)
- UIUC ECE 498/598: Deep Generative Models. <https://rrc-uiuc.notion.site/gen-models-sp26>
- Diffusion Models: From Theory to Practice (6.S982): Spring '25, MIT https://hackmd.io/@giannisdaras/mit_diffusion_class

Grading Plan :
(The table is only indicative)

Type of Evaluation	Weightage (in %)
Class Participation	5%
Scribes (2 Lectures for each Student)	10 %
Assignments (2 – 10% each)	20 %
Mid Sem Exam	15 %
End Sem Exam	30 %
Project	20 %

Mapping of Course Outcomes to Program Objectives: (1 – Lowest, 2—Medium, 3 – Highest, or a ‘-’ dash mark if not at all relevant).

CSE Program:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3	PSO4
CO1	1	1	1	2	1	1	1	3	3	2	1	3	2	2	2	3
CO2	1	1	1	3	1	1	1	3	3	2	1	3	2	2	2	3
CO3	2	2	1	3	3	1	1	3	3	2	2	3	2	2	2	3
CO4	2	2	1	3	3	1	1	3	3	2	2	3	2	2	2	3
CO5	3	3	1	3	3	1	1	3	3	2	2	3	3	3	3	3

ECE Program:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PSO1	PSO2	PSO3	PSO4
CO1	1	1	1	2	1	2	2	3	2	2	2	3
CO2	1	1	1	3	1	2	2	3	2	2	2	3
CO3	2	2	1	3	3	3	3	3	2	2	2	3
CO4	2	2	1	3	3	3	3	3	2	2	2	3
CO5	3	3	1	3	3	3	3	3	3	3	3	3

Teaching-Learning Strategies in brief (4-5 sentences):

The lectures will involve both slide presentations and boardwork. All the concepts and the theoretical results in the course will be illustrated through examples and/or applications whenever possible so that the students can comprehend them easily. To encourage students' participation in the class, 5% weightage of the total marks is reserved for it. To encourage students to explore deeper into a research problem in this area, a project component is kept which will be discussed and valued time to time during the semester.

Title of the Course : Digital Image Processing

Name of the Faculty : Raghavendra GS

Course Code : CS7.404

L-T-P : 3-0-1.

Credits : 4

(L=Lecture hours, T=Tutorial hours, P=Practical hours)

Name of the Academic Program: B.Tech. in Computer Science and Engineering

1. Prerequisite Course / Knowledge:

Programming, Data Structures, Algorithms

2. Course Outcomes (COs)

After completion of this course successfully, the students will be able to:

CO-1. Give examples of how images are stored and represented in digital machines.

CO-2. Apply basic techniques for improving subjective perception of images.

CO-3. Apply basic techniques for filtering images in spatial and frequency domain.

CO-4. Apply basic techniques for morphological and geometric transformations of images.

CO-5. Apply techniques for color image processing.

CO-6. Apply basic techniques for high-level image processing (Image Segmentation, Image Restoration, Image Compression)

Mapping of Course Outcomes (COs) with Program Outcomes (POs) and Program Specific Outcomes (PSOs) – Course Articulation Matrix

	P O1	P O2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	P O 10	PO 11	PO 12	PS O1	PS O2	PS O3	PS O4
CO1	2	2	2	2	3	-	1	-	3	1	3	2	3	3	2	3
CO2	3	3	3	1	3	-	1	-	2	2	2	3	3	2	1	1
CO3	3	2	2	1	2	2	1	-	2	2	2	1	3	3	3	2
CO4	3	2	2	1	2	2	1	-	2	2	2	1	3	3	3	2
CO5	3	2	2	1	2	2	1	-	2	2	2	1	3	3	3	2
CO6	3	3	3	3	2	2	1	-	2	2	3	2	3	3	3	3

'3' in the box denotes 'High-level' mapping, 2 for 'Medium-level' mapping, 1 for 'Low'-level' mapping

3. Detailed Syllabus:

Unit 1: Introduction and Digital Image Fundamentals (3 hours)

Unit 2: Methods for Improving Subjective Perception of Images (4.5 hours)

Unit 3: Spatial and Transform Domain Image Processing (9hours)

Unit 4: Morphological and Geometric Image Processing (4.5 hours)Unit 5: Color

Image Processing (3 hours)

Unit 6: High-level Image Processing and Advanced Approaches (15 hours)

References:

- Gonzalez and Woods, Digital Image Processing, Pearson Education, 2003

4. Teaching-Learning Strategies in brief:

Lectures are dominated by pictorial content (images, animations, videos) to explain concepts in image processing. Simulation of algorithms are used to enhance understanding. Learning by writing code is highly promoted and encouraged. Students understand difficult mathematical concepts and abstraction by coding using state of the art software, simulation frameworks, libraries and solvers. More concretely, students also learn by doing assignments designed to achieve course outcomes and collaboratively working on a final project.

5. Assessment methods and weightages in brief:

Assignments: 30 marks, Mid Semester Examination in Theory: 20 marks, End Semester Examination in Theory: 20 marks, Project: 30 marks

Title of the Course : Digital Systems and Microcontrollers (DSM)

Name of the Faculty : Anshu Sarje + Priyesh Shukla + Anil Kumar V + Madhava Krishna +
Sourav Garg

Course Code : EC2.101

L-T-P : 3-1-3

Credits : 5

(L= Lecture hours, T=Tutorial hours, P=Practical hours)

Name of the Academic Program..... B. Tech in ECE

1. Prerequisite Course/Knowledge:

Understanding of basic algebra concepts taught up to the 10+2 level

2. Course Outcomes (COs):

After completion of this course successfully, the students will be able to...

CO-1: Solve problems pertaining to the application of Boolean algebra, number systems, simplification of logic expressions using Karnaugh maps.

CO-2: Develop a simplified combinational circuit as a solution for a given problem.

CO-3: Analyze a real-world problem to develop a digital design solution using sequential circuits to solve the problem.

CO-4: Describe the working of a basic 8-bit von Neumann architecture processor.

CO-5: Develop skills for simulating circuits using basic components on online simulation tools (example, Tinker CAD).

CO-6: Design, implement and test a given logic circuit using basic electronic components such as bread boards, ICs etc.

3. Mapping of Course Outcomes (COs) with Program Outcomes (POs) and Program Specific Outcomes (PSOs)–Course Articulation Matrix

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3	PSO4
CO1	3	3	1	1	2	1	1	1	1	3	1	2	1	2	1	1
CO2	1	2	3	2	1	1	1	1	1	1	1	2	2	2	1	1
CO3	1	2	3	2	1	1	1	1	1	2	1	2	3	2	1	1
CO4	2	2	1	2	1	2	1	1	1	2	2	3	3	2	3	3
CO5	1	2	3	2	3	2	1	1	3	2	3	1	2	1	3	2
CO6	1	2	3	2	3	2	1	1	3	2	3	1	3	1	3	2

Note: 3 in the box for 'High-level' mapping, 2 for 'Medium-level' mapping, 1 for 'Low'-level 'mapping

4. Detailed Syllabus:

Unit1: Number systems and inter conversions (binary, decimal, **hexa** decimal), postulates of Boolean algebra, binary logic gates, binary functions

Unit 2: Simplification of binary expressions using K-maps, logic function implementation, combinational circuits

Unit3: Latches and flip-flops, types of **flip**-flops, internal circuit design and operation

Unit 4: Sequential circuits, state diagrams, state tables, state equations, applications of sequential

circuits Unit5: Registers and counters, memory and processor architecture

Reference Books:

1. M. Morris R. Mano and Michael D. Ciletti (2013), *Digital Design*, 6th Ed, Pearson.

5. Teaching-Learning Strategies in brief(4 to 5 sentences):

The course instruction is delivered through lectures with examples of real-world application of electronic systems to foster student understanding and interest. The course is structured as a theory and laboratory course, such that the concepts and circuits introduced in the theory classes can be experimentally applied and understood by the students. Assignments are designed to encourage students to critically think about the concepts discussed in the class and to learn to independently solve problems.

6. Assessment method and weightages in brief(4 to 5 sentences):

Continuous evaluations: Assignments–10%

MCQ Quizzes–20%

Lab reports–20%

Comprehensive evaluations:

Lab exam–15%

End semester exam **in Theory**–35%

Title of the Course : Digital VLSI Design

Faculty Name : Zia Abbas

Course Code : EC2.408

L-T-P : 3-1-0

Credits : 4

Prerequisite Course / Knowledge:

Basic knowledge of digital design.

Course Outcomes (COs):

After completion of this course successfully, the students will be able to.

CO-1: Understand the background that drive to the development of state-of-the-art VLSI digital circuits, the importance of low power, high-performance and power-delay optimal designs, state of the art design issues in digital circuits, understand the CMOS digital IC design process.

CO-2: Design and Synthesis of Verilog/VHDL codes, test benches to meet specifications, to synthesize Verilog/VHDL onto hardware using required EDA tools.

CO-3: design and analyze CMOS circuits using both analytically and SPICE tools, derive analytical circuit equations to estimate performances (e.g., power) of a VLSI design. Able to identify the impact of Process, Voltage and Temperature on circuit's performance.

CO-4: Analyze the design flow to design complex CMOS digital circuit using required CAD tools. Create a cell library to be used in other designs.

CO-5: Create a low-power digital design, estimate static and dynamic power dissipation in CMOS circuits. Impact of CMOS technology scaling. Low power design methodologies.

CO-6: Design of high-performance circuits, and power-delay optimal designs.

Mapping of Course Outcomes (COs) with Program Outcomes (POs) and Program Specific Outcomes (PSOs)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3	PSO4
CO1	3	2	1	1	1	1	2	2	2	1	1	2	3	2	2	2
CO2	3	3	2	2	2	1	1	2	1	1	1	1	3	1	1	1
CO3	3	3	2	2	2	1	1	2	2	2	1	2	3	1	1	1
CO4	3	2	3	3	3	1	1	2	2	2	1	3	3	1	1	1
CO5	2	3	3	3	3	1	1	2	3	3	3	3	3	1	2	1
CO6	2	3	3	3	3	1	1	2	3	3	3	3	3	2	1	1

Each Course Outcome (CO) may be mapped with one or more Program Outcomes (POs). Write '3' in the box for 'High-level' mapping, 2 for 'Medium-level' mapping, 1 for 'Low'-level' mapping. Mapping with PSOs, where applicable.

Detailed Syllabus:

Unit 1: Introduction to digital design, Digital design metrics (Performance, Power, Functionality, Robustness, etc.) and their discussion in general, why low power, why high performance, Power-delay optimal designs, why technology scaling, issues in state-of-the-art digital designs i.e.,

making modern digital circuits, corner-based nanoscale design, statistical circuit design.

Unit 2: Combinational IC design, Sequential IC design, Role of CAD tools, RTL design, Logic Synthesis, Logic Simulations, Static Timing Analysis.

MOS Capacitor, Electrical Characteristics of MOS Transistors, Threshold Voltage, Transconductance(gm), Body Effect, Channel-Length Modulation, MOS Transistors as Switch, MOS Inverter, Switching Characteristics, Driving Large Capacitive Loads, CMOS Realization, Switching Characteristics, CMOS NAND, NOR and other basic combinational/sequential circuits, CMOS Complex circuits, CMOS technology scaling, CMOS Gate sizing-logical effort, Complementary CMOS, Pass transistor logic, Dynamic CMOS design, Transmission gate, Layout basics, Floor Planning, Introduction to Fin FET technology.

Unit 3: Digital Design - From Power perspective: Introduction, Dynamic power dissipation (Short-Circuit and Switching), Dynamic Power in the Complex Gate, Switching Activity, Switching Activity of Static CMOS Gates, Transition Probability in Dynamic Gates, Power

Dissipation due to Charge Sharing, Static i.e. Leakage Power Dissipation (leakage mechanism): p–n Junction Reverse-Biased Current, Band-to-Band Tunnelling Current, Tunnelling through and into gate oxide, Injection of hot carriers from substrate to gate oxide, GIDL, Punch-through, Subthreshold Leakage Current including DIBL. Impact of technology scaling on leakage currents/power, need for technology scaling, factors effecting the leakage current especially in scaled technology nodes (input pattern dependency, stacking effect, loading effect, etc.), Impact of process, temperature and supply voltage variations on leakage currents. Internal node voltage impact.

Unit 4: Digital Design - From Performance (i.e., delay) perspective: Computing the Capacitances, Propagation delays, Factors affecting the propagation delays, Mathematical formulation of the delays in CMOS circuits, Technology scaling impact on propagation delays, Mean and variance of the delays in a gate, Impact of process variations on delays in CMOS circuits, Impact of operating (temperature and supply voltage) variations on delays.

FinFET technology will also be discussed in parallel. Such delay/leakage estimation techniques will also be applied to Fin FET circuits.

Reference Books:

1. Jan M. Rabaey, A. Chandrakasan, B. Nikolic “Digital Integrated Circuits - A Design Perspective, PHI.
2. Douglas A. Pucknell, K. Eshraghian, “Basic VLSI Design”, 3rd Edition, Prentice Hall of India.
3. Neil H. E. Weste, K. Eshraghian, “Principles of CMOS VLSI Design”, A Systems Perspective, 2nd Edition, Pearson Education Pvt. Ltd.

Teaching-Learning Strategies in brief (4 to 5 sentences):

The course will start with the background that drive us to the development of state-of-the-art digital VLSI designs, then fundamental and core topics of the course will be discussed in detail broadly at logic and transistor level with hands-on with related CAD tools. Circuit simulations, layout, RTL coding, synthesis, etc. will be highly encouraged throughout the course. The broad approach of the course is to discuss the digital VLSI design from three perspectives; power, performance, and power-delay optimal designs to understand the different design approaches. Students will be exposed to state-of-the-art scaled technology node to better understand the issues related to scaled nodes. Regular assignments will be given to reinforce the concepts. Weekly tutorials will involve students in active learning by applying the lecture discussion. Quizzes will be designed to test students' understandings of the discussed concepts. Projects will be carried out in groups, thereby developing the students' abilities to work in teams.

Assessment methods and weightages in brief (4 to 5 sentences):

- Home Assignments: 20%
- Quiz: 10%
- Mid Semester Exam: 15%
- End Semester Exam: 30%
- Project: 25%

Title of the Course	: Discrete Structures
Name of the Faculty	: Praveen P + Suryajith Ch
Course Code	: MA5.101
L-T-P	: 3-1-0
Credits	: 4

(L= Lecture hours, T=Tutorial hours, P=Practical hours)

1. Prerequisite Course / Knowledge:

Basic abstract algebra, High School Mathematics

2. Course Outcomes (COs)

After completion of this course successfully, the students will be able to

CO-1: Demonstrate critical thinking, analytical reasoning, and problem-solving skills

CO-2: Apply appropriate mathematical and probabilistic concepts and operations to interpret data and to solve problems

CO-3: Identify a problem and analyze it in terms of its significant parts and the information needed to solve it

CO-4: Formulate and evaluate possible solutions to problems, and select and defend the chosen solutions

CO-5: Construct graphs and charts, interpret them, and draw appropriate conclusions

CO-6: **Apply** the concepts of group theory, ring and field in various applications in computer science

3. Mapping of Course Outcomes (COs) with Program Outcomes (POs) and Program Specific Outcomes (PSOs) – Course Articulation Matrix

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3	PSO4
CO1	1	2	2	2	2	2	2	2	3	2	2	2	1	2	2	3
CO2	3	3	3	2	2	2	2	2	2	2	1	2	1	2	2	3
CO3	2	2	2	2	2	2	2	2	2	2	1	2	1	1	2	3
CO4	2	2	2	3	2	2	2	2	2	2	2	2	1	1	2	3
CO5	1	1	1	1	1	1	2	2	1	1	1	1	1	1	2	3
CO6	2	2	2	2	2	2	2	2	3	2	2	3	1	1	2	3

Note: Each Course Outcome (CO) may be mapped with one or more Program Outcomes (POs) and PSOs. Write '3' in the box for 'High-level' mapping, 2 for 'Medium-level' mapping, 1 for 'Low'-level' mapping

4. Detailed Syllabus:

- **Unit 1:** Sets, relations, functions, permutations, combinations. Applications to relations. Logic, Propositional Equivalences, Predicates and Quantifiers Sets, Proof Techniques, Contradiction. Mathematical induction, pigeonhole principle. Cardinality of sets, finite and infinite sets, countable and uncountable sets, Cantor's numbering.
- **Unit 2:** Group, subgroup/normal subgroup, homomorphism/automorphism/isomorphism/epimorphism, kernel, cosets, quotient group, product set in a group, center of a group, order/conjugate of an element, commutator. Coding theory (Application to group theory).
- **Unit 3:** Ring, Field, Finite field over a prime. Applications to finite fields.
- **Unit 4:** **Recurrence** relations, generating functions, numeric functions. Application store Currence relations.
- **Unit 5:** Basics of probability theory, birthday attacks. Applications on hash functions.
- **Unit 6:** **Graphs**, Adjacency, Special Graphs, Isomorphic Graphs, Paths, Cycles and Circuits, Connected Graphs, Eulerian Graphs, Hamiltonian Graphs and Planar Graphs.

Reference Books:

1. Thomas Koshy, "Discrete Mathematics with Applications", Elsevier Press, 2004.
2. C.L.Liu and D. P. Mohapatra, "Elements of Discrete Mathematics: A Computer Oriented Approach," Tata McGraw-Hill Edition, 2017.
3. D.S. Malik and M.K. Sen, "Discrete Mathematical Structures: Theory and Applications," Thomson, 2004.
4. Joseph Gallian, "Contemporary Abstract Algebra", 9th Edition, CENGAGE publications, 2019.
5. Gross and Yellen, "Graph Theory and its Applications", 2nd Edition, Chapman and Hall, 2005.
6. Kenneth H. Rosen, "Discrete Mathematics and Its Applications", Sixth Edition, 2006

5. Teaching-Learning Strategies in brief (4 to 5 sentences):

This course supports the expected characteristics, capabilities and skills for computer science graduates in the following ways:

- * Mastery of Computer Science technical foundations
- * Recognition of common Computer Science themes and principles
- * Recognition of interplay between theory and practice
- * Effective problem solving and critical thinking skills

6. Assessment methods and weightages in brief (4 to 5 sentences):

- Assignments: 10%
- In-Class Tests: 20%
- Mid Semester Examination: 30%
- End Semester Examination: 40%

Title of the Course : Distributed Systems

Faculty Name : Kishore Kothapalli

Course Code : CS3.401

L-T-P : 3-1-0

Credits : 4

(L= Lecture hours, T=Tutorial hours, P=Practical hours)

Name of the Academic Program B. Tech. in Computer Science and Engineering

1. Prerequisite Course / Knowledge:

An understanding of operating systems, networks, and algorithms

2. Course Outcomes (COs):

After completion of this course successfully, the students will be able to..

CO-1: Explain the challenges faced by distributed systems in terms of lack of global time, synchrony, faults, programming support, etc.

CO-2: Employ standard distributed programming frameworks to write distributed programs for problem solving

CO-3: Explain the properties and design principles of various real-world and practical distributed systems

CO-4: Interpret the impact of faults in distributed systems in the context of important problems such as distributed agreement, distributed consensus, and distributed transaction processing

CO-5: Analyze distributed algorithms for graphs with respect to correctness, round complexity, and message complexity.

CO-6: Analyze the limitations of distributed systems and assess the operational scope of large-scale distributed systems

3. Mapping of Course Outcomes (COs) with Program Outcomes (POs) and Program Specific Outcomes (PSOs) – Course Articulation Matrix

	PO 1		PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PO 12	PS O1	PS O2	PS O3	PS O4
CO1	1		1	1	1	1	1	1	1	1	2	1	2	3	3	3	3
CO 2	1		2	2	2	3	2	1	1	2	1	2	2	3	3	3	3
CO 3	1		1	2	2	1	2	1	1	1	2	1	2	3	3	3	3
CO 4	1		2	2	2	1	2	1	1	2	2	1	2	3	3	3	3
CO 5	2		2	2	2	1	1	1	1	2	1	2	2	3	3	3	3
CO 6	2	2	2	2	2	1	2	1	1	2	2	2	2	3	3	3	3

Note: '3' in the box for 'High-level' mapping, 2 for 'Medium-level' mapping, 1 for 'Low'-level' mapping

4.Detailed Syllabus:

- Unit 1
 - Introduction
 - Communication models
 - Time and Synchronization
 - Practice: MPI/Map-Reduce
- Unit 2
 - Distributed file systems
 - Consensus, Agreement, Locking
 - Practice: GFS, Chubby
- Unit 3
 - Distributed Database systems
 - Practice: NoSQL, MongoDB
- Unit 4
 - Limitations of distributed computing
 - Self-Stabilization
 - CAP Theorem
- Unit 5
 - Distributed algorithms for graphs
 - Advanced Topics such as Blockchain, Distributed Storage, and Distributed Program Verification

Reference Books:

1. A.D. Kshemkalyani, M. Singhal, (2011) Distributed Computing: Principles, Algorithms, and Systems, ISBN: 9780521189842, paperback edition, Cambridge University Press, USA.
2. N. Lynch, 1996. Distributed Algorithms, Morgan Kauffman, USA, Chapter 5.
3. Other significant papers from conferences such as OSDI, USENIX, NSDI, for material that is not part of textbooks

5.Teaching-Learning Strategies in brief:

Lectures of the class use the active learning methodology and allow students to learn concepts thoroughly in class along with practicing small examples. HomeWorks assigned as part of the course are useful to impart knowledge of using practical distributed programming tools and

libraries. To promote teamwork, some of the homework's are done in a team of two students. The overall learning from the course is enhanced by doing a substantial practice-based project – usually in a team of two students. The course will also have a summative assessment in the form of a final/end-semester exam.

6.Assessment methods and weightages in brief:

- In-class Quiz Exams (Cumulative over several): 15%
- HomeWorks: 20%
- Project: 25%
- End Semester Examination: 40%

Title of the course : **Dynamical processes in complex networks**
Faculty Name : Chittaranjan hens, CCNSB
Course Code : **SC1.440**
Credits : **4**
L - T - P : **L=2, T+P=1**
 (L - Lecture hours, T-Tutorial hours, P - Practical hours)
Semester, Year : Monsoon 2026
Pre-Requisites: Linear Algebra, Differential Equations, asic of Statistical Physics

Course Outcomes

After completion of the course, the students will be able to..

1. *Interpret and quantify* the connectivity patterns of real-world systems.
2. *Model* how the infection spreads in a well-mixed population setup and how chaos develops in the double pendulum.
3. *Identify* how structure influences synchronization and the spread of disease in metapopulation networks.
4. *Implement* the use of various appropriate pharmaceutical or non-pharmaceutical intervention strategies for reducing the severity of infectious diseases in actual circumstances.
5. *Design* cutting-edge problems regarding the diverse dynamical process in complex networks.

Course Topics :

- A. Networks:** A general introduction. Dynamical processes in networks (1 Lecture)
- B. Fundamentals of Network theory:** Mathematics of networks, Measures and metrics(3 Lectures)
- C. Network Models:** Erdos-Renyi, Scale-free networks, and small-world model. Statistical properties and degree distribution. Random graphs with general degree distributions. Percolation Process in complex networks. (4 Lectures)
- D. Nonlinear Dynamics:** From fixed points to chaos theory. Application: Epidemics (SIR/SIS dynamics) and double pendulum (4 Lectures).
- E. Dynamical Processes in Networks:** Epidemics in complex networks, Role heterogeneity in disease spreading. Pharmaceutical and nonpharmaceutical intervention strategy. Epidemics in metapopulation networks, epidemics in time- varying networks, and epidemics in multiplex networks, Activity-driven networks (4 Lectures). Synchronization in coupled nonlinear oscillators (2 lectures).
- F. Recent literature survey, and problem design** (4 lectures).

Preferred Text Books :

1. Barrat, Alain, Marc Barthelemy, and Alessandro Vespignani. *Dynamical processes on complex networks*. Cambridge university press, 2008.
2. Networks: An Introduction, Mark Newman

Reference Books :

E-book/ review paper Links:

1. Boccaletti, Stefano, et al. "Complex networks: Structure and dynamics." *Physics reports*, 424.4-5 (2006): 175-308.
2. Arenas, Alex, et al. "Synchronization in complex networks." *Physics reports* 469.3 (2008): 93-153.
3. Pastor-Satorras, Romualdo, et al. "Epidemic processes in complex networks." *Reviews of modern physics* 87.3 (2015): 925.
4. Perra, Nicola. "Non-pharmaceutical interventions during the COVID-19 pandemic: A review." *Physics Reports* 913 (2021): 1-52.

Grading Plan :

(The table is only indicative)

Type of Evaluation	Weightage (in %)
Quiz-1	10
Mid SemExam	20
Quiz-2	10
End Sem Exam	0
Assignments	
Project	30
Term Paper	20
Other Evaluation	10

Mapping of Course Outcomes to Program Objectives: (1 – Lowest, 2—Medium, 3 – Highest, or a ‘-’ dash mark if not at all relevant). Program outcomes are posted at

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3	PSO4
CO1	3	3	2	2	3	2	2	1	2	-	-	3	2	2	3	1
CO2	2	2	2	2	3	2	2	1	2	-	-	3	2	1	3	1
CO3	3	3	2	2	3	3	3	1	2	-	-	3	3	2	2	3
CO4	3	2	3	2	2	3	2	3	1	-	-	3	3	3	2	1
CO5	2	3	3	2	2	3	2	3	1	-	--	3	3	2	3	3

Teaching-Learning Strategies in brief (4-5 sentences) :

This course is specially tailored for students who are eager to learn and gain access to cutting-edge information on complex networks and disease propagation. This curriculum requires active participation from the students, who must come up with fresh ideas, present them by forming small groups, and examine related research. Comprehensive computer simulation is a requirement.

Title of the course : **Early-Stage Funding for a Startup**

Faculty Name : Sridhar Kalyanasundaram

Course Code : PD2.402

Credits : 4

L - T - P : L=2, T+P=1

(L - Lecture hours, T-Tutorial hours, P - Practical hours)

Name of the Program : M.Tech – Product Design and Management

Pre-Requisites : Keen interest in startups and exploring funding options

Course Objectives:

This is an introductory course for understanding popular funding options available for an early- stage startup, steps required to make the startup fundable, typical expectations of startup investors and an approach to valuation for such funding.

Course Outcomes :

CO1: Understand various modes of funding a startup

CO2: Compare and contrast funding options and it's impact

CO3: Learn how to prepare a startup to become fundable

CO4: Understand typical investor expectations while investing in a startup

Course Topics :

M1: Introduction	M2: Modes of Startup Funding	M3: Making the startup Fundable
Funding Needs Basic Funding Options	Investors Instrumen ts Myths	Business Model Team Compliance Data Room
M4: Terms and Valuation	M5: Project	
Approaching Valuation Negotiating Term Sheet Understanding agreements	Prepare investor pitch	

Session Plan

Sl.	Module	Session
1	Introduction	Funding Needs for a Startup
2		Financial Ecosystem for a Startup
3	Modes of Startup Funding	Instruments: Equity, Debt and Convertibles
4		Investors: Angels, VCs, and Strategic Investors
5		5 Myths about Startup Funding
6	Making a Startup Fundable	Business Plan
7		Team, ESOPs
8		Compliance and Data Room
9	Funding Terms and Valuation	How is valuation done
10		Understanding key terms
11		Negotiating Investment Agreement
12	Project	Prepare an Investor Pitch for your Startup
13		Project Presentation

Preferred Textbooks: NIL

Reference Books :

"The Lean Startup" by Eric Ries Startup

Finance 360°

Grading Plan :

(The table is only indicative)

Type of Evaluation	Weightage (in %)
Quiz-1	20 %
Mid Sem Exam	20 %
End Sem Exam	20 %
Assignments/ Project	40 %

Mapping of Course Outcomes to Program Objectives: (1 – Lowest, 2—Medium, 3 – Highest, or a ‘-’ dash mark if not at all relevant). Program outcomes are posted at

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PSO 1	PSO 2	PSO 3	PSO 4
CO1	1	1	2	2	2	3	3	3	2	2	2	2	2	3
CO2	1	1	2	2	2	3	3	3	2	2	2	2	2	3
CO3	1	1	2	2	2	2	3	2	2	3	2	2	3	3
CO4	1	1	2	2	2	3	3	3	2	2	2	2	2	3

Teaching-Learning Strategies in brief (4-5 sentences):

The course will be delivered on a live, interactive basis. The delivery will involve lectures, presentations, videos, case studies and class discussions.

There will be periodic assignments which will enable participants to reflect and respond to situations based on their learning in the program. The final project will bring all the learnings together to a real life situation of funding a startup.

Title of the Course : Earth System Science & Modelling

Faculty Name : Kiran Chand Thumaty

Course Code : GS3.603

L-T-P : 3-1-0

Credits : 4

Elective for DD Program

Prerequisite Course / Knowledge:

Basics of Physics, Math and computational knowledge

Course Outcomes (COs)

CO-1: Understanding how earth functions as a unit, different earth systems processes, forcings and feedback.

CO-2: Explain the structure and components of Earth system models.

CO-3: Basic understanding of field measurements protocols for different climate variables.

CO-4: Handling, interpretation and processing of analysis ready remotely sensed geospatial data

CO-5: Identify critical gap areas of existing models and use AI/ML for surrogate modeling and prediction.

Mapping course outcomes to project outcomes and project specific outcomes

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3	PSO4
CO1	2	3	1	3	1	-	1	-	3	-	-	3	2	-	-	-
CO2	2	3	1	3	1	-	1	-	3	-	-	3	2	-	-	-
CO3	2	1	1	2	3	-	3	-	3	-	3	3	-	2	3	-
CO4	3	3	3	3	3	3	3	2	3	-	3	3	3	3	3	3
CO5	3	3	3	3	3	3	3	3	3	-	3	3	3	3	3	3

Detailed Syllabus:

Unit-1: Introduction – Evolution of earth system science

- Systems approach to understanding ‘earth as a system’
- Earth system processes - Geosphere-biosphere interactions; coupling physical and biogeochemical cycles and human intervention– forcings and feedbacks
- Role of international programs and organizations – WCRP, IGBP, IPCC, IHDP, ESSP and Future Earth.

Unit-2: Geospatial data for earth system science

- Top-down approach - earth observation from space; satellite sensors; essential climate variables
- Bottom-up approach – inventory from ground observational network and scaling up to regional, national and global context using remote sensing proxies and modelling.

Unit-3: Modelling the earth system (Vegetation, Atmosphere, Ocean, Land Surface, Hydrological)

- Simple models - Energy balance approach
- Evolution of process-based models
- Complex models – General circulation models (GCMs)
- Models with human dynamics – Integrated Assessment Models (IAMs)

Unit-4: AI/ML for earth systems

- Deep learning for climate data
- Spatiotemporal models
- Surrogate modeling

Teaching-Learning Strategies in brief:

- Students will be taught on how earth functions as a system by introducing the concept of different process-based models, role of ground and satellite observations in model calibration and validation, and enhancing current modelling capability through AI/ML
- Presentations by students on chosen topics
- Assignments based on (a) literature survey and (b) handling of satellite derived open source ‘analysis ready data sets’
- Course project with periodical evaluation

Assessment methods and weightages in brief :

- **Theory (%):** Quiz (15), Assignments (10), Mid exam-1(20), End exam (35) = **80%**
- **Project (%):** Literature Survey, Preliminary and final presentation along with report = **20%**

Reading

- Andrew Gettelman & Richard B. Rood. Demystifying Climate Models A Users Guide to Earth System

Models,

- Martins, H., (Eds) (2020) Matching new demands of Climate Services with evolving Earth system modelling and prediction capabilities.
- Merritts, D., Dewet, A., and Menking, K., Environmental Geology: An Earth System Science Approach, 1998.
- Jacobson, M.C., Charlson, R.J., Rodhe, H., and Orians, G.H., Earth System Science, Academic Press, 2000.
- Dewet, A., and Menking, K., Environmental Geology: An Earth System Science Approach, 1998

Title of the Course : Earthquake Resistant Design of Masonry Structures

Name of the Faculty : P. Pravin Kumar Venkat Rao

Course Code : CE1.607

L-T-P : 3-1-0

Credits : 4

Name of the Academic Program: **M.Tech. - Computer-Aided Structural Engineering**

1. Prerequisite Course / Knowledge: Strength of Materials, Structural Analysis, Structural Design (RC or Steel), and Seismic Design of Structures (preferably)

2. Course Outcomes (COs):

After completion of this course successfully, the students will be able to:

CO 1: Understand the seismic vulnerability of existing masonry structures against seismic forces.

CO 2: Acquainted with principles of earthquake-resistant design of masonry structures

CO 3: Understand the failure modes and complete behaviour of masonry under different actions like: compression, tension, shear, and bending

CO 4: Analyze the seismic safety of masonry buildings and suggest retrofit measures using codal provisions.

CO 5: Design the strengthened masonry components of a building using different techniques including advanced composites.

3. Mapping of Course Outcomes (COs) with Program Outcomes (POs) and Program Specific Outcomes (PSOs) – Course Articulation Matrix

Note: Each Course Outcome (CO) may be mapped with one or more Program Outcomes (POs) and PSOs. Write '3' in the box for 'High-level' mapping, '2' for 'Medium-level' mapping, '1' for 'Low'-level' mapping; and '-' for not relevant.

	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO1 0	PSO 1	PSO 2	PSO 3	PSO 4
CO 1	3	3	1	2	3	3	3	3	3	3	3	1	1	3
CO 2	3	3	1	1	3	2	3	3	3	3	3	1	2	3
CO 3	3	3	2	2	3	3	3	3	3	3	3	1	1	3
CO 4	3	3	1	1	3	3	3	3	3	3	3	1	1	3
CO 5	3	3	1	1	3	2	3	3	3	3	3	1	2	3

4. Detailed Syllabus:

Unit 1: Introduction to masonry: Masonry buildings in India, Material properties, Masonry units - stones, brick and concrete blocks, hollow and solid units, Manufacturing process, Mortar, Grout, and reinforcement; Masonry assemblages, Masonry systems, Various tests, and standards.

Unit 2: Masonry under compression: Prism strength, Failure mechanism, types of construction and bonds, Eccentric loading, Slenderness – effective length and effective height, effect of openings, Code provisions, masonry in tension, flexural strength of masonry, shear and bending capacity of masonry.

Unit 3: Behaviour of masonry structures during past earthquakes: Common modes of failures, effect of roof and floor systems, Masonry under lateral loads: In-plane and out-of-plane loads, bending parallel and perpendicular to bed joints, Shear and flexure behaviour of piers, Test and standards, lateral force distribution for flexible and rigid diaphragms, Combined axial and bending actions.

Unit 4: Earthquake Resistant Measures: Analysis for earthquake forces, role of floor and roof diaphragm, Pier analysis using equivalent stiffness approach, Concept and design of bands, splints and bandages, Vertical reinforcement at corners and jambs, Code provisions.

Unit 5: Retrofitting of masonry building: Repair techniques of masonry buildings, IS: 13935 provisions for retrofitting, Retrofitting methodologies including advanced composites like textile reinforced mortar (TRM), fibre-reinforced polymer (FRP) composites using man-made and natural fabric textiles, Masonry strengthening case-studies, Experimental investigations – mechanical characterization of fibres/textile in direct tension, compressive, shear, and bending properties of masonry prisms and wallettes strengthened using TRM and FRP, Retrofitting design framework using ACI 549.6R-20 and ACI 440.7R-22.

Reference Books:

- ACI 440.7R-22. (2022). Externally bonded fibre-reinforced polymer systems design and construction for strengthening masonry structures - Guide. American Concrete Institute, Farmington Hills, MI, USA.
- ACI 549.6R-20. (2020). Guide to design and construction of externally bonded fabric-reinforced cementitious matrix (FRCM) and steel-reinforced grout (SRG) systems for repair and strengthening masonry structures. American Concrete Institute, Farmington Hills, MI, USA.
- Amrhein, J. E., “Reinforced Masonry Engineering Handbook,” Masonry Institute of America, CRC Press (1998).
- Donald Anderson and Svetlana Brzev, “Seismic Design Guide for Masonry Buildings”, Canadian Concrete Masonry Producers Association (2009).
- Drysdale, R. G., Hamid, A. H. and Baker, L. R., “Masonry Structure: Behaviour and Design”, Prentice Hall, Englewood Cliffs (1994).
- FEMA 356, “Prestandard and Commentary for the Seismic Rehabilitation of Buildings”, Federal Emergency Management Agency, Washington, D.C.9 (2000).
- Hendry, A. W., “Structural Masonry”, Macmillan Press Ltd. (1998).
- Paulay, T. and Priestley, M. J. N., “Seismic Design of Reinforced Concrete and masonry Buildings”, John Wiley & Sons (1995).
- Schneider, R.R. and Dickey, W. L., “Reinforced Masonry Design”, 3rd Ed, Prentice Hall (1994).
- Tomazevic, M., “Earthquake Resistant Design of Masonry Buildings”, Imperial Colleges Press (2000).
- Triantafillou, T.C., “Textile Fibre Composites in Civil Engineering”, Woodhead Publishers, UK (2016).

5. Teaching-Learning Strategies in brief (4 to 5 sentences):

The course aims at elucidating theories on the mechanical behaviour of masonry assemblages under different actions and introduces the working stress and limit state approaches to the analysis and design of masonry structures for gravity and lateral loads due to earthquakes. The course will also address structural safety assessment and retrofitting of existing masonry structures.

6. Assessment methods and weightages in brief (4 to 5 sentences):

Assignments and Quizzes - 30%

Mid Semester Exam - 25%

End Semester Exam - 25%

Project – 20%

Title of the Course : Electronic Workshop-1

Name of the Faculty: Praful Mankar + Saikiran B

Course Code : EC2.102

L-T-P : 2-0-3

Credits : 2

Name of the Academic Program: B. Tech. in ECE

Pre-requisite Course/ Knowledge:

Basics of Circuit Analysis

Introductory C Programming

Course Outcome EW1:

CO1 - Familiarization and demonstration of skill in handling electronic equipment and components such as Power Supplies, Signal Generator, CRO, breadboards, soldering iron, passive components and active devices.

CO2 - Design and implementation of electronic circuits that involve analog and digital components, on breadboard and further observing, recording, analyzing and interpreting the results therein.

CO3 - Demonstration of psycho-motor skills in the form of connecting components on a breadboard, wiring, soldering circuits, and understanding of electronic hazards.

CO4 - Understanding and demonstration of tool usage in the form of multi-Sim/LTSpice for simulation, verification and analysis of circuits

CO5 - Understanding the role of software – hardware interface in the form of software implementation on controller boards and their interface to electronic circuits. Demonstrate proficiency on the same

	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PO 12	PS O 1	PS O 2	PS O 3	PSO 4
CO1	1	1	1	1	3	3	1	1	1	1	1	3	3	1	3	3
CO2	3	2	1	3	2	1	1	1	3	3	1	3	3	3	1	1
CO3	1	1	3	2	3	3	3	3	1	1	1	3	2	1	3	3
CO4	1	1	1	1	3	1	1	1	3	1	1	3	3	3	3	1
CO5	1	1	1	1	3	1	1	1	3	1	1	3	1	3	2	3

Detailed Syllabus:

1. **Know your equipment and components** - Lab Equipment and components familiarization such as Power supply, Signal Generator, Oscilloscope, Breadboard, Transistor, Resistor etc...
2. **Design, Implementation and Analysis** - Implement circuits such as Voltage Regulator record, analyse and interpret the results. Around 3-4 circuits will be dealt with in this section.
3. **Electronic Circuit Design Simulation Software** - Learning to install and use Multisim. Design one of the earlier experiments on Multisim and compare hardware and simulation results
4. **The Art of Soldering** - Solder one of the implemented circuits now on a general-purpose PCB/Vector Board, record results, compare with the previous implementation on the bread board
5. **Hardware Software Symbiosis** - Use of controller boards to interface with electronic circuits and actuators, showcase the need for software-hardware interplay

Teaching-Learning Strategies in Brief:

Learning by Implementation and Verification of Theoretical Understanding on Hardware, Individual learning through Experimentation, Participatory Learning and Learning by Interaction and Teamwork through Final Project. The experiments and projects are designed to materialize the above learning strategies. Individual experiments teach and enable real world understanding of concepts of electronic and circuit theory. Quizzes provoke the students towards the connections between theoretical understandings and their actual realization on hardware, often not touched in the regular coursework. Final project materializes an integrated and application driven understanding of the learnings acquired from the experiments.

Reference Books:

1. Hayt, Kemmerly and Durbin, "Engineering Circuit Analysis"
2. Sedra and Smith, "Microelectronic Circuits",
3. Atmel, ATmega2560, User Manual

Grading:

1. Assessment of Lab Performance in 5 Experiments: 30%
2. Quizzes/Viva on Assessment of Theoretical Foundations: 30%
3. Final Project Performance: 40%

Title of the Course Name	: Embedded Systems Workshop
Name of the Faculty	: Deepak Gangadharan + Sachin Chaudhari + Aftab Hussain
Course Code	: EC3.202
L-T-P	: 1-0-3
Credits	: 3
(L= Lecture hours, T=Tutorial hours, P=Practical hours)	
Name of the Academic Program: B.Tech. in Computer Science and Engineering	

1. Prerequisite Course /Knowledge:

10+2 level physics
 CS0.101: Computer programming
 CS3.303: Introduction to IoT

2. Course Outcomes (COs)

After completion of this course successfully, the students will be able to (Create)

CO-1: Develop and implement an IoT-based solution for a real-life problem (Evaluate)

CO-2: Assess system designs from IoT application point of view
 (Understand)

CO-3: Explain the working on microcontrollers, peripherals and its programming (Analyze)

CO-4: Compare and select the sensors and actuators based on the system requirement (Analyze)

CO-5: Compare different communication protocols for use in IoT systems

(Apply)

CO-6: Employ techniques pertaining to the security, privacy and inoperability of IoT data (Analyze)

CO-7: Examine various available solutions for data storage and cloud computing

(Create)

CO-8: Design and fabricate a functional PCB and mechanical enclosure for their IoT project

3. Mapping of Course Outcomes (COs) with Program Outcomes (POs) and Program Specific Outcomes (PSOs) – Course Articulation Matrix

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3	PSO4
CO1	2	2	3	2	2	3	2	2	3	1	1	3	1	3	3	2
CO2	2	3	3	3	2	2	1	2	2	1	3	2	1	3	2	1
CO3	2	3	1	1	1	1	2	1	1	3	2	1	3	1	3	3
CO4	3	3	1	1	1	3	3	1	1	3	1	1	3	1	2	3
CO5	2	1	2	3	3	2	1	1	3	1	2	2	1	2	2	1
CO6	2	3	2	2	2	3	2	3	1	2	2	3	3	3	2	2
CO7	3	2	2	3	1	2	2	2	1	2	2	2	3	3	2	2
CO8	1	2	3	1	3	2	2	1	3	2	3	1	1	2	1	3

‘3’ in the box denotes ‘High-level’ mapping, 2 for ‘Medium-level’ mapping, 1 for ‘Low’-level’ mapping

4. Detailed Syllabus:

1. Sensing/Actuators and Interfacing

1. Sensor/Actuator selection (using datasheets)
2. Physics of sensors and actuators related to projects
3. Interfacing: Serial interfaces, Analog out, SPI, UART, I2C, “propriety” such as DHT22

2. Controller, Embedded Systems and Peripherals-

- Platform selection – ATME328, ESP32, STM8 Architecture; timers, interrupts, AVR, SAMR architectures
- Embedded Systems: power management, interrupts, memory managements, leaks, OTA firmware update, reliability, onboard debugging
- Peripherals: RTC, ADC channels, resolution, onboard memory, power, external/internal watchdog

3. Communications, Networking and IoT Architecture

1. Different IoT communication protocols: Comparison of Zigbee/WiFi/BLE/4G/5G/eSim/LoRaWAN
2. Data Protocols: MQTT/HTTPS/CoAP

4. Data Storage and Computation

1. Cloud storage and computing
2. Data retrieval optimization
3. IoT standards for interoperability: Implementation using oneM2M

5. PCB and Enclosure Design

6. Data privacy and security

7. Dashboard and Visualization

1. Software/Approaches: UI/UX and Time Series Data Visualization; Front-end and back-end technologies

8. Documentation

1. User document and developer's documentation
2. Best practices for writing the two documents
3. Referring style manual. For example, *Microsoft/Chicago manual of style*

Reference:

1. Raj Kamal, Internet of Things, McGraw Hill, 2018
2. P. Lea, Internet of Things for Architects, 2018
3. O. Hersent, D. Boswarthick, O. Elloumi, The Internet of Things, Wiley, 2016
4. D. Norris, The Internet of Things, McGraw Hill, 2015
5. A. Bahga and V. Madisetti, Internet of Things, University Press, 2016

5. Teaching-Learning Strategies in brief:

Lectures will be integrating ICT into classroom teaching, active learning by students, and project-based learning by doing an IoT-based project.

6. Assessment methods and weightages in brief:

Mid semester Quiz	10%
EndSem exam	30%
Project	60%

Title of the Course : Environmental Humanities
Name of the Faculty : Subha Chakraborty
Course Code : HS1.304
Credits : 4
L - T - P : 3-0-0
(L - Lecture hours, T-Tutorial hours, P - Practical hours)
Name of the Academic Program: Humanities Elective
Semester, Year: Monsoon, 2026

Pre-Requisites: No prior background in environmental science is required, but a foundational understanding of literary analysis is recommended.

Course Description: This interdisciplinary course explores the dynamic intersections between the environment and the humanities, focusing on how literature, culture, and philosophy mediate human-environment relationships. Through critical readings, literary texts, and media, students will engage with ecological thought, environmental justice, posthumanism, and the Anthropocene. The course cultivates ethical reflection, historical awareness, and critical imagination to understand climate change not only as a scientific issue but also as a cultural and ethical crisis.

Course Outcomes:

Upon successful completion of this course, students should be able to:

CO-1: Analyze environmental issues through humanities-based frameworks.

CO-2: Engage with key theoretical debates such as the Anthropocene, posthumanism, and environmental justice.

CO-3: Develop a sense of social responsibility, ethics, and equity and to transform into climate-conscious professionals.

CO-4: Critically interpret literature, films, and art as sites of ecological engagement.

CO-5: Articulate informed ethical and political positions on environmental issues.

Course Topics:

The course will be divided into four thematic modules:

Module 1: Introduction to Environmental Humanities

- Overview of Environmental Humanities as a field
- Intersections with literature, ethics, and politics

Core Readings:

- Oppermann, Serpil, and Serenella Iovino, editors. *Environmental Humanities: Voices from the Anthropocene*. Rowman & Littlefield, 2016. (selected chapters)
- Heise, Ursula K. "The Environmental Humanities and the Futures of the Human." *New German Critique*, vol. 41, no. 2 (122), 2014, pp. 21–31.

Literary Text:

- Ghosh Amitav. *The Hungry Tide*. Harper Collins. 2004.
- Ghosh, Amitav. *The Great Derangement: Climate Change and the Unthinkable*. Penguin Books, 2016. (Chapters 1 – 3)
- Devi, Mahasweta. “Pterodactyl, Puran Sahay, and Pirtha.” *Imaginary Maps: Three Stories*, translated by Gayatri Chakravorty Spivak. Seagull Books. 1995. pp. 1–96.

Module 2: Climate Fiction and Speculative Futures

- Understanding the broader historical and cultural contexts that shape climate change and the environment through climate narratives.

Core Readings - Literary and Cultural Texts:

- Margaret Atwood, *Oryx and Crake* (selections)
- Orijit Sen, *River of Stories* (1994)
- Kim Stanley Robinson, *The Ministry for the Future* (2020)
- Climate Dystopias and Utopias in Short Stories and Film/ Media
- *Chasing Ice* (2012) – Dir. Jeff Orlowski
- Selections from the *Dark Mountain Project*
- Short stories from *Solarpunk: Ecological and Fantastical Stories in a Sustainable World*

Module 3: Climate Imaginaries and Planetary Crisis

- Environmental history and critiques of capitalism

Core Readings:

- Chakrabarty, Dipesh. *The Climate of History in a Planetary Age*. University of Chicago Press, 2021.
- Haraway, Donna. *Staying with the Trouble: Making Kin in the Chthulucene*. Duke University Press, 2016.
- Roy, Arundhati. *The Cost of Living*. Modern Library, 1999. (Selections)

Module 4: Posthumanism, Poetry and Environmental Justice

- Environmental Justice
- Gender, Race, and Decolonial Ecologies

Core Readings:

- Fisher-Wirth, Ann, and Laura-Gray Street, editors. *The Ecopoetry Anthology*. Trinity University Press, 2012.
- Petrucci, Mario. *Heavy Water: A Poem for Chernobyl*. Enitharmon Press, 2004.
- Shiva, Vandana. *Staying Alive: Women, Ecology and Development*. Zed Books, 1988. (pp. 1–13)
- Subramanian, Meera. *A River Runs Again: India's Natural World in Crisis, from the Barren Cliffs of Rajasthan to the Farmlands of Karnataka*. PublicAffairs, 2015. (Chapter: "Water")

Readings:

Braidotti, Rosi. *The Posthuman*. Polity Press, 2013.

Demos, T. J. *Against the Anthropocene: Visual Culture and Environment Today*. Sternberg Press, 2017. Ghosh, Amitav. *Gun Island*. Penguin Random House India. 2019.

Heise, Ursula K. *Imagining Extinction: The Cultural Meanings of Endangered Species*. University of Chicago Press, 2016.

Hogan, Linda. *Power*. W.W. Norton, 1998.

Johns-Putra, Adeline, editor. *Climate Change and the Contemporary Novel*. Cambridge University Press, 2019.

Nixon, Rob. *Slow Violence and the Environmentalism of the Poor*. Harvard University Press, 2011.

Shiva, Vandana. *Soil Not Oil: Environmental Justice in an Age of Climate Crisis*. South End Press, 2008.

References:

Attenborough, David. *A Life on Our Planet: My Witness Statement and a Vision for the Future*. Grand Central Publishing, 2020.

Kolbert, Elizabeth. *The Sixth Extinction: An Unnatural History*. Henry Holt and Company, 2014. Maslin, Mark.

Climate Change: A Very Short Introduction. Oxford University Press, 2014.

Thunberg, Greta. *Our House Is on Fire: Scenes of a Family and a Planet in Crisis*. Translated by Paul Norlen, Penguin Press, 2020.

Whyte, Kyle. "Indigenous Climate Change Studies: Indigenizing Futures, Decolonizing the Anthropocene." *English Language Notes*, vol. 55, no. 1–2, 2017, pp. 153–162.

Case studies from different parts of India.

Grading Plan:

Type of Evaluation	Weightage (in %)
Mid-Semester Exam	40 (2 – 3 subjective questions; in-class writing exercises)
End-Semester Exam	40 (2 – 3 subjective questions; in-class writing exercises)
Class Presentation and Weekly Reflections	20 (Active participation in class discussion is necessary to receive full credit for the participation component)

Mapping of Course Outcomes to Program Objectives: (1 – Lowest, 2—Medium, 3 – Highest, or a ‘-’ dash mark if not at all relevant).

	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO10	PO11	PO12	PSO 1	PSO 2	PSO 3	PSO 4
CO1	3															
CO2	3															
CO3		3	3												2	3
CO4				2		2		3								3
CO5									3	3		3				

Teaching-Learning Strategies:

This course is structured in a format that blends lectures and discussions. It is crucial that students come to class on time, with required texts, well prepared to offer insightful responses to the assigned readings. To be effective as class participants, students need to complete reading and writing tasks by the assigned dates.

Title of the Course : **Environmental Science and Technology**

Name of the Faculty : Ramachandra Prasad P

Course Code : CS9.428

L-T-P : 3-1-0.

Credits : 4

(L= Lecture hours, T=Tutorial hours, P=Practical hours)

Open Elective (Monsoon)(UG and PG)

1. Prerequisite Course / Knowledge:

Basics Science (Biology, Physics, Chemistry, Earth systems) to understand environmental issues and phenomenon.

2. Course Outcomes (COs)

After completion of this course successfully, the students will be able to

- CO-1: Understand various environmental issues of concern
 CO-2: Identify various driving factors of environmental degradation
 CO-3: Evaluate different environmental technologies
 CO-4: Integrate IT in designing solutions to combat environmental problems
 CO-5: Comprehend green accounting and evaluation methods for ecosystem goods and services
 CO-6: Develop sense of environmental ethics and environmental legislation

3. Mapping of Course Outcomes (COs) with Program Outcomes (POs) and Program Specific Outcomes (PSOs) – Course Articulation Matrix

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3	PSO4
CO1	1	1	2	2	2	1	1	1	1	1	1	2	2	2	2	2
CO2	1	1	2	2	2	1	1	1	1	1	1	2	2	2	2	2
CO3	2	2	2	2	3	1	2	1	1	1	1	2	2	2	2	2
CO4	2	2	3	2	3	1	2	1	1	1	1	2	2	2	2	2
CO5	2	3	2	2	3	2	3	2	1	1	1	2	2	2	2	2
CO6	2	2	2	2	2	2	2	2	1	1	1	2	2	1	2	2

4. Detailed Syllabus:

- Unit-1: Basics of Environmental Science & Technology: Comprehend environment and its issues, Environmental problems and challenges, Environmental Events,
 Unit-2: Earth components; Climate Change: Climate system, Climate feedback loops, Climate Models, Climate impact on environment; Global Catastrophes, unexpected climate changes
 Unit-3: Role of geospatial technology: in assessing environmental degradation
 Unit-4: Carbon sequestration methods – vegetation, ocean and geological sequestration, IPCC, Clean Development Mechanisms, CO₂, Environmental Stress - Mitigation: Impact on vegetation
 Unit-5: Environmental Impact Assessment: Procedure, regulations and case studies
 Unit-6: Environment and Information technology: green computation, green energy, Green Engineering and technology, e-waste-disposal mechanism – impact on health.
 Unit-7: Green accounting- Evolution of process, history, case studies, Accounting of goods and Services, Sustainability concepts-weak and strong, Hicksian income concept
 Unit-8: Environmental movements
 Unit-9: Environmental Legislation & Impact Assessment: Important legislations related to Environment; Environmental Auditing; Environmental Ethics

References:

1. Environmental Science – The natural environment and human impact (1998): A. R. W. Jackson and J. M. Jackson, Longman
2. Environmental Science (2001): S. C. Santra, New Central Book Agency (P) Ltd
3. Environmental Science (6th ed) (1997): Jr. G. T. Miller, Wadsworth Pub. Co.
4. Dimensions of Environmental and Ecological Economics (2005): N. C. Sahu & A. K. Choudhury (Ed), Universities Press
5. Khoiyangbam, R.S., and N Gupta. 2012. Introduction to Environmental Sciences. New Delhi: TERI
6. Y.K Singh 2006. Environmental Science. New Age International (P) Ltd., Publishers (e-book).
7. Tery Sloan 2016. Introductory Climate Science: Global Warming Explained. New Age International (P) Ltd., Publishers (e-book).
8. Clifford Jones 2015 Global trends and patterns in carbon mitigation (e-book).

5. Teaching-Learning Strategies in brief:

Teaching, discussing current environmental issues, presentations by students on chosen topic, writing as well as drawing assignments, periodical evaluation of course project implemented with open data and tools to understand various environmental processes and possible solutions to combat anthropogenic driven environmental degradation and problems.

6. Assessment methods and weightages in brief:

Theory (%): Quiz (10), Assignments (15), Mid exam-1(20), End exam (30)= **75%**

Project (%): Literature Survey, Preliminary and final presentation along with report = **25%**

***PROJECT:** Simulation and modeling of environmental processes, development of open-source tools related to environmental problems / applications, replication of case studies or working on new problem using open data and tools.

Title of the Course	: Finite Element Method
Name of Faculty	: Jofin George
Course Code	: CE4.501
Credits	: 4
L - T - P	: 3-1-0
Semester, Year	: Monsoon 2026
Name of the Program: Computer-Aided Structural Engineering	

Pre-Requisites

Structural Analysis, Calculus, Linear algebra, Engineering Mechanics

Course Outcomes:

CO1: Develop a comprehensive understanding of finite element principles.

CO2: Formulate and solve boundary value problems using approximate methods and finite element techniques.

CO3: Analyze structural components using the finite element method

CO4: Apply advanced finite element method to elasticity problems.

CO5: Apply numerical integration to identify the solution of boundary value problems.

Detailed Syllabus:

Module 1: Introduction to finite element method (FEM): Need for FEM, General finite element solution procedure, Boundary value problems: types of boundary conditions, Approximate solution to boundary value problems: Method of weighted residuals,

Module 2: General one-dimensional (1D) boundary value problem and its applications: Axial deformation of bars: Approximate solution using Rayleigh-Ritz method, Galerkin method, Axial Spring element. Applications: One-dimensional heat flow, Column buckling. Higher order elements for 1D problems: Shape functions for second order problems, **Isoparametric** mapping concept, Quadratic **isoparametric** element.

Module 3: Beams, Frames and Trusses: Beam bending-governing differential equation, Two node beam element: Exact solution, Calculation of stresses in beams, Thermal stresses in beams. Analysis of structural frames-Plane frame element, space frame element, Introduction to truss analysis.

Module 4: 2D boundary value problems: triangular elements, isoparametric quadrilateral elements, **Isoparametric** mapping for quadrilateral elements, Numerical integration for quadrilateral elements, Four node quadrilateral element for 2D BVP, Eight node serendipity element. Isoparametric triangular elements: Natural coordinates for triangles, Shape functions for triangular elements, Natural coordinate mapping for triangles, Numerical integration for triangles, Six node triangular element for general 2D BVP.

Module 5: Applications based on general 2D boundary value problem: Torsion of prismatic bars. 2D elasticity-Governing differential equations, Constant strain triangular element, Four node quadrilateral element, Eight node isoparametric element. Constitutive equations, governing differential equations, stress equilibrium equations, finite element equations for plane stress and plane strain conditions, three-noded triangular elements, and axisymmetric elasticity problems.

Module 6: Numerical integration-Newton-Cotes rules, Trapezium rule, Simpson's rule, Error term, Gauss-Legendre rules, Gauss-Legendre rule, Numerical integration for quadrilateral elements, Numerical integration for triangular elements.

Reference Books:

1. Bhatti, M.A., Fundamental Finite Element Analysis and Applications: with Mathematical and **MATLAB** Computations, Wiley, 2005.
2. Reddy, J. N., An Introduction to the Finite Element Method, 3rd Edition, McGraw-Hill Science/Engineering/Math, 2005.
3. Logan D. L., A First Course in the Finite Element Method, Thomson- Engineering, 5th edition, 2010.

Additional Reading

1. Chandrupatla T. R., and Belegundu, A. D., Introduction to Finite Elements in Engineering, Pearson Education, 4th edition, 2015.
2. Buchanan, G. R., and Rudramoorthy, R., Schaum's Outline of Finite Element Analysis, McGraw Hill Education (India) Private Limited, 2005.
3. Zienkiewicz, O.C. and Taylor, R.L., The Finite Element Method for Solid and Structural Mechanics, Butterworth-Heinemann, 6th edition, 2005.
4. Zienkiewicz, O.C., Taylor, R.L., and Zhu, J.Z., The Finite Element Method: Its Basis and Fundamentals, Butterworth-Heinemann, 7th edition, 2013.

Grading Plan

Type of Evaluation	Weightage(in %)
Quiz-1	10
Mid Sem	15
Quiz 2	10
End Sem Exam	30
Assignments	15
Term Project	20

Mapping of Course Outcomes to Program Objectives: (1 – Lowest, 2—Medium, 3 – Highest, or a ‘-’ dash mark if not at all relevant).

	Program Outcomes										Program Specific Outcomes			
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PSO1	PSO2	PSO3	PSO4
CO1	3	2	3	1	3	1	2	3	3	2	3	3	3	2
CO2	3	2	3	1	3	1	2	3	3	3	3	3	3	3
CO3	3	3	3	-	3	1	2	3	2	3	3	3	3	3
CO4	3	3	3	-	2	1	2	3	2	1	3	3	3	3
CO5	3	3	3	-	1	1	2	3	3	1	3	3	3	3

Teaching-Learning Strategies in Brief:

- Classroom Lectures
- Weekly tutorials on problem solving.
- Term project involving the use of Finite element concepts to solve structural engineering problems.
- Active learning by students.

Title of the Course : Foundations for Signal Processing and Communication

Faculty Name : Santosh Nannuru + Gowtham Kurri

Course Code : EC5.412

Credits : 4

L - T - P : 3-1-0

(L - Lecture hours, T-Tutorial hours, P - Practical hours)

Semester, Year : Monsoon, 2026

Name of the Program: ECE

Pre-Requisites : 1) Basics of probability, vector, and matrices.

2) Signal and System, and Digital Communication.

This course is intended for students who wants conduct research in the field of communication or signal processing.

Course Outcomes :

(List about 5 to 6 outcomes for a full 4 credit course)

After successful completion of this course, the students will

CO1: build strong foundation in the fundamental mathematical concepts used for the design of digital communication systems.

CO2: be able to apply mathematical tools to analyze and solve complex problems related to signal processing.

CO3: be equipped with basic models/modules to conduct research in the field of communication and signal processing.

CO4: develop critical thinking and problem-solving skills to address practical problems in communication systems and signal processing.

Course Topics :

(Please list the order in which they will be covered, and preferably arrange these as five to six modules.)

Module 1: (5-6 lectures)

Rigorous Math Part: Probability Axioms, Random Variables and their distributions (CDF, PDF, PMF), Some Important Distributions, Joint and Conditional Distributions, Distributions of functions of Random Variables, Expectation of RVs and their functions
SPCOM Part: Random quantities in SPCOM, Notion of Communication Channel, Hypothesis testing, Simple Detection and Estimation problems in Signal Engineering.

Module 2 (4-5 lectures)

Rigorous Math Part: Concentration inequalities (Markov, Chebyshev, Chernoff), Central Limit Theorem, Mean vector and covariance matrix, Joint Gaussian Distribution, Random Processes
SPCOM Part: Discrete Memoryless Channels and Notion of Channel Capacity, Gaussian Noise, Wireless Channel.

Module 3: Sets with structure (4-5 lectures)

Rigorous Math Part: Groups (Cyclic groups, Abelian groups), Rings (polynomials and matrices), Fields (Finite and Infinite), Vector spaces, Inner Product on Vector Spaces.
SPCOM part: The Vector Space of Finite Energy Continuous Signals (L_2 signals, over $\mathbb{R}$) with inner product, Vector Space of Sampled Signals (over $\mathbb{R}$ and $\mathbb{C}$), Vector Space of discrete-time quantized signals (over finite fields), Distinguishing features of Digital Signal Processing and Communication

Module 4: (5-6 lectures)

Rigorous Math part: Linear Independence/dependence, Span, Column space and row space of matrices, Basis and Dimension of subspaces, Basis Extension Theorem, Rank of a matrix, Change of Basis.
SPCOM part: Fourier Series of L_2 signals as representation of signals using Fourier basis, Finite dimensionality of bounded-time, bounded-bandwidth L_2 signals, standard basis for Discrete-time signals, Standard Basis for Finite-field Signals, Discrete Fourier Transform as Change of Basis.

Module 5: (4-5 lectures)

Rigorous Math Part: Linear Transformations between vector spaces and their associated Matrices, Eigen Values and Vectors, Triangularization, Diagonalization, SVD.
SPCOM part: Continuous-time and Discrete-Time LTI systems and their Transfer Functions, Role of linear transformations in coding theory for forward-error correction, Do Estimation, MIMO Wireless Communication.

Preferred Textbooks:

1. Henry Stark and John Wood: Probability and Random Processes with Application to Signal Processing
2. Sheldon Axler: Linear Algebra Done Right

Reference Books :

1. Robert G. Gallager, "Principals of Digital Communication"
2. Lin and Costello, "Error Control Coding"
3. Steven Kay, "Fundamentals of Statistical Processing: Estimation Theory", Volume 1
4. Steven Kay, "Fundamentals of Statistical Processing: Detection Theory", Volume 2

E-book Links :

Grading Plan : Relative grading

(The table is only indicative)

Type of Evaluation	Weightage (in %)
Quiz-1	10
Mid Sem Exam	20
Quiz-2	10
End Sem Exam	40
Assignments	10
Project	-
Term Paper	10
Other Evaluation	-

Mapping of Course Outcomes to Program Objectives: (1 – Lowest, 2—Medium, 3 – Highest, or a ‘-’ dash mark if not at all relevant). Program outcomes are posted at

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PSO1	PSO2	PSO3	PSO4
CO1	5	5	5	4	3	5	1	1	1	5	1	1
CO2	5	5	5	4	3	5	1	1	1	5	1	1
CO3	5	5	5	4	3	5	1	1	1	5	1	1
CO4	5	5	5	4	3	5	1	1	1	5	1	1

Teaching-Learning Strategies in brief (4-5 sentences):

The objective of this course is to provide students with the necessary knowledge in linear algebra and probability for signal processing and crucial for analysing and designing communication systems. The course syllabus is divided into five modules, with three focusing on linear algebra and two on probability. Each module will be taught through a two-step approach. In the initial lectures, we will concentrate on establishing a solid foundation in the fundamental mathematical tools required. Subsequently, the following lectures will delve into applying these tools to communication and signal processing engineering. To foster critical thinking skills, students will be assigned innovative theoretical and simulation problems as part of their assignments. In addition, the students will be exposed to the seminal research works through the exercise of writing term papers.

Title of the Course : FPGA Based Accelerator Design

Faculty Name : Suresh Purini

Course Code : CS2.401

Credits : 4

Type-When : Elective Course, Monsoon 2026

Prerequisite: Digital Systems Design, Computer Systems Organization, C/C++ Programming

Objective: To design, develop and deploy application-specific accelerators targeting domains such as computer vision, speech processing, natural language processing and genomics.

Course Topics: Trends of Heterogeneous Computing, OpenCL Models for Heterogeneous Computing, OpenCL for FPGAs, pros and cons of various hardware specification systems (HDLs vs HLS vs OpenCL), Architectural Design Approaches (Single Cycle, Multi-Cycle and Pipelined Designs), exploiting memory hierarchies, performance analysis and tuning, design examples from machine learning, deep neural networks, image and video processing, genomics etc.

Preferred Text Books: None

***Reference Books:**

1. Manuals from Xilinx and Intel
2. Design of FPGA based Computing Systems with OpenCL

Course Outcomes

1. Describe the basics of digital systems design, architectural components of FPGAs, their application and utilization.
2. Design, analyze and evaluate alternate system architectures for a given complex domain problem using the techniques such as folded combinational and pipelined design approaches.
3. Compare and contrast multiple design alternatives with theoretical peak performance obtained using roofline analysis.
4. Able to use digital system design tools such as Verilog, VHDL, BlueSpec System Verilog, High Level Synthesis C, OpenCL etc.
5. Design accelerators for novel application areas such as deep learning, genomics, computational finance etc.

Mapping of Course Outcomes to Program Objectives: (1 – Lowest, 2—Medium, 3 – Highest, or a ‘-’ dash mark if not at all relevant). Program outcomes are posted at

	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PO 12	PSO 1	PSO 2	PSO 3	PSO 4
CO 1	3	3	3	3	2	2	2	3	3	3	3	3	3	3	3	3
CO 2	3	3	3	3	2	2	2	3	3	3	3	3	3	3	3	3
CO 3	3	3	3	3	2	2	2	3	3	3	3	3	3	3	3	3
CO 4	3	3	3	3	3	2	2	3	3	3	3	3	3	3	3	3
CO 5	3	3	3	3	3	2	2	3	3	3	3	3	3	3	3	3

Grading Plan:

Type of Evaluation	Weightage (in %)
Quizzes	10
Assignments	30
Term paper	15
Project	35
Final Oral Exam	10

Outcome: Acquire a thorough knowledge and experience in designing and developing complex digital systems for diverse applications.

Title of the Course : Gender and Society

Faculty Name : Sushmita Banerji

Course Code : HS8.306

L-T-P : 3-0-0

Credits : 4

(L= Lecture hours, T=Tutorial hours, P=Practical hours)

Name of the Academic Program: Elective

1.Prerequisite Course / Knowledge: None2.Course Outcomes (COs):

After completion of this course successfully, the students will be able to.

CO-1 Introduce students to basic concepts in gender theory and Feminist practice

CO-2 Help students question their prior opinions and think in more informed ways about the nature of gender relations, individual roles, and socio-cultural formations.

CO-3 Literature shall be taught to demonstrate the various ways in which culture establishes, represents and perpetuates

CO-4 Film shall be used to demonstrate the various ways in which popular culture establishes, represents and perpetuates

CO-5 Examples of Literature and film shall be used to discuss how culture can disrupt generic roles

3.Mapping of Course Outcomes (COs) with Program Outcomes (POs) and Program Specific Outcomes (PSOs) – Course Articulation Matrix

	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO1 0	PO1 1	PO1 2	PSO 1	PSO 2	PSO 3	PSO 4
CO1				1												
CO 2						1										
CO 3								2								
CO 4												1				
CO 5						2										

4.Detailed Syllabus:**Unit 1: Core concepts and terms:** Differences between terms like Gender, Sex, Normative and Non-normative sexuality, Trans-bodies.

Unit 2: Power, Ideology and Intersectionality's: Concepts of Power, Ideology, Patriarchy, and Privilege. What are intersectionality's, and why is it important to study them when we study gender?

Unit 3: Feminist Movements: Rights struggles, Gains and losses, women in the workplace, Women in India

Unit 4: Representation of Gender: Who writes women? Short Stories on, about, and by women.

Unit5: Popular representations of women in cinema. How does popular visual culture shape gender politics?

Reference Books:Adichie, Chimamanda Ngozi (2014). *We Should All Be Feminists*. Fourth Estate.Beauvoir, Simone (2010), *The Second Sex*. Trans. Constance Borde and Sheila Malovany-Chevallier. Vintage Books: London

Butler, Judith (1990), *Gender Trouble: Feminism and the Subversion of Identity*, New York: Routledge.
 Kumar, Radha (2002), *A History of Doing: Movements for Women's Rights and Feminism in India, 1800-1990*. India, Kali for Women.
 Menon, Nivedita (2012), *Seeing Like a Feminist*. New Delhi, Penguin.
 Tharu, Susie and K. Lalita eds. *Women Writing in India: 600 B.C. to the Present. I and II*. Delhi: Oxford University Press, New York: Feminist Press and London: Harper Collins, 1990-1993.
 Uberoi, Patricia (2006) *Freedom and Destiny: Gender, Family, And Popular Culture In India*. USA: Oxford University Press.

5. Teaching-Learning Strategies:

Students are expected to read up to 30 pages a week and attend film screenings when required. Lectures will be based on class readings and will assume that students will have read the required materials. Discussions in class, on chat and via emails shall be encouraged. Students are expected to write at least two, perhaps three papers that will be designed to encourage interpretative and creative writing.

6. Assessment methods and weightages:

Type of Evaluation	Weightage (in %)
Quiz 1	10%
Mid Sem- Exam	20%
End Sem Exam	20%
Term paper 1	25%
Term Paper 2	25%

Title of the Course : Human Sciences Lab-1

Faculty Name : Radhika Krishnan

Course Code : HS7.101

Credits : 2

L-T-P : 3-0-0

(L= Lecture hours, T=Tutorial hours, P=Practical hours)

Name of the Program: B. Tech in Computer Science and M.S. in Computing and Human Sciences by Research

1. Prerequisite Course / Knowledge: Admission to Human Sciences Dual Degree Programme

2. Course Outcomes (COs):

After completion of this course successfully, the students will be able to.

CO-1: Students will get an introduction to computational tools which can be used in social science research.

CO-2: Students will start using social theory to read, analyze and interpret various kinds of data.

CO-3: Students will be exposed to the potential of computational humanities and social sciences. They will identify not just the potential, but equally the limitations of computational methods in the social sciences.

3. Mapping of Course Outcomes (COs) with Program Outcomes (POs) and Program Specific Outcomes (PSOs) – Course Articulation Matrix

	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO1 1	PO1 2	PSO 1	PSO 2	PSO 3	PSO 4
CO 1	1	3	1	1	3	2	1	1	1	1	1	1	1	1	3	1
CO 2	1	3	1	1	3	2	1	1	1	1	1	1	1	1	3	1
CO 3	1	3	1	1	3	2	1	1	1	1	1	1	1	1	3	1

Note: Each Course Outcome (CO) may be mapped with one or more Program Outcomes (POs) and PSOs. Write '3' in the box for 'High-level' mapping, 2 for 'Medium-level' mapping, 1 for 'Low'-level' mapping

4.Detailed Syllabus:

Unit 1: Bag of Words exercise, followed by analysis of the results.

Unit 2: Identification and interpretation of correlations in social data. Statistical analysis, followed by study of the historical/sociological/political context.

Unit 3: Crawling of tweets, followed by an analysis of the results from a social science perspective.

Unit 4: Scraping of newspaper articles, followed by an analysis of the results from a social science perspective.

Unit 5: Working with Political Data: Extracting electoral data from various sites, followed by an analysis of the results from a social science perspective.

Reference Books:

5.Teaching-Learning Strategies in brief (4 to 5 sentences):

All the computational exercises are hands-on, where a teaching assistant and the faculty in-charge work together with the students. All classes are interactive. The computational exercises are followed by brief lectures on social theory which can help in the social science analysis. Discussions are highly encouraged, especially during the analysis from a social science perspective. Students will be trained develop a good grasp of distilling a research work to its core meaning while drawing out their implication and affectations.

6.Assessment methods and weightages in brief (4 to 5 sentences):

Class exercises: 75% (5 exercises, each accounting for 15%)

Project: 25%

Title of the Course	: Identity and Economic Development
Faculty Name	: Angarika Rakshit
Name of the Program	: CHD
Course Code	: HS5.402
Credits	: 4
L - T - P	: 3 – 1 - 0
(L - Lecture hours, T-Tutorial hours, P - Practical hours)	
Semester, Year	: Monsoon 2026
Pre-Requisites	: None

Course Outcomes :

CO-1

Understand Theoretical Perspectives on Identity and Development

Develop an understanding of how different schools of thought explain the relationship between social identity and economic outcomes. Students will examine how identities such as gender, caste, religion, ethnicity, and region shape access to resources, opportunities, and participation in economic life.

CO-2

Analyze the Mechanisms that Reproduce Inequality

Examine the historical, institutional, and market-based processes through which identity-based inequalities emerge and persist over time. The course will explore how disparities in education, health, social networks, and asset ownership translate into unequal economic outcomes across generations.

CO-3

Learn Methods for Measuring Inequality, Segregation, and Disadvantage

Develop familiarity with the concepts, indicators, and empirical methods used to study identity-based inequalities. Students will learn how researchers measure discrimination, segregation, representation, and economic disadvantage, and critically assess the strengths and limitations of these approaches.

CO-4

Evaluate Policy Approaches to Inclusion and Equity

Critically assess competing policy responses to identity-based inequalities. Drawing on both theoretical perspectives and empirical evidence, students will evaluate interventions aimed at reducing disparities and promoting inclusive development.

Course Description: This course examines the relationship between social identity and economic development through an interdisciplinary political economy perspective rooted in economics. Challenging the view that economic outcomes are determined solely by productivity and market forces, it explores how identities such as caste, gender, religion, ethnicity, and region shape access to opportunities, resources, and economic outcomes. The course investigates the historical, institutional, and market mechanisms through which inequalities emerge, persist, and reproduce across generations, influencing education, employment, occupations, wages, and wealth. It also introduces students to empirical approaches for measuring discrimination, segregation, and disadvantage, and evaluates competing theoretical explanations and policy responses to identity-based inequality. While drawing on global experiences, the course places particular emphasis on contemporary debates and evidence from India.

Module I: Identity and Markets

- **Identity and Economics**
 - Neoclassical economics and the abstract economic agent
 - Identity and economic behaviour
 - Equality of opportunity vs equality of outcomes
 - Markets, institutions, and social norms
- **Historical Origins of Identity-Based Inequality**
 - Race, slavery, and capitalism
 - Caste and occupational hierarchy
 - Patriarchy and gendered division of labour
 - Colonialism and labour systems
 - Historical persistence of inequality

Module II: Measuring Identity-based Inequality and Disadvantage

- **Measuring Inequality and Segregation**
 - Representation Index
 - Duncan Dissimilarity Index
 - Concentration Index
 - Mobility Matrices
 - Oaxaca-Blinder decomposition
- **Limits of Measurement**
 - Underrepresentation and invisibility
 - Challenges in measuring discrimination
 - Quantitative vs qualitative evidence

Module III: Evidence on Identity-based Disadvantages

- **Pre-labour Market Disadvantages**
 - Residential segregation, regional disparities
 - Access to infrastructure and Public Goods
 - Access to Education and Health
- **Identity and Labour Market Outcomes**
 - Labour hierarchies
 - Industrial and occupational segregation
 - Wage inequality
 - Identity and ownership of capital
 - Asset inequality

Module IV: Policy, Political Economy, and Competing Perspectives

- **Competing Perspectives on Identity and Inequality**
 - Neoclassical approaches
 - Institutional and behavioural approaches
 - Marxian and political economy perspectives
 - Feminist economics
- **Can Policy Reduce Identity-based Inequality?**
 - Affirmative action and reservation
 - Welfare and redistribution
 - Universalism vs targeting

Preferred Readings :

1. Identity Economics: How Our Identities Shape Our Work, Wages, and Well-Being, George Akerlof and Rachel Kranton, 2010, PUP
2. The Economics of Discrimination, Gary S. Becker, 1957, UCP
3. Theories of Discrimination, Kenneth Arrow, 1973
4. Blocked by Caste, Sukhdeo Thorat and Katherine Newman, 2010, OUP
5. Are Emily and Greg More Employable than Lakisha and Jamal? Marianne Bertrand and Sendhil Mullainathan, 2004, AER

Students will be referred to other relevant journal articles during the course

Grading Plan :

(The table is only indicative)

Type of Evaluation	Weightage (in %)
Quiz-1	15
Mid-Sem	30
Quiz-2	15
End-Sem	40

Mapping of Course Outcomes to Program Objectives: (1 – Lowest, 2—Medium, 3 – Highest, or a ‘-’ dash mark if not at all relevant). Program outcomes are posted at

<https://intranet.iiit.ac.in/offices/static/files/PEOs%20CPOs%26PSOs-ofAllProgrammes-Jan2023.pdf>

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3	PSO4
CO1	1	2	3	2	3	1	3	2	1	1	-	2	1	2	3	2
CO2	1	3	3	3	3	2	3	3	3	2	-	3	1	2	3	1
CO3	1	2	3	3	3	2	3	3	3	3	-	3	1	2	3	1
CO4	1	2	3	2	3	3	1	2	1	2	-	3	1	2	2	2
CO5																

Teaching-Learning Strategies in brief (4-5 sentences):

The course will combine lectures with a range of interactive and applied learning activities to strengthen both conceptual understanding and empirical skills. Lectures will introduce key theoretical frameworks, debates, and empirical findings on the relationship between social identity and economic development. Students will engage critically with assigned readings through structured reading responses and classroom discussions, examining how social identity shape economic opportunities and outcomes. The course will also incorporate applied data analysis exercises that enable students to investigate patterns of inequality and engage directly with empirical evidence. Together, these components are designed to foster critical thinking, and the ability to connect theoretical perspectives with real-world processes of inequality, inclusion, and economic change.

Title of the Course : Information Retrieval and Extraction

Name of the Faculty : Anil Nelakanti

Course Code : CS4.406

L-T-P : 3-1-1

Credits : 4

(L= Lecture hours, T=Tutorial hours, P=Practical hours)

Name of the Academic Program: B.Tech. in Computer Science and Engineering

1.Prerequisite Course / Knowledge:

Basic principles of Computer programming, Statistical Methods in Artificial Intelligence, Programming experience in Python, and Algorithms.

2.Course Outcomes (COs)

After completion of this course successfully, the students will be able to...

CO-1. Develop algorithms to retrieve information from unstructured data

CO-2. Design and architect information retrieval systems for worldwide web

CO-3: Design Web crawling systems

CO-4. Design algorithms to process noisy data in document repositories

CO-5. Develop information extraction system's

3.Mapping of Course Outcomes (COs) with Program Outcomes (POs) and Program Specific Outcomes (PSOs) – Course Articulation Matrix

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3	PSO4
CO1	3	3	3	3	3	1	1	-	2	2	2	2	3	3	2	2
CO2	3	3	3	3	3	1	1	1	2	2	2	2	3	3	2	2
CO3	3	3	3	3	2	1	1	1	2	2	2	2	3	3	2	2
CO4	3	3	3	3	2	1	1	-	3	2	2	2	3	3	2	2
CO5	3	3	3	3	2	1	1	-	3	2	2	2	3	3	2	2

'3' in the box denotes 'High-level' mapping, 2 for 'Medium-level' mapping, 1 for 'Low'-level' mapping

4.Detailed Syllabus:

Unit 1: Introduction to Information retrieval, Information Extraction and Information Access systems. (6 hours)

Unit 2: Information Retrieval Models and Evaluation of IR systems (7.5 hours);

Unit 3: Web Information Retrieval (4.5 hours)

Unit 4 Natural Language Processing in IR (7.5 hours)

Unit 5: Machine Learning in Information Retrieval Systems (12 hours)

Unit 6: Information Extraction (4.5 Hours)

Unit 7: IR Applications (12 Hours)

References:

1. Introduction Information Retrieval – Chris Manning et al (the Stanford IR Book) (ISBN-13: 978-0521865715)
2. Search Engines: IR in Practice – Bruce Craft et al (ISBN-13: 978-0136072249)
3. Research papers

5.Teaching-Learning Strategies in brief:

Lectures by integrating ICT into classroom teaching, weekly tutorials involving problem solving and active learning by students and Project-based Learning by doing one mini-project and a major project by the students.

6.Assessment methods and weightages in brief:

Assignments (theory and programming): 20 marks

Seminar: 10 marks

Project: 35 marks

Mid Semester Examination: 15 marks

End Semester Examination: 20 marks

Karma Points: Exceptional performance, active participation in class/tutorials, and help provided to other students, which a TA or instructor notices deem worthy of 3 marks.

Title of the Course	: Information, Computation, and Learning
Faculty Name	: Prasad Krishnan
Name of the Program	: ECE (Also intended as Open Elective for ECE/CSE)
Course Code	: EC5.415
Credits	: 3:1:0:4
L - T - P	:
(L - Lecture hours, T-Tutorial hours, P - Practical hours)	

Semester, Year : **Monsoon 2026**

Pre-Requisites : Basics of Probability (Discrete and Continuous Random Variables, pdfs/pmfs, Gaussian. Course is meant to be accessible to new MS/PhD students also, apart from B.Tech/DD in CSE/ECE.

Prior courses such as Mathematics of Information (UG2-half), Information and Communication (UG1-ECE) can be useful but not mandatory.

Note: Probability review, Discrete Entropy and its Properties – elaboration on these initial topics will be moved to tutorial sessions for those students who have not seen them in a course before.

Course Outcomes :

1. Explain various fundamental quantities in Information-theory and analyze their mathematical properties and utility.
2. Formulate Data Compression and Communication through Channels as Information-theory-based problems and characterize their fundamental limits. .
3. Apply information theoretic ideas to study the fundamental limits of secure computation and various related sub-areas.

4. Apply information-theoretic ideas for understanding machine-learning and improving the same.
5. Comprehend, write, and communicate academic research in Information-theory and applications, effectively as an individual or groups, while executing a project or a research paper.

Course Topics:

Unit 1: Basics of Information Theory : Entropy and Mutual Information, KL Divergence and its properties, Differential Entropy and its properties, Gaussian Entropy as the largest entropy for any RV with fixed variance,

Unit 2: Source and Channel Coding: Typical Sequences and their properties, Asymptotic Equipartition, Proof of Source Coding, Channel Coding and its proof for Gaussian Channels and DMC, Ideas on fundamental Limits in Other Channels such as Permutation Channels, Shuffling Channels, etc.

Unit 3: Secure Computation: Secret Sharing (Threshold, Ramp, General Access Structures), Two Party Secure Computation, Oblivious Transfer, Zero Knowledge Proofs, Multiparty Secure Computation

Unit 4: Learning: Rate Distortion theory, Learning as compression under distortion measure, the Information Bottleneck principle, Information theory and Fairness in Machine Learning, Hypothesis Testing basics, Introduction to bounds on Generalization in Neural Networks.

Preferred Textbooks:

1. Elements of Information Theory Author(s): Thomas M. Cover, Joy A. Thomas.
2. Information Theoretic Cryptography, Himanshu Tyagi and Shun Watanabe, Cambridge University Press, Online ISBN: 9781108670203, <https://doi.org/10.1017/9781108670203>
3. Changho Suh. (2023). Information Theory for Data Science. Boston–Delft: Now Publishers

Reference Books:

1. Information Theoretic Security, Foundations and Trends in Communications and Information Theory Vol. 5, 2009, Y. Liang, H. V. Poor and S. Shamai (Shitz) DOI: 10.1561/01000000036
2. The Information Bottleneck: Theory and Applications, Slonim's Thesis, Hebrew University, 2002.
3. Other research papers.

Grading Plan :

(The table is only indicative)

Type of Evaluation	Weightage (in %)
Mini Quizzes (3 mini-quizzes)	15
Mid Sem Exam	20
End Sem Exam	35
Tutorial Problem Solving Sessions (2 marks per session, based on participation in at least 5 tut sessions. Max of 10 marks)	10
Project / Term Paper	20

Mapping of Course Outcomes to Program Objectives: (1 – Lowest, 2—Medium, 3 – Highest, or a ‘-’ dash mark if not at all relevant). Program outcomes are posted at

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3	PSO4
CO1	3	3	3	1	1	2	1	1	1	1	2	1	2	2	3	2
CO2	3	3	3	1	1	1	1	1	1	1	1	1	2	3	2	1
CO3	3	3	3	1	1	2	1	2	1	1	2	1	2	3	2	1
CO4	3	3	3	1	1	1	2	2	1	2	1	1	2	2	2	3
CO5	3	3	3	1	1	2	1	1	1	1	1	1	3	3	2	2

Teaching-Learning Strategies in brief (4-5 sentences):

Probability review, Discrete Entropy and its Properties – elaboration on these initial topics will be moved to tutorial sessions for those students who have not seen them in a course before.

The key goal of the course is to generate good understanding of the classical role of information theory in communication systems and data compression, while also highlighting its deep impact in much more modern fields such as secure computation and machine learning. The course will tread a careful line between making sure the students who have not taken a formal course in basics of information theory, and those who have, are given equal opportunities in getting a successful grasp of the course topics. Tutorials will be planned and attendance will be mandated via a certain percentage of the grades, and in-class quizzes are planned early on so that all students arrive at a common platform by mid-sem. Term paper presentations or project are expected to ensure student is ready to latch on to modern research trends in this area.

Title of the Course : Intro to Cognitive Science

Course Code : CG1.402

Course Information

Instructor Information

Faculty Names : Bhakti Dongaonkar and Bapi Raju

TAs :

Day/Time: Tue/Th 11:40 am – 1:05 pm.

Venue: H201

Virtual Office Hours: By appointment (please email).

E-mail: vishnu.sreekumar@iiit.ac.in and raju.bapi@iiit.ac.in

First point of contact – TAs; emails: r.pooja@research.iiit.ac.in and pritha.ghosh@research.iiit.ac.in

Course Information

Course Description: Cognitive Science is a highly interdisciplinary field of study that seeks to understand how the mind works. In this course, we will discuss a diverse range of perspectives from philosophy, linguistics, psychology, neuroscience, and computer science, on how to unravel the mysteries of human cognition.

Credits: 4

L-T-P: 3-1-0 (L = lecture hours, T = tutorial hours, P = practical hours)

Prerequisite: None

Textbook & Course Materials

Recommended Texts & Other Readings: Lecture slides and supplementary readings will be posted to Moodle.

Course Technology Requirements

- You will need access to the following tools to participate in this course.
 - Laptop/desktop computer
 - webcam
 - microphone
 - a stable internet connection (don't rely on cellular)

Course Structure

This course will be delivered fully in-person in a physical classroom.

Student Expectations

In this course you will be expected to complete the following types of tasks.

- communicate via email
- complete basic internet searches
- download and upload documents to the course site on Moodle
- read documents online
- view online videos
- participate in online discussions
- complete quizzes/tests online
- upload documents to a Dropbox/Moodle
- participate in synchronous online discussions

Expected Instructor/TA Response Times

- We will attempt to respond to student emails within 24 hours. If you have not received a reply from us within 24 hours, please resend your email.
 - ***If you have a general course question (not confidential or personal in nature), please post it to the Course Q&A Discussion Forum found on the course homepage on Moodle. We will post answers to all general questions there so that all students can view them. Students are encouraged to answer each other's questions too.
- We will attempt to reply to and assess student discussion posts within 48 hours.

Course Outcomes (COs)

After successful completion of this course, students will be able to:

- CO-1: demonstrate familiarity with seminal research findings in cognitive science.
- CO-2: read, interpret, critique, and evaluate research in cognitive science.
- CO-3: critically think about the relationship between diverse fields such as AI, philosophy, neuroscience, and cognitive science.
- CO-4: identify flaws in how scientific results are communicated and critique scientific work in

terms of confounds, experimental design, etc.

- CO-5: appreciate the nature of scientific debate in cognitive science and be able to generate well-informed perspectives on these debates.

You will meet the outcomes listed above through a combination of the following activities in this course:

- Attend lectures and participate in class discussions (CO-1, CO-2, CO-3, CO-4, CO-5)
- Debate sessions (CO-1, CO-2, CO-3, CO-5)
- Quiz 1, Quiz 2, mid-semester, and end-semester exams (CO-1, CO-2, CO-3, CO-5)
- Complete a term paper/debate reaction paper (CO-1, CO-2, CO-3, CO-5)

List of topics and activities

Students are encouraged to ask questions and steer the lecture in directions that are interesting to them. We will still manage to cover a range of topics from the ones listed below, but we will learn via the Socratic Method.

- Introduction
- Evolution of Cognitive Science
- A free-form discussion on consciousness
- Empirical approaches in cognitive science
- Brain: Organization; Intro to sensation and perception
- Sensory systems
- Perception and Perceptual Learning, Cross-modal interactions
- Vision
- Attention
- Learning
- Development
- Memory
- Language and Cognition
- Knowledge Representation
- Special topics: e.g. Music, mind, and technology
- Several debate sessions with student debate teams

Grading Policies

Graded Course Activities

Description	Percentage
Quiz 1	10%
Quiz 2	10%
Debate+report or Term Paper (in class)	20%
Mid semester exam	20%
End semester exam	35%
Experiment participation and reflection paper (1.5-2h worth of experiments)	5%
Total (100 marks)	100

Quizzes

Quiz 1 will cover topics covered until Quiz 1, and Quiz 2 will cover topics taught between Quiz 1 and Quiz 2. They will contain mostly multiple choice questions.

Mid-semester exam

The mid-semester exam will cover all material taught up to that point, and may include both multiple choice and descriptive questions.

End semester exam

The end semester exam will cover material taught during the whole semester and will include both multiple choice and descriptive type questions.

Debate participation (25 marks = 15 marks for presenting + 10 marks for a short report)

We will reserve at least 3-4 lecture slots for student debates on contemporary issues in Cognitive Science. A list of representative topics are as follows:

1. Are there top-down influences on basic perception? Evidence for and against.
2. Do 3 year olds have a theory of mind?
3. Is cognition/consciousness a computational process?
4. Do we need representations for cognition?

Each debate team will have 3 members. They will read the recommended material for the chosen topic, and organize their arguments distributed across the 3 members. Each member gets 5 minutes to present their arguments (15 minutes per team). They may choose to use slides or not but the arguments must be clearly presented. At the end of both teams' presentations, each team gets 5 minutes for rebuttal when they can pick 2-3 claims made by the opposite team and present counterarguments.

The students participating in debate teams will only be required to write a short report but the

remaining students will need to write a reaction paper to any one debate session OR write a term paper on any other topic that they choose (see next main section).

For debate team students (each person writes this separately without discussion with other team members), your short report should contain the following:

The paper will first summarize the problem (2 marks), and then summarize the arguments made by both sides (3 marks), and then will provide the student's OWN opinion about where they stand on the debate and what arguments were convincing to them (5 marks).

Recommended: 3+ sides of A4 size sheets, hand-written (in class – one side A4 sheet handwritten cheat sheet will be allowed)

The debate teams will be made on a first-come first-serve basis. TAs will open sign-up forms and make announcements on the course page on Moodle. It is important to check announcements on Moodle regularly for this reason.

IMPORTANT: See the last section of this syllabus for policies about plagiarism. There will be no exceptions to those policies.

Term Paper or debate reaction papers for non-debate team students (25 marks)

1. Introduction and clarity of describing the background literature and specifying the nature of the problem – **5 marks**
2. Describing the different schools of thought that tackle the question – **8 marks**
3. Offer your own thinking on the matter (either siding with one school of thought, or offering a new insight or suggestions for experiments or investigations, providing appropriate justifications) – **7 marks**
4. Overall clarity, organization of thoughts, and originality – **3 marks**
5. Formatting (Citations, References) – **2 marks**

Recommended: 7+ sides of A4 size sheets, hand-written (in class – one side A4 sheet handwritten cheat sheet will be allowed)

Experiment participation and reflections (5%)

Participating in cognitive science studies will help you understand how we study the mind using empirical methods in cognitive science. So we require that you participate in experiments (for a total duration of 1.5h or more). Immediately after you participate in the experiment, turn in a 1 page reflection paper quickly summarizing the experiment, what it was supposed to measure, and your brief thoughts based on anything you've learned from this class or elsewhere about the cognitive phenomenon that was tested in that experiment. 1.5 h of experiments usually means at least 2 experiments and therefore 2 reflection papers.

Participation

Students are expected to participate in all activities as listed on the course calendar. *Failure to participate*

will result in students being unable to complete the term paper satisfactorily. The exams may also include questions from the in-class activities such as the debates and any resulting effect on the final grade is entirely the student's responsibility.

Complete Assignments

All assignments for this course will be submitted electronically through the course page on Moodle unless otherwise instructed. Assignments must be submitted by the given deadline or special permission must be requested from the instructor *before the opening of the submission window with documented evidence of an emergency.*

Late or missing assignments will affect the student's grade.

Late Work Policy

Be sure to pay close attention to deadlines—there will be no make-up assignments or quizzes, or late work accepted without a serious and compelling reason and instructor approval.

Viewing Grades on Moodle

Points you receive for graded activities will be posted to the course page on Moodle. Click on the Grades link to view your points.

Letter Grade Assignment

Final grades assigned for this course will be based on the percentage of total points earned. The exact cutoffs will be announced after the end semester exam is graded.

Course Policies

Netiquette Guidelines

Netiquette is a set of rules for behaving properly online. Your instructor and fellow students wish to foster a safe online learning environment. All opinions and experiences, no matter how different or controversial they may be perceived, must be respected in the tolerant spirit of academic discourse. You are encouraged to comment, question, or critique an idea but you are not to attack an individual. Working as a community of learners, we can build a polite and respectful course community.

The following netiquette tips will enhance the learning experience for everyone in the course:

- Do not dominate any discussion.
- Give other students the opportunity to join in the discussion.
- Do not use offensive language. Present ideas appropriately.
- Be cautious in using Internet language. For example, do not capitalize all letters since this suggests shouting.
- Avoid using vernacular and/or slang language. This could possibly lead to misinterpretation.
- Never make fun of someone's ability to read or write.
- Share tips with other students.
- Keep an "open-mind" and be willing to express even your minority opinion. Minority opinions have to be respected.
- Think and edit before you push the "Send" button.

- Do not hesitate to ask for feedback.
- Always assume good intentions and ask for clarification. Communication online is difficult without facial and gestural cues.

Adapted from:

Mintu-Wimsatt, A., Kernek, C., & Lozada, H. R. (2010). *Netiquette: Make it part of your syllabus*. Journal of Online Learning and Teaching, 6(1). Retrieved from http://jolt.merlot.org/vol6no1/mintu-wimsatt_0310.htm

Shea, V. (1994). Netiquette. Albion.com. Retrieved from: <http://www.albion.com/netiquette/book/>.

Build Rapport

If you find that you have any trouble keeping up with assignments or other aspects of the course, make sure you let your instructor know as early as possible. As you will find, building rapport and effective relationships are key to becoming an effective professional. Make sure that you are proactive in informing your instructor when difficulties arise during the semester so that we can help you find a solution.

Inform Your Instructor of Any Accommodations Needed

If you have a documented disability and wish to discuss academic accommodations, please contact your instructors as soon as possible.

Statement of Policy

The instructors of this course will modify requirements as necessary to ensure that they do not discriminate against qualified students with disabilities. The modifications should not affect the substance of educational programs or compromise academic standards; nor should they intrude upon academic freedom. Examinations or other procedures used for evaluating students' academic achievements may be adapted. The results of such evaluation must demonstrate the student's achievement in the academic activity, rather than describe his/her disability.

If modifications are required due to a disability, please inform the instructor

Commit to Integrity

As a student in this course (and at IIIT Hyderabad) you are expected to maintain high degrees of professionalism, commitment to active learning and participation in this class and also integrity in your behavior in and out of the classroom.

IIIT Hyderabad Academic Honesty Policy & Procedures

Student Academic Disciplinary Procedures

(1) Academic misconduct is an act in which a student:

- (a) Seeks to claim credit for the work or efforts of another without authorization or citation;
- (b) Uses unauthorized materials or fabricated data in any academic exercise;
- (c) Forges or falsifies academic documents or records;

- (d) Intentionally impedes or damages the academic work of others;
- (e) Engages in conduct aimed at making false representation of a student's academic performance; or
- (f) Assists other students in any of these acts.

(2) Examples of academic misconduct include, but are not limited to: cheating on an examination; collaborating with others in work to be presented, contrary to the stated rules of the course; submitting a paper or assignment as one's own work when a part or all of the paper or assignment is the work of another; submitting a paper or assignment that contains ideas or research of others without appropriately identifying the sources of those ideas; stealing examinations or course materials; submitting, if contrary to the rules of a course, work previously presented in another course; tampering with the laboratory experiment or computer program of another student; knowingly and intentionally assisting another student in any of the above, including assistance in an arrangement whereby any work, classroom performance, examination or other activity is submitted or performed by a person other than the student under whose name the work is submitted or performed.

We will be using plagiarism detection software. Please do not copy-paste from other papers or use AI/LLMs in ways that are not authorized by the faculty. If you use direct quotes, you have to use the quotation marks “xyz” and cite your source: e.g. (Johnson & Johnson, 1988, p. 5). Please use APA format. If plagiarism is detected, for the first violation, you will get 0 for the term paper or assignment in question. If plagiarism is detected a second time in another assignment/project write-up, then one letter grade will be deducted from the final grade (e.g if you get a B/B-, that will be changed to C/C-) and you will be reported to the appropriate authorities for further disciplinary action.

Note: This syllabus was adapted from a template provided at www.uwsp.edu

Title of the Course	:	Introduction to Biology
Name of the Faculty	:	Vinod PK
Course Code	:	SC3.101
L-T-P	:	3-1-0
Credits	:	4

(L= Lecture hours, T=Tutorial hours, P=Practical hours)

1.Prerequisite Course / Knowledge:

None

2.Course Outcomes (COs):

After completion of this course successfully, the students will be able to

- CO-1:** Identify levels of organization (in different time and length scales) in living organisms
- CO-2:** Describe the characteristics of living organisms
- CO-3:** Apply principles of physics to biology
- CO-4:** Distinguish different cellular and biomolecular structures and functions
- CO-5:** Explain different cellular and biochemical processes and their control
- CO-6:** Outline the applications of computers in biology
- CO-7:** Evaluate and synthesize information from the scientific literature

3.Mapping of Course Outcomes (COs) with Program Outcomes (POs) and Program Specific Outcomes (PSOs) – Course Articulation Matrix

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3	PSO4
CO1	3	1	1	1	1								1	3	3	1
CO2	3	1	1	1	1								1	3	3	1
CO3	3	2	1	1	1								1	3	3	1
CO4	3	1	1	1	1								1	3	3	1
CO5	3	1	1	1	1								1	3	3	1
CO6	3	2	1	1	2								1	3	3	2
CO7	3	1	1	1	1								1	3	3	1

Note: Each Course Outcome (CO) may be mapped with one or more Program Outcomes (POs) and PSOs. Write '3' in the box for 'High-level' mapping, 2 for 'Medium-level' mapping, 1 for 'Low'-level' mapping

4.Detailed Syllabus:

Unit 1: Cellular foundations: Cell organelles, Membranes and cellular compartments, Tree of life

Unit 2: Chemical foundations: Biomolecules, Structure and function

Unit 3: Physical foundations: Bioenergetics, Catalysis, Enzymes, Photosynthesis, Respiration

Unit 4: How cells obtain energy from the food - metabolism

Unit 5: Genetic foundations: DNA, Genes, chromosomes, Genomes, Mutations

Unit 6: Evolutionary foundations, Systematics

Unit 7: DNA Replication, Repair, and Recombination - an overview

Unit 8: How Cells Read the Genome: From DNA to Protein

Unit 9: Control of Gene expression

Unit 10: Cell Signaling, Cell cycle

Unit 11: Analyzing and manipulating DNA

Unit 12: Introduction to sequencing and computational challenges

Reference Books:

1. Lehninger Principles of Biochemistry by David Nelson, Michael Cox
2. Molecular biology of the cell, Sixth Edition by Alberts B by Bruce Alberts, Alexander Johnson, Julian Lewis, Martin Raff, Keith Roberts, and Peter Walter.
3. Fundamentals of Biochemistry by Voet, Voet& Pratt

5.Teaching-Learning Strategies in brief (4 to 5 sentences):

The topics are presented through examples of its applications (e.g., to the human body, disease), of the latest research developments and of the history of the subject. Tutorials are designed to show how computers can be used to tackle biological problems. Evaluations test their ability to understand the relationships between topics and synthesize information from scientific literature.

6.Assessment methods and weightages in brief (4 to 5 sentences):

- Assignments - 30%
 - Review essay - 10%
 - Quiz - 30%
 - Exams - 30%
-

Title of the Course : **Introduction to Economics**
Name of the Faculty : Anirban Dasgupta
Name of the Program : B.Tech in Computer Science and Engineering
Course Code : HS5.202
Credits : 4
L - T - P : L-3, T-1
 (L - Lecture hours, T-Tutorial hours, P - Practical hours)
Semester, Year : Monsoon 2026

Pre-Requisites : Introduction to Human Sciences

Course Outcomes : After completion of this course successfully students will be able to:

CO1: Identify the different definitions of economics

CO2: Distinguish between schools of thought in economics and understand their evolution

CO3: Understand the foundations of micro, macro and international economics

CO4: Analyze how the conditions of economic life are distinct in developing countries like India

CO5: Comprehend how other disciplines can complement economics in explaining social phenomena and relevant policy interventions

Course Modules:

1. **'Ways of doing economics:** Different views on scope and definition of economics. Alternative schools of economic thought- Neoclassical, Marxist, Keynesian
2. **Theoretical foundations of microeconomics:** Agent based understanding of decision making- individuals and firms. Market structure and game theory. Welfare economics, Information economics
3. **Theoretical foundations of macroeconomics:** Macroeconomic measurement. Aggregate demand, aggregate supply and determination of output. Inflation and Unemployment, Monetary and Fiscal Policy. Banking and the role of finance in the economy
4. **Introduction to the open economy:** Basic trade theory- absolute advantage, comparative advantage, Heckscher-Ohlin Model. Exchange rate determination. External accounts of an economy
5. **Interdisciplinary engagements with economics:** Economics and psychology. Economics and computational science, economics and climate science

Textbooks :

- CORE Econ. The Economy: A South Asian Perspective (free e-book available)
- Neva Goodwin *et.al.* Principles of Economics in Context. Routledge.
- Richard Wolff and Stephen Resnick. Contending Economic Theories: Neoclassical, Keynesian and Marxian. MIT Press.
- **Reference Books & Articles (indicative list, more will be added in the course of teaching):**
- Michelle Baddeley: Behavioural Economics: A very short Introduction. Oxford University Press
- Samuel Bromley *et.al.* (eds.): Making the International: Economic Interdependence and Political Order. Open University Press.

- International Monetary Fund: Balance of Payments Manual.
- K. Vela Velupillai & Stefano Zambelli (2015) Simulation, computation and dynamics in economics, Journal of Economic Methodology, 22:1, 1-27,

E-book Links :

Grading Plan : (The table is Indicative)

Type of Evaluation	Weightage (in %)
Mid Sem Exam	30%
Class Presentation	15%
End Sem Exam	40%
Assignment	15%

Mapping of Course Outcomes to Program Objectives: (1 – Lowest, 2—Medium, 3 – Highest, or a ‘-’ dash mark if not at all relevant).

Computer Science and Engineering

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3	PSO4
CO1	1	-	2	3	-	3	3	3	2	2	-	2	-	1	-	2
CO2	1	-	1	2	-	2	2	2	2	2	-	2	-	1	-	3
CO3	1	1	2	2	-	3	3	2	2	2	-	2	1	2	1	3
CO4	1	-	2	1	2	2	2	3	2	2	1	2	-	1	-	2
CO5	1	1	1	2	1	2	3	3	2	2	1	2	1	2	1	2

Teaching-Learning Strategies in brief (4-5 sentences):

In this course, teaching will be primarily based on lectures and will be supplemented with group discussions and class presentations. Relevant chapters from the textbooks and articles will be referred to for each lecture. As an introductory course, the assignments will be designed to evaluate the theoretical clarity attained by students. The lectures and reading material will be designed to highlight both the logical structure of economic theory as well as the plural approaches to economic problems that originate from alternative theoretical perspectives.

Title of the Course

: Introduction to Game Theory

Faculty Name

: Sujit Gujar

Course Code

:CS1.408

L-T-P

: 3-1-0

Credits

: 4

(L= Lecture hours, T=Tutorial hours, P=Practical hours)

1.Prerequisite Course / Knowledge:

Basic Knowledge in Linear Algebra, Probability Theory and comfortable in basic maths

2.Course Outcomes (COs) (5 to 8 for a 3 or 4 credit course):

After completion of this course successfully, the students will be able to
CO-1understand how to define a game and strategies in a game

CO-2 demonstrate familiarity with different solution concepts in game theory
 CO-3 write algorithms to solve many game theoretic problems
 CO-4 understand the concept of mechanism design (incentive engineering)
 CO-5 analyze given autonomous system for any strategic behavior of the agents

CO-6 design mechanism for autonomous agent systems to make them game theoretically sound
 CO-7 design agents to participate in auction-based competition

3. Mapping of Course Outcomes (COs) with Program Outcomes (POs) and Program Specific Outcomes (PSOs) – Course Articulation Matrix

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3	PSO4
CO1	1	1	3	1	1	1	1	1	1	1	1	3	3	1	1	3
CO2	1	1	3	1	1	1	1	1	1	1	1	3	3	1	1	3
CO3	1	3	3	1	3	1	1	1	1	2	1	3	3	1	1	3
CO4	1	2	3	1	2	2	1	1	1	1	1	3	3	1	1	3
CO5	2	3	3	3	2	2	1	1	1	1	1	3	3	2	2	3
CO6	3	3	3	2	2	2	1	1	1	1	1	3	3	3	2	3
CO7	3	2	3	2	3	3	1	1	3	3	1	3	3	3	2	3

Note: Each Course Outcome (CO) may be mapped with one or more Program Outcomes (POs) and PSOs. Write '3' in the box for 'High-level' mapping, 2 for 'Medium-level' mapping, 1 for 'Low'-level' mapping

4. Detailed Syllabus:

- What is game? Extensive form games vs strategic form games, two player zero sum games, mini-max theorem, dominant strategy equilibrium, Nash equilibrium and its existence. Co-operative game theory, core, imputations, Shapley value, Nash bargaining solution.
- Mini-max Theorem, Nash Theorem, Shapley's Theorem for core and algorithmic aspects of these theorems.
- Game with incomplete information, introduction to mechanism design, revelation principle, voting schemes.
- Application of the above concepts will be illustrated with use cases in wireless communication, e-Commerce, social networking, crowdsourcing and, cloud management.

Reference Books:

- "Game Theory and Mechanism Design" by Y Narahari.
- "Game Theory: Analysis of Conflict", by Roger B. Myerson.

5. Teaching-Learning Strategies in brief (4 to 5 sentences):

The course is designed mix of theory and practice. The theory part is planned to be taught with posing questions to the students to make them think how intelligent agents should behave in the give situation. The students are evaluated regularly with quizzes. To expose students to deep research aspects there are reading assignments. To enable learning practical aspects, there are programming assignment and tournament where they write their strategic agents. The the assignments are done in teams to enable peer learning. To further enhance the knowledge further, the reading assignments are peer-evaluated.

6. Assessment methods and weightages in brief (4 to 5 sentences):

Type of Evaluation	Weightage (in %)
End Sem Exam	25
Quizzes/Reading Assignment	15
Programming/Reading Assignments	15
Scribes	5
Course Participation	5
Project (Competition)	10

Title of the Course : **Introduction to History**
Name of the Faculty : Aniket Alam
Course Code : HS3.201
Credits : 4
L - T - P :
(L - Lecture hours, T-Tutorial hours, P - Practical hours)
Semester, Year : Monsoon 2026
Name of the Program: B. Tech in Computer Science and Engineering
Pre-Requisites : Introduction to Human Sciences, HS8.102

Course Outcomes :

After completion of this course successfully students will be able to:

- CO1: Define the concept of History and describe the development of the discipline.
- CO2: Explain range of academic theories relating to the discipline of History.
- CO3: Analyze features of historical writings and appreciate the importance of the past.
- CO4: Evaluate the different methods of historical analysis.
- CO5. Assess primary evidence,
- CO6. Develop their own understanding about History and the Past.

Course Topics :

- (1) **Historical Time and Space:** In this module students will be introduced to how historians have understood the flow of time and periodised time into historical ages. They will also appreciate how historians define regions and territory. Apart from this the module will also teach about the development of the idea of time, and of space; and how the modern map and watches came to define society in new ways.
- (2) **Historical Fact and Objectivity:** This module will define the historical fact, the different interpretations of what a fact is, and the debate among historians relating to historical objectivity: its possibility and desirability.
- (3) **The Main Theories of History:** In this module students will be exposed to the main theoretical models of historical interpretation. These will include, but not be limited to, Rankean and Whig history, Annales history, Marxist history, Structuralist and Post-Structuralist history, etc. Students will also be introduced to the newer theories like ecological history, black history, herstory, etc.
- (4) **The Main Methods of Historical Analysis:** This module will focus on source criticism, the advantages and limitations of the inductive and deductive methods, oral history, qualitative and quantitative methods, etc. that historians deploy in their identification of facts and their interpretations.
- (5) **Memory, heritage and public history:** This module will provide students brief introductions to the

fields of memory studies, heritage studies and public history and acquaint the with the debates around the relationship between the past and the present, memory and history, remembrance and forgetting and the opening of the discipline of history beyond the academy.

Preferred TextBooks:

- E.H. Carr: *What is History*.
- Marc Bloch: *The Historian's Craft*
- Umberto Eco: *This is not the end of the Book*.
- Shawn Graham, et al: *The Historian's Macroscopic*.

Reference Books:

- Romila Thapar: *From Lineage to State*.
- Mircea Eliade: *The Myth of the Eternal Return*.
- Edward Said: *Orientalism*.
- Sumit Sarkar: *Modern Times*.
- Vanessa Ogle: *The Global Transformation of Time*.
- Richard Eaton: *India in the Persianate Age*.
- Michael Mann: *South Asia's Modern History*.
- R.C. Majumdar: *An Advanced History of India*.
- Alfred Crosby: *The Measure of Reality*.
- Fernand Braudel: *A History of Civilization*.
- James C. Scott: *Against the Grain*.
- Ibn-e-Khaldun: *Muqadimah*.
- Barbara Freese: *Coal – A Human History*.
- Sidney W. Mintz: *Sweetness and Power – The Place of Sugar in Modern History*.
- Douglas A. Boyd, Mary A. Larson: *Oral History and Digital Humanities – Voice, Access, and Engagement*
- Anne Kelly Knowles: *Placing History*.
- Ann Laura Stoler: *Along the Archival Grain*
- David Lowenthal: *The Heritage Crusade and the Spoils of History*

ARTICLES:

- William H. McNeil: "Why Study History".
- Bernard S. Cohen: "The Command of Language and the Language of Command".
- E.P. Thompson: "Custom, Law, and Common Right".
- E.P. Thompson: "Time, Work-Discipline, and Industrial Capitalism".
- Ranajit Guha: "On Some Aspects of the Historiography of Colonial India".
- Ranajit Guha: "The Prose of Counter-Insurgency"
- Shahid Amin: "Gandhi as Mahatma"
- David Arnold: "Touching the Body: Perspectives on the Indian Plague"
- Jacques Le Goff: "Merchant's Time and Church Time in the Middle Ages"
- Marianne Hirsch: "The Generation of Post memory"
- David C. Harvey: "Heritage Pasts and Heritage Present: Temporality, meaning and the scope of Heritage Studies"
- Cameron and Gatewood: "Excursions into the Unremembered Past: What People Want from Visits to Historic Sites"
- Lucien Febvre: "Sensibility and History – How to Reconstitute the Emotional Life of The Past"
- Emmanuelle Roy Ladurie: "The History of Rain and Fine Weather"

- Philippe Aries: “Pictures of the Family”
- Maurice Aymard: “The Costs of War”
- Fernand Braudel: “History and the Social Sciences-The Longue Duree”
- Jean Meuvret: “Food Crises and Demography in France during the Ancien Regime”
- Karl Marx: Communist Manifesto, Chapter 1.
- Karl Marx: “British Rule in India, 10 June 1853”
- Karl Marx: “Future Results of British Rule in India, 22 July 1853”
- DDKosambi: “Social and Economic Aspects of the Bhagvad Gita”
- Irfan Habib: “Potentialities of Capitalist Development in the Economy of Mughal India”
- Romila Thapar: “Somnatha”
- Robert Darnton: “Peasants Tell Tales”
- Clifford Geertz: “The Balinese Cock-Fight”
- Arthur Conan Doyle: “Sign of Four”

E-book Links :

Grading Plan : (The table is only indicative)

Type of Evaluation	Weightage (in%)
Quiz-1	10
Mid Sem Exam	20
Project	30
End Sem Exam	40
TOTAL	100

Mapping of Course Outcomes to Program Objectives: (1—Lowest, 2—Medium, 3—Highest, or a ‘-’ dash mark if not a tall relevant). **Computer Science and Engineering**

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3	PSO4
CO1	-	-	2	1	-	2	1	3	1	1	1	2	-	1	1	2
CO2	-	-	2	1	-	2	1	2	1	2	1	3	-	1	1	3
CO3	-	-	2	1	1	3	2	2	2	1	1	3	-	1	1	1
CO4	-	-	2	1	-	1	1	3	3	1	1	2	-	1	1	2
CO5	1	1	2	2	2	2	2	3	2	2	1	3	1	2	2	3
CO6	-	-	2	1	-	3	1	3	3	2	1	3	-	2	1	3

Teaching-Learning Strategies in brief (4-5 sentences):

The course will be based on classroom lectures and will require intensive reading and writing. On average, each student will be required to read around 500 – 800 pages of books and articles, and work on a project that tests their understanding of the concepts, theories and methodologies of historical research taken up in the course. A major component of the project would be an oral class presentation.

Pictures, Extracts from primary sources, audio and video resources will be used to illustrate the points being taught.

The assignments and exams will focus on training students to develop their own ideas to the topics on hand.

Title of the course	:	Introduction to Linguistics 1
Name of the Faculty	:	Chiranjeevi Yerra + Radhika Mamidi
Course Code	:	CL1.101
Credits	:	3-0-1-4
Type when	:	Monsoon 2026
Prerequisite	:	None

COURSE OUTCOME:

CO-1: Students will have a good understanding of linguistic analysis

CO-2: Students will be introduced to different word and sentence level theories

CO-3: It will enable them in building text processing tools and systems

CO-4: They will explore different languages in class working in teams.

CO-5: Using real examples, they will analyse language data to understand the concepts.

COURSE TOPICS:

1. What is language? Difference between human language and Animal languages. Natural language, Formal language and Artificial language, Characteristic features of human language, what we know about language.

2. Study of Human language – the field of Linguistics

3. Looking at language from synchronic and diachronic points of view

4. Areas of Study from structural perspective

a) Syntagmatic and paradigmatic aspects of language structure,

b) Levels of structural analysis: Phonetics: Place and manner of articulation of speech sounds, IPA.

Phonology: Phone, phoneme, allophone; Distinctive features; Phonological rules; Syllable.

Morphology: Units of word's internal structure, word formation processes, inflectional and derivational morphology, compound words and how they are formed.

Syntax: Types of sentences, Sentence structures, Phrase structure grammar.

c) From evolution perspective: Historical Linguistics

d) From usage perspective: Sociolinguistics

e) From Psychological perspective: Mechanisms of language acquisition, knowing more than one language

f) Indian Grammatical Tradition: A communication model for language study. Paninian grammatical model.

g) Writing Systems: Representing language through graphic characters.

Mapping of Course Outcomes to Program Objectives

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3	PSO4
CO1	2	2							1	1					2	2
CO2	2	2							1	1					2	2
CO3	2	2							1	1					2	2
CO4	1	1							3	3					1	1
CO5	1	1							2	2					1	1

GRADING:

Assignments: 15%,

Mid Sem: 30%, End Sem: 35% and Project: 20%

PROJECT: The students will work on a hands-on project on language analysis. In the project they are expected to work with real time data and understand its nature.

PREFERRED TEXTBOOK:

Language: Nature, Psychology and Grammatical Aspects by Victoria Fromkin, Robert Rodman and Nina Hyams, Cenage Learning (Indian Edition)

REFERENCE BOOKS:

- 1.Linguistics by Jean Aitchison, 5th edition. London: Hodder Headline, Teach Your self-Books, 1999. ISBN:0-340-73733-6. Retitled and reprinted with corrections as: Linguistics: An Introduction. 2nd edition. London: Hodder Headline, 1999. ISBN:0-340-75792-2).
2. Introduction to Language by Fromkin, V.A. and Rodman, R., 1997. Harcourt Brace. 6th edition

Title of the Course : Introduction to Neural and Cognitive Modeling

Faculty Name : Bapi Raju Surampudi

Course Code : CG3.401

L-T-P : 3-1-0

(L - Lecture hours, T-Tutorial hours, P - Practical hours)

Credits : 4

Name of the Academic Programme: B.Tech in CSE

Prerequisite Course/Knowledge:

Interest in Neuroscience and Cognitive Science is desirable. Basic background in Calculus, Probability and Statistics, Linear Algebra, Ordinary Differential Equations (ODE) and aptitude for programming would be desirable.

Course Outcomes (COs):

After completion of this course successfully, the students will be able to:

CO-1: develop understanding of how principles of mathematics and computation are applied for problems in neuroscience

CO-2: identify the differences and similarities of how mathematical principles are applied for various levels of nervous system – from neuron to behavior.

CO-3: analyze and evaluate model components and relate them to the functions of the neural system

CO-4: design computational solutions to novel problems and phenomena of neuroscience and evaluate their goodness of fit to the actual empirical data from neuroscience

CO-5: analyze and compare the strengths and limitations of computational models in explaining brain/mind/behavior

CO-6: Appreciate the functional insights that computational model gives about the complex cognitive system, develop novel computational models and reflect on how these enable practical solutions

Mapping of Course Outcomes (COs) with Program Outcomes (POs) and Program Specific Outcomes (PSOs)

	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PO 12	PSO 1	PSO 2	PSO 3	PSO 4
CO 1	2	1	1	1	1	1	1	1	1	1	1	2	2	1	2	1
CO 2	3	3	1	2	1	1	1	1	1	1	1	2	2	2	3	2
CO 3	2	2	2	3	2	1	1	1	1	1	1	2	2	2	3	2
CO 4	2	3	2	3	3	1	1	1	3	3	1	2	3	2	2	3
CO 5	2	2	3	2	3	1	2	2	1	2	1	3	2	3	2	2
CO 6	2	2	3	2	1	1	2	2	2	1	1	3	3	1	3	3

Detailed Syllabus:

Part I: Single-Neuron Level Models (1/3rd of the semester)

Introduction to Neuroscience; Passive membrane models, Leaky Integrate and Fire (LIF) models, Hodgkin-Huxley model, Cable models of dendrites, Synapse model, Compartmental models.

Part II: Network Level Models (1/3rd of the semester)

Neural population codes; information representation; neural encoding and decoding; hierarchy and organization of sensory systems; Spiking Network models of sensory systems; Neuroplasticity and learning.

Part III: Abstract Models (1/3rd of the semester)

Introduction to Hebbian, Competitive and Error-driven learning rules; Reinforcement learning, Neural Network models of Perception, Attention, Memory, Language and Executive Function.

Reference Books:

1. R. O'Reilly & Y. Munakata (2000). *Computational Explorations in Cognitive Neuroscience: Understanding the Mind by Simulating the Brain*. MIT Press.
2. J. M. Bower and D. Beeman (2003). *The Book of GENESIS: Exploring Realistic Neural Models with the General Neural Simulation System*, Internet Edition.
3. Peter Dayan and L. F. Abbott (2005). *Theoretical Neuroscience: Computational and Mathematical Modeling of Neural Systems*. MIT press.
4. Thomas Trapp Enberg (2009). *Fundamentals of Computational Neuroscience* (2nd edition), OUP Oxford.
5. Daniel S. Levine (2018). *Introduction to Neural and Cognitive Modeling* (3rd Edition), Routledge, USA.
6. V. Srinivasa Chakravarthy (2019). *Demystifying the Brain: A Computational Approach* (1st Edition), Springer, Singapore.

Teaching-Learning Strategies in brief(4to5sentences):

Lectures emphasize the understanding of how modeling is used in Neuroscience to get a functional understanding of the biological nervous system. While basic background of Neuroscience, Cognitive Science, Calculus, ODEs are mentioned as desirable background, several of the relevant concepts will be revised before their applications are described in the INCM course. Basic modeling ideas are explained in detail along with mathematical derivations. Students further their understanding by doing the programming assignmentsdesignedtoachievecourseoutcomesandcollaborativelyworkingonafinalproject. The assignments and the final project are designed to give hands-on experience as to how modeling frameworks are deployed to solve neuroscience problems. The organizing principles of brain function are brought out clearly during the lectures and explored further in assignments and projects. This allows them to appreciate how a complex, dynamic system such as the brain and cognition can be understood from afunctional and computational points of view.

Assessmentmethodsandweightagesin brief(4to5sentences):

1.	Assignment	N=3	15%
2.	Mid Semester Exams	N=2	30%
4.	Project in Groups of 2 students	N=1	25%
5.	End Semester Exam	N=1	30%
TOTAL			100%

Grading Policy: Relative grading policy scheme

Project Evaluation Breakdown:

1.	Proposal/idea Submission: details of the model being taken up for replication, dataset to be used for	5%
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	modeling/validation, software system to be used for simulations, expected outcome	
2.	Implementation demonstration + Final Report	15%
3.	Final Presentation + Viva	5%
TOTAL		25%

Title of the Course : Introduction to Neuroeconomics

Name of the Faculty : Kavita Vemuri

Course Code : CG4.402

L-T-P : 3-0 -1.

Credits : 4

1.PrerequisiteCourse / Knowledge:

1. Intro topsychology
2. Cognitive Science
- 3.CognitiveNeuroscience
4. Game theory

2. Course Outcomes (COs) (5 to 8 for a 3 or 4 creditcourse):

A student introduced to the concepts in the course will be able to:

CO-1: Apply methods from neoclassical economics

CO-2: Test the theories or models with experiments

CO-3: Apply models in real-life conditions

CO-4:Analyze the Brain and decision making process

CO-5: Explain the cultural diversity role in risk-taking behavior

CO-6: Explain the cognitive process – memory, decision making, empathy, learning

CO-7:Implement the ethics of Neuroeconomics.

3.Mapping of Course Outcomes (COs) with Program Outcomes (POs) and Program Specific Outcomes (PSOs) – Course ArticulationMatrix

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3	PSO4
CO1	1	4	4	4	0	2	2	3	1	1	1	3	1	0	0	4
CO2	3	4	4	4	4	3	1	3	3	3	2	3	2	1	2	4
CO3	3	3	2	4	2	2	2	2	0	2	2	3	3	3	1	4
CO4	3	3	3	3	1	3	1	3	1	2	2	2	2	2	1	4
CO5	3	3	3	3	3	2	2	1	3	2	3	3	2	2	2	4
CO6	2	2	3	3	2	1	1	2	3	2	2	2	2	2	1	4
CO7	1	1	1	1	2	1	4	4	2	1	1	1	2	1	1	4

4. Detailed Syllabus:

OBJECTIVE :Extend the understand of brain and behavior to economics and decision making.

The course will be a seminar-style course covering the brain and behaviour pertaining to decision making as applied to economic theories. The course will begin with introduction to classical economics, the brain (brief intro to anatomy) structures attributed to learning, decision making, emotions, reward etc. The core part will be modern neuroeconomics and applications.

By the end of the course the student will be able to:

- Demonstrate knowledge of some of the main theoretical and empirical debates in Behavioural Economics.
- Understand the way in which Behavioural economics has developed and how we can relate it to traditional models of economics.
- Assess the strengths and weaknesses of different theories of Economic Behaviour.
- Demonstrate an ability to apply the main concepts to a variety of other economic fields.
- Construct and substantiate arguments on a variety of topics covered in the module.
- Present clearly and methodically in their own words, but also using equations and diagrams.

COURSE TOPICS :
(please list the order in which they will be covered)

1. A Brief History of Neuroeconomics – 1 lecture: Mostly from the introduction chapter in Glimcher's book. This lecture will introduce the conjunction of economics and psychology. The focus on decision making process – what is a rational choice, how does subjective interpretations, perceptions, individual cognitive abilities consequences on decision making and why it is important to discussions on behavioural economics. The discussion will start from Neoclassical Economics, Adam Smith's 1776 book, Keynes theory, etc. Maurice Allais (1953) – Allais paradox. But, the first lecture will also cover the restrictions of any interpretations from neuro/behav studies on decision/choices. (SG)
2. Basic Methods from Neoclassical Economics - 1 lecture
 Chapter 1 of the Glimcher & Fehr. Tentative list, it is to be noted that some topics can be expanded based on student interest and interactions.
 Rational Choice and Utility Theory: Some Beginnings Early
 Price Theory and the
 Marginal Revolution Early Decision Theory and Utility
 Maximization
 The Ordinal Revolution and the Logic of Choice
 Quantitative Tests of Qualitative Theories: Revealed Preference
 GARP
 Understanding Rationality Axiomatic
 Approaches: Strengths Axiomatic
 Approaches: Weaknesses
 Expected Utility Theory
 Defining the Objects of Choice: Probabilistic Outcomes
 Continuity Axiom
 Independence
 The Expected Utility Theorem Axioms and Axiomatic
 Reasoning
 Using Axioms: The Neoclassical Approach in Neuroeconomics
 The Reward Prediction Error Hypothesis
 The DRPE Axioms and the Ideal Data Set
3. Experimental Economics and Experimental Game Theory - 1 lecture
 In this lecture, the learnings from the game theory, will be extended with examples from economics.
 Game Theory Described
 Normal and Extensive Form Games
 Game Theory Experiments Design and Practice
 Experiments with Normal Form Games

Experiments with Extensive Form Games
 Neuroeconomics experiments
 Design and Practice
 Neuroeconomics Experiments with the Trust Game
 Neuroeconomics Experiments with the Ultimatum Game

Towards a Neuroeconomics Theory of Game Playing

4. Overview of behavioural economics- 1 lecture
 (some aspects from the below topics, examples and applications will be covered by student presentations)

(The below topics are from book: Choices, Values & Frames by Daniel Kahneman & Amos Tversky)

Prospect theory and extensions

-Prospect theory: an analysis of decision under risk
 Advances in prospect theory: cumulative representation of Uncertainty

The uncertainty effect and the weighting Function

-Compound invariant weighting functions in prospect theory

- Weighing risk and uncertainty

- Belief-based account of decision under uncertainty
 Loss aversion and the value function

- Loss aversion in riskless choice: a reference-dependent model

Anomalies: the endowment effect, loss aversion, and status

Quobias

Rational Choice and the Framing of Decisions

Framing, Probability Distortions, and Insurance Decisions

Mental Accounting Matters

5. Introduction to Neuroscience & Experimental Methods in Cognitive Neuroscience – 2 lectures

- Fundamentals of brain have been covered in Cognitive Neuroscience course . Here the functional connectivity maps attributed to decision making, memory, emotion will be reiterated. Experimental techniques like eye tracking, EEG, FMRI, MEG and physiological indexes like GSR/SCR in the context of responses will be covered. Importantly the limitations will be explained with examples.

6. Decision making - 3 lectures

Neuroscience of decision making will cover role of time, emotions, heuristics and bias, reward/punishment, trust, risk & uncertainty.

The above topics will illustrate the theory with examples and the limitations as a function of context or situational awareness.

(KV & PM)

Student presentations & discussions

In student presentations, taking off from the introduction to terminology, definitions and limitations published papers with experimental work will be discussed. The ambiguity to each case from subjective perceptions, cognitive capability and social structures will be discussed.

7. Risk /Ambiguity (student presentations) (2 lecture hours or 4 presentations)
 Choice (student presentation) (2 lecture hours or 4 presentations)

8. Time (2 lecture hours or 4 presentations)

9. Reward/Loss (student presentations or 4 presentations)

10. Reinforcement Learning ((student presentation) (2 lecture hours or 4 presentations)

11. Social behaviour (student presentation) (2 lecture hours or 4 presentations)
12. Neuro-Morality (1 lecture followed by 2 student presentation)

This topic was introduced to discuss the complexity of judgement and values, at a generic level and then its implications to economics. It's a new topic to bridge these two areas.

While moral judgement is well covered from neuroscience and cognitive science, its weight to economic theory needs to be explored.

Reference Books:

1. NEUROECONOMICS, Decision Making and the Brain, SECOND EDITION Edited by PAUL W. GLIMCHER & ERNSTFEHR
2. NEUROSCIENCE OF DECISION MAKING, Oshin Vartanian & David R. Mandel (eds)
3. Choices, Values & Frames by Daniel Kahneman & Amos Tversky

5. Teaching-Learning Strategies in brief (4 to 5 sentences):

The in-class lectures will cover basics – of classical, neoclassical economics but primary focus is on behaviour economics and neuro studies on decision making, risk-taking and the underlying complex cognitive processes. Quizzes are conducted periodically to evaluate transfer of knowledge and critical thinking of the implication of each study finding. The end-sem project will have the student design an experiment or test an existing theory on diverse set of people and analyse the differences in findings. Second, by considering a socially relevant problem the students will be exposed to the complexity of economic policy and financial systems.

6. Assessment methods and weightages in brief (4 to 5 sentences):

Type of Evaluation	Weightage (in %)
Mid semester Quizzes	20
Project	40
Open book exam or 30 minute quiz	20
Other Evaluation Class presentation & Viva	20

Title of the Course : Introduction to Psychology

Name of the Faculty : Priyanka Srivastava

Course Code : CG1.401

L-T-P : 3-0-1

Credits : 4

(L= Lecture hours, T=Tutorial hours, P=Practical hours)

Name of the Academic Program : Humanities Electives (UG3 and UG4)

1. Prerequisite Course / Knowledge: None

2. Course Outcomes (COs) (5 to 8 for a 3 or 4 credit course):

After completion of this course successfully, the students will be able to.. CO-

1: apply psychology knowledge base –

- describe and discuss major concepts, theories, models and overarching themes in psychology

- describe applications of psychology
- analyse and compare the major goals of psychological science and will be able to utilize different research methods used by psychological research.
- evaluate the challenges and merits of psychological observations and analyses, assess the brain and behaviour research complexity.
- explain the major historical landmarks in psychological science and their links to contemporary research.

CO-2: apply scientific inquiry and critical thinking –

- apply major perspectives of Psychology and levels of analyses to explain psychological phenomenon, e.g., cognitive, biological, social, health, behavioural, and cultural etc.
- analyse and evaluate the difference between the personal anecdotal incidences and scientific inquiry to our everyday psychological experiences. Students will be able to use different level of complexity to interpret psychological behaviour
- compare common fallacies like confirmation bias, causation to correlation etc.
- design, conduct, analyse, evaluate and interpret basic psychological research.
- analyse, interpret, and evaluate the individual experience and socio-cultural perspectives to explain psychological phenomenon

CO-3: Evaluate and Analyze research ethics of human/ behavioral sciences

- analyse and compare the benefits and risk of given psychological research
- apply key principles of APA Ethics guidelines for participants' right protection

CO-4: demonstrate effective communication skills

CO-5: demonstrate personal and professional development

- apply psychological learning to their personal and professional development, self-regulation, project management, coordinate teamwork, and develop life directions

3.Mapping of Course Outcomes (COs) with Program Outcomes (POs) and Program Specific Outcomes (PSOs) – Course Articulation Matrix

	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO10	PO11	PO12	PSO 1	PSO 2	PSO 3	PSO 4
CO 1	1	1	2	1	1	2	1	1	1	1	1	2	1	1	1	1
CO 2	3	1	2	3	1	3	3	1	1	1	1	2	2	2	3	2
CO 3	2	1	2	2	1	3	2	2	1	1	1	2	1	2	3	1
CO 4	1	1	1	1	1	1	1	1	2	3	1	2	1	1	1	1
CO 5	1	1	1	1	1	1	1	1	3	2	1	3	1	1	1	1

Note: Each Course Outcome (CO) may be mapped with one or more Program Outcomes (POs) and PSOs. Write '3' in the box for 'High-level' mapping, 2 for 'Medium-level' mapping, 1 for 'Low'-level' mapping

4. Detailed Syllabus:

Unit 1	Unit 2	Unit 3	Unit 4	Unit 5	Unit 6
Introduction	Methods	Health & Psychology	Social & Personality	Cognitive	Learning & Development
Psychology as a Science	Research methods for psychological observations	Psychological health and disorders	Social	Attention	Learning
Goals of Psychology	Neuroscience and behaviour	Psychological interventions and treatments	Gender	Perception	Life-span development
History of Psychology			Emotion	Memory	
			Personality	Intelligence	
3 hours	3 hours	6 hours	6 hours	6 hours	3.5 hours
2 Lectures	2 Lectures	4 Lectures	4 Lectures	4 Lectures	3 Lectures
CO 1 & 2	CO 1, 2 & 3	CO 1, 2, & 5	CO 1, 2, & 5	CO 1, 2, & 5	CO 1, 2, & 5

Reference Books:

1. Lilienfeld, S., Lynn, S. J., Namy, L., Woolf, N., Jamieson, G., Marks, A., & Slaughter, V. (2014). *Psychology: From inquiry to understanding* (Vol. 2). Pearson Higher Education AU.
2. Schacter, D., Gilbert, D., Wegner, D., & Hood, B. M. (2011). *Psychology: European Edition*. Macmillan International Higher Education.
3. Anderson, J. R. (1984). Cognitive psychology. *Artificial Intelligence*, 23(1), 1-11.
4. Elliot, A., Timothy, W., & Samuel, R.S. (2017). *Social Psychology* (9th Ed.) Pearson Education.

Journal Articles: Will be announced before a few key topics.

5. Teaching-Learning Strategies in brief (4 to 5 sentences):

The psychology course in monsoon 2021 will be primarily lecture and project-based learning course. Students will be required to make presentations for one of the assigned reading materials and project. Students will be introduced to undergraduate-level introductory topics and issues in psychology. Reading material will be assigned. Students will be required to engage in discussions, and to present topics based on the assigned reading topics. Each student will be required to do at least two presentations, one reading material and another accounted for their project. Students will be encouraged to take assignments inspired

from their everyday experiences and will be asked to evaluate the event/phenomenon/ processes critically and scientifically using psychological methods. They will be asked to perform some of the activities in team and demonstrate the individual contribution to the team activities. Students may be asked to perform peer review as well.

6. Assessment methods and weightages in brief (4 to 5 sentences):

Assessment Scheme:

1.	Assignment / Term Paper	N=1	10%
2.	Home and Class Activities (Student presentation)	N=2	5%
3.	Quiz	N=2	20%
4.	Project in Group – with 2-3 students	N=1	30%
5.	Final Exam	N=1	30%
6.	Experiment participation based on credits	N=2	5
TOTAL			100%

Project Evaluation Breakdown:

1.	Idea presentation / Proposal	4%
2.	Progress Report 1: with hypothesis, experiment design, paradigms, tasks, measures, prediction, and statistical analyses to use	8%
3.	Progress Report 2: with pilot data and preliminary analysis	8%
4.	Final Presentation + Peer evaluation (should be based on critical feedback)	8% + 2%
TOTAL		30%

Grading Policy: Absolute grading policy scheme

A	>=90
A-	>=80
B	>=70
B-	>=60
C	>=50
C-	>=45
D	>=40
F	<40

Academic Honesty: Do's: Discussion on meaning and interpretation of assignments, general approaches and strategies with other students in the course.

Don'ts: No sharing/copying of assignment with any student who is not in your group for any reason; not asking another student for help debugging your assignment code, method, or topics; no copying of code or document or assignment from any other sources (including internet).

The course will use plagiarism-detection software to check your assignments/ projects/ codes/ exam/ quiz responses. Copying from another student will be treated equally to plagiarism. Violation of any of the above policies, whether you are the giver or receiver of help, will result in zero on the assignment or the respective assessment components and fail the course in case of repetition

Project Evaluation – Rubric (100)

S.No.	Topic Description	Marks
1.	Clarity in Problem Statement, Method, Result, and Discussion	15
2.	Critical understanding of Literature – motivation for your research project	20
3.	Method: Participants, material, stimuli, procedure, task, measure of performance, sampling	20
4.	Results (Statistics), and Discussion and conclusion	20
5.	Future direction: Limitation and Scope of the current research/ objective, and Impact	10
6.	Citations and Reference (APA style)	10
7.	Organization	5

Assignment/ Term Paper Evaluation – Rubric (50 marks each)

S.No.	Topic Description	Marks
1.	Clarity and coherence in describing topic	10
2.	Summary and Critical Evaluation to find the gaps in the given literature	15
3.	Future direction: Limitation and Scope of the current research/ objective, and Impact	15
4.	Citations and Reference (APA style)	10
5.	Organization	5

Title of the Course : Introduction to Quantum Field Theory

Name of the Faculty : Subhadip Mitra + Monalisa Patra

Course Code : SC1.421

L-T-P : 3-1-0.

Credits : 4

(L= Lecture hours, T=Tutorial hours, P=Practical hours)

Name of the Academic Program: CND

1.Prerequisite Course / Knowledge:

Quantum Mechanics, Special Theory of Relativity.

2.Course Outcomes (COs):

After completing this course successfully, the students will be able to

CO-1 Explain the need for a quantum field theoretic description of nature

CO-2 Recognize the basic differences between wave function and quantum field.

CO-3 Discuss the idea of second quantization

CO-4 Apply second quantization to scalar, spinor, and electromagnetic field

CO-5 Calculate transition amplitudes and scattering cross-section for different processes

CO-6 Recognize the conceptual challenges in the quantum field theory

3.Mapping of Course Outcomes (COs) with Program Outcomes (POs) and Program Specific Outcomes (PSOs) – Course Articulation Matrix

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3	PSO4
CO1	1	1										1		3	1	1
CO2	1	1										1		3	1	1
CO3	1	2										1		2	1	1
CO4	2	3												1		3
CO5	3	1												1		3
CO6	3	3										1		2	2	1

Note: '3' in the box for 'High-level' mapping, 2 for 'Medium-level' mapping, 1 for 'Low'-level' mapping

4.Detailed Syllabus:

Unit 1: Introduction: review of backgrounds, motivations for QFT

Unit 2: Elements of Classical Field Theory: symmetries and No ether's theorem

Unit 3: Functional Formalism: path integral formalism, functional quantization, Feynman diagrams, quantization of scalar field, ϕ^4 theory

Unit 4: S-matrix: scattering cross-section and decay rates, from Feynman diagrams to S-matrix

Unit 5: Dirac Field: Dirac equation and its solutions, gamma matrices, quantization, Green's function

Unit 6 Quantum Electrodynamics (QED): Feynman rules for QED, cross-section of simple QED processes

Unit 7 Introduction to Renormalization

Reference Books:

1. A. Zee: Quantum Field Theory in a Nutshell

2. Ashoke Das: Field Theory—A Path integral Approach

3. Michio Kaku: Quantum Field Theory—A Modern Introduction

4. Lewis H. Ryder: Quantum Field Theory

5. Amitabha Lahiri&Palash B. Pall: A First Book of Quantum Field Theory

6. David Tong: Quantum Field Theory

7. Michael E. Peskin & Daniel V. Schroeder: An Introduction to Quantum Field Theory

5. Teaching-Learning Strategies in brief (4 to 5 sentences):

Aim of the course is to introduce to the students the main concepts and mathematical framework of Quantum Field Theory. A typical class consists of discussion of a new concept and its underlying mathematical structure. To make teaching more interactive in the online mode, instead of showing slides, mathematical derivations are done live during the class. Refined versions of the class materials are then circulated to the students. Students are encouraged to go through the materials and work out the mathematical derivations for better understanding of the concepts. Assignments are given on a regular basis. The assignments are designed in such a way that students can apply the concepts to solve problems. At the end of the course students will acquire several tools of Quantum Field Theory.

6. Assessment methods and weightages in brief (4 to 5 sentences):

Assignments: 30%, Quizzes (Mid-Sem exams): 30%, End Semester: 40%

Title of the Course	: IoT Workshop
Name of the Faculty	: Shruthi K + Sachin Chaudhari
Course Code	: CE9.609
L-T-P	: 1-0-3
Credits	: 4
(L= Lecture hours, T=Tutorial hours, P=Practical hours)	
Name of the Academic Program: MTech CASE	

1. Prerequisite Course / Knowledge:

Basic computer programming (C), 10+2 level physics, basics of structural engineering

2. Course Outcomes (COs)

After completion of this course successfully, the students will be able to

CO-1(Understand) : Explain the basic elements of an IoT system and the application of IoT for structural engineering.

CO-2(Analyze): Analyze and solve basic electrical circuits using Kirchhoff's laws

CO-3(Understand) : Describe the working principle of commonly available sensors and actuators.

CO-4(Understand) : Explain the working on microcontrollers, peripherals and its programming.

CO-5(Apply) : Write simple embedded programs and interface common sensors and actuators with Arduino and ESP 32 boards

CO-6(Remember) : State and identify different technologies related to Communications and Networking, Cloud Computing and Data Analysis, Interoperability Standards and security, Dashboard and Visualization

CO-7(Create) : Assess simple designs from IoT application point of view

CO-8(Create) : Develop and implement an IoT-based solution for a real-life problem in the domain of structural engineering

3. Mapping of Course Outcomes (COs) with Program Outcomes (POs) and Program Specific Outcomes (PSOs) – Course Articulation Matrix

Mapping of Course Outcomes to Program Objectives: (1 – Lowest, 2—Medium, 3 – Highest, or a ‘-’ dash mark if not at all relevant). Program outcomes are posted at

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PSO1	PSO2	PSO3	PSO4
CO1	2	1	3	-	-	1	-	2	-	-	-	3	1	1
CO2	2	-	-	-	-	-	-	2	-	1	-	-	-	-
CO3	2	1	1	-	-	1	-	2	-	1	-	-	-	-
CO4	2	-	2	-	-	-	-	2	-	-	-	1	1	-
CO5	1	-	2	-	2	1	1	2	-	-	-	2	2	-
CO6	1	-	3	-	-	1	-	2	-	-	-	2	2	-
CO7	2	-	3	1	2	-	2	2	2	2	-	2	2	-
CO8	3	3	3	3	3	1	3	2	3	3	1	3	3	3

4. Detailed Syllabus:

1. Basic properties of electricity and electrical circuits - DC, Voltage, Current, Power, Energy, Resistance, Ohm's Law, Circuit Diagrams
2. Kirchoff's voltage and current laws, series and parallel resistance, Voltage and Current divider
3. Online Simulations using TinkerCAD
4. Basic Circuits, Mesh analysis, Node analysis.
5. What is IoT, Embedded Systems, Components of Embedded Systems, Microcontrollers, Sensors and Actuators, Analog/Digital conversion, Internet structure, Protocols, TCP/IP
6. Arduino Environment, C Programming, Arduino programming and debugging, UART Protocol
7. Interfacing with Arduino, Pulse Width Modulation, Arduino Libraries, I2C communication, Arduino Shields
8. Peripherals: RTC, ADC channels, resolution, onboard memory, power, external/internal watchdog
9. Communications and Networking in IoT
10. Cloud Computing and Data Analysis
11. Interoperability Standards and security
12. Dashboard and Visualization
13. Documentation and Productization

Reference:

1. Raj Kamal, Internet of Things, McGraw Hill, 2018
2. P. Lea, Internet of Things for Architects, 2018
3. O. Hersent, D. Boswarthick, O. Elloumi, The Internet of Things, Wiley, 2016
4. D. Norris, The Internet of Things, McGraw Hill, 2015
5. A. Bahga and V. Madiseti, Internet of Things, University Press, 2016
6. Neil, Storey. Electronics: A Systems Approach, 4/E. Pearson Education India, 2009.

5. Teaching-Learning Strategies in brief:

Lectures will be integrating ICT into classroom teaching, active learning by students, and project-based learning by doing an IoT-based project.

6. Assessment methods and weightages in brief:

Mid Semester exam	20%
End Semester exam	30%
Lab work	20%
Mid Project Evaluation	10%
Final Project Evaluation	20%

Title of the Course : Introduction to Sociology

Name of the Faculty : Rajorshi Ray

Course Code : HS2.201

L-T-P : 3-0-0

(L= Lecture hours, T=Tutorial hours, P=Practical hours)

Credits : 4

Name of the Academic Program: Humanities Elective offered to UG3/UG4.

1. Prerequisite Course / Knowledge: Introduction to Human Sciences (Core course for UG2).

2. Course Outcomes (COs) (5 to 8 for a 3 or 4 credit course):

After completion of this course successfully, **the students will be able to.**

CO-1:

Students will have a sense of how the discipline of Sociology developed; its colonial roots, its core concerns and the impact of its roots and concerns on its methodologies.

CO-2:

Students will have an understanding of core conceptual frameworks and debates in Sociology.

CO-3:

Students will have a sense of major theoretical frameworks in Sociology. They will be introduced to the Durkheimian, Marxist and Weberian frameworks and methodologies, as well as to later theorists.

CO-4:

Students will be able to employ a sociological approach to the study of social institutions in India, such as caste, class, gender and tribe.

CO-5:

Students will be able to employ a sociological approach to the study of social, political and economic processes in India, such as development, industrialization, urbanization and migration.

3. Mapping of Course Outcomes (COs) with Program Outcomes (POs) and Program Specific Outcomes (PSOs) – Course Articulation Matrix

Matrix for CSE

	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO1 0	PO1 1	PO1 2	PSO 1	PSO 2	PSO 3	PSO 4
CO 1			1	1		3	3	3				2				
CO 2			1	1		3	3	3				2				
CO 3			1	1		3	3	3				2				
CO 4			1	1		3	3	3				2				
CO 5			1	1		3	3	3				2				

Note: Each Course Outcome (CO) may be mapped with one or more Program Outcomes (POs) and PSOs. Write '3' in the box for 'High-level' mapping, 2 for 'Medium-level' mapping, 1 for 'Low'-level' mapping

Matrix for ECE

	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO1 0	PO1 1	PO1 2	PSO 1	PSO 2	PSO 3	PSO 4
CO 1			1	1		3	3	3				2				
CO 2			1	1		3	3	3				2				
CO 3			1	1		3	3	3				2				
CO 4			1	1		3	3	3				2				
CO 5			1	1		3	3	3				2				

Note: Each Course Outcome (CO) may be mapped with one or more Program Outcomes (POs) and PSOs. Write '3' in the box for 'High-level' mapping, 2 for 'Medium-level' mapping, 1 for 'Low'-level' mapping

4.Detailed Syllabus:

Unit 1:

Development of Sociology as a Discipline: Sociology's emergence in Europe, its roots and causes. The connection between anthropology and colonialism. Enduring influences in the practice of sociology.

Unit 2:

Sociological Concepts: Social Structures, Individuals and Groups, Socialisation, Structure/Agency, Status and Roles, Rules, Values, Norms, Culture, Discourse, Deviance.

Unit 3:

Sociological Frameworks and Theories: Understanding social stratification through the structural functionalist, Marxist and the Weberian methods. Introduction to key ideas of Emile Durkheim, Karl Marx and Max Weber.

Unit 4:

Study of Social Institutions in India: Caste, Class, Tribe and Gender

Unit 5:

Political Sociology: Introduction to methods in Political sociology. Case studies from Indian politics to understand and identify shifting sociological and political narratives, political formations and political cultures. Introduction to concepts of power elites and ruling class.

Unit 6:

Study of Social Processes: Development, Industrialisation, Urbanisation and Migration

Reference Books:

Anthony Giddens, *Sociology* (Malden: Polity Press, 2009).

Alpa Shah, *In the Shadows of the State: Indigenous Politics, Environmentalism, and Insurgency in Jharkhand, India* (Durham, NC: Duke University Press, 2010).

Carol Upadhyay, *Reengineering India: Work, capital, and class in an offshore economy* (Delhi: Oxford University Press, 2016).

Gail Omvedt, *Dalit Visions: the Anticaste movement and Indian Cultural Identity* (New Delhi: Orient Blackswan, 2006).

M.N. Srinivas, *Social Change in Modern India* (New Delhi: Orient Longman, 1985).

5. Teaching-Learning Strategies in brief (4 to 5 sentences):

Students are introduced to theories and concepts through lectures. Relevant photographs, short 4-5 minutes videos, short films and clips from films are used during the lectures. Discussions and interventions in the classroom are highly encouraged. Students are given reading-based assignments through the course, which will help them to firstly understand the concepts in some depth and secondly apply the concepts in specific situations. Each assignment involves 30-40 pages of intense reading. Quizzes are designed to test analysis of a specific case.

6. Assessment methods and weightages in brief (4 to 5 sentences):

Type of Evaluation	Weightage (in %)
Mid Sem- Exam	15%. Questions designed to evaluate understanding of basic concepts.
End Sem Exam	30%. Questions designed to evaluate understanding of basic concepts. One long

	question which involves application of concepts discussed in the course.
Assignment 1	20%. Related to Unit I, II III (Essay question)
Assignment 2	20%. Related to Units IV V (Essay Question)
Quiz 1	7.5% Related to Unit I, III, III (Analysis of a case)
Quiz 2	7.5% Related to Unit IV, V (Analysis of a case)

Title of the Course : Introduction to UAV Design

Name of the Faculty : Harkumar K

Course Code : EC4.402

Credits : 4

L-T-P : 3-1-0

(L= Lecture hours, T=Tutorial hours, P=Practical hours)

1. Prerequisite Course / Knowledge:

Basics of Linear Algebra, Laplace transform and Vector calculus.

2. Course Outcomes (COs) (5 to 8 for a 3 or 4 credit course):

After completion of this course successfully, the students will be able to..

CO-1 Determine the design specifications of the Unmanned Aerial Vehicle (UAV) used for a particular application.

CO-2 Explain the various design phases involved in the UAV design.

CO-3 Perform the conceptual design and preliminary design for multi-rotor, fixed-wing and hybrid UAVs.

CO-4 Perform the stability and flight performance analysis for the designed UAV.

CO-5 Able to manufacture a prototype UAV.

CO-6 Perform the flight simulation and flight testing of the prototype UAV and verify its stability and performance characteristics.

3. Mapping of Course Outcomes (COs) with Program Outcomes (POs) and Program Specific Outcomes (PSOs) – Course Articulation Matrix

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3	PSO4
CO1	1	2	1	1	1	1	1	1	1	1	1	1	1	1	1	3
CO2	2	2	3	1	1	1	1	1	1	1	1	1	1	1	1	3
CO3	3	3	3	1	1	1	1	1	1	1	1	1	1	1	1	3
CO4	3	3	2	1	1	1	1	1	1	1	1	1	1	1	1	3
CO5	2	2	3	1	2	1	1	1	3	1	1	1	1	1	1	3
CO6	2	2	2	1	1	1	1	1	3	1	1	1	1	1	1	3

Note: Each Course Outcome (CO) may be mapped with one or more Program Outcomes (POs) and PSOs.

Write '3' in the box for 'High-level' mapping, 2 for 'Medium-level' mapping, 1 for 'Low'-level' mapping

4. Detailed Syllabus:

Unit 1: Types of UAVs--- Multi-rotors, fixed wing (FWUAV), Hybrid VTOLs

Unit 2: Multi-rotor design--- Concept of operation (CONOPS), design specifications, different reference frames, axis conventions, forces and moments, sizing and assembly, sensors and control.

Unit 3: FWUAV Flight mechanics and control--- wing, fuselage, stabilizer and control surfaces, propulsion system, forces (lift, drag, thrust, side force), moments (roll, pitch, yaw), trim conditions, longitudinal static stability, lateral and directional stability, PID control through successive loop closure.

Unit 4: FWUAV design--- Concept of operation (CONOPS), design specifications, preliminary sizing, airfoil selection, wing planform selection, control surface sizing, stabilizer sizing, selection of propulsion system (battery, motor/engine, propeller), stability and performance analysis, design trade-offs.

Unit 5: Different configurations (tilt-rotor, tail sitter), transition dynamics, design specifications,

sizing, stability and control.

Reference Books:

1. Daniel P Raymer, Aircraft Design: A Conceptual Approach, second edition, AIAA USA, 1992.
2. John D. Anderson, Introduction to flight, third edition, McGraw Hill USA, 1989.
3. R. W. Beard and T. M. McClain, Small Unmanned Aircraft: Theory and Practice, first edition, Princeton University Press USA, 2012

5. Teaching-Learning Strategies in brief (4 to 5 sentences):

Weekly lectures based on the course syllabus and based on the latest design technologies available in the literature and other industrial resources. Tutorials covering the use of software for UAV design and performance analysis. Detailed student assignment for practicing the different elements of conceptual design phase. Open book exam followed by detailed project submission including simulation studies, prototype development and flight testing.

6. Assessment methods and weightages in brief (4 to 5 sentences):

Type of Evaluation	Weightage (in %)
Quizzes	10
Assignments	40
Project	50

Title of the Course : **Language and Society**

Name of the Faculty : Aditi Mukherjee

Name of the Program : CLD

Course Code : CL2.203

Credits : 4

L - T - P : 3-1-0

(L - Lecture hours, T-Tutorial hours,

P - Practical hours)

Semester, Year : Monsoon 2026

Pre-Requisites : **None**

Course Outcomes :

1. The course intends to familiarize students with the social dynamics of language in use.
2. After doing the course, the students should be able to identify and recognize various phenomena which are at play. They should develop an understanding of social behavior such as identity assertion, attitudes in language use and choices that people make while using variants of linguistic items depending on social circumstances.
3. The students are expected to be able to classify these phenomena and explain some of their consequences. For example, code mixing and code switching are very common in multilingual societies such as India. How people use code mixing for better communication or how they use code switching for social reasons are some of the concepts the students should be able to explain given a social context.
4. At the end of the course, the students are expected to be able to analyze language data and employ basic concepts learned during the course for interpreting language data for computational models.
5. The course should give them the confidence to be able to design and develop computational models in real case scenarios.

6. The students will be working on real data projects in teams which will give them experience of working as teams to solve a real problem.

Course Topics :

1. Language Variation: Sociolinguistics and sociology of language, Variation in language, Linguistic variables and social variables. Speech Communities, Language, Dialects and Varieties. Standard language and Standardization, Regional and social dialects. Register, styles. Dialectology.

2. Language Contact: Bilingualism/Multilingualism, borrowing, code mixing/switching, pidginization and creolization, convergence, language maintenance/shift, language acquisition in a multilingual setting. Diglossia with or without bilingualism.

3. Social stratification of language. Inherent variability. Sociolinguistic variables: indicator, marker, stereotype. Style shifting. Hypercorrection. Language variation and language change. Social motivation for sound change.

Sociolinguistic devices for effective communication. Communicative competence. Politeness strategies. Pronouns of power and solidarity.

4. Language and Culture: Directions of influence. The Whorfian hypothesis

5. Critical Sociolinguistics. Language and power. Language and social attitudes: gender, race, education.

6. Language planning: codification and elaboration.

Preferred Textbooks:

- Ronald Wardaugh: *Introducing Sociolinguistics*
- R. A. Hudson: *Sociolinguistics*
- Suzanne Romaine: *Language in Society*

Reference Books :

- J.B. Pride and J. Holmes (ed): *Sociolinguistics*
- Paolo Giglioli (ed): *Language and Social Context*
- Robert Bayley and Ceil Lucas (ed): *Sociolinguistic Variation*

E-book Links :

1. <http://staffnew.uny.ac.id/upload/132107096/pendidikan/Book+for+Sociolinguistics.pdf>
2. <https://pdfcoffee.com/cambridge-textbooks-in-linguistics-r-a-hudson-sociolinguistics-cambridge-university-press-1996pdf-3-pdf-free.html>

Grading Plan : (The table is only indicative)

Type of Evaluation	Weightage (in %)
Quiz-1	No
Mid Sem Exam	20
Quiz-2	10
End Sem Exam	25
Assignments	15
Project	20

Seminar	10
Other Evaluation	None

Mapping of Course Outcomes to Program Objectives: (1 – Lowest, 2—Medium, 3 – Highest, or a ‘-’ dash mark if not at all relevant). Program outcomes are posted at

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3	PSO4
CO1	X															
CO2				X												
CO3																
CO4		X														
CO5			X													
CO6									X							

Teaching-Learning Strategies in brief (4-5 sentences):

There will be regular classes with interactive sessions to cover the theory. Since actual learning happens through practical work, each student will take up a project which will involve some field work, literature survey and working with real data. Classic research papers will be distributed for the students to read critically and present them in class. So, the idea is that learning happens through listening and discussions (classes), reading (seminar papers) and working with data (project). For each topic some assignment will be given for the students to get a better grip on the topic.

Title of the Course : Language Models and Agents

Faculty : Vasudeva Varma

Course Code : CL3.410

Credits : 4

L-T-P : 3-1-0

Semester, Year : Monsoon 2026

Computer Science Elective

Pre-requisites:

Students should have completed at least one graduate-level introductory course in AI/ML, such as Introduction to NLP, Statistical Methods in AI, Advanced NLP, or Information Retrieval and Extraction.

Course Objective

Large Language Models (LLMs) are transforming how tasks are automated and personalized across domains. As LLMs evolve into interactive agents, they are set to drive the next major breakthrough in AI. This course introduces the foundations of LLMs, key capabilities for task automation, and essential infrastructures for building LLM agents. We will explore applications such as generation, tool augmentation, web automation, healthcare, and scientific research, while also addressing current limitations and future directions.

Course Outcomes

By the end of the course, students will be able to:

- **CO1: Articulate language technologies and transformer concepts.** Explain the foundations of natural language processing, word/vector embeddings, self-attention mechanisms, transformer architectures and their variants, as well as probabilistic and causal language modeling.
- **CO2: Prepare and curate high-quality training data for LMs.**
Design scalable pipelines for collecting, cleaning, formatting, and balancing datasets across domains and languages to ensure quality and diversity in language model training.
- **CO3: Train small decoder-only language models from scratch.**
Understand and implement the end-to-end pipeline for training small-scale autoregressive language models, including tokenizer training, model initialization, and loss optimization.
- **CO4: Adapt and evaluate language models for specific tasks.** Apply pre-training, fine-tuning (including supervised and RLHF alignment), prompting techniques, and evaluation benchmarks to customize LLMs for domain-specific applications and reliably measure their performance (e.g., reasoning, hallucination, memorization).
- **CO5: Build augmented and agentic LLM systems.**
Design and implement retrieval-augmented and tool-augmented workflows, and assemble agentic frameworks that leverage LLMs to solve real-world problems through interactive, task-oriented pipelines.
- **CO6: Address responsible deployment and future extensions.** Critically assess ethical, safety, bias-mitigation, privacy, quantization/inference-optimization concerns, and explore multimodal integration and emerging research trends to responsibly deploy and extend language models.

Course Topics

Module 1: Background

- **Course Logistics** – Overview of schedule, grading, tools.
- **Introduction to Language Technologies** – Basics of natural language processing tasks.
- **Word Vectors & Language Models** – From word embeddings to classical LMs.
- **Self-Attention & Transformers** – Core mechanism behind modern Language Models.
- **Transformer Variants** – Improvements like BERT, GPT, T5, etc.

Module 2: Building Language Models (Transformers and Pre-training)

- **Pre-training Language Models** – Embedding, Objectives like masked LM, autoregression.
- **Language Generation** – Basics of text generation tasks.
- **Foundations of LMs** – Probabilistic modeling and training setups.
- **Causal LMs: Architecture & Hyperparameters** – GPT-like models, decoding, configs.
- **Scaling LLMs** – Challenges and benefits of scaling parameters and data.
- **Long Sequence Modeling** – Handling long context with efficient attention.

Module 3: Using Language Models

- **Prompting** – Techniques like few-shot, chain-of-thought.
- **Post-training Data for Fine-tuning** – Data curation for model alignment.

Module 4: Enhancing Language Model Capabilities

- **Supervised Fine-tuning** – Task-specific training after pretraining.
- **Preference Alignment** – Reinforcement Learning from Human Feedback (RLHF).
- **Reasoning, Planning & Understanding (1 & 2)** – Emergent capabilities of LLMs.
- **RAG (1 & 2)** – Integrating retrieval for grounding outputs.

Module 5: Orchestrating Language Models: Language Agents

- **Tool Augmentation** – Using tools/APIs within model outputs.
- **LLM Agents: A Case Study** – Building interactive task-oriented agents.
- **Hallucination & Memorization** – Model reliability and factuality issues.

Module 6: Misc and Advanced Topics

- **LLM Evaluation** – Benchmarks and human evaluation techniques.
- **Safety & Ethics** – Fairness, misuse prevention, and responsible AI.
- **Quantization & Inference Optimization** – Speeding up and compressing LLMs.
- **Vision Transformers** – LLMs extended to visual inputs.
- **Multimodal Pre-training** – Training with mixed input types (text, vision).
- **What Next?** – Emerging trends, research frontiers, and challenges.

Textbooks and Resources

- **Preferred Resources:** Due to the flipped-classroom format, curated interactive materials (blogs, video essays, and expert lectures from academia and industry) will be shared regularly.
- **Reference Books:** Not Applicable
- **E-book Links:** Not Applicable

Grading Plan

Component	Weightage (%)
Mini Project (Proposal, Prototype, Presentation)	20%
Major Project (Proposal, Prototype, Presentation, Team work, Learning)	40%
Quizzes, Mid sem and assignments	30%
Class Participation and Activities	10%
No End Sem exam	-

Mapping of Course Outcomes to Program Objectives

Cour se Outc ome	P O 1	P O 2	P O 3	P O 4	P O 5	P O 6	P O 7	P O 8	P O 9	P O 10	P O 11	P O 12	PS O1	PS O 2	PS O 3	PSO4
CO1	3	2	1	1	2	-	-	-	-	-	-	1	3	2	2	1
CO2	2	3	2	2	2	1	1	-	2	2	2	2	3	3	2	1
CO3	2	3	2	3	3	2	2	2	3	2	2	2	2	3	3	2
CO4	2	3	3	3	3	2	3	2	3	2	2	2	3	3	3	2
CO5	2	2	3	3	3	2	2	2	3	3	2	2	2	3	3	2
CO6	1	2	1	2	-	1	3	1	1	1	1	3	1	2	1	-

Lecture Planning

#	Module	Topics/Lectures	Date / Week	Course Materials (To be Added)	Events
*	Background	Course Logistics			
1		Introduction to Language Technologies Word vectors and Language models			
2		Transfer Learning Self Attention & Transformers			
3	Building Language Models	Pre-training Language Models			
4		Natural Language Generation Foundations of Language Models			Mini-Project
5		Data and its processing for Large Language Model		Tokens & Tokenization Length Limit	
6		Causal Language Models - Architecture & hyperparameters ***Transformer Variants			
7		Scaling Large Language Models Long Sequence Modeling & LLM beyond English			

8	Using Language Models	Prompting - 1			
9	Enhancing the large language model capabilities	Data for FineTuning Supervised FineTuning Preference Alignment with reinforcement learning - 1		- Efficient Adaptation (Prompting + PEFT) - Training language models to follow instructions with human feedback	

				- Deep reinforcement learning from human preferences	
10		Supervised FineTuning Preference Alignment with reinforcement learning - 2		- Self-Instruct: Aligning Language Models with Self-Generated Instructions - Direct Preference Optimization: Your Language Model is Secretly a Reward Model	Major-Project
11		LLM Reasoning, Planning and Understanding - 1 Data construction for LLM reasoning		- Learning to Reason: from Supervised to Reinforcement Fine-tuning - Chain-of-Thought Reasoning Without Prompting - LARGE LANGUAGE MODELS CANNOT SELF-CORRECT REASONING YET - Premise Order Matters in Reasoning with Large Language Models	

12		LLM Reasoning, Planning and Understanding - 2		- Reinforcement Learning for Reasoning in Large Language Models with One Training Example - Cognitive Behaviors that Enable Self-Improving Reasoners, or, Four Habits of Highly Effective STaRs	
13		Case Study : Deepseek, qwen			
14		Retrieval-Augmented Generation (RAG) with Large Language Model - 1			
15		Retrieval-Augmented Generation (RAG) with Large Language Model - 2			
16		Retrieval-Augmented Generation (RAG) with Large Language Model - 3			
17	Orchestrating Language Models: Language Agents	Agentic AI Frameworks & AutoGen -1			
18		Agentic AI - 2 Tool Augmentation Agentic RAG		- TOOLFLOW: Boosting LLM Tool-Calling Through Natural and Coherent Dialogue Synthesis	
19		LLM agents : A case Study			
20		EVALUATING TOOL : LLM TOOL SANDBOX & TOOLTALK			
21		LLM Hallucination & Memorisation		- (Im)possibility of Automated Hallucination Detection in Large Language Models	

22	Misc and Advanc ed Topics	Safety and ethics			
23		Large Multimodal Models -1			
24		Large Multimodal Models -2			

Teaching and Learning Strategies

This course emphasizes assignment-driven, project-based, and presentation-oriented learning. Students will engage through regular hands-on assignments, peer presentations, and thematic projects that encourage exploration beyond the classroom. Selected topics will follow a flipped-classroom model to promote self-paced and collaborative learning. The course culminates in a capstone project that creates or applies language models for multiple languages to real-world challenges in domains such as education, healthcare, and governance, enabling students to gain practical experience and deepen their understanding independently.

Title of the Course : Learning and Memory
Faculty Name : Bhaktee Dongaonkar
Name of the Program : Cognitive Science
Course Code : CG1.403
Credits : 4
L - T - P : 3-1-0
(L - Lecture hours, T-Tutorial hours, P - Practical hours)
Semester, Year : Monsoon 2026

Pre-Requisites : not applicable

Course Overview

This course is designed for students to learn the core concepts of learning and memory mechanisms in the brain. The course will go in-depth and discuss important scientific experiments and theories, and neural models that have helped to shape the understanding of learning and memory behaviour. The content is a mix of cognition, neuroscience, and neural network models.

Course Outcomes :

- CO1- Understand the basic principles of learning and memory in the brain
- CO2- Apply the fundamentals of behaviour to brain network models
- CO3- Examine the experimental results from research in the field of learning and memory

- CO4- Evaluate a chosen topic, understand its current status and propose new ideas
- CO5- Develop an experimental design that can propel the field ahead

Course Topics :

Introductory Module

- Psychology of Learning and Memory
- Neuroscience of Learning and Memory

Learning Module

- Habituation, Sensitization, and Familiarization: Learning About Repeated Events
- Classical Conditioning: Learning to Predict Significant Events
- Operant Conditioning: Learning the Outcome of Behaviors
- Generalization, Discrimination Learning, and Concept Formation

Memory Module

- Episodic Memory and Semantic Memory
- Skill Memory
- Complementary learning systems in the brain /Memory network in the brain
- Working Memory and Cognitive Control

Integrative Module

- Emotional/Stress Influences on Learning and Memory
- Social Learning and Memory: Observing, Interacting, and Reenacting
- Development and Aging: Learning and Memory Across the Lifespan

Preferred Text Books : Learning and Memory- From Brain to Behavior (3rd edition, 2020) - Mark A. Gluck, Eduardo Mercado, Catherine E. Myers, Worth Publishers (Macmillan, New York)

Reference Books :

E-book Links : <https://www.macmillanlearning.com/college/ca/product/Learning-and-Memory/p/1319107389>

Grading Plan :

(The table is only indicative)

Type of Evaluation	Weightage (in %)
In class quizzes	40% (8 quizzes x 5%)
Mid Sem-Exam	15%
End Sem Exam	30%
In-class discussions & presentations	15%

Mapping of Course Outcomes to Program Objectives: (1 – Lowest, 2—Medium, 3 – Highest, or a ‘-’ dash mark if not at all relevant).

PO1- Demonstrate conceptual knowledge of cognition at brain and behaviour level

PO2 - Evaluate and analyze scientific work done in the field

PO3 – Apply the knowledge to address important unanswered questions in the field

PO4 - Demonstrate ability to think of potential experiments

PO5 – Apply the scientific ability to work on real-world problems in the field of cognitive science

	PO1	PO2	PO3	PO4	PO5
CO1	2	1	1	1	1
CO2	1	2	2	1	1
CO3	1	3	3	2	1
CO4	1	3	3	3	3
CO5	1	2	2	3	3

Teaching-Learning Strategies in brief (4-5 sentences) :

The textbook will be used as a reference to cover the important topics and basics in the field of learning and memory. Published experimental results will be discussed in class with students to understand how experimental work is conducted and analyzed.

Students will then choose a topic of their interest, understand it in-depth, design a study that fills a gap and explain it to the class.

Title of the Course : **Making of Contemporary India**
Name of the Faculty : Aniket Alam+ Aakansha Natani
Course Code : HS4.102
L-T-P : 3-1-0
Credits : 4
Name of the Academic Program: B.Tech In Computer Science And M.S. In Computing And Human Sciences By Research

1.Prerequisite Course / Knowledge: Admission to CHD programme

2.Course Outcomes (COs):

On successful completion of this course, students will be able to

CO-1: Identify and explain major political, social, and economic trends and milestones that have made India what it is today.

CO-2: Understand and describe major frameworks and methods that scholars have used to study India.

CO-3: Compare and assess the potential as well as limitations of these frameworks and methods;

CO-4: Apply the essential conceptual foundations taught in this course to other courses that offer in-depth study of related topics and themes; and

CO-5: Develop a critical vocabulary and perspective that will contribute to the growth of their individual research voice and expertise at the confluence of computing and human sciences.

3.Mapping of Course Outcomes (COs) with Program Outcomes (POs) and Program Specific Outcomes (PSOs) – Course Articulation Matrix

	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PO 12	PS O1	PS O2	PS O3	PS O4
CO 1	1	1	1	1	1	2	3	1	1	1	1	3	1	3	2	3
CO 2	1	2	1	2	1	1	3	2	1	3	1	3	1	3	3	3
CO 3	1	2	1	2	1	1	3	2	1	3	1	3	1	3	3	3
CO 4	1	3	1	2	1	2	3	2	1	3	1	3	1	3	3	3
CO 5	1	3	1	3	1	2	3	2	2	3	1	3	1	3	3	3

Note: Each Course Outcome (CO) may be mapped with one or more Program Outcomes (POs) and PSOs. Write '3' in the box for 'High-level' mapping, 2 for 'Medium-level' mapping, 1 for 'Low'-level' mapping

4.Detailed Syllabus:

Unit 1:

Colonial Background: overview of the main features of colonial rule and of India's independence movement; important social and economic trends beginning in the late 19th and early 20th centuries.

Unit 2:

Independence, Partition, Constitution of the New Nation-State: detailed study of Independence and the making of the Constitution of India.

Unit 3:

1950s to 2000s: overview of how India's polity and society passed through transition and faced new challenges; major landmarks of independent India's political, social, economic, and development journey.

Unit 4:

Long-term Processes: literacy and education, infant mortality and sex-ratios, migration and urbanization, and travel and communication.

Reference Books:

1. Sugata Bose and Ayesha Jalal, *Modern South Asia* (Second Edition, 2004)
2. Michael Mann, *South Asia's Modern History: Thematic Perspectives* (2014)
3. Paul Brass, *The Politics of India since Independence* (Second Edition, 2001)
4. Pranab Bardhan, *The Political Economy of Development in India* (Expanded Edition, 1998)

5. Teaching-Learning Strategies in brief (4 to 5 sentences):

The teaching-learning strategy in this course will consist of lectures, which will incorporate prompts for classroom discussion and activities to enable active learning and critical thinking. The tutorial slots enable students to undertake small in-class assignments related to assigned readings. This learning will be further consolidated through assessments that will be designed to test and develop students' knowledge and skills in conducting research and writing. Students will be expected to read about 2000 pages of academic literature and write about 12,000 words of essays and answers over the semester.

6. Assessment methods and weightages in brief (4 to 5 sentences):

There will be one writing assignment, worth 15% of the total grade in this course, for each of the four Units. The project, amounting to 25% of the total grade, will consist of a research essay that students will write and then present. Class participation will account for the remaining 15% of the grade.

Title of the Course	: Market Research & Validation
Name of the Faculty	: Ratan K Putla (Guest Faculty)
Course Code	: PD2.404
Credits	: 2 (L-T-P: 1.5-0-3)
(L=Lecture hours, T=Tutorial hours, P=Practical hours)	
Duration	: 12 Weeks (Monsoon/Spring Semester)

1. Prerequisite Course / Knowledge

- **Systems Thinking:** Ability to view products as ecosystems.
- **Basic Statistics:** Understanding of distributions, sampling, and significance.
- **Design Thinking Foundations:** Empathy mapping and user persona basics.

2. Course Objectives & Outcomes (COs)

This course equips Product Management students with the frameworks, tools, and practical skills to design and execute rigorous market research programs, synthesize qualitative and quantitative evidence, and build validated business cases that de-risk product investment decisions.

- To move beyond "asking users what they want" to "observing what they need."
- To master quantitative and qualitative validation frameworks specific to tech products.
- To leverage AI and Big Data tools for modern market intelligence.

Students will graduate able to lead end-to-end discovery — from framing market hypotheses through to communicating validated insights to executive stakeholders.

Learning Outcomes

This course provides knowledge about leveraging AI/ML to solve business problems and deliver customer value.

1. Design primary and secondary research programs aligned to product strategy questions
2. Select and apply qualitative methods: in-depth interviews, contextual inquiry, diary studies
3. Design, instrument, and analyse quantitative surveys with statistical rigour
4. Map competitive landscapes and identify market whitespace using systematic frameworks
5. Build and run structured customer validation experiments (smoke tests, landing pages, concierge MVPs)
6. Synthesise mixed-methods data into actionable product insights and business cases
7. Communicate research findings persuasively to diverse executive audiences
8. Apply ethical research practices and navigate bias in data collection and interpretation

3. Mapping of Course Outcomes (COs) with Program Outcomes (POs) and Program Specific Outcomes (PSOs) – Course Articulation Matrix

	P O 1	P O 2	P O 3	P O 4	P O 5	P O 6	P O 7	P O 8	P O 9	PO 10	PO 11	PS O1	PS O2	PS O3	PS O4
CO 1	3	3	3	2	3	2	1	2	3	2	3	3	2	2	2
CO 2	3	3	3	2	3	2	1	2	3	2	2	3	2	2	2
CO 3	3	3	3	2	3	2	1	3	3	2	2	3	2	2	2
CO 4	3	2	3	2	3	2	1	3	3	3	2	3	2	2	2
CO 5	2	2	2	2	2	2	1	2	3	3	3	3	2	2	2

4. Detailed Syllabus

Unit	Topic	Week	Content Focus
Unit 1	Foundations of Evidence-Based PM Market Segmentation & Targeting	Week 1	The Mom Test; Confirmation Bias in Research; The Lean Startup Loop. Jobs-to-be-Done (JTBD) Framework; Psychographics vs. Demographics.
Unit 2	Qualitative Deep-Dives Competitive Intelligence	Week 2	Problem Interviews; Contextual Inquiry; Ethnographic Research in Tech. Feature Benchmarking; Blue Ocean Strategy; Analysing Technical Moats.
Unit 3	Quantitative Validation Market Sizing & Economics	Week 3	Survey Design (avoiding leading questions); MaxDiff & Conjoint Analysis. Top-down vs. Bottom-up TAM/SAM/SOM; Value-based Pricing research.
Unit 4	Pre-prototyping & Demand Validation Landing Page & Ad-Spend Tests	Week 4	Fake Door Testing; Wizard of Oz MVPs; Concierge MVPs. Using Performance Marketing metrics (CTR, CPC) to validate intent.
Unit 5	AI in Market Research Usability & UX Validation	Week 5	Sentiment Analysis of Reviews; Topic Modelling for Trend Spotting. RITE Method (Rapid Iterative Testing); Hallway Testing; Eye-tracking basics.
Unit 6	Product-Market Fit (PMF) Metrics The Validation Pitch	Week 6	The Sean Ellis Test; Retention Curves; Cohort Analysis. Presenting "Proof of Problem" and "Proof of Solution" to Stakeholders.

5. Reference Books & Case Studies

1. **Book:** *The Mom Test* by Rob Fitzpatrick (Essential for interview techniques).
2. **Book:** *The Right It* by Alberto Savoia (Focus on Prototyping).
3. **Book:** *Testing Business Ideas* by David J. Bland & Alexander Osterwalder.
4. **Case Study: Dropbox:** The Video MVP (Validating demand for a technical impossibility).
5. **Case Study: Zappos:** The Concierge MVP (Validating e-commerce without inventory)

6. Teaching-Learning Strategies

- The course is divided into 12 lectures/practical sessions including classroom

exercises and role-play simulations.

- It includes "Learning by Doing" via fieldwork, where students interview senior PMs on stakeholder friction.
- Students will analyse real-world case studies focused on product failures caused by poor alignment.
- A short-term project offers the opportunity to practice the full influence cycle from mapping to board-level presentation.
- Students are expected to spend 3 hours per week on home/field assignments.

7. Assessment Methods and Weightages

Component	Weightage	Description
1. Class/Home Activities	10%	Assignments – Class and Home.
2. Field Research Report	25%	Synthesis of 10+ qualitative interviews and JTBD analysis
3. Validation Experiment	25%	Running a "Fake Door" or Landing Page test and analysing data.
4. The Validation Dossier	40%	A comprehensive document proving/disproving the viability of a new product concept.

Title of the Course	:	MCS 1 – Probability and Statistics
Name of the Faculty	:	Naresh Manwani
Course Code	:	MA6.301
L-T-P	:	3-1-0
Credits	:	2

Prerequisite Course / Knowledge:

Knowledge of UG (BTech) course in Discrete Maths.

Course Outcomes (COs):

After completion of this course successfully, the students will be able to...

CO-1: Understanding the basic probability concepts sample space, events, probability mass function, conditional probability, Bayes Rule, Random Variables, Probability Mass and Density functions, Cumulative distribution function, Expectation, Variance, Bernoulli Binomial, Gaussian, Geometric, Exponential, Poisson distributions.

CO-2: Demonstrate familiarity with use of Linearity of Expectation, Markov's and Chebyshev's Inequalities, Law of Large Numbers, Central Limit Theorem.

CO-3: Apply principles of Tail bounds and Central Limit Theorem to real world problems in Estimation, Randomized Algorithms, etc.

CO-4: Derive formulas for finding Maximum Likelihood Estimates (MLE) and Maximum Apriori Estimates (MAE) for Probability Models.

CO-5: Create mathematical models using principles of Probability and analyze them.

Mapping of Course Outcomes (COs) with Program Outcomes (POs) and Program Specific Outcomes (PSOs)

	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PO 12	PSO 1	PSO 2	PSO 3	PSO 4
CO1	2	3	1	3	1	1	1	1	2	2	1	3	1	1	1	2
CO2	1	2	2	2	1	1	1	1	2	2	1	1	3	1	1	2
CO3	2	3	1	3	1	1	1	1	2	2	1	3	2	1	1	2
CO4	1	3	2	2	1	1	1	1	2	2	1	1	3	1	2	2
CO5	1	3	2	3	1	1	1	1	2	2	1	1	3	1	1	3
CO6	1	2	1	2	2	1	1	1	2	2	1	2	3	2	2	2

'3' for 'High-level' mapping, 2 for 'Medium-level' mapping, 1 for 'Low'-level' mapping.

Detailed Syllabus:

Unit 1: Sample Spaces, Counting, Uniform Probability, Axioms of Probability, Continuous Probability Spaces, Conditional Probability, Bayes Rule, Independence.

Unit 2: Random Variables, PMFs, Discrete Probability Distributions, Multiple Random Variables, Expectation, Variance, Covariance, Standard distributions of Bernoulli, Binomial, Geometric, Gaussian, Exponential, Poisson.

Unit 3: Continuous Probability Distributions, Tail Bounds (Markov, Chebyshev, Chernoff), Law of Large Numbers, Central Limit Theorem.

Unit 4: Bayesian Statistics, Maximum A Posteriori Estimation, Maximum Likelihood Estimation, Confidence Intervals.

Reference Books:

1. Introduction to Probability, 2nd Edition by Dimitri P. Bertsekas and John N. Tsitsiklis.
2. Introduction to Probability, Statistics and Random Processes. by Hossien Pishro-Nik. Textbook available online: <https://www.probabilitycourse.com/>
3. Introduction to Probability and Statistics for Engineers and Scientists by Sheldon M. Ross.
4. An Introduction to Probability Theory and Its Applications, Volume 1 by William Feller.
- 5.

Teaching-Learning Strategies in brief (4 to 5 sentences):

Lectures will initially introduce the motivations, concepts, definitions along with simpler examples. This will be followed by assignments and quizzes that will make sure that the students have understood the concepts. These will be followed by deeper lectures and assignments which lead the students to the bigger questions in the area. These will also be supplemented with real world engineering problems so that they can apply the concepts learned by them.

Assessment methods and weightages in brief (4 to 5 sentences):

- Light In-class Quizzes: 15%
- Assignments: 15%
- Class Test 1: 20%
- Class Test 2: 20%
- End Exam: 30%

Title of the Course	:	MCS 2 - Linear Algebra
Name of the Faculty	:	Pawan Kumar
Course Code	:	MA6.302
L-T-P	:	3-1-0
Credits	:	2

Prerequisite Course / Knowledge:

This is one of the first math courses and only assumes school knowledge of maths.

Course Outcomes (COs):

After completion of this course successfully, the students will be able to...

CO-1: Understanding the basic mathematical concepts like vector space, Basis, Linear Transformation, Rank Nullity Theorem, Matrix Representation of Linear Transformations, System of Equations, Determinants.

CO-2: Demonstrate familiarity with Eigenvalues, Eigenvectors, Orthogonality and Matrix Decomposition theorems.

CO-3: Synthesize proofs of theorems related to Matrices and Vector Spaces using clear mathematical and logical arguments.

CO-4: Apply principles of Spectral Decomposition and Singular Value Decompositions to real world problems in Image Compression, Principal Component Analysis etc.

CO-5: Design dimension reduction techniques with approximation guarantees using Best Fit Subspaces.

CO-6: Create mathematical models using principles of Linear Algebra and analyze them.

Mapping of Course Outcomes (COs) with Program Outcomes (POs) and Program Specific Outcomes (PSOs)

	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PO 12	PSO 1	PSO 2	PSO 3	PSO 4
CO1	2	3	1	3	1	1	1	1	2	2	1	3	1	1	1	2
CO2	1	2	2	2	1	1	1	1	2	2	1	1	3	1	1	2
CO3	2	3	1	3	1	1	1	1	2	2	1	3	2	1	1	2
CO4	1	3	2	2	1	1	1	1	2	2	1	1	3	1	2	2
CO5	1	3	2	3	1	1	1	1	2	2	1	1	3	1	1	3
CO6	1	2	1	2	2	1	1	1	2	2	1	2	3	2	2	2

'3' for 'High-level' mapping, 2 for 'Medium-level' mapping, 1 for 'Low-level' mapping.

Detailed Syllabus:

Unit 1: Vector spaces, subspaces, Linear dependence, Span, Basis, Dimension, Finite dimension vector spaces Linear transformation, Range and Null space of linear transformation, Rank Nullity Theorem, Sylvester's Law, Matrix representation of a linear transformation for finite dimensional linear spaces, Matrix operations, change of basis, Rank of a Matrix, Range and Null Space of a matrix representing a linear transformation.

Linear spaces with inner product [inner product example over space of functions: orthogonality and orthogonal functions in L₂].

Unit 2: System of Linear Equations, Row-echelon form, reduced row-echelon form. Gauss-Jordan elimination, Solution of linear systems using Gauss-Jordan elimination, matrix inversion by Gauss-Jordan elimination, Understanding Range Space and Solution Space using Rank-Nullity Theorem.

Unit 3: Eigenvalues and Inner product: Eigenvalues & Eigenvectors, Norms, Inner Products and Projections, Applications like Analysis of Random Walks.

Unit 4: Advanced Topics: Spectral & Singular Value Decomposition Theorems, Applications of SVD and Best Fit Subspaces

Reference Books:

1. Linear Algebra, 2nd edition, K. Hoffman and R. Kunze.
2. Finite Dimensional Vector Spaces, P. Halmos.
3. Introduction to Linear Algebra, Gilbert Strang.
4. Linear Algebra Done Wrong, Sergei Treil.

Teaching-Learning Strategies in brief (4 to 5 sentences):

Lectures will initially introduce the motivations, concepts, definitions along with simpler examples. This will be followed by assignments and quizzes that will make sure that the students have understood the concepts. These will be followed by deeper lectures and assignments which lead the students to the bigger questions in the area. These will also be supplemented with real world engineering problems so that they can apply the concepts learned by them.

Assessment methods and weightages in brief (4 to 5 sentences):

- Light In-class Quizzes: 15%
 - Assignments: 15%
 - Class Test 1: 10%
 - Class Test 2: 10%
 - Mid Exam: 20%
 - End Exam: 30%
-

Title of the course : Micro & Nano Fabrication Technology for ICs

Name of the Faculty : Prof. G. Rajaram (Emeritus Professor) and Dr. L. D. Varma Sangani

Course Code : EC2.416

L-T-P : 3+1+0

Credits : 4

Prerequisite Course / Knowledge (If any): Nil

Course Outcomes (COs)

After completion of this course successfully, the students will be able to

CO-1: Describe the structure and fabrication process steps for the most popular devices.

CO-2: Identify the technological problems and their resolution in IC fabrication technology.

CO-3: Distinguish between various processes required for Micro-electronic device fabrication

CO-4: Design and simulate fabrication process steps for specific devices.

CO-5: Evaluate different technologies for selecting a suitable one for a specific device.

Mapping of Course Outcomes (COs) with Program Outcomes (POs) and Program Specific Outcomes (PSOs)

	PO1	PO2	PO3	PO4	PO5	PO6	PSO1	PSO2	PSO3	PSO4	PSO5
CO1	3	0	2	2	2	2	1	2	2	3	2
CO2	3	1	2	2	2	1	2	2	2	3	1
CO3	3	2	3	3	2	2	1	2	1	3	2
CO4	3	2	3	3	3	3	2	2	2	3	2
CO5	3	2	3	3	3	3	2	2	2	3	3

Note: '3' in the box for 'High-level' mapping, 2 for 'Medium-level' mapping, 1 for 'Low'-level' mapping

Detailed Syllabus:

Unit 1 (10 hours):

- Basic Device structures: BJT, MOSFET, MESFET, capacitors, resistors
- Design rules, self-alignment, device isolation, oxide breakdown and protection
- Process stream/steps: Bipolar, NMOS, CMOS, GaAsFET, ULSI issues

Unit 2 (8 hours):

- Wafer fabrication- EGS, Czochralski technique for Si and GaAs, Wafer fab.
- Epitaxy: MBE, CVD

Unit 3 (6 hours):

- Lithography- Photoresists, optical and electron beam lithography

Unit 4 (16 hours):

- Ion implantation
- Oxidation
- Thin films deposition- vacuum techniques, sputtering techniques, e-beam and resistive heating evaporation
- Etching: wet etching, RIE, RIBE, etchants

Unit 5 (4 hours):

- Packaging- Die-bonding, wire-bonding, flip-chip technology
- Testing and Yield estimation
- Clean room methodologies

Unit 6:

- Self study and seminars (optional as determined by course instructor) :
Structure and fabrication technology for memory elements, sensors, opto-electronic devices, etc

Books:

1. Science and engineering of microelectronic Fabrication by Stephen Campbell (Oxford University Press; Second edition (2012))
2. VLSI Technology by S.M.Sze (McGraw Hill Education; 2 edition (2017))
3. VLSI Design Techniques for Analog and Digital Circuits by R. L. Geiger, P.E. Allen, and N. R. Strader (McGraw-Hill College (1989))

Type of Evaluation	Weightage (in %)
Assignments	20%
Course project	20%
Quiz-1	15%
Quiz-2	15%
End semester exam	30%

Title of the Course : Mechanistic Interpretability of Language Models

Name of the Faculty : Hrishikesh Terdalkar

Course Code : CL3.412

L-T-P : 3-1-0

Credits : 2 (L=Lecture hours, T=Tutorial hours, P= Practical hours)

Half-semester (H2, starting October), Monsoon 2026

B.Tech. in Computer Science and Master of Science in Computational Linguistics by Research (CLD)

1. Prerequisites

Required: Introduction to NLP; understanding of transformer architecture including attention mechanisms, residual-stream representation, and weight composition (OV/QK structure); backpropagation; PyTorch at a level sufficient to write and run training loops.

Helpful: Advanced NLP; linear algebra and probability at the undergraduate level.

About. Standard NLP courses treat language models as black boxes to be prompted or fine-tuned. This course opens the box. Mechanistic interpretability asks which specific components of a trained model are causally responsible for a given behaviour, and then tries to verify that claim experimentally. The main methods covered are probing classifiers, logit lens, activation patching, sparse autoencoders, and automated circuit discovery. Indian and multilingual models appear throughout the course and are required in all project directions.

2. Course Outcomes (COs)

Students completing this course will be able to:

CO-1 Distinguish behavioural, representational, and mechanistic approaches to model analysis; use dedicated interpretability tooling to inspect activations, attention patterns, and residual-stream states in publicly available transformer checkpoints.

CO-2 Design and evaluate probing classifiers for linguistic properties; apply control tasks to avoid the probing-accuracy trap; interpret probing results with appropriate caution about what correlation does and does not imply.

CO-3 Apply activation patching, attribution patching, and path patching to trace causal pathways in transformer language models; interpret logit-lens and tuned-lens trajectories as correlational evidence; articulate why intervention-based evidence is required to establish causal claims about model behaviour.

- CO-4** Use pre-trained sparse autoencoders to identify and causally test monose-mantic features in a model's residual stream (including activation steering as a causal intervention); reason about superposition and what it implies for direct neuron-level analysis.
- CO-5** Discover and validate attention circuits via automated circuit discovery and edge attribution patching; assess circuit faithfulness and completeness; criti-cally read circuit claims, including the limits of small-model findings.
- CO-6** Apply mechanistic interpretability methods to model reliability problems — hallucination localisation, faithfulness of chain-of-thought, and model debug-ging — in multilingual and Indian-language settings; evaluate the limits and transferability of small-model findings in these contexts.

3. Mapping of Course Outcomes (COs) with Programme Outcomes (POs) and Programme Specific Outcomes (PSOs): Course Articulation Matrix

	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PO 12	PSO 1	PSO 2	PSO 3	PSO 4
CO-1	3	2	1	2	3	1	—	1	1	2	—	2	3	1	2	2
CO-2	2	3	1	3	2	1	—	2	1	2	—	2	2	1	3	3
CO-3	2	3	2	3	3	1	—	1	1	2	—	2	2	1	3	3
CO-4	2	2	1	3	3	1	—	1	1	2	—	2	2	1	3	3
CO-5	2	3	2	3	3	1	—	2	2	2	1	2	2	1	3	3
CO-6	2	2	3	2	2	3	1	3	1	3	—	3	2	2	3	3

'3' = High; '2' = Medium; '1' = Low; '—' = Not mapped.

- CO -1 through CO-5 → PO-4, PO-5, PSO-3, PSO-4: the core interpretability methods are investigative (PO-4) and hands-on with modern ML tooling (PO-5); they feed directly into research-level skills expected of CLD graduates (PSO-3, PSO-4).
- CO-5 → PO-9: circuit discovery projects involve collaborative replication and peer review of results.
- CO-6 → PO-3, PO-6, PO-8, PO-12: applying interpretability to safety and faith-fulness engages design responsibility, societal impact, ethics, and the habit of reading primary literature.

4. Detailed Syllabus

The course runs over 7 weeks with 2 lecture sessions per week (14 sessions total), plus weekly tutorial slots. Each module corresponds to one week. Tutorials are used for paper discussion, assignment troubleshooting, and project check-ins.

Module 1: Foundations and Toolbox

(Week 1 – 2 sessions)

Introduces the three levels of model analysis and the distinction between correlation and

causation in interpretability research. Students set up TransformerLens and NNSight and run their first activation hooks on GPT-2.

Class 1: What Is Mechanistic Interpretability? and Transformer Internals

Class 2: Toolbox Setup and First Experiments

Module 2: Probing Classifiers

(Week 2 – 2 sessions)

Covers linear probes, control tasks, and the probing-accuracy trap. Students probe for linguistic features including POS, case, and tense in multilingual models, and critically examine what probe accuracy does and does not establish.

Class 3: Linear Probes, Control Tasks, and Structural Probes

Class 4: Probing Multilingual and Indian Language Models

Module 3: Lens-Based Analysis

(Week 3 – 2 sessions)

Traces how token predictions evolve through a model's layers using the logit lens and tuned lens. Introduces direct logit attribution and examines why attribution methods, however informative, cannot establish causal responsibility on their own.

Class 5: Logit Lens, Tuned Lens, and Direct Logit Attribution

Class 6: From Attribution to Causation

Module 4: Causal Interventions

(Week 4 – 2 sessions)

Establishes causal claims about model behaviour through activation patching and path patching. Students learn to read patch-effect heatmaps, understand the clean/corrupt/patch protocol, and trace causal flow along specific edges of the computational graph.

Class 7: Activation Patching

Class 8: Attribution Patching and Path Patching

Module 5: Sparse Autoencoders and Feature Interpretation

(Week 5 – 2 sessions)

Motivates dictionary learning as a response to polysemanticity from superposition. Students load and explore pre-trained SAEs via SAE Lens, identifying and labelling monosemantic features and comparing them against direct neuron analysis.

Class 9: Superposition, Dictionary Learning, and SAE Training

Class 10: Identifying Features and Practical SAE Exploration

Module 6: Attention Analysis and Circuit Discovery

(Week 6 – 2 sessions)

Characterises attention head roles and uses ACDC and edge attribution patching to discover minimal circuits responsible for specific behaviours. Covers circuit faithfulness and completeness evaluation, and honestly addresses the limits of small-model findings.

Class 11: Attention Head Roles and Circuit Discovery

Class 12: Circuit Replication Lab

Module 7: Applications, Open Problems, and Project Demos (Week 7 – 2 sessions)

Connects mechanistic interpretability to hallucination diagnosis, faithfulness verification, and model debugging, with examples from Indian language LLMs. Students survey open research directions and present their mini-projects.

Class 13: Applications and Open Problems

Class 14: Mini-Project Presentations

5. Teaching-Learning Strategies

Each session combines conceptual development with a worked example or short hands-on exercise. Small models (GPT-2, Pythia-160M) are used throughout so students can run experiments on a laptop or Google Colab without specialised hardware. Claims about frontier models are explicitly flagged as extrapolation; the course presents mechanistic interpretability as a young field with genuine open problems, not a settled toolkit.

Each module has one assigned paper; students submit a short critical note before the corresponding tutorial to seed discussion. Each student presents one paper during Weeks 6 and 7, scheduled in tutorial slots so that lecture time is preserved for core content.

Three programming assignments cover probing classifiers, causal interventions, and SAE feature exploration.

The mini-project runs the full seven weeks. Groups of one or two students; all projects must include a component on an Indian language or multilingual model.

Mini-project directions:

Circuit replication and extension: Replicate a published circuit (IOI, greater-than, or similar) and test whether the same circuit structure exists in an Indian language or multilingual model.

Probing study for Indian language structure: Design and run a probing experiment for a morphological or syntactic property specific to an Indian language (case suffixes in Tamil, postpositions in Hindi, etc.) using mBERT or IndicBERT.

SAE feature analysis: Using a pre-trained SAE or by training one on a small model,

identify features that activate on script-specific or language-specific tokens in an Indian language LLM; compare feature geometry across languages.

Hallucination localisation: Use causal tracing to localise where a specific factual error is introduced in an Indian language LLM; compare with results for the same architecture on English.

Project proposal due end of Week 4; progress check-in in the Week 5 tutorial; demo in Session 14.

Tools: TransformerLens, NNSight, SAELens, BertViz, CircuitsVis, Hugging Face Transformers & Datasets, PyTorch, Google Colab (GPU).

6. Assessment Methods and Weightages

Type of Evaluation	Weightage
In-class quizzes (2, end of Sessions 6 and 9)	20%
Programming assignments (3, due end of Weeks 3 and 5; Assignment 3 due start of Week 7)	30%
Paper presentation and discussion	15%
Mini-project (report and demo)	35%

Total	100%
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Assignments are graded on code correctness and the quality of the written interpretation. There is no end-semester examination; the mini-project demo in the final week serves that function.

7. References

Foundations

- Bereska, L. & Gavves, S. (2024). Mechanistic interpretability for AI safety - a review. *TMLR*. <https://openreview.net/forum?id=ePUVetPKu6>
- Elhage, N. et al. (2021). A mathematical framework for transformer circuits. *Transformer Circuits Thread*. <https://transformer-circuits.pub/2021/framework>
- Olsson, C. et al. (2022). In-context learning and induction heads. *Transformer Circuits Thread*. <https://transformer-circuits.pub/2022/in-context-learning-and-induction-heads>
- Olah, C. et al. (2020). Zoom in: an introduction to circuits. *Distill*. <https://distill.pub/2020/circuits/zoom-in>

Probing

- Hewitt, J. & Liang, P. (2019). Designing and interpreting probes with control tasks. *EMNLP*.
- Hewitt, J. & Manning, C. D. (2019). A structural probe for finding syntax in word representations. *NAACL*.
- Pimentel, T. et al. (2020). Information-theoretic probing for linguistic structure. *ACL*.
- Marks, S. & Tegmark, M. (2023). The geometry of truth: emergent linear structure in large language model representations of true/false datasets. *arXiv*.

Causal Interventions and Circuits

- Geva, M. et al. (2021). Transformer feed-forward layers are key-value memories. *EMNLP*.
- Meng, K. et al. (2022). Locating and editing factual associations in GPT (ROME). *NeurIPS*.
- Meng, K. et al. (2023). Mass-editing memory in a transformer (MEMIT). *ICLR*.
- Wang, K. et al. (2023). Interpretability in the wild: a circuit for indirect object identification in GPT-2 Small. *ICLR*.
- Conmy, A. et al. (2023). Towards automated circuit discovery for mechanistic interpretability (ACDC). *NeurIPS*.
- Syed, A., Rager, C. & Conmy, A. (2024). Attribution patching outperforms automated circuit discovery. *BlackboxNLP Workshop*.
- Hanna, M. et al. (2023). How does GPT-2 compute greater-than? Interpreting mathematical abilities in a pre-trained language model. *NeurIPS*.

Sparse Autoencoders and Superposition

- Elhage, N. et al. (2022). Toy models of superposition. *Transformer Circuits Thread*. https://transformer-circuits.pub/2022/toy_model
- Bricken, T. et al. (2023). Towards monosemanticity: decomposing language models with dictionary learning. *Transformer Circuits Thread*. <https://transformer-circuits.pub/2023/monosemantic-features>
- Cunningham, H. et al. (2023). Sparse autoencoders find highly interpretable features in language models. *arXiv*.

Lens-Based Analysis

- Nostalgebraist (2020). Interpreting GPT: the logit lens. <https://www.lesswrong.com/posts/AcKRB8wDpdaN6v6ru>
- Belrose, N. et al. (2023). Eliciting latent predictions from transformers with the tuned lens. *arXiv*.

Applications and Faithfulness

- Turpin, M. et al. (2023). Language models don't always say what they think: unfaithful explanations in chain-of-thought prompting. *NeurIPS*.

Multilingual and Indian Language Perspectives

- Pires, T., Schlinger, E. & Garrette, D. (2019). How multilingual is multilingual BERT? *ACL*.
- Karthikeyan, K. et al. (2020). Cross-lingual ability of multilingual BERT: an empirical study. *ICLR*.
- Doddapaneni, S. et al. (2021). A primer on pretrained multilingual language models. *arXiv*. (Covers *IndicBERT* and *AI4Bharat* models.)

Software and Curricula

- TransformerLens
<https://github.com/neelnanda-io/TransformerLens>
- TunedLens
<https://github.com/AlignmentResearch/tuned-lens>
- NNSight
<https://nnsight.net>
- SAE Lens
<https://github.com/jbloomAus/SAELens>
- CircuitsVis
<https://github.com/alan-cooney/CircuitsVis>
- ARENA Interpretability curriculum
<https://arena-chapter1-transformer-interp.streamlit.app>

Title of the Course : **Mobile Robotics**
Name of the Faculty : Madhava Krishna K
Course Code : CS7.503
L-T-P : 3-1-0
Credits : 4
Name of the Academic Program: B. Tech. in CSE, BTech in ECE

Prerequisite Course / Knowledge:

Should have completed Computer Programming – 1 course. Knowledge of Linear Algebra, Optimization and Probability Theory is helpful.

Course Outcomes (COs):

After completion of this course successfully, the students will be able to...

CO-1: Demonstrate familiarity with different modalities of robotic perception

CO-2: Analyze robotic perception algorithms in the context of mapping an environment and localizing the robot in the environment

CO-3: Explain the significance of mathematical frameworks of functional optimization and probabilistic reasoning in robotic perception and localization tasks.

CO-4: Apply principles of functional optimization and visual/lidar based sensing to propose analytical frameworks, algorithms for solving real world problems in robotic perception and navigation

CO-5: Create and simulate the algorithms using state of the art software and libraries and evaluate its performance on specified tasks

Mapping of Course Outcomes (COs) with Program Outcomes (POs) and Program Specific Outcomes (PSOs)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3	PSO4
CO1	3	2	1	1	1	1	1	2	2	1	1	2	1	1	1	2
CO2	3	3	1	2	1	1	1	2	2	1	1	1	1	1	1	2
CO3	2	3	1	2	1	1	1	2	2	2	1	2	1	1	1	3
CO4	3	2	3	2	2	1	1	2	2	2	1	3	1	1	1	3
CO5	2	2	3	2	3	1	1	2	3	3	3	3	1	1	1	3

Each Course Outcome (CO) may be mapped with one or more Program Outcomes (POs). Write '3' in the box for 'High-level' mapping, 2 for 'Medium-level' mapping, 1 for 'Low'-level' mapping

Mapping with PSOs, where applicable.

Detailed Syllabus:

Unit 1: Representation of Coordinate Frames, Rotation Matrices, Homogenous Transforms, Quaternions and Axis Angle Representations

Unit 2: LIDAR based Mapping and Localization

Unit 3: Principles of Computer Vision: Camera Modelling, Calibration, Reconstruction and Resection

Unit 4: Backend Optimization for LIDAR Based SLAM, Bundle Adjustment

Reference Books:

1. Introduction to Robotics: Mechanics and Control by John J Craig
2. Invitation to 3D Vision: Ma, Soatto, Koseca and Shastry
3. Multiple View Geometry in Computer Vision: Richard Hartley and Andrew Zisserman

Teaching-Learning Strategies in brief (4 to 5 sentences):

Classes invoke rich graphical content in the form of images, representations, videos to elucidate difficult concepts in robotic vision. Code walkthroughs, simulation of algorithms used to enhance understanding. Learning by doing, coding and simulation is highly promoted and encouraged. Students understand difficult mathematical concepts and abstraction by coding it using state of the art software, simulation frameworks, libraries and solvers.

Assessment methods and weightages in brief (4 to 5 sentences):

- Programming Assignments: 60%
- Mid Sem: 20%
- End Exam: 25%

Title of the Course : Modern Complexity Theory

Name of the Faculty : Suryajith Ch

Course Code : CS1.405

L-T-P : 3-1-0

Credits : 4

Name of the Academic Program: B.Tech in Computer Science

Prerequisite Course / Knowledge:

Should have taken Introduction to Algorithms, and Formal Languages, or equivalent courses

Course Outcomes (COs):

After completion of this course successfully, the students will be able to..

CO-1: Understand different models of computation including Turing Machines, Boolean Circuits and complexity measures of time, space, depth.

CO-2: Demonstrate familiarity with various complexity classes including P, NP, PSPACE, NC and problems like Halting Problem, 3SAT.

CO-3: Design reductions between problems to show hardness of solving a problem in a complexity class.

CO-4: Synthesize proofs of upper and lower bounds of resources required for solving a computational problem using clear mathematical and logical arguments.

CO-5: Apply principles of NP-Completeness and NP-Hardness to avoid intractability in design of computational problems.

CO-6: Create mathematical models and complexity measures for novel computational models.

Mapping of Course Outcomes (COs) with Program Outcomes (POs) and Program Specific Outcomes (PSOs)

	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PO 12	PS O 1	PS O 2	PS O 3	PS O 4
CO 1	2	2	2	3	1	1	1	1	2	2	1	1	3	1	1	1
CO 2	1	2	2	2	1	1	1	1	2	2	1	1	3	1	1	2
CO 3	1	3	2	3	1	1	1	1	2	2	1	1	2	1	1	2
CO 4	1	3	2	2	1	1	1	1	2	2	1	1	3	1	2	2
CO 5	1	3	2	3	1	1	1	1	2	2	1	1	3	1	1	3
CO 6	2	3	2	3	1	1	2	1	2	3	1	1	2	2	3	3

'3' for 'High-level' mapping, 2 for 'Medium-level' mapping, 1 for 'Low'-level' mapping.

[etailed Syllabus:](#)

Unit 1: Models of Computation and Impossibility Results: Turing Machines, Circuits, Encoding of Problems, Halting Problem, Shannon's Counting Lower bound.

Unit 2: Complexity Measures and Classes: Time, Space, Depth measures of complexity, Time, Space hierarchy theorems, Nondeterminism, Savitch's theorem, P, NP, P/poly, PSPACE, EXP, L, NL. **Unit 3:** Completeness and Hardness Reductions: 3SAT, Cook-Levin Theorem, NP-Complete, NL- Complete, Hardness reductions for common problems like Vertex Cover, Independent Set, Knapsack etc.

Unit 4: Advanced Topics: Definitions and relationships between PH, RP, BPP, NC including theorems like Karp-Lipton, Adleman's theorem, Derandomization Techniques.

Reference Books:

- S. Arora and B. Barak (2000), Computational Complexity: A Modern Approach, Cambridge University Press.
- C. Moore & S. Mertens (2011), The Nature of Computation, Oxford University Press.
- M. Sipser (2014), Introduction to Theory of Computation, Cengage Learning.
- C. Papadimitriou (1994), Computational Complexity, Addison Wesley Longman.

Teaching-Learning Strategies in brief (4 to 5 sentences):

Lectures will initially introduce the motivations, concepts, definitions along with simpler examples.

This will be followed by assignments and quizzes that will make sure that the students have understood the concepts. These will be followed by deeper lectures and assignments which lead

the students to the bigger questions in the area. The students will be given an advanced topic and will be required to summarize it in a presentation or a term paper.

This will encourage self-

exploration and lead the student to do research on fundamental questions.

Assessment methods and weightages in brief (4 to 5 sentences):

- Light In-class Quizzes: 20%
- Assignments: 20%
- Deep Quiz 1: 10%
- Deep Quiz 2: 10%
- Mid and End Exam: 30%
- Student Presentation and Scribe notes: 10%

Title of the Course : Music Workshop

Name of the Faculty : T K Saroja

Course Code : **HS1.210**

L-T-P : 3-0-1

Credits :4

(L= Lecture hours, T=Tutorial hours, P=Practical hours)

1.Prerequisite Course / Knowledge: Maximum students: 30

2.Course Outcomes (COs) (5 to 8 for a 3 or 4 credit course):

After completion of this course, the students will be able to

CO-1Understand the scope of research in music

CO-2Realise the potential of a thought as a seed to a productive work

CO-3 Understand the relevance of music as an integral part of human life

CO-4Develop an inter disciplinary perspective on music

CO-5 Come up with a work of their own, related to music

3.Mapping of Course Outcomes (COs) with Program Outcomes (POs) and Program Specific Outcomes (PSOs) – Course Articulation Matrix

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3	PSO4
CO1	1	-	1	1	-	2	1	2	2	2	-	2	-	2	1	3
CO2	1	-	1	1	-	1	1	2	2	2	-	2	-	2	1	3
CO3	1	-	1	1	-	1	1	2	2	2	-	2	-	2	1	3
CO4	1	-	1	1	-	1	1	2	2	2	-	2	-	2	1	3
CO5	1	-	1	1	-	1	1	2	2	2	-	2	-	2	1	3

Note: Each Course Outcome (CO) may be mapped with one or more Program Outcomes (POs) and PSOs. Write '3' in the box for 'High-level' mapping, 2 for 'Medium-level' mapping, 1 for 'Low'-level' mapping

4.Detailed Syllabus:

1. Text reading: The students will be introduced to a wide range of topics through the writings of different musicians, musicologists and music rasikas
2. Introduction to all the basic concepts of music with special focus on Indian music
3. Music listening: Music of different genres will be played for the students to observe and understand the importance of 'Sadhana' in music
4. Practical exercises: Various exercises based on notes, rhythm, scales, ragas and language
5. Discussions on topics that students want to take up as their course project (individual sessions relevant to their topic)

Reference Books:

As this course is a research/project oriented one, the references are not limited to the given list.

The list given below is to cite a few books that cover broad areas.

1. The Hindu Speaks on Music - compilation of 232 selective music articles by The Hindu --- Publishers: Kasturi and Sons Ltd, December 1999.
2. A Southern Music (The karnatic story) by T.M. Krishna, Published by Harper Collins, January 2013
- 3 South Indian Music (volumes I to VI) by P.Sambamurthy, The Indian Music Publishing House, 1994
4. Nuances of Hindustanu Classical Music by Hema Hirlekar, Unicorn books Pvt Ltd, 2010
5. The two faces of Beauty Science and Art by The Quest for Music Divine Pushpa Mittra Bhargava, Chandana Chakrabarti, map in Publishing, 2014
6. The Quest for Music Divine by Suresh Chandra Dey, Ashish Publishing House, 1990
7. History of South Indian Music (Carnatic) Music, R. rangaramanujaAyyangar, JAK Printers, 1972
8. Hindustani Music- A tradition in transition by Deepak Raja, D.K.Print World Ltd, 2005
9. Indian Aesthetics and musicology- The Art and Science of Indian music by Prof. Prem latha Sharma, Amnaya Prakasana, Bharatha Nidhi, Varanasi, 2000
10. Elements of Western music for the students of Indian music by Prof. P.Sambamoorthy, KMBC, 1961
11. Ragacikitsa (Music Therapy) by Suvarna Nalapat, Readworthy publications Pvt Ltd, 2008

Videos and audios on the YouTube and other platforms.

5.Teaching-Learning Strategies in brief (4 to 5 sentences):

This is a course designed to encourage research related to music. The students would identify some research topic to work on throughout the semester and work in groups of

three or four. There would be four to five common classes to all the students to equip them with the basic knowledge of music and direct them towards diverse prospects. Suggesting relevant resources, monitoring the projects regularly, encouraging projects in the confluence of music, science and technology would be the strategy of the course. There would be individual and group wise attention on students to make them complete a project successfully. The students would thoroughly learn the topic of their interest and finally submit a project related to it.

6.Assessment methods and weightages in brief (4 to 5 sentences):

Assignments: 40%

Class participation 10%

Project: 50%

Title of the Course	:	Networks, Signals & Systems
Name of the Faculty	:	Prasad Krishnan+ Aftab M Hussain
Course Code	:	EC5.101
L-T-P	:	3-1-0
Credits	:	4
(L= Lecture hours, T=Tutorial hours, P=Practical hours)		
Name of the Academic Program:	B. Tech in ECE	
Semester, Year	:	Monsoon, 2026

Pre-Requisites : A prior knowledge of calculus and complex numbers is required.

Course Outcomes :

CO-1 Describe various circuit elements (R, C, L), supply (current, voltage), devices (op amp, diode) and Explain the operation and characteristics of each circuit element, behavior in specific circuit configuration (DC, AC, series, parallel, mixed).

CO-2 Calculate equivalent circuit parameters (Thevenin, Norton), node voltages, branch currents etc. using reduction, KCL, KVL and reduction techniques.

CO-3 Calculate circuit response (steady state, transient) to various input stimulation. Calculate and understand the concept of time constant for RC, RL and RLC circuits. Demonstrate understanding of and calculate Power, Energy, Loss and phasors w.r.t. circuit.

CO-4 Describe signals using various representations including Fourier series representation for periodic signals

CO-5 Describe systems abstractly using block diagrams and differential equations and apply convolution operation and impulse responses for system analysis

CO-6 Analyze signals and systems in time and s-domains.

Course Topics :

Unit 1: Signals, representation, sinusoids, and Fourier series
 Unit 2: Systems and representations – differential equations, block diagram, operator, and functional form
 Unit 3: Impulse Response and Transfer functions – Laplace transform, poles and zeros
 Unit 4: Circuit elements and Network theorems
 Unit 5: Transient and Steady state analysis
 Unit 6: Sinusoidal input and phasors, two port network

Preferred Textbooks:

1. Engineering Circuit Analysis by Hyatt, Kimmerley & Durbin
2. Signals and Systems by A.V. Oppenheim, A.S. Willsky and S.H. Nawab (2015)

Reference Books :

E-book Links :

Grading Plan : (The table is only indicative)

Type of Evaluation [3 credit- lecture]	Weightage (in %)
Quiz 1	10
Quiz 2	10
Mid	25
End Exam	35
Assignments	20

Mapping of Course Outcomes to Program Objectives: (1 – Lowest, 2—Medium, 3 – Highest, or a ‘-’ dash mark if not at all relevant)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3	PSO4
CO1	2	3	1	1	1	1	1	1	1	1	1	1	3	1	1	1
CO2	1	3	3	3	1	1	1	1	1	2	1	2	3	1	1	1
CO3	2	2	3	3	1	1	1	1	1	1	1	3	3	1	1	1
CO4	3	3	3	3	1	1	1	1	1	1	1	3	3	3	1	1
CO5	3	3	3	3	1	1	1	1	1	1	1	3	3	3	1	1
CO6	3	3	2	3	1	3	1	1	3	3	1	3	3	3	1	1

Teaching-Learning Strategies in brief (4-5 sentences):

Students will be applying the lecture discussion to solve examples shared with them in the class. The assignments given will reinforce the concepts and to promote their application to difficult problems. Classroom learning will be done in interactive method as much as possible. A short question may be posted at beginning of class to gauge understanding of previous lecture. Occasionally self-assessment tests (1minute paper) will be given. In tutorial class, students will make simple circuits using basic components and solve problems. The course project is done in a way to encourage collaborative problem solving, team participation, and coming up with solution as a team.

Title of the Course : NLP for Healthcare

Name of the Faculty : Parameswari Krishnamurthy & Dipti Misra Sharma

Course Code : C L 3 . 4 1 1

Credits : 4 (Elective)

L-T-P : 3-1-0-4

(L - Lecture hours, T-Tutorial hours, P - Practical hours)

Semester, Year : Monsoon 2026

CLD and Computer Science Elective

Pre-Requisites :

- Students should have completed at least one graduate-level introductory course in CL/AI/ML:
 - Computational Linguistics 1 or 2
 - Introduction to NLP / Advanced NLP
 - Statistical Methods in AI
 - Machine Learning
 - Information Retrieval and Extraction
- Python programming (preferred for practical applications)

Course Objectives :

The course **NLP for Healthcare** aims to provide students with a comprehensive understanding of how Natural Language Processing (NLP) techniques are applied to healthcare. The course will cover essential topics including medical text processing, clinical data analysis, patient-provider communication, and multimodal data integration. Students will learn to apply NLP methods to process clinical texts, electronic health records (EHRs), and patient notes, as well as build models for disease prediction, chatbots, and question answering systems. The course will also address key challenges in healthcare NLP such as privacy, bias, and ethical considerations. By the end of the course, students will be equipped with the skills to evaluate and develop NLP models tailored to healthcare applications, ensuring their effectiveness and clinical impact.

Course Outcomes (COs) :

After completing this course, students will be able to:

1. Understand healthcare systems and how NLP improves healthcare delivery and patient care.
2. Use NLP techniques to process clinical texts, EHRs, and patient notes.
3. Build NLP models for healthcare tasks like disease prediction, chatbots, and question answering.
4. Extract insights from biomedical data to support healthcare decision-making.

5. Address challenges in medical NLP, such as privacy, bias, and ethical issues.
6. Evaluate NLP models in healthcare, including text and multimodal models, with a focus on fairness and clinical impact.

Course Topics :

Unit 1: Introduction to Healthcare and NLP

- **Role of NLP in Healthcare:** Overview of NLP applications in clinical settings, patient-centered care, education, and public health.
- **Overview of Healthcare Systems:** Understanding Subjects, Clinical Practice, Understanding EHRs, clinical workflows, and the healthcare ecosystem.
- **Medical Terminologies:** Introduction to medical vocabularies (ICD-10, SNOMED CT, UMLS, MeSH).
- **Challenges in Medical NLP:** Ambiguity, privacy issues, regulatory concerns, and domain-specific language.
- **Ethics, Privacy, and Bias in Healthcare NLP:** Introduction to healthcare data privacy (HIPAA, GDPR), and ethical considerations in NLP.

Unit 2: Medical Data and Text Processing

- **Medical Data Collection:** Data mining, data cleaning and preparation, de-identification, and federated learning.
- **Text Preprocessing:** Tokenization, stemming, lemmatization, and stop-word removal for clinical and biomedical data.
- **Medical Term Recognition and Linking:** Recognizing and linking medical entities such as diseases, drugs, symptoms, and treatments from clinical texts.
- **Clinical Text Classification:** Classifying clinical texts for disease detection, risk prediction, and patient stratification.

Unit 3: Medical Data Summarization and Question Answering (QA)

- **Text Summarization:** Techniques for automatic summarization of clinical and biomedical texts using both traditional and LLM-based approaches.
- **Question Answering (QA):** Building QA systems for extracting answers from clinical documents, leveraging both traditional methods and LLMs.
- **Unstructured to Structured Data Conversion:** Converting unstructured clinical data (e.g., text, tables) to structured formats using NLP techniques and models.

Unit 4: Healthcare Communication and Conversational AI

- **Patient-Doctor Conversations:** Analyzing doctor-patient dialogues to

extract insights, identify emotional cues, and improve communication efficiency.

- **Conversational AI in Healthcare:** Developing chatbots and virtual assistants to assist patients in navigating healthcare services, providing basic medical information, and improving access to healthcare.
- **Speech-to-Text and Dialog Systems:** Applying speech recognition and NLP techniques to transcribe and analyze spoken doctor-patient interactions for better healthcare documentation and decision support.

Unit 5: Multimodal NLP in Healthcare

- **Understanding Multimodal Data:** Text, speech, images, and their integration in healthcare applications.
- **Multimodal Data for Healthcare:** Leveraging clinical text, medical imaging, and speech data for comprehensive healthcare models.
- **Challenges in Multimodal Healthcare:** Data Alignment and Fusion, Data Quality and Availability, Computational Complexity

Unit 6: Evaluation in Healthcare NLP

- **Evaluation Metrics for Healthcare NLP:** Key metrics such as precision, recall, F1-score, accuracy, AUC, and ROUGE for evaluating text-based healthcare tasks like clinical classification, medical term recognition, and summarization.
- **Evaluation of Multimodal Healthcare Models:** Assessing models that integrate text, speech, and image data using multimodal metrics, evaluating the impact on diagnostic accuracy and clinical decision-making.
- **Ethical and Human-in-the-Loop Evaluation:** Evaluating fairness, bias, and privacy concerns in healthcare NLP models, and incorporating clinician feedback to continuously improve model performance and usability.

Preferred Textbooks: Cohen, K. B., & Demner-Fushman, D. (2014). *Biomedical natural language processing* (Vol. 11). John Benjamins Publishing Company.

Reference Books/Articles :

(will be updated and shared regularly)

Chabner, D. E. (2014). *The language of medicine* (10th ed.). Elsevier Health Sciences.

Wang, B., Xie, Q., Pei, J., Chen, Z., Tiwari, P., Li, Z., & Fu, J. (2023). Pre-trained language models in the biomedical domain: A systematic survey. *ACM Computing Surveys*, 56(3), 1-52.

Jinhyuk Lee, W., Yoon, W., Kim, S., Kim, D., Kim, S., So, C. H., & Kang, J. (2020).

BioBERT: A pre- trained biomedical language representation model for biomedical text mining. *Bioinformatics*, 36(4), 1234–1240.

Chen, I. Y., Pierson, E., Rose, S., Joshi, S., Ferryman, K., & Ghassemi, M. (2021). Ethical machine learning in healthcare. *Annual Review of Biomedical Data Science*, 4(1), 123-144.

Yuan, H., Yuan, Z., Gan, R., Zhang, J., Xie, Y., & Yu, S. (2022). BioBART: Pretraining and evaluation of a biomedical generative language model. *BioNLP 2022@ ACL 2022*, 97.

Abisha, D., Mahalakshmi, M., Pritiga, T., Thanusiya, M., Sahaya Sherin, A. P., & Evanjalin, R. N. (2024). Revolutionizing rural healthcare in India: AI-powered chatbots for affordable symptom analysis and medical guidance. In 2024 *International Conference on Inventive Computation Technologies (ICICT)*, 181-187. IEEE.

Tinn, R., Cheng, H., Gu, Y., Usuyama, N., Liu, X., Naumann, T., Gao, J., & Poon, H. (2021). Fine-tuning large neural language models for biomedical natural language processing. arXiv:2112.07869. Retrieved from <https://arxiv.org/abs/2112.07869>

Sun, C., Yang, Z., Wang, L., Zhang, Y., Lin, H., & Wang, J. (2020). Biomedical named entity recognition using BERT in the machine reading comprehension framework. arXiv:2009.01560. Retrieved from <https://arxiv.org/abs/2009.01560>

Usuyama, N., Wong, C., Zhang, S., Naumann, T., & Poon, H. (2025). Biomedical natural language processing in the era of large language models. *Annual Review of Biomedical Data Science*, 8.

AlSaad, R., Abd-Alrazaq, A., Boughorbel, S., Ahmed, A., Renault, M.-A., Damseh, R., & Sheikh, J. (2024). Multimodal large language models in healthcare: Applications, challenges, and future outlook. *Journal of Medical Internet Research*, 26, e59505.

Sarella, P. N. K., & Mangam, V. T. (2024). AI-driven natural language processing in healthcare: Transforming patient-provider communication. *Indian J Pharmacy Practice*, 17(1), 21-26.

E-book Links : -NIL-

Grading Plan :

Type of Evaluation	Weightage (In %)
Assignments	40%
Projects	25%
Quiz	15%
Seminars and Participation	10%
End-Sem exam	10%

Mapping of Course Outcomes to Program Objectives: (1 – Lowest, 2—Medium, 3 – Highest, or a ‘-’

dash mark if not at all relevant).

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3	PSO4
CO1	2	2	2	2	1	2	1	2	1	2	1	2	2	2	2	3
CO2	3	3	1	3	3	1	-	2	2	1	1	2	3	2	2	2
CO3	3	3	2	3	3	2	1	3	3	3	2	3	3	2	3	2
CO4	3	3	1	3	2	2	2	3	2	2	2	2	2	1	2	3
CO5	2	2	3	2	1	3	-	3	2	3	2	3	2	3	3	3
CO6	2	3	3	3	2	3	1	3	3	2	2	3	3	3	2	3

Teaching-Learning Strategies in brief (4-5 sentences):

The **NLP for Healthcare** course will focus on a hands-on, interactive approach, with the following strategies:

1. **Lectures:** Core NLP techniques and their healthcare applications will be covered, providing students with essential theoretical knowledge.
2. **Assignments and Project-Based Learning:** Students will apply concepts through assignments and develop their own models to solve healthcare problems, enhancing practical skills.
3. **Seminars and Case Studies:** Discussions on ethical issues, privacy concerns, and biases in healthcare NLP will be integrated, encouraging critical thinking and real-world application.
4. **Quizzes:** Regular quizzes will assess student progress and provide feedback to guide learning.
5. **Guest Lectures:** Healthcare professionals and researchers will be invited to give guest lectures, offering insights from real-world applications and expert perspectives in the field.

This approach will help students build problem-solving, critical thinking, and practical skills, preparing them to evaluate and implement NLP solutions in healthcare.

Lecture Planning:

Day	Topics	Lectures	Deadlines
Day 1	Role of NLP in Healthcare	Overview of NLP applications in clinical settings, patient-centered care, education, and public health.	
Day 2	Overview of Healthcare Systems	Understanding EHRs, clinical workflows, and the healthcare ecosystem. Medical Terminologies: Introduction to medical vocabularies (ICD-10, SNOMED CT, UMLS, MeSH).	
Day 3	Challenges in Medical NLP	Ambiguity, privacy issues, regulatory concerns, and domain-specific language. Ethics, Privacy, and Bias in Healthcare NLP: Introduction to healthcare data privacy (HIPAA, GDPR), and ethical considerations in NLP.	Assignment 1 release
Day 4	Medical Data Collection	Data mining, data cleaning and preparation, de-identification, and federated learning.	
Day 5	Text Preprocessing	Tokenization, stemming, lemmatization, and stop-word removal for clinical and biomedical data.	
Day 6	Medical Term Recognition and Linking	Recognizing medical entities such as diseases, drugs, symptoms, and treatments from clinical texts.	
Day 7	Medical Term Recognition and Linking (continued)	Fine-tuning pre-trained models for healthcare applications.	
Day 8	Clinical Text Classification	Classifying clinical texts for disease detection, risk prediction, and patient stratification.	Assignment 2 release
Day 9	Text Summarization	Techniques for automatic summarization of clinical and biomedical texts using both traditional and LLM-based approaches.	
Day 10	Text Summarization (continued)	Advanced methods and real-world applications of summarization in healthcare.	Project list release

Day 11	Question Answering (QA)	Building QA systems for extracting answers from clinical documents, leveraging both traditional methods and LLMs.	
Day 12	Question Answering (QA) (continued)	Evaluating and improving QA systems with healthcare-specific datasets.	
Day 13	Unstructured to Structured Data Conversion	Converting unstructured clinical data (e.g., text, tables) to structured formats using NLP techniques and models.	Assignment 3 release
Day 14	Patient-Doctor Conversations	Analyzing doctor-patient dialogues to extract insights, identify emotional cues, and improve communication efficiency.	Seminar list release
Day 15	Conversational AI in Healthcare	Developing chatbots and virtual assistants to assist patients in navigating healthcare services, providing basic medical information, and improving access to healthcare.	
Day 16	Conversational AI in Healthcare (continued)	Practical applications and case studies on conversational AI in healthcare.	
Day 17	Speech-to-Text and Dialog Systems	Applying speech recognition and NLP techniques to transcribe and analyze spoken doctor-patient interactions for better healthcare documentation and decision support.	
Day 18	Speech-to-Text and Dialog Systems (continued)	Implementing real-time transcriptions and analyzing spoken data for healthcare applications.	Assignment 4 release
Day 19	Understanding Multimodal Data	Text, speech, images, and their integration in healthcare applications.	Mid-project evaluation
Day 20	Multimodal Data for Healthcare	Leveraging clinical text, medical imaging, and speech data for comprehensive healthcare models.	
Day 21	Challenges in Multimodal Healthcare	Data alignment and fusion, data quality and availability, computational complexity.	
Day 22	Evaluation Metrics for Healthcare NLP	Key metrics such as precision, recall, F1-score, accuracy, AUC, and ROUGE for evaluating text-based healthcare tasks like clinical classification,	

		Term recognition, and summarization.	
Day 23	Evaluation of Multimodal Healthcare Models	Assessing models that integrate text, speech, and image data using multimodal metrics, evaluating the impact on diagnostic accuracy and clinical decision-making.	
Day 24	Ethical and Human-in-the-Loop Evaluation	Evaluating fairness, bias, and privacy concerns in healthcare NLP models, and incorporating clinician feedback to continuously improve model performance and usability.	
			Seminars/ Final project viva

Title of the Course : **Open Quantum systems and Quantum Thermodynamics**
Name of the Faculty : Samyadeb Bhattacharya
Course Code : SC1.310
Credits: : 4
L - T - P :
(L - Lecture hours, T-Tutorial hours, P - Practical hours)
Semester, Year : Monsoon semester, 2026
Name of the Program : Open Quantum systems and Quantum Thermodynamics

Pre-Requisites : Basic understanding of Quantum mechanics (graduate level), Hilbert space and linear algebra.

Course Outcomes :

1. After the course, the students will have a fundamental understanding about kinematics and dynamics of noisy quantum systems.
2. Students will also have basic ideas on current research directions in quantum thermodynamics, a little of quantum communication, quantum information and quantum entanglement detection.
3. Students will have basic training on research in related topics through projects.
4. Students will have a firm background for pursuing a MS thesis in quantum information, communication and thermodynamics.
5. The course can be understood as a pre-PhD course related to quantum information science in general.

Course Topics :

1. Review on basic linear algebra; Metric space, Dual space, Hilbert space and bra-ket algebra.
2. Review on unitary quantum mechanics; Unitary evolution, state vectors, uncertainty principle, Schroedinger equation etc.
3. Pure and mixed states, basics on measurement theory, projective measurements, positive operator valued measures etc.
4. non-unitary evolution, tensor product space, bi-partite quantum systems, global evolutions, non-unitary dynamics, completely positive trace preserving maps, operator-sum representation.
5. Monotones under completely positive operations, basics of distance measures and entropic measures.
6. Selected topics on complete positivity, positivity, Choi-Jamiolkowski isomorphism, Stinespring dilation, entanglement detection.
7. General quantum dynamical equations, master equation, Lindblad equation and its derivation from a few different perspectives, properties of Lindblad dynamics.
8. Basic idea on Markovianity and non-Markovianity from quantum mechanical perspectives.
9. Basic ideas on quantum heat engines and a few other selected topics on thermodynamics.

Preferred Textbooks: 1. Theory of open quantum systems by H.P. Breuer and F. Petruccione, 2. Modern quantum mechanics by J J Sakurai.

Reference Books : Lecture notes on quantum dynamical semigroups and applications by R. Alicki and K. Lendi (2nd edition 2007)

E-book Links : 1.

https://books.google.co.in/books/about/The_Theory_of_Open_Quantum_Systems.html?id=oYx5VzaMYm8C&redir_esc=y

2. <https://link.springer.com/book/10.1007/3-540-70861-8>

Grading Plan : (The table is only indicative)

Type of Evaluation	Weightage (in %)
Quiz-1	10%
Mid SemExam	20%
Quiz-2	10%
End Sem Exam	30%
Assignments	10%
Project	20%
Term Paper	NA
Other Evaluation	NA

Mapping of Course Outcomes to Program Objectives: (1 – Lowest, 2—Medium, 3 – Highest, or a ‘-’ dash mark if not at all relevant). Program outcomes are posted at

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3	PSO4
CO1	3	3	1	1	1	2	2	1	2	1	1	1	1	1	1	1
CO2	1	1	2	1	1	1	2	1	1	3	1	1	1	1	1	1
CO3	1	2	2	2	1	1	1	1	1	1	1	1	1	3	1	2
CO4	1	1	1	1	1	1	2	1	1	1	1	1	2	1	1	1
CO5	1	1	1	1	2	1	1	3	3	1	2	2	1	2	1	2

Teaching-Learning Strategies in brief (4-5 sentences) :

The course is also self-evolving. Since this course is a pre-PhD level course, it is heavily dependent on the evolution of current research in said topics. Therefore, I have to modify and upgrade the course structure in regular intervals of a few years.

Title of the Course : Operating Systems and Networks

Name of the Faculty : Deepak Gangadharan

Course Code : CS3.301

Name of the Academic Program: B.Tech. in CSE

L-T-P : 3-1-0.

Credits : 4

(L= Lecture hours, T=Tutorial hours, P=Practical hours)

1.Prerequisite Course / Knowledge:

Programming languages, Digital Logic Design, Computer Organization

2.Course Outcomes (COs)

After completion of this course successfully, the students will be able to,

CO-1. Extend the concepts of layering and modularity to build new software systems

CO-2. Develop appropriate scheduling/synchronization/memory management/ virtual memory/protection module for a new task-specific operating system.

CO-3: Implement an application on the top of given operating system in an efficient manner based on process and thread framework available in the given operating system.

CO-4. Architect the given system on the top of operating systems by exploiting the system calls of the given operating system services as far as possible.

CO-5. Develop a network-based application by exploiting networking related system calls.

3.Mapping of Course Outcomes (COs) with Program Outcomes (POs) and Program Specific Outcomes (PSOs) – Course Articulation Matrix

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3	PSO4
CO1	3	2	1	1	2	-	-	-	3	2	2	3	3	2	1	2
CO2	3	3	3	2	2	-	1	-	2	2	2	3	2	1	2	2
CO3	3	3	3	2	2	2	1	-	3	2	2	3	3	2	1	2
CO4	2	2	3	2	2	3	2	-	3	2	2	2	3	3	2	3
CO5	3	2	1	1	2	-	-	-	3	2	2	3	3	2	1	2
CO6	3	3	3	3	2	2	1	-	2	2	3	2	3	3	3	3

4.Detailed Syllabus:

Unit 1: Introduction, Process and Memory Virtualization – Scheduling, Memory addressing and Paging, and Networking Overview (10 hours);

Unit 2: Concurrency – Threads and locking mechanisms, Common concurrency problems, Data transmission and Network Technologies (10 hours);

Unit 3: Persistence – File Systems, Protection, Network File Systems and basics of Network Security (6 hours);

Four mini projects and one overall project related to the above syllabus will be done by students in the laboratory

Reference Books:

1. Operating systems in three easy pieces by Andrea Arpaci-Dusseau and Remzi Arpaci-Dusseau, 2018 (<https://pages.cs.wisc.edu/~remzi/OSTEP/>)
2. Computer Networks (5th Edition) Andrew S. Tanenbaum, David J. Wetherall Prentice Hall, 2013
3. William Stallings, Operating Systems, Prentice-Hall, 2018.
4. Tanenbaum, A., Modern Operating Systems, Prentice-Hall, Second Edition (latest edition, 2015).

5. Teaching-Learning Strategies in brief

Lectures by integrating ICT into classroom teaching, weekly tutorials involving problem solving and active learning by students on a Unix-based OS like xv6 and Project-based Learning by doing 4 mini-projects and one overall project.

6.Assessment methods and weightages in brief (Tentative)

Component	Weightage
Final Exam	35%
Mid-term Exam	15%
Quizzes	10%
Mini-projects	25%
In-class activities	5%
Final Project	10%

Note: Instructor reserves the right to make any changes in the above distribution based on the progress of the course

Title of the Course : **Optoelectronics & Integrated Photonics**
Faculty Name(s) : Tanmay Bhowmik
Name of the Program : B.Tech in ECE, M.Tech in VLSI and Semiconductor Technologies, B.Tech and MS in ECE (Dual Degree), MS in ECE, PhD in ECE
Course Code : **EC2.415**
Credits : **4**
L - T - P : **3 - 1 - 0** (L - Lecture hours, T-Tutorial hours, P - Practical hours)
Semester, Year : **Monsoon 2026**
Pre-Requisites : Semiconductor Devices, Electromagnetic Theory (Basic level)

Course Outcomes :

- **CO1:** Understand the concepts behind mode propagation and dispersion in optical fibers and Silicon-on-Insulator (SOI) waveguides.
- **CO2:** Investigate the LED and laser performance limits using direct/indirect bandgap semiconductors and stimulated emission process.
- **CO3:** Analyze semiconductor photodiodes and receiver architectures for minimal signal-to-noise ratio (SNR).
- **CO4:** Understand electro-optic effect in high-speed on-chip optical modulators (Mach-Zehnder, microring, etc.) and semiconductor optical amplifiers.
- **CO5:** Understand the concept and importance of optical interconnects in data centers, energy efficiency (fJ/bit), and electronic-photonics integration for high throughput.

Course Topics :

Module 1: Introduction to Optics and Waveguides

Review of Electromagnetic field theory and Maxwell's equations, optical wave propagation in isotropic and anisotropic media. Symmetric planar dielectric slab waveguide, step-index and graded-index optical fibers, waveguide dispersion and attenuation.

Module 2: Semiconductor Light Sources I : LEDs

Review of semiconductor physics and energy bands, direct vs. indirect bandgap semiconductors, homojunction & heterojunctions, recombination lifetime, quantum efficiency, and LED structures.

Module 3: Semiconductor Light Sources II : Laser Diodes

Stimulated emission & Einstein coefficients, population inversion, cavity resonators, Erbium-doped fiber amplifiers, quantum wells, and laser diode equation.

Module 4: Photodetectors and Noise Analysis

Principle of the p-n junction photodiode, p-i-n and avalanche photodiodes, quantum efficiency, responsivity. shot noise, thermal noise, and signal-to-noise ratio.

Module 5: Integrated On-Chip Optical Modulators and Amplifiers

Electro-optic effect, Mach-Zehnder modulators, electro-absorption modulators, and coupled waveguide modulators. Semiconductor optical amplifiers (SOAs).

Module 6: Integrated Photonics and Applications

Introduction to silicon photonic integrated circuits, optical interconnects in data centers, electronic-photonic integration and photonic transceivers.

Preferred Textbooks:

1. S O Kasap, Optoelectronics and Photonics: Principles and Practices, 2nd Edition, Pearson Prentice Hall, 2013
2. J B. Saleh, M. Teich, Fundamentals of Photonics, Wiley-Interscience, 2nd edition, 2007
3. Richard Osgood jr.Xiang Meng, Principles of Photonic Integrated Circuits, Springer Nature, 2021

Reference Books :

1. Pallab Bhattacharya, Semiconductor Optoelectronics Devices, 2nd Edition, PHI, 2009
2. Gerd Keiser, Optical Communications Essentials, Mc Graw Hill 2003
3. Govind P. Agrawal, Fiber-Optic Communication Systems, 5th Edition, John Wiley & Sons, 2021
4. Larry A. Coldren, Scott W. Corzine, Milan L. Mašanović, Diode Lasers and Photonic Integrated Circuits, Wiley Series in Microwave, 2012

E-book Links :

Review articles and relevant journal papers can be accessible via IIIT-H digital library (IEEE Digital Xplore, Optica, AIP publishers, etc.)

Grading Plan :

Type of Evaluation	Weightage (in %)
Quiz-1	10%
Mid Sem Exam	20%
Quiz-2	10%
End Sem Exam	35%
Assignments	10%
Project+Term Paper	15%

Mapping of Course Outcomes to Program Objectives: (1 – Lowest, 2—Medium, 3 – Highest, or a ‘-’ dash mark if not at all relevant).

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3	PSO4
CO1	3	2	2	1	2	1	1	-	1	1	1	2	3	1	1	1
CO2	3	3	2	2	1	1	1	-	1	1	1	2	3	2	1	1
CO3	3	3	3	2	2	1	1	-	1	1	1	2	3	2	1	1
CO4	3	2	2	2	2	1	1	1	3	2	-	2	3	1	1	2
CO5	3	3	3	3	2	1	1	1	3	2	-	2	3	3	1	1

Teaching-Learning Strategies in brief (4-5 sentences):

The course will be delivered through a combination of structured, interactive lectures, digital presentations, and intensive board work for mathematical derivations. Regular analytical assignments and problem-solving goals will be integrated into the course delivery schedule to enhance the fundamental ideas. As part of a unique Group Project milestone, student clusters choose, assess, and deliver peer-reviewed journal publications on high-speed modulators and integrated silicon photonics. In order to offer a practical perspective, illustrations of modern commercially available optoelectronic components and research-level photonic simulation tools will be presented. Furthermore, carefully chosen peer discussions on cutting-edge semiconductor photonic technologies that address current advancements in optical interconnects and electronic-photonic integration will broaden the comprehension of students.

Title of the Course : Physics of Early Universe
Name of the Faculty : Diganta Das
Name of the Academic Program : CNS, HSME
Course Code : SC1.415
L-T-P : 3-1-0.
Credits : 4
 (L= Lecture hours, T=Tutorial hours, P=Practical hours)

1.Prerequisite Course / Knowledge:

Differentiation and integration, classical mechanics, electricity and magnetism

2.Course Outcomes (COs) (5 to 8 for a 3 or 4 credit course):

After completing this course successfully, the students will be able to

CO-1 Explain the large-scale structure of the universe and its observational components

CO-2 Demonstrate understanding of how mass, radiation distribution shapes the dynamics of the universe

CO-3 Apply their knowledge and **calculate** dynamical properties of few model universe

CO-4 Discover the thermal history of the early universe

CO-5 Familiarize themselves with several unsolved problems in the research of cosmology

3.Mapping of Course Outcomes (COs) with Program Outcomes (POs) and Program Specific Outcomes (PSOs) – Course Articulation Matrix

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3	PSO4
CO1	2	1										2		3	1	2
CO2	2	1										2		3	1	2
CO3	1	1												2	3	1
CO4	2	2										1		2		3
CO5	3	1										2		3		3

Note: Each Course Outcome (CO) may be mapped with one or more Program Outcomes (POs) and PSOs. Write '3' in the box for 'High-level' mapping, 2 for 'Medium-level' mapping, 1 for 'Low'-level' mapping

4.Detailed Syllabus:

Unit-1: Universe Observed: Expansion. Isotropy and homogeneity. Age. Cosmic microwave background

Unit-2: Geometry and Dynamics: Universe in the eyes of Newton. Geometry. Mass and curvature. Friedmann equations. Model universes: empty universe, matter or radiation dominated universe, multi-component universe

Unit-3: Black-body radiation and the early history: Observation of CMB. Recombination and decoupling. Last scattering. Temperature fluctuations

Unit-4: Very early history of the Universe: Thermal history. Nucleosynthesis. Cold dark matter

Unit-5: Inflation: Flatness, horizon, and monopole problem. Physics of inflation

Reference Books:

1. **Barbara Ryden: Introduction to Cosmology**
2. **Matts Roos: Introduction to Cosmology**

5. Teaching-Learning Strategies in brief (4 to 5 sentences):

This is an introductory course to cosmology. The course is for students who do not have any knowledge of cosmology. It is also designed to be taught to students from diverse backgrounds of science. In each lecture session, the focus will be on building concepts and intuition about physics. It will be followed by a hands-on session where application of the concepts to simple problems will be practiced.

6. Assessment methods and weightages in brief (4 to 5 sentences):

Assignments: 30%,

Quizzes : 30%,

End Semester: 35%,

Attendance: 5%

Title of the Course : Principles of Programming Languages (PoPL)

Name of the Faculty: Venkatesh Choppella

Course code: CS1.402

1 Course structure

Name Principles of Programming Languages

Credits 4, Lectures-Tutorials-Practicals=3-1-0 (hours/week)

2 Prerequisite courses

1. Computer Programming
2. Discrete Mathematics (with some exposure to writing proofs)
3. Automata Theory

3 Course outcomes

A student graduating from a PoPL course should be able to perform each of the following sample tasks:

1. **CO1: Document Abstract Syntax** Document and critique the abstract syntax of industrial scale programming language like C or Java.
2. **CO2: Design domain specific languages** Design a small, domain purpose languages like a language for propositional logic and implement them.
3. **CO3 Design object small, oriented language** Design a small object-oriented language implement it either using an interpreter or by embedding it into a base language.
4. **CO4: Compare languages** Compare and analyse the semantic expressibility (in terms of first-class values) between imperative languages like C and functional languages Racket, and object-oriented languages like Java and Python.
5. **CO5: Specify application interfaces** Specify the structure of a software application like a spreadsheet or a word processor in terms of its interface as a language of user operations and its internal structure as an abstract machine.

4 Mapping to Programme and Programme Specific Outcomes

Table 1: Mapping of Course Outcomes to programme and programme specific outcomes

Programme Outcome (PO/PSO)	CO1	CO2	CO3	CO4	CO5
PO1 Engg. Knowledge	3	3	3	3	3
PO2 Problem Analysis	3	3	3	3	3
PO3 Design/Develop	1	3	3	2	2
PO4 Complex Problems	2	3	3	3	3
PO5 Modern tool usage	3	3	3	2	2
PO6 Engr. & Society	1	2	2	2	2
PO7 Environment & Sustainability	1	1	1	1	1
PO8 Ethics	2	1	1	1	1
PO9 Team work	2	3	3	2	2
PO10 Communication	3	3	3	3	3

Programme Outcome (PO/PSO)	CO1	CO2	CO3	CO4	CO5
P11 Project Mgmt & Finance	1	1	1	1	1
P12 Life learning	3	2	2	2	3
PSO1 Specialised knowledge	3	3	3	3	2
PSO2 Roadmap for technologies	3	3	3	3	3
PSO3 Research & Development Skills	2	3	3	3	3
PSO4	3	3	3	3	2

Potential for PG study

5 Syllabus

Functional Programming: Abstract vs. Concrete Syntax. Racket syntax, Functions, recursion, syntactic extensions, higher-order functions, map, reduce and other combinators.

Operational Semantics: Abstract Reduction Systems, Transition Systems, Reduction, Simplification and Evaluation relations. Judgements and Rules. Rule Induction.

Scope: Identifiers, Scope and extent, Lexical scope, Environments, ‘Dynamic scope’ and parameters. Closures

State: Stores and imperative constructs, explicit and implicit store references, objects, invariants and safety, interfaces and constructors, inheritance, Parameter passing. Call-by-value, call-by-name and lazy evaluation.

Control: Tail calls, Contexts, continuations, continuation passing style, exceptions, threads.

Types: Types syntax, type safety theorems. Type inference

Special Topics (if time permits): Monads, Concurrency.

6 Texts and References

Textbook

EOPL *Essentials of Programming Languages* 3rd Edition. Friedman and Wand. This is the main text for the course. Available on Amazon.in.

References

HtDP *How to Design Programs*. Felleisen et al. Available online.

SICP *Structure and Interpretation of Programs*. Abelson and Sussman. Available online. Accompanying video lectures also available online.

TRaAT *Term Rewriting and All That*. Baader and Nipkow. Chapters 1 and 2.

PLAI *Programming Languages: Application and Interpretation* 2nd Edition. Available online.

SSICS *Simply Scheme: Introducing Computer Science*. Brian Harvey and Matthew Wright. Available online.

RG *Racket Guide*. Available as part of Racket language documentation.

7 Teaching and Learning strategies

Lectures will cover the theoretical aspects of operational semantics but will have plenty of examples explaining interpreters of programming languages visually and interactively. Question-answer discussion will accompany each class. Quizzes each week will test student’s attention diligence, and concept recall, understanding and application. Summative assessments will be through a mid-semester and a final exam or project. Reading assignments will precede each lecture. Homework (programming) assignments will mostly involve implementation of interpreters discussed in the class and the textbook. Tutorials will walk-through abstract syntax tree annotation, components of the interpreter implementation, and inductive proofs of properties in operational semantics.

8 Assessment (Tentative)

Item	Weight (%)
Quizzes (1 per week)	25
HW	25
Mid-semester exam	20
Final exam/Project	30

Appendix: Programme and Programme Specific Outcomes

Programme Outcomes

- PO1 :: Engineering knowledge** Use concepts from varied disciplines including Computer Science, Electronics, Mathematics, and the Sciences, to engineer and develop systems of varying scale.
- PO2 Problem analysis** Identify, formulate and analyze complex engineering problems reaching substantial conclusions using first principles of Mathematics, Natural Sciences and Engineering Sciences.
- PO3 Design/Development of solutions** Identify and bring to fore the necessary concepts from Computer Science and arrive at creative ways to solve problems that take into account the societal, cultural, and ethical considerations.
- PO4 Conduct investigations of complex problems** Interpolate and extrapolate based on existing knowledge base and self-learning skills to investigate the dynamics of complex problems and find solutions.
- PO5 Modern tool usage** Demonstrate requisite hands-on skills to work with a variety of software packages, libraries, programming languages, and software development environment tools useful in engineering large scale systems
- PO6 The engineer and society** Make judicious use of resources and understand the impact of technology across the societal, ethical, environmental, and economic aspects.
- PO7 Environment and sustainability** Find technological solutions by considering the environmental impact for sustainable development
- PO8 Ethics** Practice principles of professional ethics and make informed decisions after a due impact analysis.
- PO9 Individual and team work** Work efficiently in individual and team-oriented projects of varying size, cultural milieu, professional accomplishments, and technological backgrounds.
- PO10 Communication** Effectively communicate and exchange ideas and solutions to any individual including peers, end-users, and other stakeholders.
- PO11 Project management and Finance** Apply the principles of project management in general and software project management in particular with focus on issues such as the life cycle, scoping, costing, and development.
- PO12 Life-long learning** Exhibit the aptitude for independent, continuous, and life-long learning required to meet their professional and career goals.
- Programme Specific Outcomes (PSOs)**
- PSO1** Exhibit specialized knowledge in some sub-areas of Computer Science and Engineering such as Theoretical Computer Science, Computer Systems, Artificial Intelligence, Cyber-physical Systems, Cyber-security and use this specialized knowledge base to solve advanced problems
- PSO2** Perform gap analysis in terms of systems and technologies and prepare roadmaps for incorporating state-of-the-art technology into system analysis, design, implementation, and performance.
- PSO3** Demonstrate research and development skills needed to define, scope, develop, and market futuristic software systems and products.
- PSO4** Demonstrate knowledge and skills at the required depth and breadth to excel in post-graduate and research programs.
-

Title of the Course : Principles of Semiconductor Devices

Course Code : EC2.409

Name of the Faculty : Aftab Hussain

L-T-P : 3-1-0

Credits : 3

(L= Lecture hours, T=Tutorial hours, P=Practical hours)

Name of the Academic Program B. Tech in ECE

1.Prerequisite Course / Knowledge:

EC, EW1 & EW2

2.Course Outcomes (COs) (5 to 8 for a 3 or 4 credit course):

After completion of this course successfully, the students will be able to..

CO-1 Describe quantum mechanics basics: Heisenberg's principle, energy band (conduction & valance bands, energy gap).**CO-2** Explain the basic physics for PN junctions, MOS, MS junctions, MOSFET & BJT**CO-3** Calculate basic semiconductor device parameters and solve problems related to design of above-mentioned semiconductor devices.**CO-4** Design very simple diode & MOSFET circuits**3.Mapping of Course Outcomes (COs) with Program Outcomes (POs) and Program Specific Outcomes (PSOs) – Course Articulation Matrix**

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3	PSO4
CO1	3	1	1	1	1	1	1	1	1	1	1	3	2	2	1	3
CO2	3	3	1	1	1	1	1	1	1	3	1	3	2	3	3	3
CO3	2	2	3	2	1	1	1	1	1	1	1	3	2	2	3	3
CO4	2	1	2	3	3	1	2	1	1	1	1	3	2	3	1	3

Note: Each Course Outcome (CO) may be mapped with one or more Program Outcomes (POs) and PSOs. Write '3' in the box for 'High-level' mapping, 2 for 'Medium-level' mapping, 1 for 'Low'-level' mapping

4.Detailed Syllabus:

Unit 1: Semiconductor Properties

Unit 2: Quantum Mechanics and Energy Band Theory

Unit 3: Carriers in equilibrium, G-R processes

Unit 4: Carrier Transport

Unit 5:PN Junction physicsUnit6: MOS & MOSFETUnit7: BJT

Reference Books: 1. Advanced Semiconductor Fundamentals by Robert Pierret

1. Semiconductor Device Fundamentals by Pierret

5. Teaching-Learning Strategies in brief (4 to 5 sentences):

Students will be applying the lecture discussion to solved examples shared with them in the class. The assignments given will reinforce the concepts. Classroom learning will be done in interactive method as much as possible. Occasionally self-assessment test (1minute paper) will be given. In lab class, students will make simple circuits using simple basic components.

6. Assessment methods and weightages in brief (4 to 5 sentences):

Type of Evaluation [3 credit- lecture]	Weightage (in %)
Mid Sem Exam 1	15*
Mid Sem Exam 2	15*
End Exam	25*
Assignments	15
Mini Project	25
1 minute paper (in class) [weekly prescheduled]	5

Title of the Course : Probability and Random Processes

Name of the Faculty : Praful Mankar

Course Code : MA6.102

L-T-P : 3-1-0

Credits : 4

(L= Lecture hours, T=Tutorial hours, P=Practical hours)

Course: ECE

Name of the Academic Program: B. Tech in Electronics and Communication Engineering

1. Prerequisite Course / Knowledge:

Basic idea of set theory, counting

2. Course Outcomes (COs):

After completion of this course successfully, the students will be able to:

CO-1: Describe the probability space associated with an experiment, conditional probability and Bayestheorem

CO-2: Give examples of discrete and continuous random variables and their distributions

CO-3: Calculate conditional and marginal distributions, distributions of functions of random variables, expectation and variance

CO-4: Analyze the properties of independent random variables, sums of random variables

CO-5: Interpret the tail bounds, law of large numbers and central limit theorem

CO-6: Evaluate the real-world applications of random variables and random processes

3. Mapping of Course Outcomes (COs) with Program Outcomes (POs) and Program Specific Outcomes(PSOs) – Course Articulation Matrix

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3	PSO4
CO1	3	3	2	2	1	-	-	-	-	-	-	-	-	3	2	3
CO2	2	3	2	2	1	-	-	-	-	-	-	-	-	3	2	3
CO3	3	3	2	2	1	-	-	-	-	-	-	-	-	3	2	3
CO4	3	3	2	2	1	-	-	-	-	-	-	-	-	3	2	3
CO5	3	3	2	2	1	-	-	-	-	-	-	-	-	3	2	3
CO6	3	3	2	2	1	-	-	-	2	2	-	2	-	3	2	3

4. Detailed Syllabus:

Unit 1 : Sets and set operations, Probability space, Conditional probability and Bayes theorem.

Unit 2: Discrete random variables, probability mass function, probability distribution function, example random variables and distributions, Continuous random variables, probability density function, probability distribution function, example distributions.

Unit 3: Joint distributions, functions of one and two random variables, expectation and variance, Conditional distribution, densities, conditional expectation, moment generating functions, characteristic functions.

Unit 4: Markov, Chebyshev and Chernoff bounds. Random sequences and modes of convergence, Strong and weak laws of large numbers, central limit theorem.

Unit 5: Random processes, Mean and covariance functions, Stationary processes and wide-sense stationary processes, power spectral density, linear filtering of random processes.

Reference Books:

1. Bertsekas, Dimitri P., and John N. Tsitsiklis. Introduction to Probability. Vol. 1. Belmont, MA: Athena Scientific, 2002.
2. Henry Stark, John. W. Woods, Probability and Random Processes with Applications to Signal Processing, 3rd edition, Prentice Hall, 2002.
3. Gallager, R. (2008). Principles of Digital Communication. Cambridge: Cambridge University Press. doi:10.1017/CBO9780511813498.
4. Sheldon Ross, Introduction to Probability Models, Academic Press, 2010.

5. Teaching-Learning Strategies in brief (4 to 5 sentences):

The course has lectures supported by tutorials. In tutorials, problems related to the concepts presented in the class are solved by teaching assistants. Quizzes and group learning activities are conducted periodically so that students can actively engage with the course material. An assignment is given towards the end of the course, which requires the students to understand various applications of the theory and prepare a report.

6. Assessment methods and weightages in brief (4 to 5 sentences):

Type of Evaluation	Weightage (in %)
Mid Semester Exam	15%
Quizzes/Viva	35%
End Semester Exam	25%
Home Assignments	25%

Title of the Course : Probability and Statistics
 Name of the Faculty : Tajas Bodas
 Course Code : MA6.101
 L-T-P : 3-1-0.
 Credits : 4
 (L= Lecture hours, T=Tutorial hours, P=Practical hours)
 Course : CSE
 Name of the Academic Program: **B.Tech. in Computer Science and Engineering**

1. Prerequisite Course / Knowledge:

Linear Algebra, Real Analysis

2. Course Outcomes (COs)

After completion of this course successfully, the students will be able to –

CO-1. Explain the axioms of probability and rules, discrete and continuous random variables.

CO-2. Derive the density function of transformations of random variables and use these to generate data corresponding to various distributions.

CO-3: Derive marginal and conditional distributions of multivariate random variables and probability bounds.

CO-4. Discuss the classical and Bayesian inference theory and applications.

CO-5. Discuss the basic random processes and their applications.

CO-6. Outline a proof of stated theorem and write the logically derived proof.

3. Mapping of Course Outcomes (COs) with Program Outcomes (POs) and Program Specific Outcomes (PSOs) – Course Articulation Matrix

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3	PSO4
CO1	1	3	1	1	-	-	-	-	2	2	-	3	3	-	-	3
CO2	1	3	1	3	1	-	-	-	2	2	-	3	3	-	-	3
CO3	1	3	1	3	1	-	-	-	2	2	-	3	3	-	-	3
CO4	1	3	2	3	1	-	-	-	2	2	-	3	3	-	-	3
CO5	2	3	2	3	1	-	-	-	2	2	-	3	3	-	-	3
CO6	1	3	1	1	-	-	-	-	2	2	-	3	3	-	-	3

‘3’ in the box denotes ‘High-level’ mapping, 2 for ‘Medium-level’ mapping, 1 for ‘Low’-level’ mapping

4. Detailed Syllabus:

Unit 1: Discrete Random Variables: Probability mass function, expectation and variance. Bernoulli distribution, Geometric distribution, Binomial distribution. Derivation of mean and variance. Sampling with and without replacement. Hypergeometric distribution and Poisson distribution. Derivation of the Poisson distribution as limiting form of Binomial. Derivation of mean and variance. Bounding probabilities, tail sum formula, Markov's inequality and Chebyshev's inequality. Probability generating functions and moment generating functions. (9 hours)

Unit 2: Continuous Random Variable: Probability density function, cumulative distribution function, expectation, mean and variance. Moment generating functions and uniqueness theorem. Chebyshev's inequality. The uniform distribution on (a, b), the normal distribution. Mean and variance of the normal distribution. The Cauchy distribution. The exponential distribution, moments, memoryless property, hazard function. Gamma distribution, moments, Chi-square distribution. (9 hours)

Unit 3: Multivariate Distributions: Cumulative distribution function method for finding the distribution of a function of random variable. The transformation rules. Discrete bivariate distributions, marginal and conditional distributions, the trinomial distribution and multinomial distribution. Continuous bivariate distributions, marginal and conditional distributions, independence of random variables. Covariance and correlation. Mean and variance of linear combination of two random variables. The joint Moment generating function (MGF) and MGF of the sum. The bivariate normal distribution, marginal and conditional distributions, conditional expectation and variance, joint MGF and marginal MGF. Linear combinations of independent random variables. Means and variances. Sequences of independent random variables and the weak law of large numbers. The central limit theorem, normal approximation to the binomial distribution. (9 hours)

Unit 4: Statistical Inference and Random Processes: Point Estimation, Interval Estimation, Hypothesis Testing, Linear Regression, Bayesian Inference. Point Processes, Poisson Process, Discrete Time Markov Chains. (9 hours)

References:

- Sheldon Ross, Introduction to Probability Models, Academic Press, 2010.
- Online resource: <https://www.probabilitycourse.com/>
- A first course in Stochastic Models, H. C. Tijms, Wiley, 2003.
- Introduction to Probability and Statistics for Engineers and Scientists, S. Ross, 5th Edition, Academic Press, 2014.

5. Teaching-Learning Strategies in brief:

Lectures by integrating ICT into classroom teaching, weekly tutorials involving problem solving and active learning by students and Project-based Learning by doing one mini-project.

6. Assessment methods and weightages in brief:

Assignments in theory: 15 marks, Mid Semester Examination-1: 25 marks, Mid Semester Examination-2: 30 marks, End Semester Examination: 30 marks

Title of the Course: **Product Management 101**
Faculty Name: Ramesh Swaminathan + Guest Faculty
Course Code : PD2.401
L-T-P: 3 -1- 0
Credits: 2
(L= Lecture hours, T=Tutorial hours, P=Practical hours)
Name of the Program: M Tech Product Design & Management
Semester, Year : Monsoon, 2026

Pre-Requisites : No prerequisite

Course Outcomes :

CO1: Understand the key role of a product manager in ideating & developing technical products for the Digital World

CO2: Create a Product Strategy by using various market research techniques

CO3: Develop a product mindset to create innovative product & solutions that solve complex technical problems that is required by the Market

CO4: Execute product strategy through Roadmaps & Release Plans

CO5: Learn various product development methodologies that can be applied to enable faster Go to Market

Course Topics :

Module 1: Introduction to Product Management – the Art & Science of Product Management

1. What is product management
2. Types of Product Management
3. Product life cycle

Module 2: Product Strategy

1. Market Research
2. Product Value Proposition
3. Product Strategy
 - a. Market Needs
 - b. Key Differentiators
 - c. Business Goals

Module 3: Product Ideation & Market fit

1. Product Ideation
2. Product Feasibility
3. Product market Fit
4. User Journeys

Module 4: Adopting a Product Mindset

1. Prioritization
2. People
3. Process

4. Progress

Module 5: Product Roadmap

1. Top-Down Product Strategy
2. Featureless Roadmaps to feature roadmaps
3. Roadmaps to Release plans

Module 6: Product release execution

1. Product Prototyping & Market Validation
2. Product Development Process
 - a. Agile development process
 - b. Lean product development
3. Product release to market

Preferred Textbooks: None

Reference Books :

1. Inspired: How to create Tech Products Customers Love by Marty Cagan
2. The Lean Product Playbook by Dan Olsen

E-book Links :

Grading Plan :

(The table is only indicative)

Type of Evaluation	Weightage (in %)
Quiz-1	10%
Quiz-2	10%
End Sem Exam	10%
Assignments	10%
Mini Project	40%
Other Evaluation (Product Workshop)	20%

Mapping of Course Outcomes to Program Objectives: (1 – Lowest, 2—Medium, 3 – Highest, or a ‘-’ dash mark if not at all relevant). Program outcomes are posted at

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO 10	PSO 1	PSO 2	PSO 3	PSO 4
CO1	3	3	3	2	3	3	3	3	2	2	3	3	3	3
CO2	3	3	3	3	3	3	3	3	2	3	3	2	3	2
CO3	3	2	2	2	3	3	2	3	2	2	2	2	2	2
CO4	3	3	3	3	3	3	3	2	2	3	3	2	2	3
CO5	2	2	3	2	3	3	2	3	2	3	2	2	3	3

Teaching-Learning Strategies in brief (4-5 sentences):

This Course will be taught through a Hands-on train model wherein the students will learn by doing. Theory will be taught through Power Point presentation and will be followed up by Assignments. The Assignments will conclude in a mini project that needs to be submitted at the end of semester. A Role play driven workshop at the end of the Semester will be a major evaluation factor for this Semester.

Title of the Course	: Product Marketing
Faculty Name	: Ravi Warriar
Course Code	: PD2.501
Credits	: 4
L - T - P	: 3-1-0
(L - Lecture hours, T-Tutorial hours, P - Practical hours)	
Semester, Year	: Monsoon 2026

Pre-Requisites : No pre-requisites

Course Outcomes :

The outcome of the Product Marketing course is to ensure that participants acquire skills and understand the processes involved in defining a complete and functional strategy to take their product or service to the market.

CO1- Learn methods, models and frameworks that can be employed to developing strategy artifacts relevant to product marketing.

CO2- Apply the concepts and tools in the development of go-to-market strategy and sales playbooks

CO3- Learn the principles of effective marketing, specifically those of pricing, positioning, narratives with respect to targeted segments and types of customers in each segment.

CO4- Understand the synthesis between product management and product marketing as to design and build better products from the beginning

Course Topics:

Unit 1 – Product Marketing and Product Management

Sub-topics covered: Recap of 1) how marketing is an integral part of Product Management, 2) Product Management fundamentals, 3) Product Market Fit – problem and solution validation

Unit 2 – Marketing Mix

Sub-topics covered: 1) Value Model (4Cs), 2) Marketing Mix (4Ps) and how to use them in tandem to strategize product launch

Unit 3 – Value Proposition and Promotion

Sub-topics Covered – 1) Segmentation and User Personas, 2) Customer Journey Mapping, 3) Pricing Mix, 4) Positioning, and 5) Product Branding

Unit 4 – Customer Engagement

Sub-topics – 1) Introduction to Marketing Communication, 2) Building a comprehensive narrative for marketing, and 3) Customer Engagement at all stages of acquisition

Unit 5 – Sales Playbooks

Sub-topics covered: 1) What are playbooks? and 2) how to develop an actionable playbook for sales

Unit 6 – Go-to-Market Strategizing

Sub-topics covered: 1) Customer Acquisition Strategies, 2) Defining the GTM Strategy, and 3) Executing the GTM Strategy successfully

Preferred Textbooks: None

Reference Books:

There are a few books I will recommend during the course. Will make a list and add them here shortly.

E-book Links: TBD

Grading Plan:

Type of Evaluation	Frequency	Weightage (in %)
Assignments	10	50
Final Submissions	1	25
Final Exam	1	25

Mapping of Course Outcomes to Program Objectives: (1 – Lowest, 2—Medium, 3 – Highest, or a ‘-’ dash mark if not at all relevant). Program outcomes are posted at

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO1	PSO2	PSO3	PSO4
CO1	3	3	3	2	3	2	2	2	2	3	3	2	2	2	2
CO2	3	3	3	2	3	2	2	2	2	3	3	2	2	2	2
CO3	3	3	3	2	3	3	2	2	2	2	3	3	2	2	2
CO4	3	3	3	2	3	2	2	3	3	2	3	3	2	2	2

Teaching-Learning Strategies in brief (4-5 sentences):

1. Each class will have prerequisite reading material that will be announced or provided at the end of the previous seminar. Discussions will cover subjects of interest or concern from the pre-reads with respect to that topic of the class.
2. Seminars will always cover the fundamentals and cover detailed concepts only when the participants need additional assistance grasping those concepts.
3. Assignments will be given in each seminar session and will be followed by a discussion session to review and discuss the assignment
4. The course leans more on practice and completion of assignments which will form a larger chunk of their efforts and grading.

Title of the Course	: Programming AI Accelerators
Name of the Faculty	: Girish Varma, P J Narayanan
Name of the Program	: B.Tech./MS/PhD in CS/ECE
Course Code	: CS3.406
Credits	: 4
L-T-P	: 3-0-1
Semester, Year	: Monsoon 2026
Pre-Requisites	

Computer Programming (in C/C++), Data Structures and Algorithms, Computer Organization and Architecture, Operating Systems (memory management fundamentals), Basic Linear Algebra, Statistical Methods in AI

Course Outcomes

Upon successful completion of the course, students will be able to:

CO1: Analyse GPU execution behaviour, memory hierarchy, and parallel programming models to evaluate the performance characteristics of deep learning workloads.

CO2: Design and implement tensor abstractions and runtime systems using modern C++ and CUDA for efficient management of multidimensional data.

CO3: Develop computational graph frameworks and reverse-mode automatic differentiation engines to support gradient-based learning.

CO4: Construct and optimize neural network training pipelines including loss functions, optimization algorithms, and numerical stability mechanisms.

CO5: Design high-performance GPU kernels for matrix multiplication, normalization, and attention operations using memory-aware optimization techniques.

CO6: Develop a complete Transformer-based inference framework capable of loading pretrained weights, managing KV caches, and generating text using custom CUDA kernels.

Course Topics

Module I: GPU Architecture and Performance Engineering

Introduction to Bare-Metal AI Systems, GPU Architecture Fundamentals, CUDA Programming Model, GPU Memory Hierarchy, Memory Optimization Techniques, GPU Profiling and Performance Analysis, GPU Design Patterns, Tensor Mathematics and Memory Layouts

Module II: Tensor Runtime and Neural Network Training Systems

Tensor Runtime Design in Modern C++, Tensor Operations and Shape Inference, Reverse-Mode Automatic Differentiation, Dynamic Computational Graphs, Backpropagation Engine Design, Neural Networks and Matrix Calculus, Loss Functions and Numerical Stability, Optimization Algorithms (SGD, Momentum, AdamW)

Module III: High-Performance GPU Kernels for Deep Learning

Matrix Multiplication Fundamentals, Shared-Memory Tiled GEMM, Kernel Fusion Techniques, Runtime Optimization Strategies, Softmax and Reduction Kernels, LayerNorm and RMSNorm, Attention Kernel Engineering, CUDA Graphs and Execution Optimization

Module IV: Transformer Systems and LLM Inference

Language Modelling Foundations, Tokenization and Embeddings, Self-Attention Mathematics, Multi-Head Attention, Transformer Block Architecture, Rotary Positional

Embeddings (RoPE), KV Cache Management, Checkpoint Loading and Weight Serialization, Autoregressive Inference, LLM Deployment and Optimization

Preferred Textbooks

1. David B. Kirk and Wen-mei W. Hwu, *Programming Massively Parallel Processors: A Hands-on Approach*, Morgan Kaufmann.
2. John Cheng, Max Grossman, Ty McKercher, *Professional CUDA C Programming*, Wrox.
3. Ian Goodfellow, Yoshua Bengio, Aaron Courville, *Deep Learning*, MIT Press.

Reference Books

1. Chip Huyen, *Designing Machine Learning Systems*, O'Reilly.
2. Christopher Manning, Jacob Eisenstein, *Introduction to Natural Language Processing*.
3. Aston Zhang et al., *Dive into Deep Learning*.
4. Bryant and O'Hallaron, *Computer Systems: A Programmer's Perspective*.
5. Patterson and Hennessy, *Computer Organization and Design*.

Grading Plan

Type of Evaluation	Weightage (%)
Quiz-1	7.5
Mid-Semester Examination	15.0
Quiz-2	7.5
End-Semester Examination	20.0
Programming Assignments	10.0
Laboratory Evaluations (Proctored)	20.0
Final Project	20.0
Total	100.0

Mapping of Course Outcomes to Program Outcomes

CO/PO	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PO 12	PSO1	PSO2	PSO3	PSO4
CO1	3	3	2	2	3	-	-	-	-	-	-	1	3	2	-	-
CO2	3	2	3	2	3	-	-	-	-	-	-	1	3	3	-	-
CO3	3	3	3	2	2	-	-	-	-	-	-	1	3	3	-	-
CO4	3	3	2	2	2	-	-	-	-	-	-	1	3	2	-	-
CO5	3	3	3	3	3	-	-	-	-	-	-	1	3	3	2	-
CO6	3	3	3	3	3	-	-	-	1	1	1	1	3	3	3	2

Teaching-Learning Strategies in Brief

The course follows a project-driven and systems-oriented pedagogy in which students progressively build a deep learning framework from first principles using C++ and CUDA. Concepts from GPU architecture, numerical optimization, machine learning, and software systems are introduced through interactive lectures and reinforced through structured laboratory exercises. Every two lectures are followed by a hands-on laboratory session that contributes directly to the semester-long MiniTensor-LM project. Assessment emphasizes implementation, profiling, debugging, and oral code defense to ensure deep understanding and discourage dependence on AI-generated code. By the end of the course, students develop an end-to-end Transformer inference system capable of loading pretrained weights and generating text using their own runtime.

Title of the Course : Quantum aspects of cryptography

Faculty Name : Atul Singh Arora

Course Code : CS1.505

Credits : 4

L - T - P : 3-1-0

(L - Lecture hours, T-Tutorial hours, P - Practical hours)

Name of the Program: Computer Science Elective (UG3/UG4/Dual degree)

Semester, Year : Monsoon 2026

Pre-Requisites: Mathematical maturity from courses like Discrete Structures or Linear Algebra course.

Desirable: Familiarity with the basics of (classical) cryptography, and quantum computation/information

Course Outcomes: The primary goal of this course is to enable students to perform research in this exciting newly emerging discipline. The students will be able to do the following:

CO.1 (Understand level) — Demonstrate familiarity with the basic security definitions in the area, including those of verification of quantum computations, homomorphic computation, including those based on complexity theoretic assumptions such as the hardness of the learning with errors problems.

CO.2 (Analyse) — Analyse whether a formal security definition is meaningfully capturing the relevant notion of security, and construct security proofs given explicit candidate constructions or show impossibility given a model of computation.

CO.3 (Evaluate level) — Review the literature at the frontier of quantum cryptography and related areas.

CO.4 (Evaluate level) — Evaluate the runtime/resource requirements of cryptographic protocols, derive reductions to show protocols satisfy appropriate security definitions.

Course Topics:

Depending on the maturity of the students, the course will be adapted. The following is an ambitious and therefore tentative course outline. The basic goal will be to cover Unit 1 and one key result from Units 2, 3 and 4. The remaining units will be covered, depending on the pace of the course and interest of the students.

Unit 1: Review:

Primer on quantum formalism. [VW '16] Lay of the land:

Impagliazzo's worlds: in particular MiniCrypt and Cryptomania Introduce the main directions:

(T1) post-quantum cryptography (make classical constructions secure against quantum adversaries), (T2) quantum analogues of classical functionalities,

(T3) functionalities impossible without quantum (excluding those in T2), (T4) basing cryptography on quantum complexity (T3) Information Theoretic: key distribution (BB84, Ekert), proof of security, secret sharing, impossibility of bit commitment, impossibility of strong coin flipping, achieving optimal strong coin flipping using weak coin flipping, self-testing CHSH, all bipartite states can be self- tested [VW '24]

Unit 2: (T2) Verification:

Regev's quantum reduction [Regev'09]

Weak quantum verifier: based on MBQC [GKK17]

Classical verifier: assuming LWE is hard, Mahadev [Mahadev 18, Vidick 22] Classical

verifier: two non-communicating provers [RUV 12]

Unit 3: (T2) QFHE: construction and its applications: Mahadev's QFHE construction

[Mahadev 17] Compiling non-local games [KLVY 23] Verification assuming QFHE [NZ 23]

Unit 4: (T4) Minimal assumptions: Minicrypt and below OT is in Miniqcrypt [GLSV 20, BCKM 20]

Crypto despite having $NP=P$ or similar [Kre 21, KQST 23] Microcrypt Primitive Zoo [Or Sattah]

Unit 5: (T1/T2) Multi Party Computation

(T1) Post-quantum Commitments (collapse-binding commitments) [Unruh'16] (T1) Post-quantum MPC [HSS 11]

(T2) Quantum 2-PC [DNS '10 and '12]

(T2) Quantum MPC w/ quantum communication [Dulek, Grilo, Jeffery, Majenz, Schaffner]

(T2) Quantum MPC w/ classical communication [Bartusek '21]

Unit 6: (T3) Quantum-only functionalities—I

Unclonable encryption: construction in the Random Oracle Model [AKLLZ '22] Certified deletion [BK '22]

Quantum Pseudorandom unitaries [MH '24]

Unit 7: (T3) Quantum-only functionalities—II Quantum Money and Lightning [Zhandry '17]
iO for pseudo-deterministic functions [BKNY '23]

Unit 8: Bonus/extra reading (references will be provided later) Other key topics in cryptography:

Interactive proofs Zero knowledge Quantum rewinding

Quantum Random Oracle Model Everlasting security Connections to physics:

Self-testing using a single quantum device Non-locality => Proof of quantumness + rigidity

Black hole radiation decoding and commitments Cryptographic tests of python's lunch conjecture Computationally bounded theory of entanglement

References:

GKK 17: [Verification of quantum computation: An overview of existing approaches](#) VW 24:

[Introduction to Quantum Cryptography](#) (book)

VW 16: [Quantum Proofs](#) (survey)

RUV 12: [Classical command of quantum systems](#)

Vidick 22: [Course FSMP, Fall'20: Interactions with Quantum Devices](#) Mahadev 18: [Classical verification of quantum computations](#) Mahadev 17: [Classical Homomorphic Encryption for Quantum Circuits](#) KLVY 23: [Quantum Advantage from Any Non-Local Game](#)

NZ 23: [Bounding the Quantum Value of Compiled Nonlocal Games: From CHSH to BQP verification](#)

BCKM: [One-Way Functions imply Secure Computation in a Quantum World](#) MH24: [How to Construct Random Unitaries](#)

Or Sattah: [MicroCrypt Zoo](#)

Kre21: [Quantum pseudorandomness and classical complexity](#) KQST21: [Quantum cryptography in algorithmica](#)

DNS '10: [Secure two-party quantum evaluation of unitaries against specious adversaries](#)

DNS '12: [Actively Secure Two-Party Evaluation of any Quantum Operation](#)

DGJMS '19: [Secure Multi-party Quantum Computation with a dishonest majority](#) Bartusek '21: [Secure Quantum Computation with Classical Communication](#)

BK 22: [Cryptography with Certified Deletion](#)

HSS11: [Classical Cryptographic Protocols in a Quantum World](#) Unruh'16: [Collapse-binding quantum commitments w/o random oracles](#) AKLLZ '22: [On the Feasibility of Unclonable Encryption and more](#) Zhandry '17: [Quantum Lightning Never Strikes the same state twice](#) MH '24: [How to Construct Random Unitaries](#)

BKNY '23: [Obfuscation of pseudo-deterministic quantum circuits](#)

Regev '09: [On Lattices, Learning with Errors, Random Linear Codes, and Cryptography](#) Video Tutorials:

[Dakshita Khurana: Cryptography with certified deletion](#) Mark Zhandry: [Security Reductions](#) (multi- part series)

Preferred/reference Textbooks:

There are no textbooks for this course as the material is at the frontier of current research in quantum cryptography. Relevant lecture notes/tutorials/survey articles have already been referenced above.

Resources to review basics of quantum info/computation

- [Lecture notes on Quantum Computation by John Preskill](#) (Caltech)
- [Introduction to Quantum Information and Computation](#), MA Nielsen and IL Chuang

Resources to review classical cryptography

- Introduction to Modern Cryptography. Jonathan Katz and Yehuda Lindell.
- Foundations of Cryptography (Volumes 1 and 2). Oded Goldreich.

Other resources

Grading Plan:

Type of Evaluation	Weight (%)
Assignments/Scribe	40
Term Paper	20
Two Exams (mid-semester and final)	40 (20+20)

Assignments. Approximately, every two lectures. Will largely consist of completing proofs that could not be carried out in class. Resources will be provided for finding the proofs. Deadlines for assignments will be decided by consensus. Every student is allowed to miss the deadline by at most 48 hours, at most once per month.

Exams. The final exam would not overlap (to the extent reasonable) with the material assessed in the mid-semester exam. Two previously assessed assignments can be resubmitted, one during the mid-term and one during the final, to improve one's score on those assignments. Depending on the need, one may also offer to scribe a lecture, to improve one's score on one of the assignments.

Term Paper.

The goal here is for each student to read and understand a related paper. To this end, the course requires each student to submit a short four-page summary of the paper (page limit is not strict) that also identifies one potential direction for further research, together with some ideas on how to make progress towards achieving it. The research aspect does not have to be anything too ambitious. The emphasis will be on understanding the main result. Depending on the number of students, there may be a presentation in addition/instead of the written report.

The choice of paper must be finalised before the mid-term. A list will be provided but students are welcome to submit papers they find interesting.

The term paper will be due at least one month after the mid-term. The exact date will be announced after the mid-term.

Mapping of Course Outcomes to Program Objectives: (1 – Lowest, 2—Medium, 3 – Highest, or a ‘-’ dash mark if not at all relevant). Program outcomes are posted at

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3	PSO4
CO1	2	2	2	1	1	1	-	-	2	2	1	2	3	2	3	3
CO2	2	2	3	1	2	-	-	-	2	2	1	2	3	2	3	3
CO3	2	2	3	1	1	-	-	-	2	2	1	2	3	2	3	3
CO4	2	2	3	1	1	1	-	-	2	2	2	1	3	2	3	3

Teaching-Learning Strategies in brief (4-5 sentences):

The course will facilitate inter-student and faculty student interactions by periodic gaps for completing in-class exercises and discussing their solutions. Proofs will be broken down into smaller steps and some steps will be given as exercises after each class. Spaced repetition of exercises will be used to encourage retention of key ideas and concepts. The focus will primarily be on understanding and asking questions, preparing the students to conduct research. The course explicitly has an exploratory component that will allow students to study cutting edge research papers and formulate new research directions and crystallise ideas on how to make progress along these directions. Students will submit written reports and, time permitting, also present their work. Both being crucial skills for conducting effective research.

Title of the Course : Quantum Mechanics
Name of the Faculty : Harjinder Singh
Course Code : SC1.203
L-T-P : 3-1-0.
Credits : 4
(L= Lecture hours, T=Tutorial hours, P=Practical hours)
Name of the Academic Program: CND

1.Prerequisite Course / Knowledge:

Basic linear algebra, complex numbers.

2.Course Outcomes (COs):

After completing this course successfully, the students will be able to

CO-1 Recognize the basic differences between the inherently probabilistic description in quantum mechanics with the deterministic description in the classical theories.

CO-2 Discover the role of linear algebra, complex analysis and probability theory in quantum mechanics and modern physics.

CO-3 Calculate and **solve** simple 1D quantum problems like particle in a box, the simple harmonic oscillator, and the free particle, etc.

CO-4 Apply their knowledge of basic problems in more complicated problems like the Hydrogen atom and **discover** advanced techniques.

CO-5 Recognize the conceptual challenges in quantum mechanics

3.Mapping of Course Outcomes (COs) with Program Outcomes (POs) and Program Specific Outcomes (PSOs) – Course Articulation Matrix

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3	PSO4
CO1	3											1		2		1
CO2	3	2		1								1		2		3
CO3	1	2	2	2									1	2		2
CO4	2	3	2	3									2	2		3
CO5		3		3								1	2	2		3

Note: Each Course Outcome (CO) may be mapped with one or more Program Outcomes (POs) and PSOs. Write '3' in the box for 'High-level' mapping, 2 for 'Medium-level' mapping, 1 for 'Low'-level' mapping

4.Detailed Syllabus:

Unit 1: Introduction: The Schrödinger equation and the uncertainty principle

Unit 2: Mathematical Formalism: Review of linear algebra, complex functions, Fourier transformation etc. and the generalized statistical interpretation, Heisenberg picture

Unit 3: Time independent Schrödinger equation: Infinite square well, harmonic oscillator, free particle, delta function potential, finite square well

Unit 4: 3D Problems: Spherical coordinates - Hydrogen atom, angular momentum, spin, two-particle systems, atoms

Unit 5: Advanced topics: Time independent perturbation theory, the variational principle, Bell's theorem

Reference Books:

1. Introduction to Quantum Mechanics by David J Griffiths
2. Molecular Quantum Mechanics by P W Atkins and R S Friedman
3. Principles of Quantum Mechanics by R Shankar
4. Modern Quantum Mechanics by J J Sakurai
5. Quantum Physics by Stephen Gasiorowicz

5.Teaching-Learning Strategies in brief:

This is the first course on Quantum Mechanics. The students will see most of the topics for the first time. The focus would be on concepts and intuition building with reasonable stress on the mathematics of Quantum Mechanics.

6.Assessment methods and weightages in brief:

Assignments + Quizzes – (30%), Mid-term evaluation (30%), Final exam (40%)

Title of the Course : **Quantum Error-Correcting Codes**

Name of the Faculty : Lalitha Vadlamani

Name of the Program : ECE/CSE

Course Code : EC5.413

Credits : 4

L - T - P : 3-1-0

(L - Lecture hours, T-Tutorial hours, P - Practical hours)

Semester, Year : Monsoon, 2026

Pre-Requisites : Mathematical maturity, basics of probability theory, linear algebra. Background in quantum mechanics/quantum computation will be helpful but are not necessary.

Course Outcomes : After successful completion of the course, the students will be able to

1. Understand the framework of quantum computation, noise models and why error correction is central to quantum computation.
2. Apply the stabilizer formalism and calculate the parameters of quantum codes.
3. Implement and evaluate the encoding and decoding algorithms of topological codes like surface codes and color codes.
4. Design a fault tolerant quantum circuit by applying the principles of quantum error correction.
5. Explore advanced topics in the area of quantum LDPC codes and topological codes, demonstrating critical thinking and problem-solving skills.

Course Topics :

The course is organized into four modules. A basic introduction to quantum computation will be provided followed by basic concepts of quantum error correcting codes. Stabilizer codes, CSS codes and topological codes will be dealt with in detail. Some main results of fault-tolerant quantum computation will be discussed.

Module 0: Basics of Quantum Computation: qubits, quantum states, quantum gates, measurement, density matrices, trace, partial trace; the no-cloning theorem (5 Lectures)

Module 1: Basics of Quantum Error Correction: Quantum noise model - bit flips and phase flips, depolarizing channel, amplitude damping, Shor's code, Pauli error basis; discretization of quantum errors, Knill-Laflamme conditions;

(approx. 5 Lectures)

Module 2: Stabilizer and CSS Codes: Stabilizer Formalism, dimensions and minimum distance of stabilizer codes, Symplectic representation, CSS codes, Hypergraph product codes (approx. 5 Lectures)

Module 3: Topological Codes: Surface Codes, Color Codes, MWPM Decoder, Union Find Decoder (approx 4 Lectures)

Module 4: Fault-Tolerant Quantum Computation: Universal quantum computation, Threshold theorem, Eastin-Knill theorem, Magic state distillation (approx. 5 Lectures)

Preferred Textbooks/Reference Material: The following are the preferred textbooks for the course.

- Daniel Gottesman's book on Surviving as a Classical Computer in a Classical World, online copy available
- Dan Browne, [Lectures on topological codes and quantum computation](#), University College London
- M.A. Nielsen and I.L. Chuang, [Quantum Computation and Quantum Information](#), Cambridge University Press, 2010 (10th Anniversary Edition)
- D.A. Lidar and T.A. Brun (eds.), [Quantum Error Correction](#) Cambridge University Press, 2013.

Grading Plan :
(The table is only indicative)

Type of Evaluation	Weightage (in %)
Scribes	10 %
Assignments	20 %
Mid Sem Exam	20 %
End Sem Exam	30 %
Project	20 %

Mapping of Course Outcomes to Program Objectives: (1 – Lowest, 2—Medium, 3 – Highest, or a '-' dash mark if not at all relevant).

	P O 1	P O 2	P O 3	P O 4	P O 5	P O 6	P O 7	P O 8	P O 9	PO10	PO11	PO12	PS O 1	PS O 2	PS O 3	PS O 4
CO1	3	2	2	2	1	3	2	1	3	3	1	2	2	1	3	2
CO2	2	3	2	2	2	2	2	2	3	3	1	3	3	1	3	2
CO3	3	3	2	3	2	2	2	1	3	3	1	3	3	1	3	3
CO4	3	3	2	3	2	3	2	3	3	3	2	3	3	2	3	3
CO5	3	3	3	3	2	3	2	2	3	3	3	3	3	2	3	3

Teaching-Learning Strategies in brief (4-5 sentences):

There will be some lectures reviewing the concepts from quantum systems. This is an advanced course and hence some relevant material in terms of video lectures of researchers in the field will be provided. The lectures of the course will be recorded for easy reference of students. Problem solving sessions will be conducted for better understanding of course material.

Title of the Course	: Quantum Information Science
Name of the Faculty	: Siddhartha Das
Course Code	: SC4.430
Name of the Program	: Science Elective (Maximum: 30 students)
Credits	: 4
(3 lecture hours-1 tutorial hour-0 practical hour)	
[Course level: 3 without project and 4 with project]	
Semester, year	: Monsoon, 2026
Course outcomes:	

After completion of this course, the students will be able to

1. **Analyze** the deep, foundational interconnection between physical laws and information-theoretic limits.
2. **Reformulate** the physical postulates of quantum mechanics through an information-theoretic framework.
3. **Quantify** non-classical correlations in quantum systems and evaluate their physical ramifications.
4. **Design and evaluate** quantum systems and processes for practical information processing tasks.
5. **Apply and derive** information-theoretic tools to characterize, study, and solve novel problems in quantum systems.

Course topics: This is an advanced level course on quantum information science.

Unit 1: Overview of why the need of quantum information science: In this part, we will overview the need for quantum mechanics. Outline briefly what Shannon information theory means and why and how can we study information theoretical aspects of physical systems obeying quantum theory.

Unit 2: Information-theoretic formulations of the postulates quantum mechanics: In this part, we will study the postulates of non-relativistic quantum mechanics from an information-theoretic viewpoint. We will mainly describe postulates in the form of informational statements and rules. We will deal with both closed (unitary evolution) and open (noisy evolution) quantum systems. We will discuss standard theorem statements (e.g., Choi-Jamiołkowski isomorphism) after introducing quantum states, quantum channels, and quantum measurements that form the preliminaries of quantum mechanics.

Unit 3: Quantum correlations: In this part, we will discuss different kinds of quantum correlations that have no classical counterpart. We will mainly focus on quantum correlations like entanglement and nonlocal quantum correlations. We will discuss measures and tests to quantify or test these correlations. We may touch upon local hidden variable theories to derive Bell tests for nonlocal quantum correlations. These correlations are pertinent for the study of many-body quantum systems.

Unit 4: Information-theoretic tools for quantum physics: In this part, we will introduce and discuss some of the important tools that are employed widely in the domain of quantum information science. Some of these tools are trace-distance, relative entropy, entropy, fidelity, etc. We will discuss formulas and bounds for quantum entropy, quantum conditional entropy, quantum mutual information, quantum conditional mutual information. We will discuss quantum data-processing inequality and strong subadditivity.

Unit 5: We will apply the information-theoretic tools to study quantum systems and processes. We will inspect memory aspects of open quantum dynamics and implications on quantum correlations present in many-body quantum systems.

Unit 6: If time permits, we will apply information-theoretic tools to develop thermodynamic concepts for quantum processes.

Suggested textbooks:

1. Quantum Information Theory (Second Edition) by Mark M Wilde (<https://www.markwilde.com/qit-notes.pdf>)
2. Information for Physics 219/Computer Science 219: Quantum Computation by John Preskill (<https://www.preskill.caltech.edu/ph229/>)
3. Theory of Quantum Information by John Watrous (<https://cs.uwaterloo.ca/~watrous/TQI/>)
4. Quantum Information Meets Quantum Matter – From Quantum Entanglement to Topological Phase in Many-Body Systems (arXiv:2507.09229)
5. As the field is rapidly growing, we will make use of research papers as well for the course content.

Grading Plan :

(The table is only indicative)

Type of Evaluation	Weightage (in %)
Quiz-1	10
Mid Sem Exam	20

Quiz-2	10
End Sem Exam	30
Assignments	10
Project	10
Other Evaluation (Scribing notes)	10

Mapping of Course Outcomes to Program Objectives: (1 – Lowest, 2—Medium, 3 – Highest, or a ‘-’ dash mark if not at all relevant).

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3	PSO4
CO1	3	2	2	2	2								2	1	1	1
CO2	2	3	2	2	3								2	2	1	1
CO3	3	2	2	2	2								2	1	1	1
CO4	2	2	3	2	3								2	1	1	1
CO5	2	2	2	2	3								1	1	1	1

Teaching-Learning Strategies in brief (4-5 sentences):

The course integrates interactive lectures with mathematical derivations to build a strong conceptual foundation in quantum information science. Problem-solving tutorials, analytical assignments, and classroom discussions reinforce the information-theoretic formulation of quantum mechanics and quantum correlations. Students engage with case studies and computational exercises to analyze quantum systems and information-processing protocols, fostering both theoretical understanding and practical application.

Title of the Course : Readings from Hindi Literature
Faculty Name : Harjinder Singh
Course Code : HS1.401
Credits : 4
L - T – P : (24-0-0)
(L - Lecture hours, T-Tutorial hours, P - Practical hours)
Semester, Year : Monsoon 2026

Pre-Requisites : Ability to read and comprehend Hindi Assignments and exams may be done in English or any of the Indian languages comprehensible to the instructor.

Course Outcomes :

1. Greater interest and appreciation of literature in general
2. Reasonable amount of knowledge of contemporary writing.
3. Motivation to continue reading and writing in creative literature in future
4. Comparative understanding of literature in different languages
5. Elementary understanding of literary criticism

The action verbs to be used for writing the course outcomes can be found on slide 22 in the following presentation. You may remove this line and the following link after the course outcomes are formulated.

<https://intranet.iiit.ac.in/offices/static/files/HelponpreparationofCoursedescriptions-2022.pdf>

Course Topics :

1. History of Hindi literature – 2 L
 2. Fiction: Nirmal Varma, Bhishm Sahani, Manto, Gyanranjan, Kamaleshwar, Uday Prakash, Ismat Chughtai, etc. - 7-8 L.
 3. Poetry: Ajnyeya, Muktibodh, Nagarjun, Shamsheer Bahadur Singh, Raghuveer Sahay, Faiz Ahmed Faiz, Nand Kishore Acharya, Sarveshwar Dayal Saxena, Manglesh Dabral, Vinod Kr Shukla, Kumar Vikal, Gagan Gill, Katyayani, Anamika, etc. - 8-9 L,
 4. Play: Mohan Rakesh (Aadhe-adhoore) – 2 L.
 5. Non-fiction and criticism: (i) Pleasure readings; (ii) aesthetics versus committed literature; (iii) Sociology of literature; - 4L.
- (Please list the order in which they will be covered, and preferably arrange these as five to six modules.)

Preferred Textbooks: Works of writers available in IIIT library. Additional handouts may be given.

Reference Books : Same as above

E-book Links : Material available from KavitaKosh.org, rekhta.org, hindwi.org and hindisamay.com and other sites may be used.

Grading Plan : (The table is only indicative)

Type of Evaluation	Weightage (in %)
Mid Sem Exam	20
End Sem Exam	35
Assignments	25
Project	20

Mapping of Course Outcomes to Program Objectives: (1 – Lowest, 2—Medium, 3 – Highest, or a ‘-’ dash mark if not at all relevant). Program outcomes are posted at <https://intranet.iit.ac.in/offices/static/files/PEOs%2CPOs%26PSOs-ofAllProgrammes-Jan2023.pdf>

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3	PSO4
CO1																
CO2																
CO3																
CO4																
CO5																

Teaching-Learning Strategies in brief (4-5 sentences):

Title of the Course : Real Analysis
Faculty Name : Indranil Chakrabarthy + Lalitha V+ Uttam Singh
Course Code : MA4.101
L-T-P : 3-1-0.
Credits :4
(L= Lecture hours, T=Tutorial hours, P=Practical hours) Course: **MA**

Course Outcomes (COs) :

After completion of this course successfully, the students will be able to:

CO-1: Explain the definition of cardinality of sets and real numbers.

CO-2: Interpret and apply the concept of convergence of sequences and limit points of sequences.

CO-3: Employ convergence tests to determine the convergence of an infinite series.

CO-4: Analyze the continuity of functions and apply intermediate value theorem for continuous functions.

CO-5: Analyze the differentiability of functions and apply mean value theorems.

CO-6: Calculate the Riemann integral of functions by partitioning the domain and taking limits.

Unit 1 [6 hours]

Peano Axioms and Natural Numbers, Axiom of Choice and Set Theory, Russel's Paradox, Cardinality of Sets, Integers, Rationals, Real Numbers, Functions of Real Numbers.

Unit 2 [16.5 hours]

Sequences of Rational and Real Numbers, Convergence of Sequences, Limits of Sequences, Limit Laws, Suprema and Infima of Sequences, Limit Superior, Limit Inferior and Limit Points of Sequences, Some Convergence Tests of Sequences, Cauchy Sequences, Bounded and Monotone Sequences, Sub sequences and Bolzano-Weierstrass Theorem.

Finite and Infinite Series, Convergence and Absolute Convergence of Infinite Series, Series Laws, Rearrangements of Infinite Series, Some tests of Convergence of Infinite Series: Root Test, Ratio Test, Raabe's Test, etc.
Countable and Uncountable Sets, Summation on Countable Sets and Fubini's Theorem, Cantor's Theorem, Ordered Sets and Zorn's Lemma, Closure of Sets, Limit Points of a Set, Bounded and Unbounded Sets, Heine-Borel Theorem.

Unit 3 [16.5 hours]

Continuity of Function of Real Numbers, Maxima, Minima, and Maximum Principle for Functions of Real Numbers, Intermediate Value Theorem, Monotonic and Uniformly Continuous Functions, Limits at Infinity.

Differentiability of Functions, Differential Calculus, Local Maxima and Minima of Functions, Mean Value Theorems (Rolle's Theorem, Cauchy Mean Value Theorem, Lagrange's Mean Value Theorem), Indeterminate Forms and L'Hôpital's Rule

Partition of Sets, Upper and Lower Riemann Integrals, Riemann Sums, Some Integrable and Riemann non Integrable Functions, Two Fundamental Theorems of Calculus

References:

Textbook:

- Analysis I by Terrence Tao Companion Reference Books:
 - Principles of Mathematical Analysis by Walter Rudin
 - Introduction to Real Analysis by Robert G. Bartle and Donald R. Sherbert
 - A Basic Course in Real Analysis by Ajit Kumar and S. Kumaresan.
 - Introduction to Real Analysis by S. K. Mapa

	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PO 12	PSO 1	PSO 2	PSO 3	PSO 4
CO 1	2	3	3	3	2	1	2	2	2	3	1	3	3	3	2	3
CO 2	2	3	3	3	2	1	2	2	2	3	1	3	3	3	2	3
CO 3	2	3	3	3	2	1	2	2	2	3	1	3	3	3	2	3
CO 4	2	3	3	3	2	1	2	2	2	3	1	3	3	3	2	3
CO 5	3	3	3	3	2	1	2	2	2	3	1	3	3	3	2	3
CO 6	3	3	3	3	2	1	2	2	2	3	1	3	3	3	2	3

Teaching-Learning Strategies in brief (4-5 sentences):

The course materials will be delivered through a systematic set of lectures, assignments and tutorials. The lectures will be highly interactive, where students will be encouraged to participate and ask questions the class. In order to evaluate overall understanding of students, there will be short quizzes with multiple choice questions during the lectures. There will be one tutorial session per week and periodic set of assignments consisting of practice questions throughout the semester.

Title of the Course : Research in Information Security

Name of the Faculty : Srinathan Kannan + Ashok Kumar Das

Course Code : CS8.501

L-T-P : 3-1-0

Credits : 4

(L= Lecture hours, T=Tutorial hours, P=Practical hours)

Name of the Academic Program B.Tech. In CSE / M.Tech. in CSE/CSIS

1. Prerequisite Course / Knowledge:

Cryptography, Network Security, System Security, Programming Languages

2. Course Outcomes (COs) :

After completion of this course successfully, the students will be able to

CO-1: Demonstrateskills in solving research problems and critical thinking skills

CO-2: Demonstrate security protocols practically

CO-3: Analyse various techniques for security protocols against different potential attacks

CO-4: Demonstrate the knowledge ofFormal security verification using automated software validations tools

CO-5:Survey the literature in detail on existing security protocols to enable oneself to design, analyse and implement new security protocols

3. Mapping of Course Out comes (COs)with Program Outcomes (POs) and Program Specific Outcomes (PSOs) – Course Articulation Matrix

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3	PSO4
CO1	1	1	2	2	3	2	2	2	2	2	2	2	2	2	2	3
CO2	1	1	2	2	2	2	2	2	2	2	2	2	2	2	3	3
CO3	1	1	2	2	3	2	2	2	1	2	2	2	2	2	3	3
CO4	1	1	2	2	3	2	2	2	1	1	2	2	2	2	3	3
CO5	1	1	2	2	3	2	3	2	1	1	2	2	2	2	2	3

Note: Each Course Outcome (CO) may be mapped with one or more Program Outcomes (POs) and PSOs. Write '3' in the box for 'High-level' mapping, 2 for 'Medium-level' mapping, 1 for 'Low'-level' mapping

4. Detailed Syllabus:

- **Unit 1:** Elliptic-Curve Cryptography (ECC), Key management in hierarchical access control, Key management, user authentication and access control, Proxy signature
- **Unit2: Security in vehicular adhoc networks/Internet of Vehicles (IoV), Security in smart grid/smart home, Security in Cloud/Fog computing**
- **Unit 3:** Wireless Sensor Networks (WSNs) and Internet of Things (IoT)security
- **Unit 4:** Intrusion detection and prevention
- **Unit 5:** Blockchain and its security and privacy issues, Blockchain-based AI/ML security

Reference Books:

1. Top research papers (journals and conferences) from the IEEE Transactions, ACM Transactions, Elsevier, Springer, Wiley, etc.
2. William Stallings, “Cryptography and Network Security: Principles and Practices”, Pearson Education, 2010.
3. Bernard Menezes, “Network Security and Cryptography”, Cengage Learning, 2010.
4. Behrouz A. Forouzan, “Cryptography and Network Security”, Special Indian Edition, 2010.

5. Teaching-Learning Strategies in brief (4 to 5 sentences):

- * Recognition of interplay between theory and practice
- * Design of efficient and secure research problems
- * Various security analysis techniques against potential attacks
- * Automated software validations tools based formal security verification

6. Assessment methods and weightages in brief (4 to 5 sentences):

- In-Class Tests:20%
- Assignments:20%
- Research Project: 40% (including report and presentation)
- End Semester Examination:20%

Title of the Course : Retrofit of Existing Infrastructure

Faculty Name : Shubham Singhal

Course Code : CE1.621

Credits : 4

L - T - P : 3-1-0

(L - Lecture hours, T-Tutorial hours, P - Practical hours)

Name of the Academic Program: M.Tech. - Computer-Aided Structural Engineering

1. Prerequisite Course / Knowledge: Solid Mechanics, Reinforced Concrete Design, Earthquake Engineering

2. Course Outcomes (COs):

After completion of this course successfully, the students will be able to:

- CO 1: Explain the theory and concepts of structural retrofit and identify the damage mechanism and distress in buildings.
- CO 2: Propose suitable retrofit techniques depending upon the extent and type of damage and the type of building.
- CO 3: Quantify damage, stresses, and seismic parameters for structurally deficient and retrofitted buildings through computational methods.
- CO 4: Apply the concepts of solid mechanics and structural analysis in the analysis and design of retrofitted structural elements.
- CO 5: Conduct experimental investigations on structural elements retrofitted using textile or fibre-reinforced composite materials.

3. Mapping of Course Outcomes (COs) with Program Outcomes (POs) and Program Specific Outcomes (PSOs) – Course Articulation Matrix

Note: Each Course Outcome (CO) may be mapped with one or more Program Outcomes (POs) and PSOs. Write '3' in the box for 'High-level' mapping, '2' for 'Medium-level' mapping, '1' for 'Low-level' mapping; and '-' for not relevant.

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PSO1	PSO2	PSO3	PSO4
CO1	3	2	2	1	1	1	-	-	-	1	2	-	2	2
CO2	3	2	2	2	2	2	-	-	-	2	3	1	2	3
CO3	3	3	3	1	2	1	-	-	-	3	3	3	3	2
CO4	3	3	3	2	3	1	-	-	-	2	3	3	3	3
CO5	3	3	3	2	3	2	1	2	1	3	3	1	2	3

4. Detailed Syllabus:

Unit 1: Introduction: Retrofit and its concept; Basic principles of seismic retrofit; Requirement of retrofitting; Advantages and challenges; Retrofit materials; Difference between repair, retrofit, strengthening and rehabilitation; Overview of retrofit and rehabilitation of civil infrastructure; Standards and guidelines

Unit 2: Retrofitting Techniques: Local and global methods of retrofit; Retrofit of super-structure and sub-structure; Strengthening of existing components - RC jacketing, steel jacketing, shotcrete; Addition of new components - shear walls, frames, bracing; Use of textile reinforced composites in retrofitting - poly-propylene band, industrial geo-grid, fibre reinforced polymers (FRP) laminates/sheets/bars; Engineered cementitious composites (ECC) using steel and textile fibres; Introduction to supplemental energy dissipation and base isolation, shape memory alloys; Retrofit of foundation system - micro-piles, non-woven geo-textiles, textile bracing for soil reinforcement; Modelling of various retrofit techniques in software.

Unit 3: Evaluation of Retrofitted Building: Qualitative and quantitative methods of seismic evaluation; Computational dynamic analysis of structurally deficient and retrofitted building; Comparison between structurally deficient and retrofitted building; Check for capacity/drift and strength; Codal provisions and acceptance criteria.

Unit 4: Experimental Investigations and Case Studies: Tensile testing of textile materials and composites applicable for retrofit; Compressive tests on textile-reinforced concrete columns; Bending tests on beams with textile/fibre reinforced concrete; Selected case studies for different retrofit techniques.

Reference Books:

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- Tomazevic, M., “Earthquake Resistant Design of Masonry Buildings”, Imperial Colleges Press (2000).
 - Amarnath Chakrabarti et al. “Handbook on Seismic Retrofit of Buildings”, Indian Building Congress.
 - “NEHRP Handbook for the Seismic Evaluation of Existing Buildings”, Federal Emergency Management Agency, Building Seismic Safety Council, Washington, D.C. FEMA 178.
 - “Building Seismic Safety Council”, Federal Emergency Management Agency, Washington, D.C, FEMA 356, 2000, FEMA 440 / ATC 55, 2005, FEMA 310.
 - <https://nehrpsearch.nist.gov/static/files/FEMA/PB2009105376.pdf>
-

5. Teaching-Learning Strategies in brief (4 to 5 sentences):

Lectures on theoretical concepts and principles, followed by analysis and design examples using manual approach as well as software. Students will be encouraged to apply concepts taught in class to solve problem statements through software and experiments. Project-based learning through application of computer software and experimental investigations.

6. Assessment methods and weightages in brief (4 to 5 sentences):

Assignments and Quizzes - 30%
 Mid Semester Exam - 20%
 End Semester Exam - 30%
 Project – 20%

Title of the Course	Robotics: Dynamics and Control
Faculty Name -	Antony Thomas
Name of the Program:	B.Tech in Electronics and Communication Engineering
Course Code:	EC4.401
Credits:	4
L - T - P:	3-1-0
(L - Lecture hours, T-Tutorial hours, P - Practical hours)	
Semester, Year	: Monsoon, 2026

Pre-Requisites : Must be familiar with Linear Algebra, Differential Calculus, and Numerical methods for solving system of nonlinear equations.

Course Outcomes :
 After completion of this course successfully, the students will be able to

CO-1 Describe coordinate frames, spatial transformations, and mathematical representation of joints and links.

CO-2 Describe the kinematics and dynamics of rigid body systems - serial manipulator.

CO-3 Formulate and analyze the forward and inverse model using analytical and numerical methods.

CO-4 Develop software programs to generate trajectory and control the robot to track the commanded trajectory.

CO-5 Apply the learned robot model for joint space control and end-effector control of the manipulator and analyze them.

Course Topics :

Unit 1: Introduction to robotics and rigid body motion – robot structure and workspace, transformation matrices, mathematical representation of joints and links, mobility analysis and constraints

Unit 2: Manipulator forward and inverse kinematics – position analysis and velocity analysis, singularities

Unit 3: Robot Dynamics – Euler-Lagrangian formulation and simulation of equations of motion

Unit 4: Trajectory planning and generation

Unit 5: Robot control – joint space control and operation space control

Preferred Textbooks:

1. J.J. Craig, Introduction to Robotics: Mechanics and Control, 3rd Ed., Prentice Hall, 2004.
2. Kevin M. Lynch and Frank C. Park, Modern Robotics, Cambridge Press 2017.

Reference Books :

1. M. Spong, S. Hutchinson, M. Vidyasagar, Robot Dynamics and Control, John Wiley & Sons, 2004.
2. R.M. Murray, Z. Li, S.S. Sastry, A Mathematical Introduction to Robotic Manipulation, CRC Press, 1994.
3. Ashitava Ghosal, Robotics: Fundamental Concepts and Analysis, Oxford University Press, 2006.

Grading Plan : (The table is only indicative)

Type of Evaluation	Weightage (in %)
Quiz-1	10
Quiz-2	10
End Sem Exam	40
Assignments	30
Other Evaluation (Seminar presentation-critique a research paper and submit a one-page report)	10

Mapping of Course Outcomes to Program Objectives: (1 – Lowest, 2—Medium, 3 – Highest, or a ‘-’ dash mark if not at all relevant). Program outcomes are posted at

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3	PSO4
CO1	3	3	2	2	2	1	1	2	1	2	1	2	3	1	2	3
CO2	3	3	2	3	2	2	2	2	2	3	2	3	3	1	3	3
CO3	3	3	3	3	3	1	1	2	2	3	2	3	3	2	3	3
CO4	3	3	2	2	2	1	2	2	3	2	2	2	3	3	2	3
CO5	3	3	3	3	2	3	2	2	2	3	2	3	3	3	2	3

Teaching-Learning Strategies in brief (4-5 sentences):

The course focuses on imparting knowledge, developing problem-solving skills, and motivating students for continued learning on various aspects of robot dynamics and control. Each student must critique a research article related to advanced topics of mechanisms and robotics to hone their knowledge in the current state of the art and presentation skills by giving a seminar. Exams will include similar problems encountered in the assignments and cover the content from the lectures.

Title of the Course	: Science I
Name of the Faculty	: Prabhakar Bhimalapuram + Samyadeep Bhattacharya
Name of the Academic Program	: B. Tech. (CSE)
Course Code	: SC1.110, SC1.110a and SC1.110b
L-T-P:	3-1-0
(L= Lecture hours, T=Tutorial hours, P=Practical hours)	
Credits	: 4

1.Prerequisite Course / Knowledge: NA

2.Course Outcomes (COs):

Outcomes of the Second Half (Introduction to Biology):

After completion of this course successfully, the students will be able to

CO-1: Analyze the aims, methodology of science and technology, and their impact on society

CO-2: Explain Special Theory of Relativity and compute its consequences for typical scenarios of relevance

CO-3: Demonstrate familiarity with Lagrangian and Hamiltonian formulations of mechanics, by formulating the equations of motion from basic principles for mechanical systems

CO-4: Explain connections between thermodynamics and statistical mechanics and their use in modern chemical computations

Co-5: Infer the stability of molecules using the concepts of hybridization and molecular orbital theory

CO-6: Recognize the role of symmetry in nature

CO-7: Demonstrate problem solving skills up to a level that allows application to research topic of their interest

3.Mapping of Course Outcomes (COs) with Program Outcomes (POs) and Program Specific Outcomes (PSOs) – Course Articulation Matrix

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3	PSO4
CO1	2	2	1	2	2	2	2	1	1	2	1	2	2	1	1	1
CO2	2	2	2	2	1	1	2	1	1	1	1	2	0	3	2	1
CO3	2	2	1	2	1	1	2	1	1	1	1	2	0	3	2	1
CO4	2	2	1	2	1	1	2	1	1	1	1	2	1	3	2	1
CO5	2	2	1	2	1	1	2	1	1	1	1	2	1	3	2	1
CO6	2	2	1	2	1	1	2	1	1	1	1	2	2	2	2	2
CO7	2	2	1	2	1	1	2	1	1	1	1	2	1	1	1	2

Note: '3' in the box for 'High-level' mapping, 2 for 'Medium-level' mapping, 1 for 'Low'-level' mapping

4.Detailed Syllabus:

Unit 1: Mathematical modeling in sciences, (i) geometry and linear algebra, (ii) change and calculus and (iii) chance and probability. Simple models can have complicated behavior: logistic map demonstrates deterministic chaos

Unit-2: Forms in nature. Scales of length, time and energy in nature.

Unit 2: Special theory of relativity: postulates, Lorentz Transformation, Length Contraction, Time dilation, Doppler effect, relative velocity determination, twin paradox, relativistic momentum and energy. Space time graphs, and relativity of simultaneity.

Unit 3: Review of Newtonian Mechanics and its difficulties / failures. Introduction to Lagrangian and Hamiltonian formulations, and application to mechanical problems.

Unit 5: Need for Quantum Mechanics. Schrodinger equation for time-dependent and time-independent scenarios. Application to atoms and molecules; provide qualitative picture of orbital hybridization to explain the molecular structures

Unit 6: Review of Thermodynamics and introduction Statistical Mechanics and applications to problems of relevance. Lasers. Fermi statistics, bandgap in semiconductors.

Reference Books:

1. "Concepts in Modern Physics" A Beizer (6th Edition)
2. "Classical dynamics of particles and systems" by Stephan Thornton and Jerry Marion (5th edition)
3. "Physical Chemistry" Atkins and de Paula (8th Edition)

5.Teaching-Learning Strategies in brief (4 to 5 sentences):

The objective of the course is to give the CSE/ECE students a good understanding of the concepts in Modern Physics and modern chemistry. To familiarize the students with available web-based resources, and problem solving (whenever possible with scientific programming).

6.Assessment methods and weightages in brief (4 to 5 sentences):

Assignments – (20%), Class notes (10%) Preannounced and surprise In-class quizzes (25%), End semester exam (35%)

Title of the Course: Science Lab -I
Name of the Faculty: Chittaranjan Hens, Tapan K Sau
Course Code: SC4.110
L-T-P 0-0-3
Credits 2
 (L= Lecture hours, T=Tutorial hours, P=Practical hours)
Name of the Academic Program Dual Degree CNS...

1.Prerequisite Course / Knowledge:

None

2.Course Outcomes (COs):

After completion of this course successfully, the students will be able to

CO-1: Setup and perform optics experiments to measure properties of material like optical rotation, wavelength of monochromatic light etc

CO-2: Setup and perform chemistry experiments to measure properties like pH, concentration of chemicals and

CO-3: Perform physical measurements to measure properties like frequency of oscillator, young's modulus etc.

3.Mapping of Course Outcomes (COs) with Program Outcomes (POs) and Program Specific Outcomes (PSOs) – Course Articulation Matrix

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3	PSO4
CO1	3	1	2	2	1	1	1	2	1	3	1	3	1	2	1	2
CO2	2	1	2	2	1	1	1	2	1	3	1	2	1	2	1	2
CO3	2	1	2	2	2	2	1	2	1	3	1	2	2	2	2	1

Note: Each Course Outcome (CO) may be mapped with one or more Program Outcomes (POs) and PSOs. Write '3' in the box for 'High-level' mapping, 2 for 'Medium-level' mapping, 1 for 'Low'-level' mapping

4.Detailed Syllabus:

- Unit-1: Determination of the Specific Rotation of Sucrose and Composition of Sucrose Solution by a Polarimeter
- Unit-2: Potentiometric Titration of a Mohr Salt Solution with Standard $K_2Cr_2O_7$
- Unit-3: Kinetic Study of the Decomposition of H_2O_2 in the Presence of $FeCl_3$ Solution and the Effect of the Catalyst on the Rate Constant
- Unit-4: Verification of the Beer-Lambert's Law with a given solution and the determination of the concentration of a solution
- Unit-5: Determination of pKa of a Weak Acid by pH-Metric Titration Method
- Unit-6: Newtons Ring Method to measure Radius of Curvature of Plano-Convex Lens

7. Unit-7: Determination of Young's Modulus of Material of a Beam by Method of Flexure
8. Unit-8: To determine the number of rulings per unit length of a diffraction grating
9. Unit-9: Stewart and Gee's Method for Determining the Magnetic Field of the Earth
10. Unit-10: To Measure Slit-Width and the Separation between two Slits of Double Slit, by Observing Diffraction and Interference Fringes and to Compare them by Microscopic Measurement

Reference Books:

1. Introduction to Protein by Branden and Tooze
2. Fundamentals of Biochemistry by Voet, Voet and Pratt

5.Teaching-Learning Strategies in brief (4 to 5 sentences):

Course is a hands-on laboratory course requiring student to perform experiment after showing some prerequisite preparation. Then, the student's setup of the experiment is checked, before allowing to proceed to experimental measurements. After completion of all measurements, student will perform required calculations for drawing the conclusions. Finally, a viva voce examination is conducted for the experiment to check a broad level knowledge of the experiment.

6.Assessment methods and weightages in brief (4 to 5 sentences):

☐Laboratory record- 40%

☐Quiz - 30%

☐Exams - 30%

Title of the Course	: Scientific Visualisation
Faculty Name(s)	: Raghavendra GS
Name of the Program	: Elective
Course Code	: CS5.501
Credits	: 4
L - T - P	: 3:1:0
(L - Lecture hours, T-Tutorial hours, P - Practical hours)	
Semester, Year	: Monsoon 2026
Pre-Requisites	: DSA, Linear Algebra, Computer Graphics (optional)

Course Outcomes :

CO-1: Understand the geometric nature of the domain, various ways to represent the domain, differentiate between scalar, vector, and tensor fields both in continuous and piecewise linear cases.

CO-2: Understand classical pipelines and design new pipelines to analyse and visualise scientific data, both in theory and practice.

CO-3: Understand computational topology based methods in detail to analyse and visualise scientific data.

CO-4: Extension of these methods to high-dimensional data, applications to learning.

CO-5: Ability to implement some of the pipelines along with expertise in tools like Paraview, VTK, TTK etc.

Course Topics:

Module I: Introduction

Introduction to scientific visualisation. Types of data (scalar, vector, tensor, time-varying, high dimensional). Domain representations.

Module II: Classical methods

Classical methods to visualise

1. Scalars (slicing, iso-surfaces, volume rendering).
2. Vectors (glyphs, stream-lines, streak-lines, line integral convolution).
3. Tensors (Eigen analysis, super quadric glyphs).

Module III: Topology based methods

Motivation, definition, algorithms, and applications of

1. Voronoi diagrams, Delaunay triangulations, alpha/Cech/Rips complexes,
2. Simplicial homology, persistent homology,
3. Merge trees, contour trees, and Reeb graphs,
4. Morse, and Morse-Smale complexes, extremum graphs.

Module IV: Vis for Learning and Learning for Vis

Mapper graphs, visualising loss landscapes, topological inference, applications to high dimensional data, persistence kernels, biomolecular visualisation.

Preferred Textbooks:

1. The Visualisation Handbook, Charles Hansen and Chris Johnson.
2. Computational topology: An introduction, Herbert Edelsbrunner and John Harer.
3. Computational topology for data analysis, Tamal Dey and Yusu Wang.
4. Recent and current literature.

Reference Books :

E-book Links (Preprints):

1. <https://my.eng.utah.edu/~cs6635/VH.pdf>
2. <https://webhomes.maths.ed.ac.uk/~v1ranick/papers/edelcomp.pdf>
3. <https://www.cs.purdue.edu/homes/tamaldey/book/CTDAbook/CTDAbook.pdf>

Grading Plan :

(The table is only indicative)

Type of Evaluation	Weightage (in %)
Quiz-1	5%
Mid Sem Exam	25%

Quiz-2	5%
End Sem Exam	
Assignment	5%
Project	45%
Term Paper	
Other Evaluation (evaluation of vis systems)	15%

Mapping of Course Outcomes to Program Objectives: (1 – Lowest, 2—Medium, 3 – Highest, or a ‘-’ dash mark if not at all relevant). Program outcomes are posted at

<https://intranet.iiit.ac.in/offices/static/files/PEOs%20CPOs%26PSOs-ofAllProgrammes-Jan2023.pdf>

	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PO 12	PSO 1	PSO 2	PSO 3	PSO 4
CO 1	3	1	1	1	2	-	-	-	-	2	-	1	-	-	-	-
CO 2	1	3	1	1	2	-	-	-	-	2	-	-	-	-	-	-
CO 3	1	3	1	1	2	-	-	-	-	2	-	-	-	-	-	-
CO 4	1	3	3	3	3	1	-	3	3	2	-	2	3	2	3	-
CO 5	3	3	3	3	3	1	-	3	3	2	-	3	3	3	2	3

Teaching-Learning Strategies in brief (4-5 sentences):

This course would be an interdisciplinary course, target audience being students with varying interests and backgrounds, theory, systems, data sciences, and natural sciences (CVIT, CSTAR, RRC, CCNSB). The learning will be tested through a combination of assignments, exams and a project where the students (may be a small team) will build a visualisation system from scratch. They'll also be tested for their understanding of existing systems and tools.

Title of the Course : **Service Design**
Name of the Faculty : Raman Saxena
Course Code : PD1.402
L-T-P : 1.5-0-3
Credits : 2
(L=Lecture hours, T=Tutorial hours, P=Practical hours)
Semester, Year : 3rd Sem – Year 2

1. Prerequisite Course / Knowledge

- **UX Research Methods:** Foundational understanding of interviews, ethnography, and persona building.
- **Systems Thinking:** Ability to see beyond a single touchpoint.
- **Business Basics:** Familiarity with value propositions and stakeholder mapping.

2. Course Objectives & Outcomes (COs)

1. To transition students from "feature-focused" thinking to "ecosystem" thinking.
2. To master tools for visualizing invisible service layers (backstage vs. frontstage).
3. To learn how to design for "omnichannel" experiences where digital and physical worlds collide.

Learning Outcomes: Upon completion, students will be able to:

- **Analyse** complex service ecosystems using Stakeholder Maps and GIGA-mapping.
- **Synthesize** user research into Service Blueprints that identify operational bottlenecks.
- **Prototype** service "moments" through role-playing and experience prototyping.
- **Measure** service success using holistic metrics (Customer Effort Score, Service Recovery Rate).

3. Mapping of Course Outcomes (COs) with Program Outcomes (POs) and Program Specific Outcomes (PSOs) – Course Articulation Matrix

	P O 1	P O 2	P O 3	P O 4	P O 5	P O 6	P O 7	P O 8	P O 9	PO 10	PO 11	PS O1	PS O2	PS O3	PS O4
CO 1	1	2	2	2	2	3	2	2	3	3	3	1	1	2	1
CO 2	3	3	3	3	3	2	2	2	3	3	3	1	2	2	1
CO 3	2	2	2	2	3	2	1	2	3	3	2	1	2	1	2

4. Detailed Syllabus

Week	Unit Topic	Key Concepts / Class Activities
1	The Service Logic	Goods-Dominant vs. Service-Dominant Logic; The 4 Ps to the 7 Ps; Introduction to the "Service Encounter."
2	Mapping the Ecosystem	Stakeholder Mapping; Value Exchange Maps; Identifying the "Service Constellation" (Partners and Providers).
3	The Journey & The Blueprint	The Core Tool: Differentiating between Customer Journey Maps and Service Blueprints. Frontstage, Backstage, and Support Processes.
4	Designing Touchpoints	Omnichannel orchestration; Designing for the "Wait"; Managing Service Evidence (Physical and Digital cues).
5	Service Prototyping	Experience Prototyping; Service Staging; Role-playing "Critical Incidents"; Testing for Service Recovery.
6	Service Implementation	Integrating SD with Agile Product Management; Scaling services; Measuring Service ROI and impact.

5. Reference Books & Case Studies

Books:

- *This is Service Design Doing* – Marc Stickdorn, et al. (The "Bible" of SD practitioners).
- *Service Design: From Insight to Implementation* – Andy Polaine, Lavrans Løvlie, and Ben Reason.
- *The Design of Business* – Roger Martin.

Case Studies:

- **Healthcare:** Redesigning the Mayo Clinic patient experience (Systems-level change).
- **FinTech:** How Neobanks (like Jupiter or Monzo) use Service Design to disrupt traditional banking.
- **Urban Mobility:** The orchestration of Uber/Ola—mapping the driver, rider, and algorithm interaction.

6. Teaching-Learning Strategies

- **The "Shadowing" Assignment:** Students must pick a local service (e.g., a local hospital, a metro station, or a cloud kitchen) and shadow the process for 3 hours.
- **Live Lab:** Classes function as design studios. We move from theory to "mapping on walls" within 30 minutes.
- **Role-Play Simulation:** Testing service failures (e.g., "The server is down," "The delivery is late") to design service recovery protocols.

7. Assessment Methods and Weightages

Component	Weightage	Description
1. Service Audit (Individual)	20%	Deconstruct an existing service using a Stakeholder Map.
2. The "Gaps" Analysis (Group)	20%	Identifying failures in a real-world service journey.
3. Final Service Blueprint	50%	A comprehensive blueprint and a "future-state" prototype for a new service venture.
4. Class Participation	10%	Classroom activities and assignments

Title of the Course:	Signal Detection and Estimation Theory
Name of the faculty :	Praful Mankar + Arti Yardi
Course Code:	EC5.406
L-T-P:	3-1-0
Credits:	4
Course :	ECE
Name of the Academic Program	<u>B. Tech. in Electronics and Communication Engineering</u>

Prerequisite Course / Knowledge: Probability Theory and Random Processes

Course Outcomes (COs):

After completion of this course successfully, the students will be able to..

CO-1: Describe the various detection methods for detecting/classifying the deterministic/random signals with perfect or statistical knowledge of their parameters.

CO-2: **Discuss** the various estimation methods for estimating the parameters of linear and non-linear signal models in the presence of Gaussian and non-Gaussian noise.

CO-3: Analyze and design an optimal detector for a given false alarm rate to detect deterministic/random signals.

CO-4: Analyze and design a minimum variance unbiased estimator, if it exists, for estimating the parameters of a signal.

CO-5: Implement and perform numerical analysis of the estimation and detection methods using Matlab.

CO-6: Apply a suitable method for the estimation/detection problems in the diverse engineering fields.

Mapping of Course Outcomes (COs) with Program Outcomes (POs) and Program Specific Outcomes (PSOs)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3	PSO4
CO1	3	2	0	0	0	0	0	0	0	1	0	2	0	2	1	0
CO2	3	2	0	0	0	0	0	0	0	1	0	2	0	2	1	0
CO3	3	3	3	2	0	0	0	0	0	0	0	2	0	3	2	1
CO4	3	3	3	2	0	0	0	0	0	0	0	2	0	3	2	1
CO5	3	2	2	1	0	0	0	0	0	0	0	1	0	2	1	1
CO6	3	2	2	1	0	0	0	0	0	0	0	1	0	2	3	2

Note: '3' in the box for 'High-level' mapping, 2 for 'Medium-level' mapping, 1 for 'Low'-level' mapping

Detailed Syllabus:

Unit 1: Estimation Theory - Minimum Variance Unbiased (MVU) Estimation, Cramer Rao lower bound, Fisher Information, Neyman-Fisher factorization, Sufficient and complete test statistics, Rao-Blackwell-Lehmann-Scheffe theorem.

Unit 2: Estimation methods - Best linear unbiased estimation (BLUE), Least square estimation (LSE), Maximum likelihood estimation (MLE), Bayesian Approach, Numerical methods - Newton Raphson and Expectation maximization (EM) methods.

Unit 3: Detection Theory - Hypothesis testing, Neyman-Pearson (NP) theorem, Likelihood ratio test (LRT), Receiver operating characteristic (ROC), Minimum probability of error, Bayes Risk, Minimum Bayes risk detector, MAP detector.

Unit 4: Detection methods—Detection of deterministic signals - Matched filter for WGN and non-WGN, Binary and M-array signal detection using matched filter.

Detection of random signals - Estimator-correlator and linear model;

Detection of deterministic signal with unknown parameters - Composite hypothesis testing, Generalized LRT (GLRT), Bayesian approach, Rao test, Wald test.

Reference Books:

1. **Steven M. Kay, *Fundamentals of Statistical Signal Processing: Estimation Theory*, Vol. 1, Prentice-Hall, 1993**
2. **Steven M. Kay, *Fundamentals of Statistical Signal Processing: Detection Theory*, Vol. 2, Prentice-Hall, 1998**

Teaching-Learning Strategies in brief

This course includes the topics on theoretical understanding and the optimal designs of the detection and estimation methods. The lectures are designed to teach complex theoretical concepts using simplistic examples while assuming that students have prerequisite knowledge in probability theory and random processes. The tutorials are focused on applying estimation/detection methods learned in class to more complex signal processing and communication engineering problems. The grading plan of this course includes **one mid semester exam and one end semester exam** along with the homework assignments and term paper presentation. While **50%** of the weightage is given for the **mid semester and end semester exams**, the remaining weightage is reserved for the term paper presentation

and the homework assignment for engaging students in research-oriented thinking. The assignments problems are designed to compel students to creatively apply the complex concepts learned in the class for the designing optimal estimation/detection methods for various problems. Besides, the assignments also include the MATLAB/Python programming problems for implementing some of the estimation methods learned in the class. Students in the group of two are encouraged to choose their term paper presentation topics based on the seminal research articles on estimation and detection theory and its applications.

Assessment methods and weightages in brief

- Home Assignments: 30%
 - Term Paper: 30%
 - Mid Semester Exam: 20%
 - End Semester Exam: 30%
-

Title of the Course:	Signal Processing
Course Code:	EC5.201
Name of the Faculty:	Santosh Nannuru
L-T-P	3-1-3
Credits	5
Name of the Academic Program	B. Tech. in ECE

Prerequisite Course / Knowledge:

Should have taken the course Network Signals and Systems.
A prior knowledge of calculus and complex numbers is required.

Course Outcomes (COs):

After completion of this course successfully, the students will be able to..

CO-1: Describe continuous-time and discrete-time signals using various representations

CO-2: Apply various transforms including Fourier transform, DTFT, and Z-transform to study signals and systems

CO-3: Apply sampling theorem to do analog-to-digital conversion of signals, perform ideal and non-ideal reconstruction of signal from its samples

CO-4: Examine computational complexity of efficient DFT implementations using FFT

CO-5: Design digital filters with specified requirements to process signals

CO-6: Analyze systems and real-world signals using signal processing tools in MATLAB software

CO-7: Analyze a signal processing application or problem by reading research papers and performing simulations as part of the course project

Mapping of Course Outcomes (COs) with Program Outcomes (POs) and Program Specific Outcomes (PSOs)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3	PSO4
CO1	3	3	1	1	2	-	-	-	-	-	-	1	-	3	-	-
CO2	3	3	2	2	2	-	-	-	-	-	-	1	1	3	-	-
CO3	3	3	3	2	2	-	-	-	-	-	-	1	-	3	1	1
CO4	3	3	3	3	2	-	-	-	-	-	-	1	-	3	1	1
CO5	3	3	3	3	2	-	-	-	-	-	-	1	1	3	1	1
CO6	3	3	3	3	3	-	-	-	3	2	-	1	1	3	1	1
CO7	3	3	3	3	3	-	-	-	3	2	2	1	-	3	1	1

Detailed Syllabus:

Unit 1: Fourier transform (FT) of continuous-time signals, analysis of linear and timeinvariant (LTI) systems using Fourier transform

Unit 2: Sampling and reconstruction of bandlimited signals, analog-to-digital conversion, aliasing, quantization

Unit 3: Analysis of discrete-time signals and systems using Fourier transform (DTFT) and Z-Transform

Unit 4: Discrete Fourier transform (DFT) for finite length sequences, efficient implementation of DFT using radix-2 fast Fourier transform (FFT) algorithms

Unit 5: Digital filter design, techniques for FIR and IIR filter design

Reference Books:

1. Signals and Systems by A.V. Oppenheim, A.S. Willsky and S.H. Nawab (2015)
2. Digital Signal Processing: Principles, Algorithms and Applications by J.G. Proakis and D. Manolakis (2007)
3. Digital Signal Processing: A Computer Based Approach, S.K. Mitra (2013)
4. Principles of Signal Processing and Linear Systems, B.P. Lathi (2009)

Teaching-Learning Strategies in brief (4 to 5 sentences)

Lectures are used to explain the core concepts in signal processing and work out a few problems. Detailed handwritten notes are shared along with book sections and practice problems. A short question is posted at beginning of class to gauge understanding of previous lecture. Tutorials are used mainly for doubt clarifications and problem solving. Assignments are given to promote application of concepts to difficult problems. The weekly lab sessions supplement the course lectures with MATLAB software-based signal analysis which are evaluated through short viva. The course project exposes students to advanced concepts and real-world applications in the domain. The lab sessions and final course projects are done in teams of two to encourage collaborative problem solving and team participation.

Assessment methods and weightages in brief (4 to 5 sentences):

Continuous evaluations:

- Quizzes: 30%

- Assignments: 15%
 - Lab viva and evaluations: 20%
- Comprehensive evaluation:
- Project: 15%
 - End Exam: 20%

Title of the Course : Social Science Perspective on Human-Computer Interaction

Course Code :CS9.425
 Faculty Name : Nimmi Rangaswamy
 Name of the Academic Program: B. Tech. in CSE
 L-T-P : 4-0-0.
 Credits : 4
 (L= Lecture hours, T=Tutorial hours,
 P=Practical hours)

Prerequisite Course / Knowledge:

UG3 and above – no other prerequisite knowledge

2. Course Outcomes (COs)- After completion of this course successfully, the students will be able to do the following”

CO-1. Develop understanding of the fundamentals of Human Computer Interaction [HCI] and Human sciences

CO-2. Understand and apply main theoretical foundations of HCI from a human centric perspective

CO-3: Grasping concepts and application of Design in HCI systems

CO-4. Deep analysis of case studies of HCI systems

CO-5. Develop a research project around a HCI platform, system or theory

3.Mapping of Course Outcomes (COs) with Program Outcomes (POs) and Program Specific Outcomes (PSOs) – Course Articulation Matrix

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PSO1	PSO2	PSO3	PSO4
CO1	2	2	3	3	3	1	1	1	2	3	3	2	2
CO2	2	2	3	3	3	1	1	1	2	3	3	2	2
CO3	3	3	3	3	2	1	1	1	2	3	3	2	2
CO4	3	3	3	3	3	2	3	3	1	2	2	2	2
CO5	3	3	3	3	2	1	1	2	3	3	3	2	2

‘3’ in the box denotes ‘High-level’ mapping, 2 for ‘Medium-level’ mapping, 1 for ‘Low’-level’ mapping

Course Structure in Detail

Overview of Course

Quote: “A sushi restaurant puts sensors on its plates to assess, in real time, what’s being eaten so it can adjust its food offerings” [Goodman, The Atomic Age of Data, 2015] Radically different ways of interacting with computationally based systems are possible, ranging from the visual [surfaces, input devices] to the invisible [sensor technologies, backend processors] and importantly social [which means non-technological] affectations triggering diverse ways of interfacing with technology. Human-Computer Interaction [HCI] is a vision for a world of interconnected devices, that have acquired smartness due to computing power. As computational technologies continue to ‘disappear’ and merge with the physical world, becoming increasingly tangible, embedded and embodied in a range of environments, architectures and artifacts, new research agendas and design approaches are called for [Nansen et al, 2014].

Broad Objectives:

To introduce Human-Computer Interaction as an inter-disciplinary domain of study to students of Engineering and the Social Sciences

To bring a social perspective and the importance of lived contexts in the framing and understanding of man-machine interaction

To get a grasp of the theoretical and applied frameworks supporting the domain of HCI

Importantly, to introduce the idea of cross-fertilisation of academic domains, especially computer sciences and humanities to originate Human-Computer Interaction as a fertile research and academic science

COURSE TOPICS/OUTLINE/CONTENT

Introducing seminal topics and key concepts: 10 Hours

This course is an introduction to the field of Human-Computer interaction research with a focus on ‘human’ and how the HCI domain interfaces with the social sciences. The course begins with a selection of seminal work that establish the HCI domain: interactive systems/techniques, design and user interfaces. We will then move on to topics including social and context aware computing, design research and evaluation methods.

Role of Objects in the Social sciences: 4 Hours

The course will also present a perspective based on the importance and role of objects in social relations. We situate this work in relation to a conceptual understanding of objects and social relations, suggest effective methodological and theoretical tools to study of a more object-centered sociality and suggest design opportunities to make better products.

Idea and application of Design in HCI: 6 Hours

The course will center on the processes and challenges of ideating, designing and evaluating technologies as products, their usability and immersion into the social contexts of users. We will study contextual design as a field that emerged in response to the challenges of designing for context and usability.

Context and Mediation in HCI systems- 6 Hours

Another important strand in this course will dwell on the sociological aspects of HCI and explore the ‘mediation’ of technology use by a range of contextual situations: socio-cultural obligations, habits, values, infrastructure, material objects and not in the least family, kinship and human bonds. Some examples of the above are:

Case Studies- 14 Hours

Understanding social interactions with a webcam as an important new development in communication interfaces and its widespread adoption in the real world supporting family relationships, business work flows and social networking.

A deep look at social networking as everyday HCI- Facebook; Twitter; Messaging applications

Another example will be looking at technologies driven by data science, like mobile marketing analytics, and their consequences for society

A third example will be studying real world application of big data to social situations: real time traffic; real world geographic navigation; geo-location-based services [food delivery; friendship; dating]; Consumer-centric health care services [monitoring parameters; precision medicine; Health care platforms]

A close look at the impacts of peer to peer sharing platforms [Uber, AirBnB]

This class has no pre-requisite requirements and open to students from any background. Students are expected to do all the readings. Students will be evaluated with a quizzes, research project design, research report, and 2 class presentations. The students will also be evaluated on the ability in engaging with and comprehending the course readings in classroom discussions. The quiz/test and the research projects will be based on the class lectures and readings assigned for the course

PREFERRED TEXTBOOKS:

***REFERENCE BOOKS:**

Lucy Suchman Human-machine reconfigurations, Cambridge University Press

Norman, D. A. (1990). *The design of everyday things*. New York: Doubleday.

Miller, D and Sinanan, J, *Webcam*, Polity Press

Sterling, B. *The Epic Struggle Of The Internet Of Things*, Moscow: Strelka Press

Rogers, Y. *HCI Theory: Classical, Modern, and Contemporary*. Morgan & Claypool

Blomberg, J., Burrell, M., and Guest, G. *An Ethnographic Approach to Design*, Human-Computer Interaction Handbook, L. Erlbaum Associates Inc. Hillsdale, NJ, USA

***REFERENCE ARTICLES:**

Bell, G., Blythe, M., and Sengers, P. 2005. Making by Making Strange: Defamiliarization and the Design of Domestic Technology. *ACM Trans. Computer-Human Interaction*, 12(2), 149-173.

Dourish, P. 2006. Implications for Design. *Proc. ACM Conf. Human Factors in Computing Systems CHI 2006* (Montreal, Canada), 541-550.

O'Brien, J., Rodden, T., Rouncefield, M., and Hughes, J. 1999. At Home with the Technology: An Ethnographic Study of a Set-Top Box Trial. *ACM Trans. Computer-Human Interaction*, 6(3), 282- 308.

Kelson, J.A.S. (1982). The process approach to understanding human motor behavior: An introduction. In J.A.S. Kelso (Ed.), *Human Motor Behavior: An Introduction*, 3-19, Hillsdale, N.J.: Lawrence Erlbaum Associates.

Bell, G., Blythe, M., Gaver, B., Sengers, P., and Wright, P. Designing culturally situated technologies for the home. *Ext. Abstracts CHI 2003*. ACM Press (2003), 1062-1063.

GRADING PLAN:

Type of Evaluation	Weightage (in %)
Quizzes/exam	40%
Individual Research Project	50%
Class participation	10%

OUTCOME:

Students will be able to identify and apply a sociological lens to a human-computer interaction context. This will mean applying informed ways to draw boundaries to an HCI context, use the right theoretical tools of study and processing appropriate data to conduct an independent academic study of selective HCI situations in the real world

Title of the Course:	Software Systems Development
Name of the Faculty:	Sai Anirudh K + Abhishek Singh
Course Code :	CS6.302
Credits:	4
L - T - P :	3-0-2
(L - Lecture hours, T-Tutorial hours, P - Practical hours)	
Semester, Year:	Monsoon 2026
Name of the Program:	PG Program (M.Tech I year I Semester - CSE & CSIS)
Pre-Requisites	No

Course Outcomes : The aim of this course is to

1. Make students to have working knowledge of tools and technologies to build software systems.
2. They can analyze and evaluate the ideas to develop the applications.
3. Comfortable enough to work with various Unix-like computing environments.
4. Able to write simple to complex scripts/programs.
5. In a position to build small to medium sized software applications using various tools and technologies to automate tasks/solve problems.
6. Students will be able to extend the existing applications/software's to add more functionalities.

Course Topics : Linux and Shell Scripting, HTML, CSS, JavaScript and related libraries, Python, Basics of SDLC, Simple Queries, Networking and Security concepts.

1. Shell Programming - Linux basic commands, script writing, Swiss-army- knife tools (vi, grep, awk, sed ..)
2. Web Programming - Intro to basic concepts of the World Wide Web (WWW) and tools used to develop web apps. -Client-side & server-side scripting (HTML, XHTML, CSS, JavaScript, Python,)
3. Database Programming, Networking and Security

Reference Books :

- Mastering Linux Shell Scripting: A practical guide to Linux command-line, Bash scripting, and Shell programming, by Mokhtar Ebrahim, Andrew Mallett
- Learning Python: Powerful Object-Oriented Programming, by Mark Lutz
- JavaScript: The Definitive Guide, by David Flanagan

E-book Links :

Grading Plan :

Type of Evaluation	Weightage (in %)
Quiz-1	10%
Quiz-2	10%
End Sem Exam	20%
Assignments	20%
Project	25%
Other Evaluation (Lab activities + Class activities)	15%

Mapping of Course Outcomes to Program Objectives: (1 – Lowest, 2—Medium, 3 – Highest, or a ‘-’ dash mark if not at all relevant). Program outcomes are posted at

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3	PSO4
CO1	3	3	3	3	3	3	2	3	3	3	3	3	3	3	3	3
CO2	3	3	3	3	3	3	2	3	3	3	3	3	3	3	3	3
CO3	3	3	3	3	3	3	2	3	3	3	3	3	3	3	3	3
CO4	3	3	3	3	3	3	2	3	3	3	3	3	3	3	3	3
CO5	3	3	3	3	3	3	2	3	3	3	3	3	3	3	3	3
CO6	3	3	3	3	3	3	2	3	3	3	3	3	3	3	3	3

Teaching-Learning Strategies in brief (4-5 sentences):

The plan is to use the prepared slides/documents in general to explain the problem and methods. This would include the handwritten/typed notes or using board to describe the topics. The outline has quite a few topics from different domains of computer science and would be taught in detail. Coding sessions would be conducted through labs and tutorials to make the topics easier to understand.

Title of the Course:

Spatial Informatics

Name of the Faculty:

RC Prasad + Kuldeep Kurte

Course Code:

GS2.401

Credits:

3-1-0-4

TYPE-WHEN:

Open Elective

PRE-REQUISITE:

Open to UG-3, UG-4, DD/MS, and PhD students

OBJECTIV:

Spatially explicit information like a map (e.g. Google Maps) informs us not just the geographical location but also the relationship between the objects in it. While mapping models focus on the Spatial (and Temporal) data collection, storage and management (Spatial DBMS) with map generation as one of the key elements; the recent advances in technology have expanded the horizon to include Spatio-temporal Analytics, 3D GIS, Ontology and GML, etc.

This course gives an introduction to the concepts of GIS, the science and algorithms behind it and how this technology can benefit many disciplines, including navigation, transportation and traffic planning, Urban planning, hydrology, environmental management, disaster response, etc.

COURSE TOPICS:

Course Structure (each of approximately 1–2-week duration):

1. What is Geographical Information Systems (GIS)?
2. Fundamental concepts of Space
3. Geospatial data and its Digital representation – Vectors and Raster

4. GIS Data collection, Editing and Data formats
5. Data structures for Spatial data and Spatial data management (Geospatial database)
6. Spatial Data Query and Analysis – Spatial Analysis, Network Analysis
7. Data compatibility - Projections and Georeferencing
8. Spatial reasoning and uncertainty
9. Web-GIS, GML and Map services
10. Geospatial applications in few areas like in Hydrology (Water flows and floods); Ecology and Environment; Land use and Land cover; Urban planning and Transportation; etc.
11. Topics in Spatial Informatics
 1. 3D GIS
 2. Open-Source Initiatives in GIS/RS

A few lectures, may be given by Invited Speakers in related areas during the course to provide the students a wider understanding of its relevance and application.

In addition, there will be a hands-on (lab tutorials) introduction to one or two GIS software and tools at relevant times during the course.

Course Outcomes :

CO-1: Describe how Spatial Data Science helps uncover patterns

CO-2: Apply Geospatial techniques to Prepare the data for analysis

CO-3: Analyze the spatial and temporal data and interpret its outcomes

CO-4: Assessment of application of Spatial data science in key domain areas

CO-5: Design research projects that helps synthesize the learning into an application

Mapping of Course Outcomes to Program Objectives:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3	PSO4
CO1	3	1	1	3	-	-	-	1	2	-	-	3	2	-	-	-
CO2	2	-	-	-	3	-	-	-	2	-	-	-	2	-	2	2
CO3	3	2	-	-	3	-	-	-	2	-	-	-	2	2	3	2
CO4	3	2	3	3	2	2	-	3	2	3	-	3	3	2	2	3
CO5	3	3	3	3	2	3	1	3	3	3	2	3	2	3	2	3

PREFERRED TEXTBOOKS:

1. Geographical information systems and science by Paul A. Longley, Michael F. Goodchild, David J. Maguire, and David W. Rhind
2. Introduction To Geographic Information Systems by Kang-Tsung Chang
3. GIS–A computing perspective by Micheal Worboys and Matt Duckham
4. Concepts and techniques of geographic information systems by C P Lo and Albert K W Yeung

Course Assessment Plan

Assignments	-	10%
Project	-	20%
Any other	-	30%
Quiz	-	20%
Open Book Exam/ 30 Min Quiz	-	20%

OUTCOME: Students will learn the basic concepts of Geospatial data representation, cartography, visualization, data manipulation and how to extract meaningful information from it. In addition, they will be exposed to the application potential of this fast-developing domain cutting across disciplinary interests.

Title of the Course : **Spatial Thinking and Practice**
Name of the Faculty : Shaik Rehana + K S Rajan
Course Code : GSO.101
Credits : 4
L-T-P : **3-1-0.**
(L= Lecture hours, T=Tutorial hours, P=Practical hours)

1. Prerequisite Course / Knowledge:

Anyone with Physics and Mathematics background

2.Course Outcomes (COs)

After completion of this course successfully, the students will be able to

CO-1: Comprehend concept of spatio-temporal analysis

CO-2: Describe various surveying methods

CO-3: Describe the significance of Global navigation systems

CO-4: Implement a range of photogrammetric measurement techniques

CO-5: Explain the basics of geodesy technology

CO-6: Criticize the relevance of surveying methods in geospatial context

3. Mapping of Course Outcomes (COs) with Program Outcomes (POs) and Program Specific Outcomes (PSOs) – Course Articulation Matrix

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3	PSO4
CO1	1	1	2	1	2	1	1	1	1	1	1	1	2	2	2	2
CO2	2	1	2	3	2	2	2	2	2	1	2	2	3	2	2	2
CO3	2	2	1	2	3	2	1	1	1	1	2	2	2	2	2	2
CO4	3	2	2	2	2	1	2	1	1	1	2	2	2	2	2	2
CO5	2	2	2	2	1	2	2	2	2	1	2	3	3	2	2	2
CO6	1	2	2	1	2	1	1	1	1	1	1	2	2	2	2	1

4. Detailed Syllabus:

Unit-1: Concept of Space and Time: Geographical entities and Phenomena, Properties of Space, Size and Scale, comprehend spatial and temporal data throughout the processes of conceptualization, representation, computation, and visualization.

Unit-2: Surveying: Principles of Surveying, Measuring equipment, Techniques, Levelling, Triangulation and Trilateration, Plane Table Surveys, Modern surveying equipment- Total Station, The mathematical model and errors, Random and systematic errors, adjustments techniques.

Unit-3: GNSS: Intro to GNSS; GNSS survey, Keplerian laws, inertial coordinate systems etc., Overview of GNSS: GLONASS, GALILEO, BIDOU, IRNSS satellite systems etc., Differential GPS: GNSS Basic Observables: GNSS Surveying Techniques. Planning and field observations: Networking, data post processing. GNSS applications to Global, Regional and Local issues

Unit-4: Digital Photogrammetry: Fundamental principles and techniques of Photogrammetry surveying, camera calibration, image orientation, stereoscopic viewing, and digital image processing, photogrammetric methods for terrain modeling, Digital elevation model (DEM) generation Integration with other geospatial technologies (LiDAR, UAVs).

Unit-5: Geodesy: Introduction to geodesy & its development; Earth and its gravity field. Earth and its atmosphere – physical properties, Introduction to GPS- its components, Instruments & processing software, GPS signals. GPS data collection, Data Handling-, GPS data processing, Errors in GPS data – Satellite Geometry, Multipath errors & corrections; Accuracy of GPS data. Datum transformation. Geoid modelling.

References:

1. Arora, K.R., “Surveying”, Vol. 1, 2 & 3, Standard Book House. 2005
2. Chandra, A.M., “Higher Surveying”, New Age International Publications. 2002
3. Subramanian, R., “Surveying and Levelling”, Oxford University Press. 2007
4. Gopi, S, Sathi kumar, R. and Madhu, N., “Advanced Surveying, Pearson Publication
3. Torge, W., “Geodesy: An Introduction”, Walter de Gruyter, Berlin. 1980
5. Vanicek, Peter and Krakiwsky, E.J., “Geodesy: The Concepts”, Elsevier. 1

6.Assessment methods and weightages in brief:

Teaching, discussing current surveying methods, their practical implications. Research papers presentations by students on chosen topic, writing assignments, periodical evaluation of course project implementation.

Assessment methods and weightages in brief:

Theory (%): Quiz (10), Assignments (20), Mid exam (20), End exam (30) = **80%**

Project (%): Literature Survey, Preliminary and final presentation along with report = **20%**

Title of the Course:	Spectroscopy
Name of the Faculty:	Harjinder Singh
Name of the Academic Program	CND
Course Code:	SC2.304
L-T-P	3-1-0
Credits	2

1.Prerequisite Course / Knowledge:

Basic quantum mechanics and computing skills

2.Course Outcomes (COs) (5 to 8 for a 3 or 4 credit course):

After completion of this course successfully, the students will be able to

CO-1 Outline the basic principles of different spectroscopic techniques

CO-2 Analyze electronic, vibrational, and rotational spectra of molecules

CO-3 Apply classical and quantum mechanical models to spectroscopy

CO-4 Calculate the normal modes of simple molecules

3.Mapping of Course Outcomes (COs) with Program Outcomes (POs) and Program Specific Outcomes (PSOs) – Course Articulation Matrix

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3	PSO4
CO1	3	2											1	1	1	2
CO2	3	3											2	2	2	2
CO3	3	2											1	1	1	2
CO4	3	2											1	1	2	2

4.Detailed Syllabus:

Unit 1: Introduction: Classical mechanical description of spectroscopy, quantum mechanics and energy quantization, energy-level diagram, energy spectrum: electronic states, vibrational states, rotational states, excitation and relaxation, absorption and emission of electromagnetic waves by materials

Unit 2: Atomic Spectra: Spectral series of hydrogen and alkali atoms, selection rules, L-S coupling, many-electron atoms, isotope shift, hyperfine splitting of spectral lines

Unit 3: Molecular Spectra: Electronic spectra of diatomic and polyatomic molecules, Born-Oppenheimer approximation, Franck-Condon principle, absorption and emission spectra, fluorescence and phosphorescence, Jablonsky diagram, effect of solvation of electronic spectra, rotational spectrum of a diatomic molecule using a rigid rotator model, energy levels and spectrum of a non-rigid diatomic molecule, effect of isotopic substitution on rotational spectra, vibrational spectrum of a diatomic molecule using the harmonic and anharmonic oscillator models. vibrational-rotational coupling in a diatomic molecule, molecular spectra of chain molecules

Unit 4: Raman and Infrared Spectroscopy: Classical and quantum theory of Raman effect, normal vibrations of CO₂ and H₂O molecules, vibrational and rotational Raman spectra, basic concept of infrared spectroscopy, interpretation of Raman and IR spectra, identification of Raman-active and/or IR-active modes based on symmetry arguments

Unit 5: Introduction to Nuclear Magnetic Resonance (NMR), and Electron Spin Resonance (ESR) spectroscopy

Reference Books:

1. Physical Chemistry - P. W. Atkins

2. Fundamentals of Molecular Spectroscopy - C. N. Banwell

3. Molecular Spectroscopy - G. M. Barrow

4. Molecules and Radiation: An Introduction to Modern Molecular Spectroscopy - J. I. Steinfeld

5. Physical Chemistry – A Molecular Approach – D. A. McQuarrie and J. D. Simon

5. Teaching-Learning Strategies in brief (4 to 5 sentences):

6. Assessment methods and weightages in brief (4 to 5 sentences):

Quizzes (25%), Assignments (35%), Final Exam (40%)

Title of the Course: **Speech Analysis and Linguistics (SAL)**

Name of the Faculty: Chiranjeevi Yarra

Course Code: CL2.405

Credits: 4

L - T - P: 3-1-0

(L - Lecture hours, T-Tutorial hours, P - Practical hours)

Name of the Program: CLD/CSD/ECD/ECE/CSE

Semester, Year: Monsoon 2026

Pre-Requisites : No

Course Outcomes :

(list about 5 to 6 outcomes for a full 4 credit course)

CO-1: Explaining the basics of speech and linguistics.

CO-2: Analyzing the linguistics in the state-of-the-art speech applications.

CO-3: Applying computational linguistics foundations for speech analysis.

CO-4: Analyzing the speech applications using phonemic, prosodic and text modelling.

CO-5: Explaining the basics of phonemic, prosodic and text analysis.

CO-6: Designing the algorithms for phonemic, prosodic and text modelling.

Course Topics :

(Please list the order in which they will be covered, and preferably arrange these as five to six modules.)

Unit-1: Speech and linguistic basics – Description of frames, phonemes, syllables, words, phrases, sentences and its use in speech-based feature computation.

Unit-2: Linguistics in speech analysis -- Overview of speech applications (such as ASR, TTS, Speech pathology etc..), use of phonemes, graphemes, prosody and text.

Unit-3: Phonemic analysis -- Phonemes and its accents, visual phonetics (Spectrogram, articulatory videos), pronunciation variations and modelling, grapheme to phoneme conversion, phoneme accent variations and identification.

Unit-4: Prosodic analysis -- Prosodic structure, word and syllable prominence, prominence detection and its applications, Intonation and its modelling (such as ToBI etc..), pitch and prominence variations, intonation identification, pauses, disfluencies and its detection, speech rhythm and speaking rate.

Unit-5: Text analysis -- language modelling, neural language models, metrics, text normalization, character-based speech analysis.

Preferred Textbooks:

Speech and Language Processing, Daniel Jurafsky & James H. Martin (2000), Pearson Education/Prentice Hall.

Prosody in Speech Understanding Systems, Ralf Kompe, Springer, 1997.

The Music of Everyday Speech: Prosody and Discourse Analysis, Ann Wennerstrom, Oxford University Press, 2001

Grading Plan : (The table is only indicative)

Type of Evaluation	Weightage (in %)
Quiz-1	10%
Mid Sem Exam	15%
Seminar	20%
End Evaluation	15%
Assignments	15%
Project	25%

Mapping of Course Outcomes to Program Objectives: (1 – Lowest, 2—Medium, 3 – Highest, or a ‘-’ dash mark if not at all relevant)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3	PSO4
CO1	2	3	1	1	1	1	1	1	2	1	1	2	2	3	-	1
CO2	3	2	2	1	1	1	3	1	2	1	1	2	2	3	-	1
CO3	3	3	3	3	2	1	2	1	3	1	1	3	2	3	-	2
CO4	3	2	3	1	1	2	3	1	2	1	1	2	2	3	-	2
CO5	2	3	1	2	1	2	2	1	2	1	1	2	2	3	-	2
CO6	3	3	3	3	3	2	3	1	3	2	2	3	2	3	-	2

Teaching-Learning Strategies in brief (4-5 sentences):

Lectures are given by integrating ICT into classroom teaching. Regular software-oriented assignments are given to understand the concepts. Along with assignments, course projects are considered to encourage the students to learn the concepts by doing and the problem-solving ability. As a part of course, seminars are conducted to create awareness of the recent trends in the course research area.

Title of the Course : **Statistical Methods in Artificial Intelligence**

Name of the Faculty : Sai Kiran B + Anoop M Namboodiri

Course Code : CS7.403

L-T-P : 3-1-0

Credits 4

Name of the Academic Program B. Tech. in CSE

Prerequisite Course / Knowledge:

Should have taken Basic courses in maths (related topics: Linear Algebra, Probability, Differential Calculus).

Course Outcomes (COs):

After completion of this course successfully, the students will be able to..

CO-1: Demonstrate capability to model and represent physical entities as vectors (feature vectors) and carry out numerical computation.

CO-2: Formulate and solve many practical problems as classification and regression. Also appreciate other problem settings like clustering, structured prediction.

CO-3: **Explain** the fundamental mathematical ideas behind the popular machine learning algorithms

CO-4: **Discuss** the practical (computational) challenges in design and implementation of machine learning algorithms including (i) dimensionality reduction (ii) computational complexity (iii) convergence of the algorithm (iv) offline and online computation

CO-5: Apply the learnings on practical problems and real-life data. Appreciate the challenges with the real-world data sets.

CO-6: **Discuss** the nuances of conducting experiments, analyzing performances and expose the world of empirical science in computation.

Mapping of Course Outcomes (COs) with Program Outcomes (POs) and Program Specific Outcomes (PSOs)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3	PSO4
CO1	3	2	2	2	1	1	1	1	1	1	1	2	3	1	1	3
CO2	1	3	3	2	1	2	1	2	2	2	1	3	3	2	1	3
CO3	3	1	1	1	1	1	1	1	1	1	1	1	3	1	1	3
CO4	3	2	3	2	2	1	2	1	2	2	1	3	3	1	2	3
CO5	1	1	3	2	3	2	1	2	2	2	2	3	3	3	3	3
CO6	1	1	3	1	3	1	1	2	1	1	1	2	3	1	1	3

Each Course Outcome (CO) may be mapped with one or more Program Outcomes (POs). Write '3' in the box for 'High-level' mapping, 2 for 'Medium-level' mapping, 1 for 'Low'-level' mapping
Mapping with PSOs, where applicable.

Detailed Syllabus:

Unit 1: Representation, Vectors, Distributions, Dimensionality reduction, problems and challenges in machine learning

Unit 2: Basic algorithms in machine learning, PCA, Perceptron's, Decision Trees, Analysis

Unit 3: Popular algorithms and settings including unsupervised learning, Support Vector Machines, Kernels, Bias and Variance, Model Selection.

Unit 4: Neural Network Learning, Multi-Layer Perceptron, Backpropagation Algorithms, Exposure to Deep Learning.

Reference Books:

1. MDeisenroth, A. Faisal, C.Ong, Mathematics for Machine Learning, Cambridge Univ Press, 2020
2. R. Duda, P. Hart and D. Stock, Pattern Classification, Wiley, 2007
3. I Goodfellow, Y. Bengio, A. Courville, Deep Learning, MIT Press, 2016

Teaching-Learning Strategies in brief (4 to 5 sentences):

Course lectures will connect the algorithms and approaches to the real-world examples. This motivates the student and also convince the need of formal and mathematical way of approaching the real-world problem solving. Lectures also introduce the visualization skills of the data and distribution with the aim of appreciating the data. Associated sessions and components (tutorials, HomeWorks) expose the popular libraries and software infrastructure for machine learning today.

Assessment methods and weightages in brief (4 to 5 sentences):

- HomeWorks: 30%
 - In-class Objective Tests: 10%
 - Projects/Term Papers: 10%
 - Mid semester exam1: 15%
 - Mid Semester exam2: 15%
 - End Semester Exam: 20%
-

Title of the Course: **Structural Dynamics**

Name of the Faculty: Sunitha Palissery

Name of the Program: M.Tech CASE

Course Code:

CE1.501

Credits 4

L - T - P : 3-1-0

(L - Lecture hours, T-Tutorial hours, P - Practical hours)

Semester, Year : Monsoon 2026

Pre-Requisites : Mechanics of Materials & Structural Analysis

Course Outcomes :

After completion of this course successfully, the students will be able to:

CO-1. Develop knowledge and skills to mathematically express dynamics of structural systems

CO-2. Employ the computer application skills in developing structural dynamics tools for predicting structural response to dynamic loading like earthquakes

CO-3. Demonstrate problem solving skills for various scenarios of structural dynamics and worktowards a research-based approach to the course

CO-4: Develop critical thinking to help improve dynamic responses of structures

CO-5. Analyze ethical and effective structural design practices in line with good dynamic responseof structures under earthquake loading

CO-6. Reorganize inter-personal skills required to manage possible inter-disciplinary, inter-departmental collaborations in structural engineering and thus in structural dynamics

Course Topics :

Unit 1: Response of simple Single Degree of Freedom (SDOF) systems

Introduction to structural dynamics; Definition of DOF; idealization of structure as SDOF system; Formulation of equations of motion for various SDOF systems; Free

vibration of systems; Damping in structures; Undamped Systems; Forced vibration of systems; Steady state response to harmonic forces; Experimental determinations of natural frequency; Duhamel's integral and other methods.

Unit 2: Analysis of Multi-Degree of Freedom (MDOF) systems

Static force displacement relationship; Strain energy of system; Formulation of equation of motion; Evaluation of natural frequencies and modes; Free vibration of undamped systems; Forced vibration of damped systems; Review of time history & response spectrum methods.

Unit 3: Numerical Methods, and Approximate methods of computing natural frequencies

Eigen value problems and applications: Mode superposition principle; modal truncation errors; Ritz Vector approach; Direct Integration methods: Explicit methods - Central difference method; Implicit methods - Newmark- β method; Rayleigh's method; Dunkerley's method; Rayleigh-Ritz method

Unit 4: Base excited systems

Formulation of equations of motion for SDOF and MDOF systems; Concept of spectral quantities; Response spectrum; Fundamentals of earthquake engineering; Discussion on IS 1893 (1:2016) provisions for buildings.

Unit 5: Overview of dynamics of continuous systems

Vibration of flexural beams and shear beams: Equation of motion; Free vibration and Forced vibration

Preferred Textbooks:

1. Chopra, A.K., (2017), *Dynamics of Structures: Theory and Applications to Earthquake Engineering*, V edition, Pearson Education, India
2. Clough, R.W and Penzien, J., (1993), *Dynamics of Structures*, II edition, McGraw-Hill International Ed.
3. Paz, M. (2006), *Structural Dynamics: Theory & Computation*, V edition, Kluwer Academic Publishers, Amsterdam

Reference Books :

1. Craig, R.L., and Wiley, J., (1981), *Structural Dynamics: An Introduction to Computational Methods*, Wiley (Publishing)
2. Humar, J.L., (2002), *Dynamics of Structures*, II edition, Swets & Zeitlinger B.V., Lisse, Netherlands
3. Mukhopadhyay, M., (2008), *Structural Dynamics: Vibrations and Systems*, Ane Books Pvt. Ltd, New Delhi
- 4.

E-book Links :

Grading Plan : (The table is only indicative)

Type of Evaluation	Weightage (in %)
Quiz-1	5
Mid Sem Exam	20
Quiz-2	5
End Sem Exam	30
Assignments	20
Project	20

Mapping of Course Outcomes to Program Objectives:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PSO1	PSO2	PSO3	PSO4
CO1	3	3	3	2	3	2	2	3	2	2	3	3	3	3
CO2	3	3	3	2	2	2	3	3	2	2	3	3	3	3
CO3	3	3	3	2	3	2	3	2	2	3	3	3	2	2
CO4	3	3	3	2	3	1	2	3	2	3	3	3	2	3
CO5	3	3	3	2	3	2	2	2	2	3	2	2	2	2
CO6	2	2	1	2	3	1	2	2	3	2	2	2	2	2

Teaching-Learning Strategies in brief (4-5 sentences):

1. Lectures by integrating ICT into classroom teaching
2. Tutorials involving numerical examples to reinforce structural dynamics concepts
3. Assignments involving analysing structural data to understand dynamic response earthquake data for predicting earthquake response of SDOF and MDOF structural systems
4. Critical and active learning through projects, and project-based learning by doing term-projects which involves hands-on use of computer programming skills and software tools.

Title of the Course:	Structural Engineering Design Studio
Name of the Faculty:	Shubham Singhal
Course Code:	CE1.502
L-T-P:	3-1-0
Credits:	4
Name of the Academic Program:	M. Tech in Computer Aided Structural Engineering

1.Prerequisite Course / Knowledge:

B. Tech in Civil Engineering subjects i.e., Engineering Mechanics, Reinforced Concrete Design, Structural Analysis.

2.Course Outcomes (COs) (5 to 8 for a 3 or 4 credit course):

After completion of this course successfully, the students will be able to..

- CO-1** Use the understanding of the structural analysis concepts of structural components, for structural system design.
- CO-2** Write computer programs, to develop structural analysis software.
- CO-3** Analyse the structure using commercially available software
- CO-4** Design the components and systems using commercially available software
- CO-5** Appreciate the challenges in construction industry and get equipped to address some of the challenges

3.Mapping of Course Outcomes (COs) with Program Outcomes (POs) and Program Specific Outcomes (PSOs) – Course Articulation Matrix

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3	PSO4
CO1	3	3	3	1	-	1	2	-	-	-	-	-	3	3	3	3
CO2	3	2	2	1	-	1	1	-	-	-	-	-	1	3	3	1
CO3	1	1	2	2	1	2	1	-	-	-	-	-	2	1	2	1
CO4	2	2	3	2	3	2	3	-	-	-	-	-	2	2	3	1
CO5	3	3	1	3	3	3	3	-	-	-	-	-	3	3	3	3

Note: '3' in the box for 'High-level' mapping, 2 for 'Medium-level' mapping, 1 for 'Low-level' mapping

4.Detailed Syllabus:

Unit 1: Review of Structural Analysis – Analysis of Planar Trusses, Shear Force and Bending Moment Diagrams, Structural Analysis-Force Methods-Displacement Methods, Deflected Shape-Deflections, Approximate Methods of Analysis- Gravity and Lateral Loads-Beams, Frames.

Unit 2: Numerical Methods – Bi-Section Method, Gauss Elimination, Newton-Raphson

Unit 3: Introduction to MATLAB – Vectors, Arrays and Matrices, Branching Statements and Loops, Introduction to Plotting, Linear Algebra, Curve Fitting, Script and Functions, Graphics -2D and 3D, Development of GUI Tool.

Unit 4: Application of MATLAB – Analysis of Beams & Frames, Gravity and Lateral Load Analysis of Frames, Analysis of Planar Trusses.

Unit 5: Introduction to STAAD, SAP2000 & ETABS – Modelling, Analysis and Design Concepts, Elastic Analysis and Design-Reinforced Concrete Building -Bare Frame Analysis, Analysis of Frames with Unreinforced Masonry Infills, Steel Building, Elastic Earthquake Behaviour of Buildings-Modal Analysis-Linear Dynamic Analyses and Design-Response Spectrum Analysis, Time History Analysis, Introduction to Inelastic Earthquake Behaviour of Buildings-Nonlinear Static Analysis Essentials- Nonlinear Static Analysis.

Reference Books:

1. Hibbeler, R.C., (2011), Mechanics of Materials, Pearson Prentice Hall, New Jersey
2. Hibbeler, R.C., (2012), Structural Analysis, Pearson Prentice Hall, New Jersey
3. Norris, C.H., Wilbur, J.B., and Utku, S., (2003) Elementary Structural Analysis, McGraw-Hill Book Company, Auckland
3. Wang, C.K., (1983), Intermediate Structural Analysis, McGraw-Hill Book Company, Auckland
4. Computers and Structures Inc. (CSI), (2012), Structural Analysis Program (SAP) 2000, Version 16, USA
5. Chapman, S.J., (2007), MATLAB Programming for Engineers, Thomson Learning, Canada
6. Pratap, R., (2003), Getting Started with MATLAB: A Quick Introduction for Scientists and Engineers, OXFORD University Press, Oxford, UK

5.Teaching-Learning Strategies in brief (4 to 5 sentences):

A lecture on a theory concept will be preceded by its practical relevance, appreciation of field level challenges and immediately followed by on-hands-practice using manual approach as well as using appropriate scientific software. Student will be encouraged to come up with issues and how the theory and hands-on experience is helping them. Student is also encouraged to do homework and assignments individually and mini projects as a group task.

6. Assessment methods and weightages in brief (4 to 5 sentences):

The course will rely heavily on looking at problem solving capability of student and hence the assessment is divided as follows i.e..

- a) 20% weightage is given to individual assignments for checking the concepts taught in the class,
- b) 20% weightage is for group projects for checking software application
- c) 30% is quizzes & Mid exam for checking the application of concept and,
- d) 30% for end-sem exam is for overall assessment.

Title of the Course	: Structural Safety of Built Infrastructure
Name of Faculty	: Jofin George
Course Code	: CE1.625
Name of the Academic Program:	M.Tech. - Computer-Aided Structural Engineering
L-T-P	: 3-1-1 (L-T-P)
(L = Lecture hours; T = Tutorial hours; and P = Practical hours)	
Credits	: 4

1. Prerequisite Course / Knowledge: Strong fundamentals in structural analysis and design principles, Familiarity with analysis software such as ABAQUS and ETABS, and Understanding of probability concepts in a civil engineering context.

2. Course Outcomes (COs):

After completion of this course successfully, the students will be able to:

- CO 1: Evaluate the significance of structural assessment in preventing collapses and analyze structural safety through real-world examples and visual inspection techniques.
- CO 2: Develop innovative solutions for structural assessment using advanced textile-based materials.
- CO 3: Develop proficiency in using computational methods, material models, and modelling techniques to assess structural components and systems accurately.
- CO 4: Apply various non-destructive testing methods to evaluate the integrity and safety of structures through practical applications and case studies.
- CO 5: Utilize both traditional methods like thrust line analysis for stability and advanced numerical methods for static and dynamic analysis of structures.

3. Mapping of Course Outcomes (COs) with Program Outcomes (POs) and Program Specific Outcomes (PSOs) – Course Articulation Matrix

Note: Each Course Outcome (CO) may be mapped with one or more Program Outcomes (POs) and PSOs. Write '3' in the box for 'High-level' mapping, '2' for 'Medium-level' mapping, '1' for 'Low-level' mapping; and '-' for not relevant.

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PSO1	PSO2	PSO3	PSO4
CO1	2	3	1	-	1	3	1	2	1	2	2	1	1	2
CO2	3	3	2	1	2	3	1	3	1	3	3	3	2	3
CO3	3	3	3	-	3	2	1	2	2	2	3	2	3	3
CO4	2	3	3	1	2	2	1	2	1	3	3	2	3	2
CO5	2	2	2	-	2	3	2	3	3	3	3	2	3	3

. Detailed Syllabus:

Unit 1: Introduction to Structural Assessment: Definitions, Concepts, Importance, Factors Affecting the Performance of Structures, Classification of Deficiencies, Structural and Non-structural Deficiencies.

Unit 2: Visual Inspection and Condition Assessment: Methodologies, Damage Classification, Types of Defects, Preliminary Evaluation, In-situ Testing, Introduction to Non-destructive and Partial Destructive Testing Techniques.

Unit 3: Non-Destructive and Partial Destructive Testing Methods: Rebound Hammer, Ultrasonic Pulse Velocity, Core Testing, Half-cell Potential, Carbonation Depth, Impact Echo, Ground Penetrating Radar; Testing of textile-based composites for strength, durability, and impact resistance using NDT techniques.

Unit 4: Structural Health Monitoring (SHM): Introduction to SHM. Use of DIC, UTM, and NDT techniques for textile-reinforced concrete assessment, including strain mapping and crack propagation studies. Application of smart textiles and fibre optic sensors for real-time monitoring of retrofitted structures.

Unit 5: Performance-Based Assessment & strengthening: Performance criteria for structural evaluation; Performance-based seismic design principles; Use of fibre-reinforced composites and geotextiles in structural performance enhancement.

Unit 6: Advanced Techniques for Structural Assessment: Digital Image Correlation (DIC), Application of AI and Machine Learning in Structural Assessment; DIC-based failure analysis of structural elements strengthened using textile-reinforced composite materials.

Reference Books:

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- Brameshuber, J. (2006). Textile Reinforced Concrete.
 - Chapman, R. (2012). Smart Textiles for Protection.
 - FEMA P-58: Seismic Performance Assessment of Buildings.
 - Indian Railway Institute of Engineering, Pune, 2021, Non-destructive testing of bridges.
-

- International Atomic Energy Agency, Vienna, 2002, Guidebook on non-destructive testing of concrete structures.
 - IS 13311: Non-destructive testing of concrete-Methods of test.
 - Sarsby, R.W. (2007). Geotextiles in Civil Engineering.
 - SP:25-1984: Handbook on Causes and Prevention of Cracks in Buildings.
 - Sutton, M. (2018). Digital Image Correlation (DIC) Techniques: Applications in Structural Engineering.
-

Teaching-Learning Strategies in brief (4 to 5 sentences):

- Classroom Lectures
- Active learning by students.
- Field visits to understand the types and causes of structural collapse and suggest mitigation strategies.
- Term project focusing on the process of structural assessment based on available information.
- Invited lectures and discussion sessions led by industry experts on practical case studies of structural collapse, assessment and rehabilitation.

6. Assessment methods and weightages in brief (4 to 5 sentences):

Assignments and Quizzes - 30%

Mid Semester Exam - 20%

End Semester Exam - 30%

Project – 20%

Title of the Course	: System Thinking and Design
Name of the Faculty	: Raman Saxena
Course Code	: PD2.405
L-T-P	: 1.5-0-3
Credits	: 2
(L=Lecture hours, T=Tutorial hours, P=Practical hours)	
Semester, Year	: 3rd Sem – Year 2

1.Prerequisite Course / Knowledge

- **Product Management Foundations:** Understanding of product lifecycles.
- **Service Design:** Familiarity with ecosystem mapping and stakeholder interests.
- **Basic Mathematics:** Comfort with the concepts of variables, rates of change, and logic.

2.Course Objectives & Outcomes (COs)

1. To develop the ability to see "the whole" rather than just the parts.
2. To identify leverage points in complex organizational or technological systems.
3. To anticipate the second and third-order effects of product decisions

Learning Outcomes: Upon completion, students will be able to:

- **Model** complex systems using Causal Loop Diagrams (CLDs) and Stock-and-Flow diagrams.
- **Identify** systemic archetypes (e.g., "Fixes that Fail") within business models.
- **Design** interventions that address root causes rather than symptoms.
- **Evaluate** the resilience and sustainability of a product ecosystem.

3. Mapping of Course Outcomes (COs) with Program Outcomes (POs) and Program Specific Outcomes (PSOs) – Course Articulation Matrix

	P O 1	P O 2	P O 3	P O 4	P O 5	P O 6	P O 7	P O 8	P O 9	PO 10	PO 11	PS O1	PS O2	PS O3	PS O4
CO 1	3	3	3	3	3	2	2	2	1	2	3	1	1	2	1
CO 2	3	3	3	3	3	2	2	2	3	3	3	1	2	2	1
CO 3	3	3	3	3	3	2	1	2	3	3	2	1	2	1	2

4. Detailed Syllabus

Week	Unit Topic	Key Concepts / Class Activities
1	The Systems Lens	Introduction to Systems Thinking; Reductionism vs. Holism; Elements, Interconnections, and Purpose.
2	Mapping Causality	Causal Loop Diagrams (CLDs); Reinforcing (R) and Balancing (B) Loops; Identifying "Delay" in systems.
3	System Archetypes	Recognizing patterns: "Tragedy of the Commons," "Shifting the Burden," and "Success to the Successful."
4	Stocks, Flows, & Dynamics	Introduction to System Dynamics; Modeling accumulation and flow; Using software (e.g., Vensim or Insight Maker).

5	Leverage Points	Places to intervene in a system; Designing for Emergence; Understanding "Wicked Problems."
6	Systems in Practice	Platform Ecosystems (e.g., UPI, Amazon); Socio-technical systems; Ethical implications of systemic design.

5. Reference Books & Case Studies

Books:

- *Thinking in Systems: A Primer* – Donella Meadows.
- *The Fifth Discipline* – Peter Senge (Focus on Learning Organizations).
- *Systemic Design: Theory, Methods, and Practice* – Peter Jones & Kyoichi Kijima.

Case Studies:

- **The UPI Ecosystem:** Analyzing the systemic interplay between banks, NPCI, third-party apps, and merchants.
- **Climate Change & Policy:** Using the EN-ROADS simulator to understand systemic levers in global warming.
- **The "Cobra Effect":** Historical and modern examples of unintended consequences in product incentives.

6. Teaching-Learning Strategies

- **The "Iceberg" Analysis:** For every product failure discussed, students must use the Iceberg Model to uncover mental models and systemic structures.
- **Modeling Workshops:** Hands-on sessions building visual maps of complex problems like "Urban Traffic in Hyderabad" or "E-commerce Logistics."
- **Simulations:** Engaging in the "Beer Game" simulation to experience the Bullwhip Effect in supply chains.

7. Assessment Methods and Weightages

Component	Weightage	Description
1. The "System Audit"	20%	individual mapping of a common system archetype found in a tech company (e.g., Netflix's growth).
2. Modelling Lab	30%	Creating a functional Stock-and-Flow model for a hypothetical startup's growth.
3. Capstone; Intervention	40%	Identifying a "Wicked Problem" and designing a systemic intervention with a 3rd-order effect analysis.
4. Active Synthesis	10%	Contribution to class-wide system-mapping debates.

Title of the Course : Systems Thinking
Faculty Name : Spandan Roy
Course Cod : EC5.202
L-T-P : 3-1-0
Credits : 4
Name of the Academic Program B. Tech. in ECE

Prerequisite Course / Knowledge: None

Course Outcomes (COs):

After completion of this course successfully, the students will be able to.

CO-1: Apply knowledge of 1st principles from physics to develop system model

CO-2: Develop state-space formulation for systems and analyze the behavior of 1st and 2nd order systems via time-domain specification for transients and steady-state

CO-3: Design and develop proportional, derivative and integral controllers

CO-4: Demonstrate a familiarity with organization of biological system and their parts

CO-5: Apply principles of control to biological systems

CO-6: Analyze emergent properties of biological systems by mathematical modeling

Mapping of Course Outcomes (COs) with Program Outcomes (POs) and Program Specific Outcomes (PSOs)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3	PSO4
CO1	3	3	2	1	1	1	1	1	1	1	1	3	1	1	1	3
CO2	2	3	3	2	2	1	1	1	1	1	1	3	1	1	1	3
CO3	2	3	3	3	3	1	1	1	1	1	1	3	1	1	1	3
CO4	1	1	1	1	1	1	1	1	1	1	1	3	1	1	1	1
CO5	2	3	2	2	2	1	1	1	1	1	1	3	1	1	1	1
CO6	2	3	3	3	3	1	1	1	1	1	1	3	1	1	1	1

Each Course Outcome (CO) may be mapped with one or more Program Outcomes (POs). Write '3' in the box for 'High-level' mapping, 2 for 'Medium-level' mapping, 1 for 'Low'-level'mapping

Mapping with PSOs, where applicable.

Detailed Syllabus:

Unit 1: Mathematical modelling of systems via transfer functions and state-space

Unit 2: Time-domain performance and stability analysis of first and second order systems

Unit 3: Biological signals and systems (case study)

Unit 4: Design principles of biological systems and control

Unit 6: Modeling and design of biological circuits

Reference Books:

- 1) Modern Control Engineering by K Ogata, Prentice Hall.
- 2) An Introduction to Systems Biology: Design Principles of Biological Circuits, Uri Alon, Chapman & Hall.
- 3) Biomolecular Feedback Systems, Domitilla Del Vecchio and Richard M. Murray, Princeton University Press

Teaching-Learning Strategies in brief (4 to 5 sentences):

The course lectures will include activities that promote the understanding of the lecture content by using small examples that students work out during the class itself and promote active and participatory learning. A good part of the lecture will involve problem solving and finding solutions to problems. Homework assignments are designed to reiterate the material covered in class lectures and apply them via simulation.

Assessment methods and weightages in brief (4 to 5 sentences):

- Assignments: 30%
 - ☐ Quiz1: 15%
 - ☐ Quiz 2: 15%
 - End Exam: 40%
-

Title of the Course : Theory of Elasticity
Name of Faculty : P. Pravin Kumar Venkat Rao
Course Code : CE0.501
L-T-P : 3-1-0
Credits : 4
Name of the Academic Program: M. Tech in CASE

1.Prerequisite Course / Knowledge: Solid Mechanics/Strength of Materials

2.Course Outcomes (COs):

After completion of this course successfully, the students will be able to:

CO 1: Explain the basics concepts of stress, strain, tensor, vector, traction, and important properties of solids.

CO 2: Discuss the fundamental theories of elasticity.

CO3: Idealize the physical systems through mathematical equations.

CO4: Represent the state of stress and strain in a body (2D and 3D) with respect to different planes or orientations.

CO5: Analyze the boundary value problems using equilibrium, compatibility, and constitutive relations.

CO 6: Derive the governing equations and their solutions for application to problems in plane stress and plane strain state, torsion and bending.

3.Mapping of Course Outcomes (COs) with Program Outcomes (POs) and Program Specific Outcomes (PSOs) – Course Articulation Matrix

Note: Each Course Outcome (CO) may be mapped with one or more Program Outcomes (POs) and PSOs. Write '3' in the box for 'High-level' mapping, 2 for 'Medium-level' mapping, 1 for 'Low'-level' mapping

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PSO1	PSO2	PSO3	PSO4
CO1	3	3	2	1	3	2	3	3	3	3	3	1	3	3
CO2	3	3	1	2	3	2	3	3	3	3	3	1	3	3
CO3	2	3	2	1	3	2	3	3	3	3	3	2	3	3
CO4	2	3	1	2	3	2	3	3	3	3	3	1	3	3
CO5	3	3	2	1	3	2	3	3	3	3	3	2	3	3
CO6	2	3	1	2	3	2	3	3	3	3	3	2	3	3

4.Detailed Syllabus:

Unit 1: Elasticity and its types, Inelastic material, Difference between theory of elasticity and strength of material, Materials and its properties, Assumptions in elasticity, Mathematical preliminaries: cartesian co-ordinates, introduction to tensor, matrix representation, operators and symbols.

Unit 2: Concept of stresses and strains, Infinitesimal area and volume, 2D and 3D stress tensor, Stress and strain transformation at a point in an elastic body, Difference between stress and traction, Cauchy's stress, Components of traction, Strain tensors, Types of deformation, Measurement of surface strains

Unit 3: Rigid body translation and rotation of an element in space, Generalized Hook's law, Stress-strain relationships, Equilibrium equations, Strain-displacement relationships, Compatibility conditions, Constitutive relations.

Unit 4: Principal planes, Principal stresses and invariants, Octahedral plane and stresses, Deviatoric stress, Hydrostatic stress, Plain stress, Plain strain, Formulation of boundary value problems in equilibrium and compatibility, Stress functions, Biharmonic equation, Solution of 2D problems by the use of polynomials.

Unit 5: Torsion of bars, Saint venant principle, Rigid body rotation, bending of beams, Elastic stability, Factors affecting lateral stability, Analysis of beam-column with different loading conditions, Different types of buckling.

Reference Books:

1. Ugural, A. C., & Fenster, S. K. (2003). Advanced strength and applied elasticity. Pearson education.
2. Timoshenko, S. P., & Goodier, J. N. (1971). Theory of Elasticity, McGraw-Hill, New York, 1970. Fok-Ching Chong received the BS degree from the Department of Electrical Engineering, National Taiwan University, Taipei, Taiwan, in.
3. Shames, I. H. (1964). Mechanics of deformable solids.
4. Srinath, L. S. (2003). Advanced mechanics of solids. Tata McGraw-Hill.
5. Timoshenko, S. (1953). History of strength of materials: with a brief account of the history of theory of elasticity and theory of structures. Courier Corporation.
6. Boresi, A. P., Chong, K., & Lee, J. D. (2010). Elasticity in engineering mechanics. John Wiley & Sons.
7. Popov, E. P., & Balan, T. A. (1968). Mechanics of solids. Mexico City, Mexico: Pearson Education, 2000) (in Spanish).
8. NPTEL Lecture Notes: IIT, Madras.

5. Teaching-Learning Strategies in brief (4 to 5 sentences):

In this course the concept of elasticity, an important property of solids, will be discussed in a comprehensive way. Idealization of physical system, representing the idealized system through mathematical equation and finally finding solution of those equations are the key features that constitute the structure of this course. In this course emphasis will be given on both theory and applications.

6. Assessment methods and weightages in brief (4 to 5 sentences):

Assignments and Quizzes - 40%

Mid Semester Exam - 25%

End Semester Exam - 35%

Title of the Course:	Thinking and Knowing in the Human Sciences – II
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Name of the Faculty:	Rajorshi Ray + Isha Dubey
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Course Code:	HSO.202
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L-T-P:	3-1-0
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(L= Lecture hours, T=Tutorial hours, P=Practical hours)

Credits:	4
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Name of the Academic Program: CHDCore offered to UG2 (third semester of the CHD program)

1. Prerequisite Course / Knowledge:

Making of Contemporary India and Making of the Contemporary World(Core courses in the CHD program).

2.Course Outcomes (COs) (5 to 8 for a 3 or 4 credit course):

After completion of this course successfully, the students will be able to.

CO-1: Students will understand how a historical and sociological perspective helps to understand society. They will learn to see the human world appears through the lens of these two disciplines and their insights.

CO-2: Students will understand the methods, assumptions, principles, and the foundational ideas of the various schools of history.

CO-3: Students will understand key concepts and theoretical and methodological tools in sociology.

CO-4: Students will see the potential as well as the limitations of historical and sociological approaches to social analysis.

CO-5: Students will use sociological and historiographical theories and frameworks. They will gain some experience in engaging with academic texts as well as primary sources in a theoretically informed manner.

3.Mapping of Course Outcomes (COs) with Program Outcomes (POs) and Program Specific Outcomes (PSOs) – Course Articulation Matrix

	PO 1	PO 2	PO3	PO 4	PO5	PO 6	PO7	PO 8	PO 9	PO1 0	PO1 1	PO1 2	PS O1	PS O2	PS O3	PS O4
CO1	1	3	3	3	1	3	3	3	1	1	1	3	1	3	2	3
CO2	1	3	3	3	1	3	3	3	1	1	1	3	1	3	2	3
CO3	1	3	3	3	1	3	3	3	1	1	1	3	1	3	2	3
CO 4	1	3	3	3	1	3	3	3	1	1	1	3	1	3	2	3
CO5	1	3	3	3	1	3	3	3	1	1	1	3	1	3	2	3

Note: Each Course Outcome (CO) may be mapped with one or more Program Outcomes (POs) and PSOs. Write '3' in the box for 'High-level' mapping, 2 for 'Medium-level' mapping, 1 for 'Low'-level' mapping

4.Detailed Syllabus:

Unit 1: Sociological Concepts: Social Structures, Individuals and Groups, Socialisation, Structure/Agency, Status and Roles, Rules, Values, Norms, Culture, Discourse, Deviance.

Unit 2: Sociological Frameworks and Theories: Understanding social stratification through the structural functionalist, Marxist and the Weberian methods. Introduction to key ideas of Emile Durkheim, Karl Marx and Max Weber.

Unit 3: Social Institutions and Processes in India.

Unit 4: Ideas of History (Progress, Decline, Morality; Facts, Objectivity, Interpretation)

Unit 5: Concepts of Time and Space

Unit 6: Main methods of knowing the past (Inductive and deductive methods; structuralism and post-structuralism).

Reference Books:

Anthony Giddens, *Sociology* (Malden: Polity Press, 2009).

D.P. Mukerji, *Basic Concepts in Sociology* (New Delhi: Rupa and Co., 1932).
 E. H. Carr, *What is History* (London: Cambridge University Press, 1961).
 James Scott, *Against the Grain* (New Haven: Yale University Press, 2017).
 Jean-Claude Carriere and Umberto Eco, *this is Not the End of the Book* (London: Random House, 2009).
 Mircea Eliade, *The Myth of the Eternal Return: Or Cosmos and History* (New Jersey: Princeton University Press, 1992).
 Nandini Sundar, Patricia Uberoi, Satish Deshpande, *Anthropology in the East: Founders of Indian Sociology and Anthropology* (Hyderabad: Orient Black Swan, 2007).

5.Teaching-Learning Strategies in brief (4 to 5 sentences):

Students are introduced to theories and concepts through lectures. This course expects the student to read about 2000 pages of academic literature and write about 12,000 words of essays and answers over the semester. The tutorial slots are used to get students to do small in-class assignments related to assigned readings. Throughout the course, students are equipped to deploy theories and methods to a research question and to draw interconnections between the different ways in which the human world is understood and explained. The course draws connections as well as differences between the historical and the sociological methods.

6.Assessment methods and weightages in brief (4 to 5 sentences):

In the sociology module, students will do 2 assignments (worth 15% each and related to one Unit), and one exam (30%) which covers the entire syllabus of the module. The assignments are designed to test grasp over concepts and theories discussed in the lectures and is also reading based. In the history module, students will do two quizzes (worth 15% each, and covering 1 unit) and 1 reading based writing assignment worth 20%.

Title of the Course	: Thinking and Knowing in Human Sciences- 3
Name of the Faculty	: Anirban Dasgupta + Khaliq Parkar
Name of the Program	: CHD
Course Code	: HSo.216
Credits	: 4
L - T - P	: L-3, T-1
(L - Lecture hours, T-Tutorial hours, P - Practical hours)	
Semester, Year	: Monsoon 2026
Pre-Requisites	: Nil

CourseOutcomes:

After completion of this course successfully students will be able to:

CO1: Distinguish between schools of thought in economics and understand their evolution

CO2: Understand the foundations of micro, macro and international economics

CO3: Analyse how the conditions of economic life are distinct in developing countries like India

CO4: Describe the concept of politics and identify the general scope and methods of Political Science at an introductory level.

CO5: Understand the meaning and interpretation of key concepts in Political Science

CO6: Explain range of academic theories relating to key concepts in Political Science.

Course Structure

Part A: Economics

Module 1: Ways of doing economics: Different views on scope and definition of economics.

Alternative schools of economic thought- Neoclassical, Marxist, Keynesian

Module 2: Theoretical foundations of microeconomics: Agent based understanding of decision making- individuals and firms. Market structure. Welfare economics.

Module 3: Theoretical Foundations of Macroeconomics: Macroeconomic measurement.

Aggregate demand, aggregate supply and determination of output. Inflation and Unemployment, Monetary and Fiscal Policy. Banking and the role of finance in the economy

Part B: Political Science

Module 1: Introduction to various perspectives on how we define politics and its domain; Nature and scope of Political Science as a field of knowledge; Meaning and origin of State: divine theory and social contract theory; Forms and functions of government

Module 2: Meaning, interpretation and classification of the concepts of Power, Authority and Legitimacy; Definitions and Characteristics of Sovereignty; Theory of Citizenship

Module 3: Brief introduction to key concepts of Political Science; Liberty: Negative and Positive; Equality: Equality of opportunity, outcome, welfare, resources, capabilities; Justice: Procedural, Substantive, Social Justice; Rights: Legal Rights and Human Rights; Democracy: Idea and Practice

Preferred Text Books:

Economics:

Selected Chapters from-

- CORE Econ. *The Economy: A South Asian Perspective* (free e-book available)
- Neva Goodwin et.al. *Principles of Economics in Context*. Routledge.
- Richard Wolff and Stephen Resnick. *Contending Economic Theories: Neoclassical, Keynesian and Marxian*. MIT Press.

Political Science:

Selected Chapters from-

- Andrew Heywood: *Politics* (forth edition)
- Andrew Heywood: *Political Theory: An Introduction*
- Kenneth Minogue: *Politics: A Very Short Introduction*
- Rajeev Bhargava and Ashok Acharya (eds): *Political Theory: An Introduction*

Reference Books :

- Amit Bhaduri: *Macroeconomics*
- Roger Backhouse: *The Penguin History of Economics*
- Dimitris Milanokis and Ben Fine: *From Economics Imperialism to Freakonomics: The Shifting Boundaries between Economics and other Social Sciences*
- Rand Dyck: *Studying Politics: An Introduction to Political Science, Third edition*
- Larry Johnston: *Politics: An Introduction to the Modern Democratic State*
- Eric Mintz, David Close, and Osvaldo Croci: *Politics, Power and the Common Good: An Introduction to Political Science*
- Jonathan Wolff: *An Introduction to Political Philosophy*
- Matt Zwolinski: *Arguing about Political Philosophy*

Grading Plan: Economics

Type of Evaluation	Weightage (in %)
Quiz-1	
Mid Sem Exam	30%
Quiz-2	
End Sem Exam	
Assignments	20%
Project	
Term Paper	
Other Evaluation	

Political Science (The table is only indicative)

Type of Evaluation	Weightage (in %)
Quiz-1	30%
Mid Sem Exam	
Quiz-2	
End Sem Exam	20%
Assignments	
Project	
Term Paper	
Other Evaluation	

Mapping of Course Outcomes to Program Objectives: (1 – Lowest, 2—Medium, 3 – Highest, or a ‘-’ dash mark if not at all relevant).

Matrix for CSE

	P O1	P O2	P O3	P O4	P O5	P O6	P O7	P O8	P O9	PO 10	PO 11	PO 12	PS O1	PS O2	PS O3	PS O4
C O1						2	2	3		2		3				2
C O2				1		2	2	2		2		3				3
C O3						2		2				2				2

Matrix for ECE

	P O1	P O2	P O3	P O4	P O5	P O6	P O7	P O8	P O9	PO 10	PO 11	PO 12	PS O1	PS O2	PS O3	PS O4
C O1						2	2	3		2		3				2
C O2				1		2	2	2		2		3				3
C O3						2		2				2				2

Teaching-Learning Strategies in brief (4-5 sentences) :

Economics- In this course, teaching will be primarily based on lectures and will be supplemented with group discussions and class presentations. Relevant chapters from the textbooks and articles

will be referred for each lecture. As an introductory course, the assignments will be designed to evaluate the theoretical clarity attained by students. The lectures and reading material will be designed to highlight both the logical structure of economic theory as well as the plural approaches to economic problems that originate from alternative theoretical perspectives

Political Science- The course will be based on classroom lectures and in class discussion of assigned reading material. On an average, each student will be required to read between 200 to 300 pages of books and articles and submit written work between 1500-2000 words, cumulatively. The students will be expected to follow the latest news and developments on the topics to be discussed in this course. The assignments and project will focus on training students to develop their own ideas and research skills in social sciences. Audio-visual and interactive materials may be used.

Title of the Course	: Topics in AI for Healthcare
Faculty Name(s)	: Vinod PK, Bapi Raju S, Raja Poladi
Name of the Program	: CSE
Course Code	: CS9.442
Credits	: 4
L - T - P	: 3-1-0
(L - Lecture hours, T-Tutorial hours, P - Practical hours)	
Semester, Year	: Monsoon 2026

Pre-Requisites : **Completed Statistical Methods in Artificial Intelligence (SMAI)**

Course Outcomes :

(list about 5 to 6 outcomes for a full 4 credit course)

After completion of this course successfully, the students will be able to.

- **CO-1:** Analyze healthcare data modalities and formulate clinically relevant AI problems involving clinical, imaging, physiological, and multimodal data.
- **CO-2:** Apply modern AI paradigms in healthcare applications.
- **CO-3:** Design and implement a real-world healthcare AI problem using appropriate datasets, methodologies, and evaluation protocols.
- **CO-4:** Assess healthcare AI systems with respect to interpretability, robustness, fairness, generalization, and deployment considerations.
- **CO-5:** Critically evaluate methodologies in healthcare AI applications.

Course Topics :

(Please list the order in which they will be covered and preferably arrange these as five to six modules.)

Module 1: Healthcare AI Ecosystem

- Evolution of healthcare and digital health ecosystems
- Healthcare data modalities

- Challenges in real-world healthcare AI
- Problem formulation in clinical AI
- Evolution of AI paradigms in healthcare: from supervised to foundation models and multimodal learning.

Module 2: Clinical AI

- Structured and longitudinal healthcare data, Electronic Health Records (EHR), Laboratory and clinical measurements, Time-series bedside monitoring data
- Risk prediction and prognosis modeling
- Patient trajectory and temporal modelling

Module 3: Biomedical Imaging AI

- AI applications in radiology and computational pathology
- Image classification, detection and segmentation
- Representation learning for biomedical imaging
- Advanced learning paradigms for biomedical imaging
- Foundation models and generative AI for medical imaging
- Radiomics and pathomics
- Interpretation methods for biomedical imaging models

Module 4: Physiological Signals AI

- Physiological signal modalities: ECG, EEG, EMG, PPG, Wearable and remote monitoring data, ICU monitoring signals
- Temporal and sequential modeling of physiological signals
- Applications including arrhythmia detection, sleep stage classification, seizure prediction, vital sign forecasting, and patient monitoring.

Module 5: Multimodal AI for healthcare

- Multimodal learning in healthcare
- Clinical, imaging, molecular, and physiological data fusion strategies
- Multimodal foundation models for healthcare
- Cross-modal representation learning, embeddings and fusion architectures
- Challenges in multimodal healthcare AI

Module 6: Trustworthy AI in Medicine

- Interpretability & Explainability
- Fairness & Bias
- Distribution Shift
- Robustness
- Privacy-preserving AI
- Federated Learning
- Ethical & Legal Considerations
- Regulatory pathways

Module 7: Implementation and Deployment of Healthcare AI

- Validation and evaluation of healthcare AI models
- Deployment in clinical and hospital settings
- Human-in-the-loop systems
- Monitoring and maintenance after deployment
- AI governance and regulatory considerations

Preferred Textbooks :

(The library will maintain multiple copies for borrowing or arrange for e-books)

As this is a new and developing area, there are no standard textbooks or reference books. Following is a sample list of survey / reference papers/sources, more such articles on healthcare AI from leading journals and conferences including Nature Medicine, N Engl J Med, The Lancet, IEEE Transactions on Medical Imaging (TMI), and Medical Image Analysis, MICCAI etc. will be identified.

1. Rajpurkar, P., Chen, E., Banerjee, O. et al. AI in health and medicine. *Nat Med* **28**, 31–38 (2022).
2. Acosta, J.N., Falcone, G.J., Rajpurkar, P. et al. Multimodal biomedical AI. *Nat Med* **28**, 1773–1784 (2022).
3. Moor, M., Banerjee, O., Abad, Z.S.H. et al. Foundation models for generalist medical artificial intelligence. *Nature* **616**, 259–265 (2023).
4. Azad R, Kazerouni A, Heidari M. et al. Advances in medical image analysis with vision Transformers: A comprehensive review, *Medical Image Analysis*, Volume **91**, 103000 (2024).
5. Lekadir, K., et al., FUTURE-AI: International Consensus Guideline for Trustworthy and Deployable Artificial Intelligence in Healthcare, *BMJ*, 2025.

(The library will maintain one or two copies in the reference section)

1. Topol, E. (2019). *Deep medicine: How artificial intelligence can make healthcare human again*. Basic Books.

E-book Links :

(for Core courses provide e-book title)

Grading Plan : (The table is only indicative)

Type of Evaluation	Weightage (in %)
Quiz-1	10%
Mid Sem Exam	--
Quiz-2	10%
End Sem Exam	--
Assignments	20%
Project	50%
Term Paper	--
Other Evaluation: Paper Presentation + Class Participation	10%

Mapping of Course Outcomes to Program Objectives: (1 – Lowest, 2—Medium, 3 – Highest, or a ‘-’ dash mark if not at all relevant). Program outcomes are posted at

	P O 1	P O 2	P O 3	P O 4	P O 5	P O 6	P O 7	P O 8	P O 9	P O 10	P O 11	P O 12	PS O 1	PS O 2	PS O 3	PS O 4
C O 1	2	1	2	2	3	3	3	3	2	2	3	2	1	2	2	2
C O 2	2	2	2	2	1	3	3	3	2	2	3	2	1	2	2	2
C O 3	1	1	1	1	1	2	3	3	1	1	3	2	1	1	1	1
C O 4	3	2	3	1	1	2	3	3	2	2	3	2	1	3	2	1
C O 5	1	3	3	2	3	3	3	3	2	2	3	3	1	3	2	2

Teaching-Learning Strategies in brief (4-5 sentences):

This course will follow a combination of lecture-based teaching and project-driven learning. The lectures will involve faculty from both healthcare and technology domains, including clinicians from different specialties, to provide interdisciplinary perspectives on real-world healthcare problems and AI solutions. The course will emphasize critical analysis of healthcare AI systems, research literature, and clinical use cases. Students will gain hands-on experience through projects involving healthcare datasets, model development, evaluation, and presentation of clinically relevant AI solutions. Selected projects will be curated in collaboration with clinical partners and hospitals to expose students to real-world healthcare challenges. Paper discussions, case studies, quizzes, and project reviews will further encourage active learning and research-oriented thinking.

Title of the Course : Topics in Nanosciences

Name of the Faculty : Tapan K. Sau

Course Code : SC2.401

L-T-P : (4-0-0)

Credits : 4

Name of the Academic Program: CND

1.Prerequisite Course / Knowledge:

2.Course Outcomes (COs) :

After completion of this course successfully, the students will be able to..

CO-1. Define terminology used in the fields of nanoscience and nanotechnology.

CO-2. Explain the nanoscale confinement effects on various material properties.

CO-3. Discuss various methods of synthesis of nanoparticles.

CO-4. Identify the factors that need control for the preparation of stable and controlled sized and shaped nanoparticles.

CO-5. Explain the determination of the particle size and shape.

CO-6. Identify and formulate appropriate methods and experimental techniques that can be used to study various nanoscale materials and phenomena. **CO-7.** Analyze the size- and shape-dependent physical/chemical properties of nanoparticles.

CO-8. Identify various applications of nanoparticles and their future potential.

CO-9. Describe the advantages and limitations of nanostructured materials.

3.Mapping of Course Outcomes (COs) with Program Outcomes (POs) and Program Specific Outcomes (PSOs) – Course Articulation Matrix

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3	PSO4
CO1	3															2
CO2	3															2
CO3	3															2
CO4		2														2
CO5	3															3
CO6			3													3
CO7	3															3
CO8				2												2
CO9	3															3

Note: '3' in the box for 'High-level' mapping, 2 for 'Medium-level' mapping, 1 for 'Low'-level' mapping

4.Detailed Syllabus:

Unit 1. Introduction to Nanoscience. (3L)

Nanomaterials: Definitions and Scopes.

Length Scales: Size Scales, Surface and Interface, Surface Energy, Coordination Numbers.

Classification of nanomaterials: Clusters and Magic Numbers, Nanoparticles, and Colloids. Metal, Semiconductor, and Bio Nanomaterials.

Unit 2. Properties of Nanomaterials. (9L)

Electrons in Nanostructures. Discrete states vs. band structure: Effects of dimensionality and symmetry in nanostructures. Metal-to-nonmetal transition.

Thermodynamics and Kinetics of Small-Sized Systems. Capillarity, Liquid droplets, Lotus effect. Self-assembly principles. Adsorption, Electrostatic and Steric Stabilization. Classical nucleation theory. Size and shape control in nanoparticle formation.

Magnetic (Super paramagnetism), *Electrical* (quantized conduction and Coulomb Staircase), *Optical* (size and shape effects), *Thermal* (melting and conduction), *Mechanical and Catalytic properties.*

Unit 3. Making Nanostructures. (3L)

Top-down and bottom-up methods.

Unit 4. Tools for Nano systems. Microstructure/Chemistry/Defects and Structure. (5L)

AFM, SEM, TEM, XRD, SAXS, Nanoindentation.

Unit 5. Applications of Nanomaterials. (4L)

Catalysis, Band Gap Engineered Quantum Devices, Sensors, Field Effect Transistor (FET), Photoelectrochemical Cells, Photonic Crystals and Waveguides, Theragnostics (Magnetothermal Therapy), food and agriculture industries, automobile, textile, water treatment and civil applications, use in energy, space, and defense.

Unit 6. Concerns and Challenges of Nanotechnology. (2L)

Environmental, ecological and health hazards of nanoparticles. Nanotoxicology and its effect.

Reference Books:

1. S. M. Lindsay (2010), *Introduction to Nanoscience*, 1st Edition, Oxford University Press, New York.
2. E. Roduner (2006), *Nanoscopic Materials: Size-dependent Phenomena*, 1st Edition, RSC Publishing, Cambridge.
3. B. S. Murty, P. Shankar, B. Raj, B. B. Rath and J. Murday (2013), *Textbook of Nanoscience Nanotechnology*, 1st Edition, Universities Press (India) Private Limited, Bangalore.
4. G.A. Ozin, A. C. Arsenault, and L. Cademartiri (2009), *Nanochemistry*, 2nd Edition, RSC Publishing, Cambridge.
5. M. Kohler and W. Fritzsche (2004), *Nanotechnology*, 1st Edition, Wiley-VCH, Weinheim.

5. Teaching-Learning Strategies in brief (4 to 5 sentences):

The course involves lectures, quizzes, laboratory demonstrations, assignments, and finding and reading relevant scientific literature.

6. Assessment methods and weightages in brief (4 to 5 sentences):

The student assessment in the course involves written tests/quizzes/assignments to determine their learning proficiency in the course and their grades. Grading is done as follows:

- | | |
|---------------------------------------|-----|
| 1. Assignments | 20% |
| 2. Quizzes (2*10) | 20% |
| 3. Mid-Term Exam | 20% |
| 4. End-Semester Exam (whole syllabus) | 40% |

Title of the Course : **Topics in Software Engineering**
Name of the Faculty : Raghu Reddy Y
Course Code : CS6.501
L-T-P : 3-0-1
Credits : 4
Name of the Academic Program: MS/PhD in Computer Science and Engineering

1. Prerequisite Course/Knowledge:

Students must have taken an Software Engineering or equivalent course at the undergraduate level.

2. Course Outcomes (Cos)

After completion of this course successfully, students will be able to...

CO-1: Understand state-of-research in advanced areas of software engineering such as Artificial Intelligence for software engineering, self-adaptive systems, software reengineering, etc.

CO-2: Apply standard principles of software architecture to construct software systems

CO-3: Apply specialized knowledge to identify open research problems and disseminate state-of-the-art software engineering research.

CO-4: Critique the quality of published research papers with well-known assessment criteria

CO-5: Synthesize novel solutions to open research problems in software engineering by following well-established principles of software engineering research.

3. Mapping of Course Outcomes (COs) with Program Outcomes (POs) and Program Specific Outcomes (PSOs) – Course Articulation Matrix

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	P10	P11	P12	PSO1	PSO2	PSO3	PSO4
CO1	2	2	2	1	1	2	2	1	2	2	1	2	3	2	2	3
CO2	3	3	3	1	3	2	3	2	3	3	2	2	2	3	2	2
CO3	1	3	3	3	1	2	1	2	3	3	1	3	3	2	2	3
CO4	2	2	2	2	2	1	1	2	3	3	1	3	2	3	2	3
CO5	2	3	3	3	2	2	2	2	3	2	2	3	3	2	1	3

Note: Each Course Outcome (CO) may be mapped with one or more Program Outcomes (POs) and PSOs. Write '3' in the box for 'High-level' mapping, 2 for 'Medium-level' mapping, 1 for 'Low'-level' mapping

4. Detailed Syllabus:

The course will cover the following topics:

- Art of writing research papers,
- Software Aging
- Introduction to software architecture
- Software architectural framework and patterns
- Applying AI for software engineering practices,
- Introduction to self-adaptive systems
- Applying ML for self-adaptation, microservices and serverless architectures,
- Software product lines

- Software reengineering
- Software evolution

Reference Books:

1. Software Architecture in Practice. Fourth Edition, Len Bass et al.
2. Pattern-Oriented Software Architecture. Buschmann et al.
3. Design it! From Programmer to Architect by Michael Keeling
4. The Software Architect Elevator: Redefining the Architect's Role in the Digital Enterprise by Gregor Hohpe, Orielly Publications
5. Designing data-intensive applications by Martin Klepman, O'Reilly
6. Building Microservices by Sam Newman, O'Reilly
7. Modernizing Legacy Systems: Software Technologies, Engineering Processes, and Business Practices, Seacord, Plakosh, Lewis, Addison-Wesley, 2003.
8. Refactoring: Improving the Design from Existing Code, Addison-Wesley, 2000
9. Successful Evolution of Software Systems, Yang & Ward, Artech House Publishers, ISBN 1580533493, 2003.
10. Software Evolution and Feedback : Theory and Practice, Nazim H. Madhavji, M.M. Lehmann, J. Ramil, Dewayne Perry, John Wiley & Sons, 2005.

5. Teaching-Learning Strategies in Brief (4 to 5 sentences):

The course is delivered using a combination of project based and research-based learning methodology. Topics like software architecture, software architectural framework and patterns, self-adaptive systems are taught and reinforced via mini projects. The lectures emphasize on cutting edge research in the advanced areas of software engineering. The focus is on imparting knowledge of software engineering research methodologies to students through paper presentations, providing state-of-the art research papers as review assignments to students and supporting students to develop novel research proposals. Entire class is run in a working research session mode to foster advanced discussion among students as well as between students and instructors.

6. Assessment Methods and Weightage in brief (4 to 5 sentences)

Final Project	40%
3 Mini Projects (3*10)	30%
Research paper reviews (4*2.5)	10%
Research paper Presentations (2*5)	10%
Other in-class activities	10%

Title of the Course : Understanding Public and Digital History

Name of the Faculty : Isha Dubey

Name of the Program : B. Tech in Computer Science and Engineering

Course Code : HS3.402

Credits : 4

L - T - P : 3-0-0

(L - Lecture hours, T-Tutorial hours, P - Practical hours)

Semester, Year : Monsoon 2026

Pre-Requisites : Intro to Human Sciences, CHD 3rd and 4th year students

Course Outcomes

After the completion of the course students should be able to :

- CO1:** Explain the intellectual foundations of the vast and diverse field of public history and the various theoretical approaches that inform it.
- CO2:** Understand the ways in which public history is both connected to and differs from academic history
- CO3:** Critically reflect on the role played by history and historians in shaping the contours of some of the most important public debates in the world.
- CO4:** Create and communicate, engaging, academically rigorous content using appropriate tools and mediums, to promote public engagement in history through storytelling.
- CO5:** Assess the impact and implications of the 'digital turn' on the ways in which we study and disseminate knowledge about the past
- CO6:** Develop a critical understanding of the diversity of methods and sources at the disposal of public historians enabled by technology.

Course Topics :

Module 1: What is public history who are its 'publics'

This module will introduce the students to the vast field of public history and its relationship with academic history. It shall initiate discussions on the definition(s) of public history, its origins and trajectory into the present and the ways in which it has been approached by historians in theory and praxis. The module also aims to facilitate reflections on the purpose of public history by focusing the question of the 'audience'. Who is it meant for? What role does/can the civil society play in its creation and curation? Who is the 'owner' of public history? And finally what purpose should it serve?

Module 2: Public history and collective memory

This module will focus on the question of collective memory and how public historians approach the question of remembrance. It shall do this by dwelling at length on the themes of as memorialization, commemoration of collective pasts for the wider public and the role they play in constructing and institutionalizing historical narratives. In doing so, the module aims to introduce students to the foundations of interdisciplinary field of memory studies and how it engages with the discipline of history by focusing on questions such as: What role does memory play in the writing of history? Do collectivities such as communities, ethnicities, nations and so on have memories of their own? What forms can collective memory assume? How is it transmitted across generations?

Module 3: Power and politics in/of public history

This module shall be structured around questions that foreground the politics inherent in the forging and dissemination of public history. The discipline of history has always been a deeply political one. However, when specific versions of the past are constructed and curated for a much wider public consumption beyond the academy, this politics assumes added layers and significance. Using examples and case studies from across the world the module aims to equip students with the ability to read and identify the politics of public history by asking and seeking answers to questions such as: Whose history is easy to tell? Who are the people absent from history? Can public history be used as a tool of perpetuating oppression? Can it be deployed as a form of resistance?

Module 4: Practicing public history

This module shall introduce the class to the praxis of public history. Public history is more than just an academic field of study. It is about applying expert knowledge about the past into practice to help people write, create, and understand their own history. Still others believe public history should influence the formulation of public policy. So, what is it that public historians do and how do they go about doing what they do? Following from these questions, the module shall familiarize students with the workings of public history, the forms it can take and its potential for shaping public policy beyond education and academia.

Module 5: History in and of the digital age

This module shall be structured around two related questions: 1) How is the discipline of history as well as its public application impacted by the larger 'digital turn' in the humanities and social sciences? 2) How will the histories of an age whose archives shall eventually entirely be 'born digital' be documented, written and disseminated? Following from this overarching framework the lectures in this module shall focus on (but not limited to) themes such as archives and digitality, history and the question of technology, the methods of digital history, new medias and the problem to 'doing' history.

Preferred Textbooks

:

Introduction to Public History: Interpreting the Past, Engaging the Audiences by Cherestine M. Lyon, Elizabeth M. Nix, Rebecca K. Shrum (2017)

Public History: Essays from the Field edited by James B. Gardner and Peter S. LaPaglia (2004)

The Oxford Handbook of Public History edited by Paula Hamilton and James B. Gardner (2017)

Silencing the Past: Power and the Production of History by Michel-Rolph Trouillot (1995)

The Public Life of history, special issue of *Public Culture* edited by Claudion Lomnitz, Dipesh Chakrabarty and Bain Attwood

History, Memory and Public Life: The Past in the Present edited by Anna Maerker, Simon Sleight and Adam Sutcliffe

Technology of the Historian: Transformations in the Digital Age by Adam Crymble (2021)

Reference Books (and journal articles) :

"Public History and the Study of Memory" in *The Public Historian*, vol 18, no. 2 (1996):7-23 by David Glassberg

The New Berlin: Memory, Politics, Place by Karen E. Till (2005)

A Misplaced Massacre: Struggling Over the Memory of Sand Creek by Ari Kelman (2013)

History in the Digital Age by Toni Weller (2012)

Chirs Campbell, "The Historian in the Age of AI" in *Transactions of the Royal Historical Society*, (2025):1-11.

Shalini Sharma, "Conservative Public History in India." In *International Public History*, volume 8, no. 2 (2025): 89-93.

Nora, Pierre. "Between Memory and History: Les Lieux de Mémoire." *Representations*, no. 26 (1989): 7-24.

Difficult Heritage: Negotiating the Nazi Past in Nuremberg and Beyond by Sharon Macdonald (2009)

The Generation of Postmemory: Writing and Visual Culture after the Holocaust by Marianne Hirsch (2012)

Museums and Communities: The Politics of Public Culture edited by Ivan Kirp, Christine Mullen Kreamer and Steven Levine (1992)

Alison Twells et al., 'Undisciplined History: Creative Methods and Academic Practice', *History Workshop Journal*, 96 (2023), 153-175.

Computation and the Humanities: Towards an Oral History of Digital Humanities by Julianne Nyhan and Andrew Finn (2016)

Decolonizing Museums: Representing Native America in National and Tribal Museums by Amy Lonetree (2012)

Jenniger R O'Neal, "The Right to Know": Decolonizing Native American Archives," *Journal of Western Archives*: Volume 6, no. 1 (2015)

James E, Young. *The Stages of Memory: Reflections on Memorial Art, Loss, and the Spaces Between*. University of Massachusetts Press, (2016).

Online resources

The National Council on Public History Blog https://ncph.org/history-at-work/project_category/blog/

Programming Historian <https://programminghistorian.org>

Doing History in Public <https://doinghistoryinpublic.org/about-us/>

History@work <https://ncph.org/history-at-work/>

Beyond Academe <http://www.beyondacademe.com>

Indira Gandhi Centre for the Arts <https://ignca.gov.in>

Making Digital History <https://makingdigitalhistory.co.uk>

Grading Plan :

(The table is only indicative)

Type of Evaluation	Weightage (in %)
Quiz 1	10
Mid Sem Exam	20
Quiz-2	10
End Sem Exam	30
Project	30
Total	100

Mapping of Course Outcomes to Program Objectives: (1 – Lowest, 2—Medium, 3 – Highest, or a ‘-’ dash mark if not at all relevant).

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3	PSO4
CO1	1	-	3	3	2	3	-	2	2	3	-	3	1	3	3	2
CO2	1	-	3	3	2	3	-	2	2	3	-	3	1	3	3	2
CO3	1	-	3	3	2	3	-	3	2	3	-	3	1	3	3	1
CO4	1	-	3	3	2	3	-	3	3	3	-	3	3	3	3	3
CO5	2	-	3	3	2	3	-	3	3	3	-	3	2	3	3	3
CO6	2	-	3	3	2	3	-	3	3	3	-	3	3	3	3	3

Teaching-Learning Strategies in brief (4-5 sentences):

The course shall be a lecture-based one while integrating active learning strategies into its structure through encouraging engaged classroom-based discussions and group work. Further, the nature of the course is such that it needs to engage students in ways that go beyond the traditional template of purely classroom-based learning. The students shall hence be required to apply the knowledge and insights gained in the class into designing a semester-long digital public history project as a vital component of the evaluation of their performance in the course. Finally, public history seeks to inculcate an interest and investment in the preservation and dissemination of the past beyond academia. Following from this, a course on public history, for students with very limited exposure to the discipline beyond schoolbook history and the various narratives about the past circulating online, aims to provide them with a sense of the rigor that goes into any earnest and conscientious effort for the preservation, and dissemination of the past. This shall be done through planned through invited lectures and talks by practicing public historians, civil society organisations engaged in public history projects, and potentially a field visit to a local museum or archive to provide students with a more meaningful insight into the praxis or public history.

Title of the Course : VALUE EDUCATION – I

Name of the Faculty(s) : Indranil Chakraborty + Saurabh Todaria (Coordinators)

Course Code : OC3.101

L-T-P : 0-2-0-2 (Total number of hours)

Credits : 2

(L= Lecture hours, T=Tutorial hours, P=Practical hours)

Name of the Academic Program: B. Tech. in ECE, BTech in CSE

1.Prerequisite Course / Knowledge: -NIL-

2.Course Outcomes (COs) :

After completion of this course successfully, the students will be able to:

CO-1: Apply the basic framework of universal human values to the self.

CO-2: Look at larger issues that (for many reasons) most are not exposed to: social, political, community, family, individual, etc. in a sensitized way.

CO-3: Understand themselves and their own roles within the bigger context. What are really, truly important to them? What are made important by others?

CO-4: Engage and connect with others and nurture the relationships.

CO-5: Think to shape and change the world, and not be mere technologists or scientists.

3.Mapping of Course Outcomes (COs) with Program Outcomes (POs) and Program Specific Outcomes (PSOs) – Course Articulation Matrix

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3	PSO4
CO1	-	-	-	-	-	3	2	3	2	-	-	-	-	-	-	-
CO2	-	-	-	-	-	3	3	3	3	-	-	-	-	-	-	-
CO3	-	-	-	-	-	3	3	3	2	-	-	-	-	-	-	-
CO4	-	-	-	-	-	2	3	3	3	-	-	-	-	-	-	-
CO5	-	-	-	-	-	3	3	3	2	-	-	-	-	-	-	-

Note: '3' in the box for 'High-level' mapping, 2 for 'Medium-level' mapping, 1 for 'Low'-level' mapping

4.Detailed Syllabus:

Unit 1: Goal in life - short term and long-term goals; Basic aspirations - Happiness and Prosperity; Role of education and human conduct; Self-exploration; Developing a holistic view
Unit 2: Gratitude and the need to acknowledge one's gratefulness; Understanding Self and Other.
Unit 3: Living in harmony at 4 levels: self-self, self-family, self-society, self-nature
Unit 4: Understanding needs of body and self; Right understanding of physical facilities and relationships; Understanding human relationships; Trust and Respect - the foundational values in relationships.
Unit5: Harmony in Society; The sense of safety, justice and peace in society; Nature and Sustainability; Self-reliance and Gandhian thought

Reference Books:

1. R.R. Gaur, R. Sangal, G. P. Bagaria. 2009. A Foundation course in Human Values and Professional Ethics. Excel books, New Delhi.
2. Randy Pausch. 2008. The Last Lecture. Hachette Books.
3. E. F. Schumacher. 1973. Small is beautiful: a study of economics as if people mattered. Blond & Briggs, Britain.
4. P. L. Dhar, R. R. Gaur. 1990. Science and Humanism. Commonwealth Publishers.

5.Teaching-Learning Strategies in brief (4 to 5 sentences):

This is a discussed based course. The instructor shares information on a topic and guides the discussion in the class by asking the right questions. By keeping the objectives in mind, the instructor adopts different techniques including smaller group discussions, role-play/skit, use of video clips or images to analyze and some activities to keep the students engaged in class throughout. Talks by experts who made a difference are also organized for the batch. Field trips to farms, orphanages, old-age homes, villages and jails are arranged as part of the induction programme, in parallel to the classes in VE for the first year UG batch.

6.Assessment methods and weightages in brief (4 to 5 sentences):

This is a Pass/Fail course. The assessment methods include submissions of assignments and term papers. Critical thinking is expected from watching relevant short films or by reading assigned

books. The classroom participation is also taken into consideration for evaluation. There are a few community-based activities and projects also. Participation in them is also important. (weightage for each kind of assessment may be given.)

Title of the Course : **VLSI Design**
 Name of the Faculty : Abhishek Srivastava
 Course Code : EC2.201
 L-T-P : 3-1-0.
 Credits : 4
 (L= Lecture hours, T=Tutorial hours, P=Practical hours)
 Academic Program : B.Tech. in Electronics and Communication Engineering

1. Prerequisite Course / Knowledge:

Digital electronics, Network theory.

2. Course Outcomes (COs)

After completion of this course successfully, the students will be able to:

CO-1: Analyze delay and noise performances of CMOS inverter

CO-2: Apply the knowledge of delay and noise analysis of CMOS inverter for other logic styles

CO-3: Apply the knowledge of different logic styles for developing digital building blocks such as gates, multiplexors, latches and flip-flops

CO-4: Design delay optimized multistage logic circuits by using method of logical effort

CO-5: Design combinational circuits using CMOS and pass transistor logic for minimum delay and maximum noise margin performances

CO-6: Design a delay optimized sequential CMOS circuit such as 8-bit multiplier for the given load and speed requirements, while ensuring no setup time or hold time violations and verify its post layout performance using SPICE tools

3. Mapping of Course Outcomes (COs) with Program Outcomes (POs) and Program Specific Outcomes (PSOs) – Course Articulation Matrix

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3	PSO4
CO1	3	3	2	2	2	-	-	1	3	1	-	2	3	-	-	-
CO2	3	3	3	2	2	-	-	1	3	3	1	3	3	-	-	-
CO3	3	3	3	2	2	-	-	1	3	3	1	3	3	-	-	-
CO4	3	3	3	2	2	-	-	1	3	3	1	3	3	-	-	-
CO5	3	3	3	2	2	-	-	1	3	3	1	3	3	-	-	-
CO6	3	3	3	2	2	-	-	1	3	3	1	3	3	-	-	-

‘3’ in the box denotes ‘High-level’ mapping, 2 for ‘Medium-level’ mapping, 1 for ‘Low’-level’ mapping.

1. Detailed Syllabus:

Unit 1 (Introduction to VLSI design):1) Introduction to VLSI design (top-bottom approach) - flow, applications, technologies, 2) MOSFET, FinFET transistors – Geometry and model, 3) Introduction

to basic building blocks - SPICE, HDL, layout, 4) Moore's law, technology scaling, current trends(5-lectures/7.5-hours)

Unit 2 (CMOS Inverter): 1) Static characteristics- VTC, switching threshold, Noise margin, 2) Dynamic characteristics – rise time, fall time, delay, power, 3) Why CMOS Inverter, 4) CMOS inverter design flow- problem of achieving higher speeds (solution/technique discussed in the following unit), 5) From inverters to other logic - pull-up, pull-down networks, tristate inverter, Gates, Mux, Latches, Flip-flops, set-up hold time, clocked CMOS and true single phase clocked (TSPC) latches (7-lectures/10.5-hours)

Unit 3 (Multistage Logic Design and Optimization: 1) Parasitics in layout causing performance degradation – field transistor, active MOS, gate-drain overlap, latch-up, 2) Method of logical effort-fan-out, Stage effort, electrical effort, device sizing, design examples. (5-lectures/7.5-hours)

Unit 4 (Other Logic Styles): Pseudo nMOS, pass transistor logic, Cascode Voltage Switch Logic (CVSL), Dynamic logic. (3-lectures/4.5-hours)

Unit 5 (Other topics Introduction to System Design using HDL): Finite state machines – Mealy, Moore, Intro to RTL, Data path, Control unit, combinational and sequential circuit design examples (6-lectures/9-hours)

REFERENCES:

1. Neil H. E. Weste, K. Eshraghian, “Principles of CMOS VLSI Design- A Systems Perspective”, 2nd Edition, Pearson Education Pvt. Ltd.
2. J. M. Rabaey, A. Chandrakasan, B. Nikolic, “Digital Integrated Circuits - A Design Perspective”, 2nd Edition, Prentice Hall of India.
3. Stephen Brown and Z. Vranesic, “Fundamentals of Digital Logic with Verilog Design”, Tata McGraw-Hill Edition 2002.
4. Samir Palnitkar, “Verilog HDL – A Guide to Digital Design and Synthesis”, Second edition, Pearson, 2003.
5. J. Bhaskar, “Verilog HDL Synthesis- A Practical Primer”, Star Galaxy Pub; 1st edition, 2001

5.Teaching-Learning Strategies in brief:

Fundamentals of VLSI design will be discussed in the course with examples. SPICE tools will be introduced, and regular assignments will be given based on topics covered in lectures. Weekly tutorials will be conducted for problem solving and further discussions on any questions related to topics covered in lectures. A course project will be given that will involve analysis, design, layout and simulations (schematic and post-layout level) of an analog circuit for given specifications.

6. Assessment methods and weightages in brief:

Type of Evaluation	Weightage (in %)
HomeWorks	20%
Course project	20%
Mid Semester exam-1	15%
Mid Semester Exam-2	15%
End semester exam	30%

Title of the Course: **Wireless Communications**
Name of the Faculty: Rajeev Gangula
Course Code: EC5.407
L-T-P: 3-1-0.
Credits: 4
 (L= Lecture hours, T=Tutorial hours, P=Practical hours)

1. Prerequisite Course / Knowledge:

Communication Theory, Probability and Random Process

2. Course Outcomes (COs)

After completion of this course successfully, the students will be able to:

CO-1: Identify and explain the fundamental operational and design problems of wireless communication systems.

CO-2: Demonstrate understanding of evolution of different wireless communication systems and standards.

CO-3: Determine the type and appropriate model of wireless fading channels based on the system parameters and the properties of the wireless medium.

CO-4: Design appropriate receiver and transmitter diversity techniques and analyze their performance theoretically and via simulations.

CO-5: Design appropriate multiple-antenna transceivers and evaluate rate and error performance.

CO-6: Demonstrate understanding of OFDM and massive MIMO techniques and application in existing and upcoming wireless systems.

Mapping of Course Outcomes (COs) with Program Outcomes (POs) and Program Specific Outcomes (PSOs)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3	PSO4
CO1	3	2	1	1	1					1		2		3	1	
CO2	3	2	1	1						1		2		3	1	
CO3	3	3	1	2	2					1		2		3	1	1
CO4	3	3	3	2	3					1		2	1	3	1	1
CO5	3	3	3	1	3					1		2	1	3	1	1
CO6	3	3	2	1	2					1		2	1	3	1	

‘3’ in the box denotes ‘High-level’ mapping, 2 for ‘Medium-level’ mapping, 1 for ‘Low’-level’ mapping

2. Detailed Syllabus:

Unit 1: Review of digital communication, optimal detection, overview of wireless communication generations and standards

Unit 2: Channel modeling; Multipath propagation; pathloss and fading; types of fading; frequency and time selectivity

Unit 3: Diversity techniques; spatial, time and frequency diversity; performance analysis of various diversity techniques

Unit 4: MIMO communication systems; capacity analysis; MIMO receivers

Unit 5: OFDM, massive MIMO, multiuser communication

References:

Tse and Viswanath, “Fundamentals of Wireless Communications”, Cambridge University Press, 2005.

Andrea Goldsmith, “Wireless Communications”, Cambridge University Press, 2005.

5. Teaching-Learning Strategies in brief:

Lectures cover the topics in the syllabus and tutorials cover how to solve some design and analysis problems related to topics covered in the lectures. Lectures and tutorials emphasize active learning by students. Assignments will provide the students experience in software-based implementation and performance analysis of various wireless communication techniques. There is a long project which is either based on an idea the student wants to explore from the course topics or based on an existing research paper. Project evaluation involves multiple assessments, submission of project report, and a final presentation and viva.

6. Assessment methods and weightages in brief:

Quiz: 10%, Assignments: 20%; Project: 40%, End-Sem exam: 30%.
